

SingleRAN

LTE FDD and NR Dynamic Spectrum Sharing Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	Base Station Model
Modified the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing for NR SDL cells. For details, see 4.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Removed the mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with uplink-interference-based inter-frequency handover from 4.3.2.2 Mutually Exclusive Functions .	None	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Removed the mutually exclusive relationships of PSD adaptation with network slicing enhancement and percentage-based dynamic RB sharing among operators from 4.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added impact relationships of PSD adaptation with "NR DU cell resource management between network slices" and "percentage-based dynamic RB sharing among operators". For details, see 4.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
 - Added descriptions of the cell requirements for configuring the **NRDUCellGigaBand.LnrUIRbEstCorrectCoeff** parameter. For details, see [8.3.2.2 Mutually Exclusive Functions](#) and [8.3.3 Hardware](#).
 - Added descriptions of configuration requirements, network impacts, and compatible RF modules for the "LTE and NR 3-ms frame offset difference" function when it is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing. For details, see [4.1.1.3.3 Common Functions Supported by Both LTE and NR](#), [4.2.2 Impacts](#), and [4.3.3 Hardware](#).
 - Revised descriptions in this document.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 08 (2025-12-14).

- Technical changes

Change Description	Parameter Change	Base Station Model
Enabled the UMPThb to support LTE FDD and NR Flash Dynamic Spectrum Sharing. For details, see 4.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Enabled 16T16R RF modules to support LTE FDD and NR Flash Dynamic Spectrum Sharing. For details, see 4.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added incompatibility of BTS5929R with LTE FDD and NR Flash Dynamic Spectrum Sharing. For details, see 4.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added "LNR independent configuration of uplink and downlink resource percentages" to the DSS/Hybrid DSS Performance Boost feature. For details, see 6.1 Principles .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added support for simultaneous use of LTE FDD and NR Flash Dynamic Spectrum Sharing and the LTE and NR flexible frame offset function. For details, see: • 4.1.1.3.3 Common Functions Supported by Both LTE and NR • 4.1.2.3 Simultaneous Use with Other Functions • 4.1.3.3.3 Common Functions Supported by Both LTE and NR • 4.3.1 Licenses • 4.3.2.1 Prerequisite Functions • 4.3.2.2 Mutually Exclusive Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter on the NR side: Added the LNR_3MS_FRAME_OFFSET_DIFF_SW option to the NRDUCellSpctShare.LnrSpectrumShare.mShareSwitch parameter. Modified parameter on the LTE side: Added the LNR_3MS_FRAME_OFFSET_DIFF_SW option to the SpectrumCloud.SpectrumCloudExtSwitch parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support for LTE and NR Uplink Spectrum Pooling. For details, see: • 2.3 Features in This Document • 2.4 Differences • 3 Overview • 8 LTE and NR Uplink Spectrum Pooling	Added parameters on the NR side: • NRDUCellGigaBand.GigaBandCellAlgoSw • NRDUCellGigaBand.LnrPucchRbAdjProtDuration • NRDUCellGigaBand.LnrULRBEstCorrectCoeff Added parameters on the LTE side: • GigaBandCell.GigaBandCellAlgoSw • GigaBandCell.ServiceBasedOptMaxEffTime • PUCCHCfg.NrPucchExtendedRbNum Modified parameter on the LTE side: Added the GRP_NO_PREALLOCATE_TRIG_OPT_SW option to the AsParaGroup.AccelerationServOptExtSw parameter.	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Modified the impact relationship of DSS and NR Flexible Refarming with CMAS. For details, see 7.3.2.3 Function Impacts.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Revised the constraints affecting cell activation due to license insufficiency. For details, see 4.3.1 Licenses .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added compatibility of 16T16R with LTE FDD and NR Flash Dynamic Spectrum Sharing. For details, see: • 4.1.1.3.3 Common Functions Supported by Both LTE and NR • 4.1.3 LTE FDD and NR FDD Dynamic Spectrum Sharing (Multiple LTE Cells) • 4.2.1 Benefits • 4.3.2.2 Mutually Exclusive Functions • 4.3.2.3 Function Impacts • 4.3.3 Hardware • 4.3.4 Networking • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands • 6.3.3 Hardware • 7.3.2.2 Mutually Exclusive Functions • 7.3.3 Hardware	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added formal parameters to replace the reserved parameters related to the simultaneous use of LTE FDD and NR Flash Dynamic Spectrum Sharing with DRX. Specifically: - Added the BIT22 option of the eNBCellRsvdPara.RsvdSwPara11 parameter to the disuse list, and replaced it with the DSS_GAP_POS_OPT_SW option of the CellDrxPara.DrxAlgoExtSwitch parameter. - Added the BIT23 option of the eNBCellRsvdPara.RsvdSwPara11 parameter to the disuse list, and replaced it with the DSS_VOICE_NSA_SA_OD_POS_OPT_SW option of the CellDrxPara.DrxAlgoExtSwitch parameter. - Added the eNBCellRsvdPara.RsvdU16Para29 parameter to the disuse list, and replaced it with the CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod parameter. For details, see: 4.1.1.3.1 LTE Functions 4.1.3.3.1 LTE Functions 4.2.2 Impacts 4.4.1.1 Data Preparation 4.4.1.2 Using MML Commands	Modified parameter on the LTE side: Added the DSS_GAP_POS_OPT_SW and DSS_VOICE_NSA_SA_OD_POS_OPT_SW options to the CellDrxPara.DrxAlgoExtSwitch parameter. Added parameter on the LTE side: Added the CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added support for "LTE and NR PUCCH resource optimization" in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see: 4.1.1.3.2 NR Functions 4.1.3.3.2 NR Functions 4.3.2.3 Function Impacts	Modified parameter on the NR side: Added the LNR_PUCCH_RE_S_OPT_SW option to the NRDUCellPucch.PucchAlgoExtSwitch parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added an impact relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with LNR CRS muting. For details, see 4.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added configuration requirements for adjusting the compression degree of common PUCCH RBs in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see 4.3.5 Others .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a formal parameter to replace the reserved parameter that controls DSS PDCCH resource sharing policy optimization. Specifically, added the BIT26 option of the eNBCellRsvdPara.RsrdSwPara10 parameter to the disuse list, and replaced it with the DSS_PDCCH_SHR_POLICY_OPT_SW option of the CellAlgoExtSwitch.LnrCellAlgoSwitch parameter. For details, see 6.1.7 DSS PDCCH Resource Sharing Optimization and 6.3.2.3 Function Impacts .	Modified parameter on the LTE side: Added the DSS_PDCCH_SHR_POLICY_OPT_SW option to the CellAlgoExtSwitch.LnrCellAlgoSwitch parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with Static Multiple Beam 2.0. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with intelligent beam weight optimization. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with LTE flexible spectrum. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added an impact relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with accurate uplink RB scheduling. For details, see 4.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a switch for RedCap SSSG switching, which has a mutually exclusive relationship with LTE FDD and NR Flash Dynamic Spectrum Sharing. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with intra-frequency paging interference avoidance. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Modified the switch for LTE-NR multi-dimensional scheduling optimization. For details, see 4.3.2.3 Function Impacts , 4.3.3 Hardware , and 6.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship of DSS and NR Flexible Refarming with ETWS. For details, see 7.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added compatibility of LTE and NR shared spectrum allocation adaptation with uplink VoLTE optimization. For details, see: • 4.1.1.3.1 LTE Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter on the LTE side: Added the LNR_UPLINK_V OLTE_GTEE_SW option to the SpectrumCloud.SpectrumCloudExtSwitch parameter.	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added a mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with PUSCH coordinated power control. For details, see 4.3.2.2 Mutually Exclusive Functions.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with the LTE and NR flexible frame offset function. For details, see 4.3.2.3 Function Impacts.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with UL and DL Decoupling. For details, see 4.3.2.2 Mutually Exclusive Functions.	None	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship of DSS and NR Flexible Refarming with RF module fast deep dormancy. For details, see 7.3.2.3 Function Impacts.	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added an impact relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with port sparseness. For details, see 4.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added support of LTE FDD and NR Flash Dynamic Spectrum Sharing for "cross-RAT configuration check in DSS scenarios". For details, see: • 4.1.1.1 Principles • 4.1.3.1 Principles • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter on the LTE side: Added the DSS_CROSS_RAT_CFG_CHECK_SW option to the eNodeBResMod <i>eAlgo.ParamCfgProtectSw</i> parameter. Modified parameter on the NR side: Added the DSS_CROSS_RAT_CFG_CHECK_SW option to the gNB0amParam <i>CfgInterceptPolicySw</i> parameter.	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship of DSS and NR Flexible Refarming with "cell RX channel interference and noise query". For details, see 7.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with user-rate-based dynamic network slice resource management. For details, see 4.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added an impact relationship of LTE FDD and NR Flash Dynamic Spectrum Sharing with user-rate-based dynamic QCI resource management. For details, see 4.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes
 - Added descriptions of the configuration requirements for modifying the parameters related to common PUCCH RB resource optimization and ultimate PUCCH compression in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. For details, see [4.3.5 Others](#).
 - Added descriptions of the network impacts of DSS PDCCH resource sharing optimization. For details, see [6.2.2 Impacts](#).
 - Renamed fast intra-gNodeB CA as downlink fast intra-gNodeB CA. For details, see [4.3.2.3 Function Impacts](#).
 - Added MML command examples for disabling DSS PDCCH resource sharing optimization for specified UEs. For details, see [6.4.1.2 Using MML Commands](#).
 - Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding

(MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to LTE FDD and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-16022 2	LTE FDD and NR Flash Dynamic Spectrum Sharing (LTE FDD)	4 LTE FDD and NR Flash Dynamic Spectrum Sharing
NR FDD	MRFD-16026 2	LTE FDD and NR Flash Dynamic Spectrum Sharing (NR)	4 LTE FDD and NR Flash Dynamic Spectrum Sharing
NR FDD	FOFD-051250	DSS-based Flexible PDSCH Scheduling (NR FDD)	5 DSS-based Flexible PDSCH Scheduling
LTE FDD	MRFD-19022 1	DSS/Hybrid DSS Performance Boost(LTE FDD)	6 DSS/Hybrid DSS Performance Boost
NR FDD	MRFD-19026 1	DSS/Hybrid DSS Performance Boost(NR FDD)	6 DSS/Hybrid DSS Performance Boost
LTE FDD	MRFD-20022 2	DSS and NR Flexible Refarming (LTE FDD)	7 DSS and NR Flexible Refarming
NR FDD	MRFD-20026 2	DSS and NR Flexible Refarming (NR FDD)	7 DSS and NR Flexible Refarming

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-22122 5	LTE and NR Uplink Spectrum Pooling (LTE FDD)	8 LTE and NR Uplink Spectrum Pooling
NR FDD	MRFD-22126 5	LTE and NR Uplink Spectrum Pooling (NR FDD)	8 LTE and NR Uplink Spectrum Pooling

2.4 Differences

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
LTE FDD and NR Flash Dynamic Spectrum Sharing	Supported only in NR FDD	4 LTE FDD and NR Flash Dynamic Spectrum Sharing
DSS-based Flexible PDSCH Scheduling	Supported only in NR FDD	5 DSS-based Flexible PDSCH Scheduling
DSS/Hybrid DSS Performance Boost	Supported only in NR FDD	6 DSS/Hybrid DSS Performance Boost
DSS and NR Flexible Refarming	Supported only in NR FDD	7 DSS and NR Flexible Refarming
LTE and NR Uplink Spectrum Pooling	Supported only in NR FDD	8 LTE and NR Uplink Spectrum Pooling

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
LTE FDD and NR Flash Dynamic Spectrum Sharing	- Whether the NR PUCCH includes the common PUCCH differs between NSA networking and SA networking. - The setting notes for the SSB frequency-domain position and Four Frame Allocation parameters differ between NSA networking and SA networking. - A neighboring NR frequency cannot be configured as a dynamic spectrum sharing frequency in NSA networking.	4 LTE FDD and NR Flash Dynamic Spectrum Sharing
DSS-based Flexible PDSCH Scheduling	None	5 DSS-based Flexible PDSCH Scheduling
DSS/Hybrid DSS Performance Boost	None	6 DSS/Hybrid DSS Performance Boost
DSS and NR Flexible Refarming	None	7 DSS and NR Flexible Refarming
LTE and NR Uplink Spectrum Pooling	None	8 LTE and NR Uplink Spectrum Pooling

Differences Between High Frequency Bands and Low Frequency Bands

Function Name	Difference	Chapter/Section
LTE FDD and NR Flash Dynamic Spectrum Sharing	Supported only in low frequency bands	4 LTE FDD and NR Flash Dynamic Spectrum Sharing
DSS-based Flexible PDSCH Scheduling	Supported only in low frequency bands	5 DSS-based Flexible PDSCH Scheduling
DSS/Hybrid DSS Performance Boost	Supported only in low frequency bands	6 DSS/Hybrid DSS Performance Boost

Function Name	Difference	Chapter/Section
DSS and NR Flexible Refarming	Supported only in low frequency bands	7 DSS and NR Flexible Refarming
LTE and NR Uplink Spectrum Pooling	Supported only in low frequency bands	8 LTE and NR Uplink Spectrum Pooling

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

In the early stages of 5G deployment, many operators have no dedicated NR spectrum resources or face low spectrum usage. To address this issue, LTE FDD and NR Flash Dynamic Spectrum Sharing (DSS) is introduced. This feature allows LTE FDD cells and NR FDD or NR supplementary downlink (SDL) cells to share time-frequency resources on the same spectrum segment. In this feature, the NR cell bandwidth is the same as the LTE cell bandwidth, and spectrum across the entire cell bandwidth can be shared. For more details, see [4 LTE FDD and NR Flash Dynamic Spectrum Sharing](#).

On the other hand, 3GPP specifications define large-bandwidth NR FDD cells to further improve the 5G network experience. However, the use of spectrum resources of large-bandwidth NR FDD cells is low in the early stages of 5G deployment due to the low penetration rate of 5G terminals. In view of this, Hybrid DSS Based on Asymmetric Bandwidth is introduced, which is applicable to operators that have both LTE FDD and NR FDD networks. This feature allows large-bandwidth NR FDD cells to dynamically share some of their spectrum resources with LTE FDD cells. In this feature, the NR cell bandwidth is greater than the LTE cell bandwidth. The amount of the shared spectrum is equal to the LTE cell bandwidth, and the NR cell bandwidth minus the amount of the shared spectrum is the amount of the NR-dedicated spectrum. For more details, see *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*.

DSS-based Flexible PDSCH Scheduling aims to allocate more spectrum resources to NR cells. This feature applies to scenarios where LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled. For more details, see [5 DSS-based Flexible PDSCH Scheduling](#).

To further improve spectrum utilization when LTE FDD and NR share spectrum, the following features have been introduced: DSS/Hybrid DSS Performance Boost, DSS and NR Flexible Refarming, and LTE and NR Uplink Spectrum Pooling. For more details, see [6 DSS/Hybrid DSS Performance Boost](#), [7 DSS and NR Flexible Refarming](#), and [8 LTE and NR Uplink Spectrum Pooling](#).

4 LTE FDD and NR Flash Dynamic Spectrum Sharing

4.1 Principles

Some operators lack dedicated NR FDD spectrum resources in the early stages of 5G deployment, while others possess the dedicated NR FDD spectrum resources, but encounter low spectrum usage due to the low penetration rate of 5G terminals in the early stages.

LTE FDD and NR Flash Dynamic Spectrum Sharing enables the sharing of spectrum resources between LTE FDD and NR FDD or NR SDL. For those operators lacking dedicated NR FDD spectrum resources, this feature can be used to allow LTE FDD to share its existing spectrum with NR, enabling fast 5G deployment. For operators who possess sufficient dedicated NR FDD spectrum resources, this feature enables NR FDD to share spectrum with LTE FDD, improving spectrum utilization.

For ease of description, "LTE" represents "LTE FDD", and "NR" represents "NR FDD" and "NR SDL". For details about NR SDL cells, see *Cell Management* in *5G RAN Feature Documentation*.

LTE FDD and NR Flash Dynamic Spectrum Sharing works in the following scenarios:

- Scenario 1: spectrum sharing between an LTE cell and an NR cell. For details, see [4.1.1 LTE FDD and NR FDD Dynamic Spectrum Sharing \(Single LTE Cell\)](#) and [4.1.2 LTE FDD and NR SDL Dynamic Spectrum Sharing \(Single LTE Cell\)](#). The NR cell in this scenario can be an NR FDD cell or an NR SDL cell.
- Scenario 2: spectrum sharing between multiple LTE cells and an NR cell. For details, see [4.1.3 LTE FDD and NR FDD Dynamic Spectrum Sharing \(Multiple LTE Cells\)](#). The NR cell in this scenario must be an NR FDD cell.

4.1.1 LTE FDD and NR FDD Dynamic Spectrum Sharing (Single LTE Cell)

4.1.1.1 Principles

LTE FDD and NR Flash Dynamic Spectrum Sharing allows an LTE cell and an NR cell to dynamically share time-frequency resources on a shared spectrum segment based on their traffic volume requirements. The LTE and NR cells are configured with the same bandwidth and can share the entire cell bandwidth, as shown in Figure 4-1.

Figure 4-1 Working principles of flash dynamic spectrum sharing (single LTE cell)

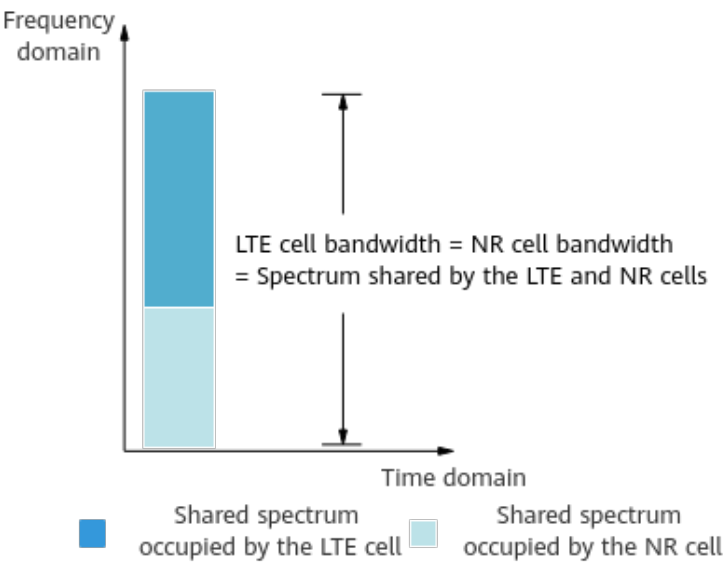


Table 4-1 lists the applicable cell bandwidths.

Table 4-1 Cell bandwidths applicable to LTE FDD and NR Flash Dynamic Spectrum Sharing

LTE Cell Bandwidth	NR Cell Bandwidth
10 MHz	10 MHz
15 MHz	15 MHz
20 MHz	20 MHz

This function is enabled by turning on function switches and configuring spectrum sharing cell groups. In addition, the LTE and NR cells sharing spectrum resources must have aligned radio frames and subframes. (The alignment is ensured by configuring frame offsets and TA offsets.) Table 4-2 describes parameter settings for enabling this function.

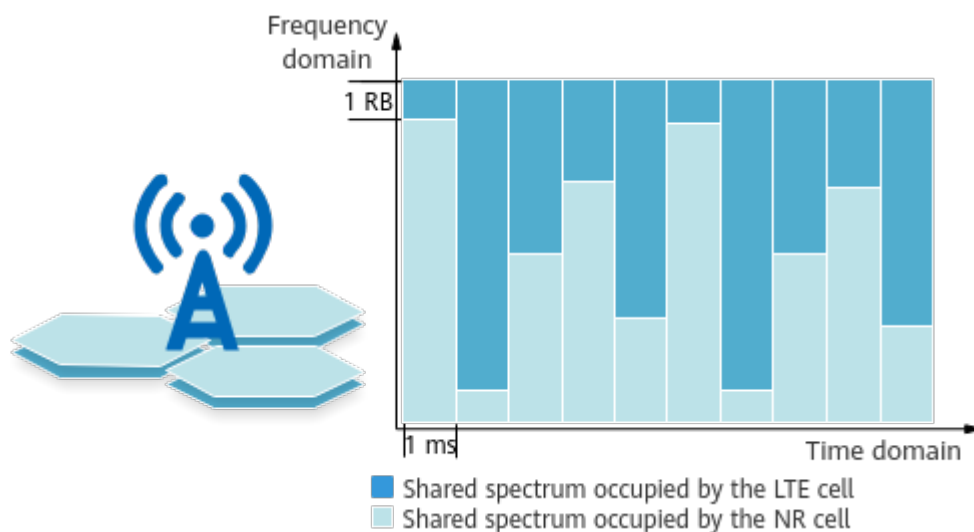
Table 4-2 Activation of LTE FDD and NR Flash Dynamic Spectrum Sharing (single LTE cell)

Operation	LTE Parameter Setting	NR Parameter Setting
Turning on function switches on the LTE and NR sides	Set the SpectrumCloud.SpectrumCloudSwitch parameter to LTE_NR_SPECTRUM_SHR .	Select the LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter.
Configuring spectrum sharing cell groups	Add a planned LTE cell to an LTE spectrum sharing cell group by setting the SpectrumCloud.LteNrSpectrumShrCellGrpId and LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId parameters.	1. Configure an association between the LTE spectrum sharing cell group and an NR spectrum sharing cell group by setting the gNBdulTeNrSpctShrCg.NrSpctShrCellGrpId and gNBdulTeNrSpctShrCg.LteSpctShrCellGrpId parameters. 2. Add a planned NR cell to the NR spectrum sharing cell group by setting the NRDUCellSptCloud.NrDUCellId and NRDUCellSptCloud.NrSpctShrCellGrpId parameters.
Configuring frame offsets	Set the CellFrameOffset.FrameOffset or ENodeBFrameOffset.FddFrameOffset parameter. If both parameters are configured, the value of the CellFrameOffset.FrameOffset parameter is used.	Set the gNodeBParam.FrameOffset or gNBFreqBandConfig.FrameOffset parameter. If both parameters are configured, the value of the gNBFreqBandConfig.FrameOffset parameter is used.
Configuring TA offsets	Set the CellFrameOffset.TaOffset parameter.	Set the NRDUCell.TaOffset parameter.

After this function is enabled, LTE and NR cells in the associated spectrum sharing cell groups can share time-frequency resources on a shared spectrum segment.

Figure 4-2 shows the sharing mechanism.

- Time domain: Flash spectrum sharing is supported on a 1 ms basis, meaning that spectrum resources can be coordinated and scheduled every 1 ms.
- Frequency domain: Dynamic spectrum sharing is performed per RB. Spectrum resources are dynamically allocated to LTE and NR cells based on their traffic volume requirements.

Figure 4-2 Time-frequency resource sharing between LTE and NR cells

You are advised to enable "cross-RAT configuration check in DSS scenarios" by selecting both the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter and the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **gNBOamParam.CfgInterceptPolicySw** parameter. This function checks whether the configurations of NR and LTE cells that are configured to be activated in the same spectrum sharing group meet all requirements.

4.1.1.2 Key Technologies

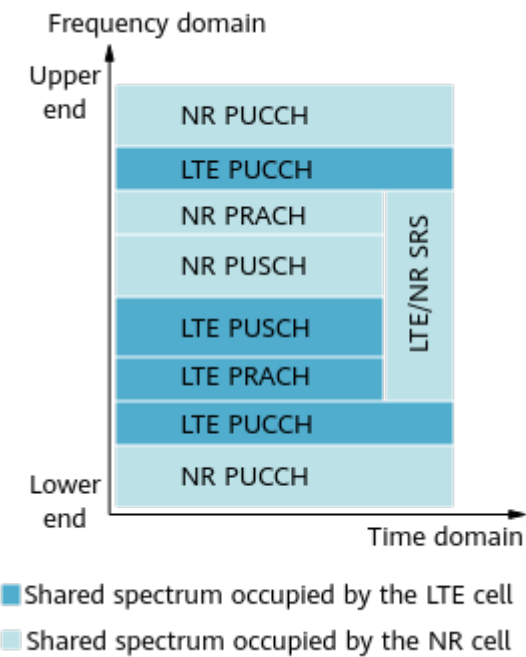
4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources

Some channels will conflict with each other during spectrum sharing between LTE and NR cells in a spectrum sharing cell group. Such conflicts can be avoided by properly coordinating uplink and downlink physical channel resource scheduling, improving spectrum utilization.

Uplink Physical Channels

Figure 4-3 shows the positions of uplink physical channels and signals after this function is enabled.

Figure 4-3 Positions of uplink physical channels and signals



Uplink physical channel resources are coordinated and scheduled as follows:

- LTE and NR sounding reference signals (SRSs): SRSs can be configured either for both LTE and NR or for NR only. [Table 4-3](#) details the two configuration modes.

Table 4-3 LTE and NR SRS configuration modes

Configuration Mode	Parameter Setting	Effect
SRSs are configured for NR only.	The LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter is selected.	SRS resources are allocated to NR but not to LTE.
SRSs are configured for both LTE and NR. ^a	The LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter is deselected.	SRS resources are allocated to both LTE and NR.

Configuration Mode	Parameter Setting	Effect
		a: In this scenario, LTE reserves subframes 0 and 5 for NR as cell-specific SRS subframes. When LTE UEs are scheduled in subframes during which NR SRSs are transmitted, the uplink MCS index decreases, affecting the uplink data rates of LTE UEs. LTE and NR SRS resource allocation optimization can be enabled by selecting the LTE_NR_SRS_ALLOC_OPT_SW option of the LteNrSpctShrCellGrp.LteNrSpctShrSwitch parameter. After the optimization, LTE reserves only subframe 0 for NR as the cell-specific SRS subframe, reducing the impact of NR SRS subframes on the uplink data rates of LTE UEs.

- NR physical uplink control channel (PUCCH): The NR PUCCH occupies symmetrical RBs at the two ends of the bandwidth. The number of RBs for the NR PUCCH is configured using the NR parameters listed in [Table 4-4](#). In NSA networking, the NR PUCCH does not include the common PUCCH. In SA networking or NSA and SA hybrid networking, the NR PUCCH always includes the common PUCCH, which occupies four RBs. For details about the common PUCCH, see descriptions of the PUCCH in *Channel Management* in *5G RAN Feature Documentation*.

Table 4-4 Parameters for configuring the number of RBs for the NR PUCCH

Parameter	Function
NRDUCellPucch.Format1RbNum	Specifies the number of PUCCH format-1 RBs.
NRDUCellPucch.Format3RbNum	Specifies the number of PUCCH format-3 RBs.
NRDUCellPucch.Format4RbNum	Specifies the number of PUCCH format-4 RBs.
NRDUCellPucch.CsiDedicatedRbNum	Specifies the number of RBs reserved for format-3 periodic CSI.
NRDUCellPucch.Format4CsiDedicatedRbNum	Specifies the number of RBs reserved for format-4 periodic CSI.

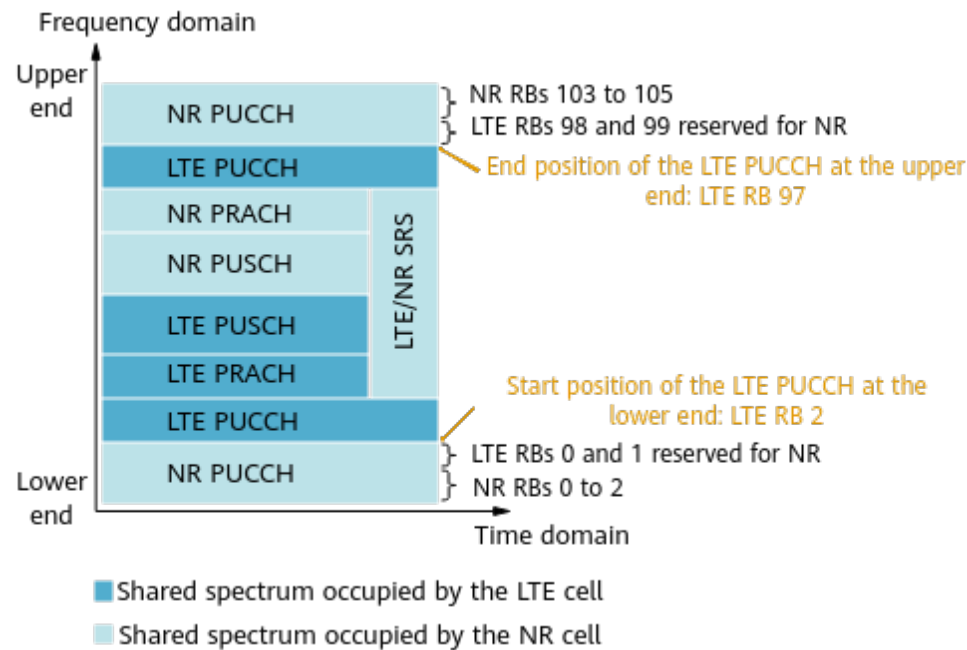
- LTE PUCCH: The LTE PUCCH is symmetrically configured adjacent to the NR PUCCH, so that the NR PUCCH and LTE PUCCH occupy consecutive RBs. The position of the LTE PUCCH depends on the position of the NR PUCCH and the number of RBs occupied by the NR PUCCH. The number of RBs occupied by the LTE PUCCH is specified by the parameters listed in [Table 4-5](#).

Table 4-5 Parameters for configuring the number of RBs occupied by the LTE PUCCH

Parameter	Function
CellRbReserve.RbRsvMode	Specifies the RB reservation mode.
CellRbReserve.RbRsvType	Specifies the type of physical-layer reserved PRBs.
CellRbReserve.RbRsvStartIndex	Specifies the start index of reserved RBs.
CellRbReserve.RbRsvEndIndex	Specifies the end index of reserved RBs.

The following description uses a 20 MHz bandwidth as an example to illustrate how to determine the start and end positions of the LTE PUCCH. In a 20 MHz bandwidth, NR has 106 available RBs in the uplink, and LTE has 100 available RBs in the uplink. Assume that 10 RBs are configured for the NR PUCCH, then the NR PUCCH occupies five RBs at each end of the frequency domain: RBs 0 to 4 at the lower end and RBs 101 to 105 at the upper end. As NR has six more RBs (106 – 100) than LTE in the frequency domain, that is, NR has three more RBs than LTE at each end of the frequency domain, LTE needs to reserve two RBs at each end (RBs 0 and 1 at the lower end, and RBs 98 and 99 at the upper end) for NR so that the NR PUCCH can occupy five RBs at each end of the frequency domain. In this case, you need to set the **CellRbReserve.RbRsvStartIndex** parameter to **2** (RBs 0 and 1 are reserved for NR), and the **CellRbReserve.RbRsvEndIndex** parameter to **97** (RBs 98 and 99 are reserved for NR). For details, see [Figure 4-4](#).

Figure 4-4 Example of the start and end positions of the LTE PUCCH



The number of RBs for the LTE PUCCH is specified by parameter configurations or is dynamically adjusted. For details, see descriptions of the PUCCH in *Physical Channel Resource Management* in *eRAN Feature Documentation*. In this function, the LTE PUCCH can occupy a maximum of 16 RBs. If the number of RBs occupied by the LTE PUCCH exceeds this value, it may interfere with SRSs, affecting network performance.

- LTE physical random access channel (PRACH): The LTE PRACH is configured adjacent to the LTE PUCCH at the lower end, and always occupies six RBs.
- NR PRACH: The NR PRACH is configured adjacent to the LTE PUCCH at the upper end. The frequency-domain start position of the NR PRACH is determined based on the configuration of the LTE PUCCH, and the NR PRACH always occupies six RBs.
- LTE and NR physical uplink shared channels (PUSCHs): The total available shared PUSCH resources for LTE and NR equal the total spectrum resources minus the resources occupied by the LTE and NR PUCCHs and PRACHs. The total available shared PUSCH resources for LTE and NR are dynamically shared between them based on traffic requirements.

Downlink Physical Channels

When a resource conflict occurs between LTE and NR, resources are allocated to the channels and signals with a higher priority through coordinated scheduling. Conflicts between channels or signals of the same type are avoided in a static or dynamic way.

Mutual avoidance of conflicts between LTE SS/PBCH and NR SSB

The LTE SS/PBCH and NR synchronization signal and PBCH block (SSB) mutually avoid conflicts with each other. The NR SSB is transmitted in fixed positions in Case A. For details, see section 4.1 "Cell Search" in 3GPP TS 38.213 V15.5.0.

Table 4-6 describes cell coverage scenarios using different numbers of SSB beams.

Table 4-6 Cell coverage scenarios using different numbers of SSB beams

Scenario	Referred to As	SSB Transmission Position ^a	Configuration Requirement
A single SSB beam is used for cell coverage.	Single-beam function	Index 3	For better performance during switching between the single-beam and dual-beam functions, the PAGING_ADJUST_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter can be used to adjust the paging time sequence when either function is used.
Two SSB beams are used for cell coverage.	Dual-beam function ^b	Index 2 and index 3	

Scenario	Referred to As	SSB Transmission Position ^a	Configuration Requirement
Four SSB beams are used for cell coverage.	Four-beam function ^b	Index 0 to index 3	The four-beam function requires the NR cell bandwidth to be 20 MHz and the SSB_ADJ_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter to be selected beforehand.
a: For details about SSB transmission positions, see <i>Channel Management</i> in <i>5G RAN Feature Documentation</i>. b: For details about the dual-beam and four-beam functions, see <i>Beam Management</i> in <i>5G RAN Feature Documentation</i>.			

In addition, the NR SSB frequency domain can be configured to improve the spectrum usage on the NR side. The details are as follows:

- **NRDUCell.SsbDescMethod**: indicates the description method of the SSB frequency-domain position. In NSA networking, it is recommended that this parameter be set to **SSB_DESC_TYPE_NARFCN**, which indicates absolute frequency channel numbers. This is because the step of absolute frequency channel numbers is shorter, which improves the spectrum usage. In SA networking, this parameter must be set to **SSB_DESC_TYPE_GSCN**, which indicates the Global Synchronization Channel Number (GSCN). This is because UEs need to determine the NR SSB frequency-domain position based on the GSCN. After the position is determined, UEs can access the cell.
- **NRDUCell.SsbFreqPos**: The frequency-domain position indicated by this parameter is determined by the value of the **NRDUCell.SsbDescMethod** parameter. The specific parameter value for this function must be confirmed by Huawei engineers.

Before configuring the preceding two parameters, you need to ensure that the number of RBs for common control resources (specified by the NR parameter **NRDUCellCoreset.CommonCtrlResRbNum**) in the cell meets the requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing. The requirements vary depending on the number of symbols occupied by the NR PDCCH (specified by the NR parameter **NRDUCellPdcch.OccupiedSymbolNum**). [Table 4-7](#) details the requirements.

Table 4-7 Required number of RBs for common control resources

Value of Occupied Symbol Number	Required Value of Common Control Resource RB Number
1SYM	RB48
2SYM	RB24 or RB48

It is recommended that LTE and NR resource allocation direction optimization be enabled to optimize resource allocation and improve NR performance. (This function is controlled by the **LNR_RES_ALLOC_DIRECTION_OPT_SW** option of the **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** parameter.)

Yielding by the NR PDSCH to LTE CRS

The NR PDSCH (excluding system information and messages transmitted before RRC reestablishment during initial access) yields to LTE CRS through CRS rate matching. NR UEs must support CRS rate matching. For details about CRS rate matching, see section 5.1.4.2 "PDSCH resource mapping with RE level granularity" in 3GPP TS 38.214 V15.5.0.

- Some NR UEs on the network may not support CRS rate matching. To reduce the impact of such UEs on spectrum sharing, the following configurations can be made:
 - (SA networking only) The **LTE_HO_TO_DSS_NR_OPT_SW** option of the NR parameter **gNodeBParam.LteHandoverAlgoSwitch** can be selected to prohibit UEs that do not support CRS rate matching from being handed over to the NR cell.
 - (SA networking or NSA and SA hybrid networking only) The **DSS_FREQ_INDICATION** option of the LTE parameter **NrNFreq.AggregationAttribute** can be selected to configure the neighboring NR frequency of the LTE cell as a spectrum sharing frequency. UEs that do not support CRS rate matching are prohibited from being handed over or redirected to this frequency.
- When the number of LTE CRS ports (specified by the LTE parameter **Cell.CrsPortNum**) is 1, the **LTE_CR_S_PORT_1_RM_SW** option of the NR parameter **gNBdLteNrSpctShrCg.FddLteNrSpctShrSwitch** can be selected so that the NR PDSCH performs CRS rate matching in the pattern of one LTE CRS port. CRS rate matching in the pattern of one LTE CRS port is supported when the cells use the independent power configuration mode (with the LTE parameter **SpectrumCloud.SpctShrMode** set to **LTE_NR_PWR_INDEPENDENT**) or in certain scenarios where the cells use the spectrum power sharing mode (with the LTE parameter **SpectrumCloud.SpctShrMode** set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**). If CRS rate matching in the pattern of one LTE CRS port is not enabled, the NR PDSCH performs CRS rate matching in the pattern of two LTE CRS ports even when only one CRS port is configured.

When two or four CRS ports are configured, the NR PDSCH performs CRS rate matching in the pattern of two or four LTE CRS ports.
- The **DL_RB_ESTIMATE_OPT_SW** option of the NR parameter **NRDUCellPdsch.DlPdschAlgoSwitch** can be selected to enable the downlink scheduler to select a more appropriate number of RBs for NR UEs to be scheduled in the downlink. This prevents UE data from being truncated and increases the downlink user-perceived rate. For details, see *Downlink Scheduling* in *5G RAN Feature Documentation*.

Mutual avoidance of conflicts between LTE CRS and NR SSB/NR PDSCH SIB1

Parameters are set such that LTE CRS yields to the NR SSB and SIB1 on the NR PDSCH. Specifically, MBSFN subframes are configured for the LTE cell in positions aligned with the NR SSB subframes and subframes occupied by SIB1 on the NR PDSCH. This ensures that the LTE MBSFN subframes overlap with the NR SSB

subframes and subframes occupied by SIB1 on the NR PDSCH in the time domain. As CRS is sent only in PDCCH symbols in an LTE MBSFN subframe, conflicts between LTE CRS and the NR SSB and SIB1 on the NR PDSCH are avoided. MBSFN subframes are configured using the LTE parameters listed in [Table 4-8](#).

Table 4-8 Parameters for configuring MBSFN subframes

Parameter	Function
CellMbsfnSfEnhConfig.SubframeAllocationMode	Specifies the MBSFN subframe allocation mode.
CellMbsfnSfEnhConfig.RadioFrameAllocPrd	Specifies the MBSFN subframe allocation period.
CellMbsfnSfEnhConfig.RadioFrameAllocOfs	Specifies the MBSFN subframe allocation offset.
CellMbsfnSfEnhConfig.OneFrameAllocation or CellMbsfnSfEnhConfig.FourFrameAllocation	Specify the positions of MBSFN subframes.

The MBSFN subframe configuration is affected by the number of SSB beams and the SSB period. For details, see the configuration suggestions in [4.4.1.1 Data Preparation](#).

For details about the concepts related to MBSFN subframes, see related descriptions in *eMBMS* in *eRAN Feature Documentation*. For details about the impact of MBSFN subframe configuration on the network, see the network analysis description in *eMBMS* in *eRAN Feature Documentation*.

Mutual avoidance of conflicts between NR PDCCH and LTE PDCCH/PCFICH/PHICH

The NR PDCCH and LTE PDCCH/PCFICH/PHICH avoid conflicts with each other. The LTE PDCCH occupies one symbol (in the case of one or two LTE CRS ports) or two symbols (in the case of four LTE CRS ports), and the number of symbols occupied by the NR PDCCH can be set to 1 or 2 by using NR parameters **NRDUCellPdcch.SpctShrStartSymbol** and **NRDUCellPdcch.OccupiedSymbolNum**. For details about configuration suggestions, see [4.4.1.1 Data Preparation](#).

Mutual avoidance of conflicts between LTE and NR PDSCHs through coordinated dynamic scheduling

Coordinated dynamic scheduling is used for LTE and NR PDSCHs to avoid conflicts with each other. In addition, the LTE PDSCH is dynamically scheduled to yield to the NR SSB/CSI-RS/TRS/DMRS/PDCCH, and the NR PDSCH is also dynamically scheduled to yield to the LTE CRS/CSI-RS/SS/PBCH/PHICH/PCFICH/PDCCH. Specifically:

- The NR PDSCH can yield to LTE CSI-RS at the RE or subframe level. When the **LTE_CSI_RS_AVOID_POLICY_SW** option of the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** and the **LTE_CSI_RS_AVOID_SW**

option of the NR parameter **gNBdULteNrSptShrCg.FddLteNrSptShrSwitch** are both selected, RE-level yielding is used. That is, the NR PDSCH avoids the REs occupied by the LTE CSI-RS. In other circumstances, subframe-level yielding is used. That is, the NR PDSCH avoids the subframes occupied by the LTE CSI-RS.

When the TX/RX mode of the LTE and NR cells is 8T8R, the NR PDSCH cannot yield to LTE CSI-RS at the RE level.

- The NR PDSCH can yield to the LTE SS/PBCH through SS/PBCH rate matching at the RB or symbol level. You can select the **LTE_SS_PBCH_RM_OPT_SW** option of the NR parameter **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** to enable the NR PDSCH to use one pattern for SS/PBCH rate matching. When this option is deselected, the NR PDSCH uses two patterns for SS/PBCH rate matching. This option can be set based on the network plan. When the NR PDSCH uses two rate matching patterns, the yielding is more accurate and at the same time more pattern resources are consumed, compared with when one rate matching pattern is used.

The position of the downlink additional DMRS on the NR PDSCH is specified by the NR parameter **NRDUCellPdsch.DlAdditionalDmrsPos**. This parameter can be set to **POS1** to improve user experience.

LTE DRX information and UE-level control information have to yield to NR CSI-RS and TRS in the time domain and may therefore fail to be scheduled in time. If this occurs, control-plane performance indicators are affected. Adaptive DRX cycle extension can address this issue. This function is controlled by the **DSS_DRX_CYCLE_ADAPT_SW** option of the **SpectrumCloud.SpectrumCloudExtSwitch** parameter. With this function, the DRX cycle is adaptively prolonged to reduce the probability of not scheduling such LTE signaling in a timely manner. This function may affect the UE power saving effect. In addition, on the LTE side, this function may decrease the packet transmission delay and change the user-perceived rate, BLER, and number of UEs scheduled in each TTI in both the uplink and downlink.

If the SSB is configured at the position indicated by SSB index 0 or 1 in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, there may be insufficient power for common channel transmission on the LTE and NR sides and insufficient bandwidth resources for the NR SSB. If the **NRDUCell.SsbTimePos** parameter is set to **SSB12_PCI_MOD2_STAGGER** or **SSB_PCI_MOD3_STAGGER**, the SSB may be configured at the position indicated by SSB index 1. Therefore, in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the **NRDUCell.SsbTimePos** parameter can only be set to **SSB23_PCI_MOD2_STAGGER**.

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the **PAGING_OSI_STAGGER_SW** option of the **NRDUCellPagingConfig.PagingSchAlgoSwitch** parameter can be selected to enable paging-OSI staggering.

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the **TIME_DOMAIN_RMSI_STAGGER_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter can be selected to enable time-domain RMSI staggering.

Avoidance of conflicts between other channels or signals

- LTE CRS yields to RRC messages on the NR PDSCH through static coordinated scheduling.
- LTE CSI-RS/CRS/DMRS and NR CSI-RS/TRS/PDCCH avoid conflicts with each other through static coordinated scheduling.

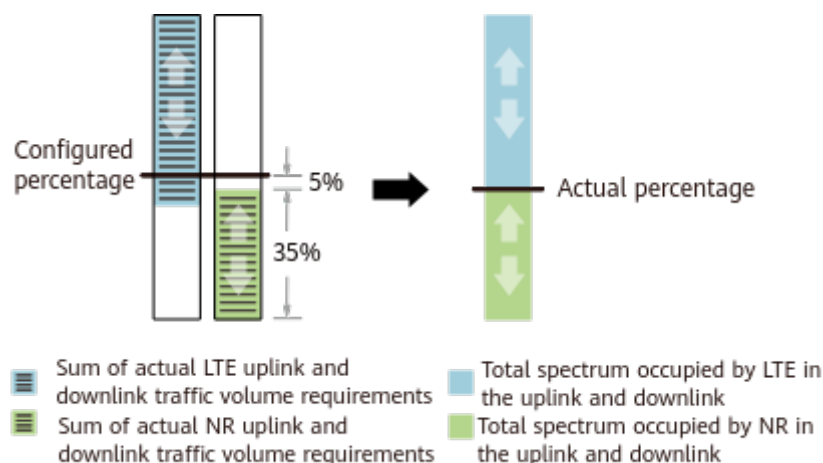
4.1.1.2.2 Flexible Spectrum Priority Mode

Configuration of the Percentages of Spectrum Resources to Be Preferentially Allocated

To ensure proper resource allocation, the percentages of resources preferentially allocated to the LTE and NR cells for MBB services can be set based on their traffic volume requirements. The resources here refer to the PUSCH and PDSCH resources. The percentage for LTE is specified by the **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** parameter, and the percentage for NR is equal to 100% minus the value of this parameter. Spectrum resources are allocated based on these percentages as follows:

- When the percentages of actual traffic volume requirements of the LTE and NR cells to the total spectrum resources are both less than, or both exceed, the percentages of resources preferentially allocated, spectrum resources are allocated based on the percentages of resources preferentially allocated to each cell.
- When the actual traffic volume requirement of the cell of one RAT (RAT 1) exceeds the percentage of resources preferentially allocated to that RAT, and the actual traffic volume requirement of the cell of the other RAT (RAT 2) does not, the unused resources for RAT 2 can be occupied by RAT 1. For example, if the configured percentage of resources preferentially allocated to the LTE cell is 60% but the LTE cell requires 70%, and the configured percentage of resources preferentially allocated to the NR cell is 40% but the NR cell only requires 35%, then the unused 5% resources for the NR cell can be used by the LTE cell. That is, the LTE cell can use 65% of the shared spectrum resources, as shown in [Figure 4-5](#).

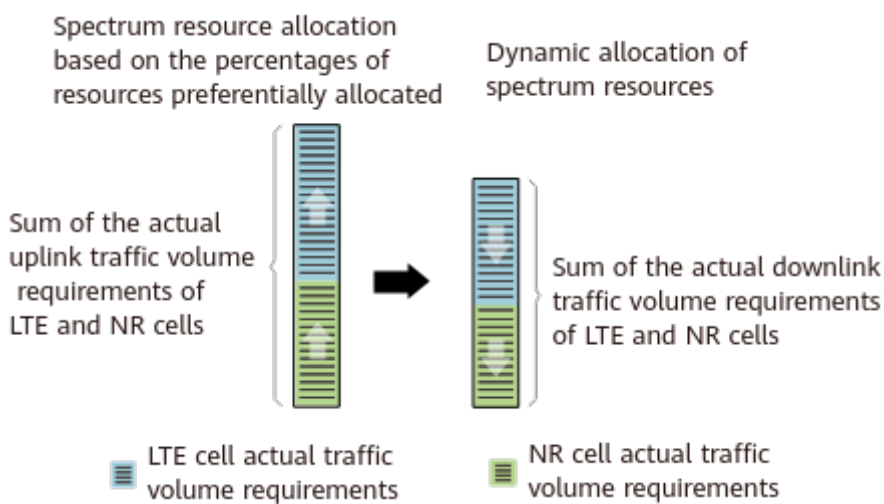
Figure 4-5 Example 1 of how the percentages of resources that are preferentially allocated work



In addition to the preceding allocation principles, the setting of the **LNR_RES_ALLOC_ADAPT_SW** option of the **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** parameter also affects the actual allocation of spectrum resources.

- If this option is selected, the system allocates spectrum resources based on the percentage specified by the **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** parameter, for whichever direction (uplink or downlink) with the larger sum of traffic volume requirements of the LTE and NR cells. For the other direction, the system dynamically allocates spectrum resources based on the actual traffic volume requirements of the LTE and NR cells. For example, if the sum of the uplink traffic volume requirements of the LTE and NR cells is greater than the sum of the downlink traffic volume requirements of the LTE and NR cells, the uplink spectrum resources are allocated to the LTE and NR cells based on the percentage specified by the **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** parameter. The downlink spectrum resources are dynamically allocated to the LTE and NR cells based on their actual traffic volume requirements, as shown in [Figure 4-6](#).

Figure 4-6 Example 2 of how the percentages of resources that are preferentially allocated work



- If this option is deselected, the system considers the uplink and downlink as a whole and allocates resources to the LTE and NR cells based on the value of the **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** parameter.

[Table 4-9](#) lists the counters used for measuring the shared spectrum resources occupied by the LTE and NR cells in the uplink and downlink.

Table 4-9 Counters for measuring the number of uplink and downlink available RBs in LTE and NR cells

Counter ID	Counter Name
1526759050	L.ChMeas.PRB.UL.Actual.Avail
1526759052	L.ChMeas.PRB.UL.PUSCH.Actual.Avail

Counter ID	Counter Name
1526759051	L.ChMeas.PRB.DL.Actual.Avail
1911827161	N.PRB.UL.Actual.Avail.Avg
1911827162	N.PRB.UL.PUSCH.Actual.Avail.Avg
1911827160	N.PRB.DL.Actual.Avail.Avg

Inter-RAT Preferential Guarantee for GBR Services

If the percentage of resources preferentially allocated to a RAT is less than the percentage required by guaranteed bit rate (GBR) services of that RAT, experience of GBR services of that RAT will be affected. For example, if the percentage of resources preferentially allocated to LTE is 40% and the percentage required by GBR services is 60%, 20% of GBR service requirements cannot be satisfied, affecting GBR service experience.

The function of inter-RAT preferential guarantee for GBR services is introduced. This function defines the priority of GBR services as follows: The priority of LTE GBR services is higher than that of NR non-GBR services, and the priority of NR GBR services is higher than that of LTE non-GBR services. In the preceding example, the unsatisfied LTE GBR services can occupy the resources of NR non-GBR services. In this way, GBR service experience is preferentially guaranteed.

This function is enabled when the **GBR_PRI_ALLOC_SW** option of the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** is selected. For details about LTE GBR and non-GBR services, see *Uplink Scheduling* and *Downlink Scheduling* in *eRAN Feature Documentation*. For details about NR GBR and non-GBR services, see *QoS Management* in *5G RAN Feature Documentation*.

4.1.1.2.3 Flexible Power Allocation Mode

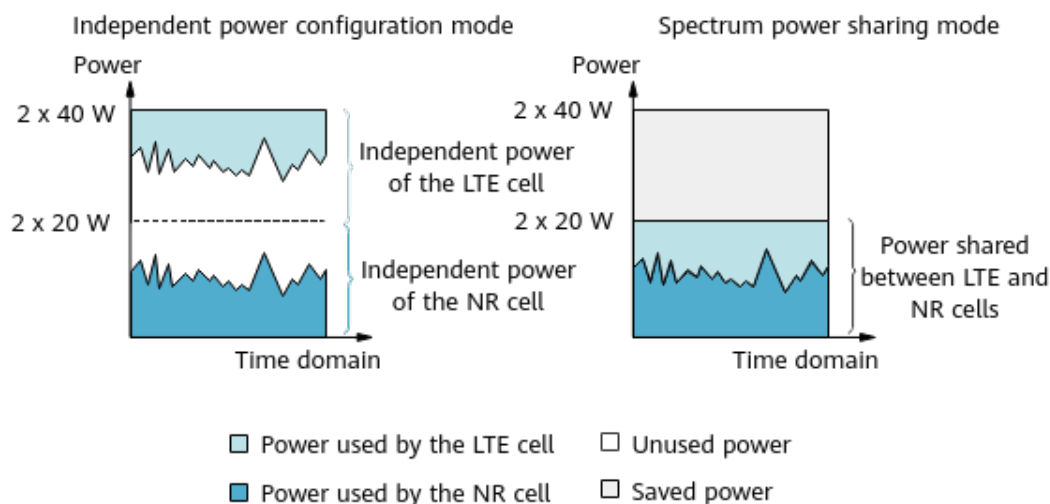
The power allocation mode can be flexibly configured using the LTE parameter **SpectrumCloud.SpctShrMode**:

- When this parameter is set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**, spectrum power sharing mode is used. The LTE and NR cells share the same power, and the percentage of power allocated to a RAT is equal to the percentage of spectrum allocated to that RAT.
- When this parameter is set to **LTE_NR_PWR_INDEPENDENT**, independent power configuration mode is used. The LTE and NR cells use independently configured power.

Figure 4-7 elaborates on these power allocation modes using a 2 x 40 W RF module as an example. In independent power configuration mode, the LTE and NR cells can each only be configured with the maximum transmit power of 2 x 20 W. In spectrum power sharing mode, the power configured for the LTE and NR cells must be greater than or equal to the power in independent power configuration mode. The LTE and NR cells can both be configured with a transmit power within the range of 2 x 20 W to 2 x 40 W, and minimally share the total power of 2 x 20 W. The spectrum power sharing mode helps save the transmit power by 2 x 20 W

(2 x 40 W – 2 x 20 W), compared with the independent power configuration mode.

Figure 4-7 Power allocation mode



Given the same cell power configuration (for example, 2x20 W for NR in both independent power configuration mode and spectrum power sharing mode), independent power configuration mode results in better capacity performance but a higher power consumption, whereas spectrum power sharing mode reduces power consumption but leads to capacity performance loss. Therefore, **when the power is sufficient, independent power configuration mode is preferred. When the power is insufficient, spectrum power sharing mode can be used.**

In spectrum power sharing mode, the power spectral density (PSD) adaptation function can be used to reduce the PSD of the NR cell when the transmit power required by the NR cell exceeds the configured power. This ensures that the NR cell can use more spectrum resources, improving the spectral efficiency of UEs in the NR cell. This function is enabled when the **PWR_SPCT_DENSITY_ADAPT_SW** option of the NR parameter **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** is selected.

When spectrum power sharing mode is used, and the LTE and NR baseband processing units need to be connected to the RRU through their respective optical fibers, the **LNR_PWR_WITH_SPCT_OPT_SW** option of the LTE parameter **SpectrumCloud.SpectrumCloudEnhSwitch** must be selected. When independent power configuration mode is used, LTE and NR can use their respective optical fibers over the CPRI port without the need of selecting this option.

4.1.1.2.4 Co-carrier Co-CPRI Data

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the CPRI bandwidth and RF module carrier specifications are occupied as follows:

- If the co-carrier co-CPRI data function is disabled, the LTE and NR cells each occupy a portion of CPRI bandwidth. In addition, LTE and NR each occupy certain units of RF module carrier specifications.
- If the co-carrier co-CPRI data function is enabled when LTE and NR share the same optical fiber (the LTE or NR baseband processing unit is connected

directly to the RRU through an optical fiber), the LTE and NR cells occupy only one portion of CPRI bandwidth. In addition, only NR-occupied units of RF module carrier specifications count toward the total units of specifications occupied by LTE and NR.

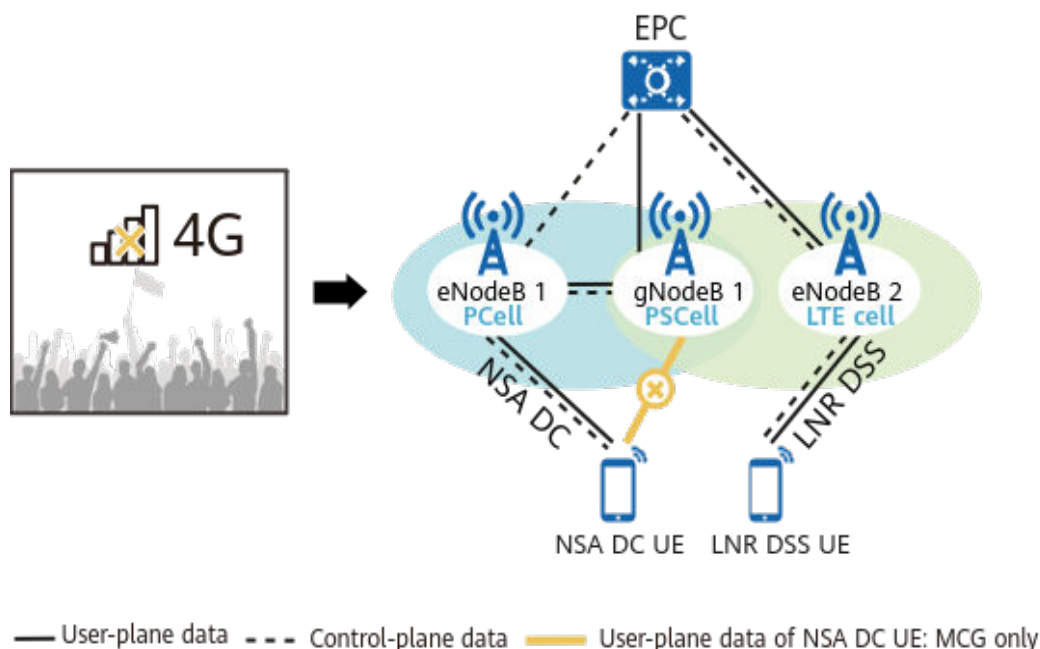
The co-carrier co-CPRI data function for LTE and NR can be enabled by setting the LTE parameter **LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw** to **ON**.

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, when the LTE and NR co-carrier co-CPRI data function is used together with the CPRI compression function, the CPRI compression ratios between LTE and NR cells must be the same; when the LTE and NR co-carrier co-CPRI data function is used together with the uplink interference suppression function on the LTE side, the uplink interference suppression value must be consistently set between LTE and NR cells. For details about the configuration requirements, see [4.1.1.3 Simultaneous Use with Other Functions](#).

4.1.1.2.5 LTE Key Event Assurance

In NSA networking, LTE cells engaged in spectrum sharing are heavily loaded in large gatherings such as concerts, sports events, and major holidays. To preferentially ensure the capacity of LTE cells, the LTE key event assurance function can be enabled so that **all shared spectrum resources are allocated to LTE cells**. As shown in [Figure 4-8](#), when this function is triggered, the uplink and downlink data split modes for the NR cell change to master cell group (MCG) only. That is, user-plane data of NSA dual connectivity (DC) UEs is distributed to the primary cell (PCell) but not to the primary secondary cell (PSCell).

Figure 4-8 LTE key event assurance



This function is enabled by selecting the **LTE_BIG_EVENT_ASSURANCE_SW** option of the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrSwitch**. When this

function is required, you need to set the

LteNrSpctShrCellGrp.LteBigEventUeNumThld and

LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset parameters to specify the LTE key event user number threshold and the threshold offset. In addition, you also need to set the NR parameter

gNBDULteNrSpctShrCg.LteBigEventAssureUeNumThld, which specifies the user number threshold for LTE key event assurance for NR cells. This function is applicable only to Option 3x architecture in NSA networking.

The LTE key event assurance function is triggered when both of the following conditions are met:

- Number of LTE UEs in RRC connected mode >
LteNrSpctShrCellGrp.LteBigEventUeNumThld +
LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset
- Number of NR UEs in RRC connected mode ≤
gNBDULteNrSpctShrCg.LteBigEventAssureUeNumThld

When the number of LTE UEs in RRC connected mode becomes less than or equal to **LteNrSpctShrCellGrp.LteBigEventUeNumThld** minus **LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset**, both the LTE and NR cells exit the LTE key event assurance state and enter the non-LTE-key-event-assurance state.

To protect the NR cell from ping-pong switchovers between these two states, a 5-minute protection timer is introduced to the NR cell.

4.1.1.3 Simultaneous Use with Other Functions

4.1.1.3.1 LTE Functions

Simultaneous Use with NB-IoT Deployed in LTE Guard Band Mode

For 3900 and 5900 series base stations, LTE FDD and NR Flash Dynamic Spectrum Sharing can work together with NB-IoT deployed in LTE guard band mode.

In this scenario, the spectrum occupied by the NB-IoT cell and the spectrum occupied by the NR cell overlap, causing severe mutual interference. As a result, the NB-IoT cell or NR cell becomes unavailable. Specifically:

- If the NB-IoT cell is configured and activated prior to the NR cell, the NR cell cannot be activated.
- If the NR cell is configured and activated prior to the NB-IoT cell, the NB-IoT cell cannot be activated.

To eliminate the interference, spectrum resources need to be reserved for the NB-IoT cell, and the reserved spectrum resources are unavailable for the NR cell. The uplink and downlink reserved RBs are configured using the parameters listed in [Table 4-10](#).

Table 4-10 Parameters for configuring uplink and downlink reserved RBs

Parameter	Function
NRDUCellRbReserve.RbRsvMode	Specifies the supported RB reservation mode.
NRDUCellRbReserve.RbRsvType	Specifies the type of RB reservation.
NRDUCellRbReserve.RbRsvStartIndex	Specifies the start index of reserved RBs.
NRDUCellRbReserve.RbRsvEndIndex	Specifies the end index of reserved RBs.

The uplink and downlink RBs reserved for NB-IoT must be confirmed by Huawei engineers. [Table 4-11](#) lists the recommended number of uplink and downlink RBs to be reserved in different LTE and NR cell bandwidth scenarios. The recommended configuration preferentially ensures the NB-IoT cell performance while considering the NR cell throughput.

For details about NB-IoT deployed in LTE guard band mode, see *NB-IoT Basics (FDD)* in *eRAN Feature Documentation*.

Table 4-11 Recommended reserved spectrum resources for the NB-IoT cell in typical scenarios

LTE and NR Cell Bandwidth	Number of Uplink and Downlink RBs Reserved for NB-IoT
20 MHz	3
15 MHz	2
10 MHz	1

Simultaneous Use with LTE Uplink Interference Suppression

For the DBS3900 LampSite and DBS5900 LampSite, when the co-carrier co-CPRI data function in spectrum sharing and the uplink interference suppression function are both enabled, the **ULInterfSuppressCfg.UlIntrfSuppressionValue** parameter for the LTE cell must be set to the same value as that of the **NRDUCellTrp.UlRxPowerAttenuationValue** parameter for the NR cell. For details about LTE uplink interference suppression, see *pRRU Uplink Interference Suppression* in *eRAN Feature Documentation*.

Simultaneous Use with LTE Downlink Semi-persistent Scheduling

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with LTE downlink semi-persistent scheduling, the upper limit on the proportion of RBs that can be allocated for LTE downlink semi-persistent scheduling is configurable (through the **SpectrumCloud.DlSpsRestrictRatio** parameter). This proportion

together with the percentage of spectrum resources to be preferentially allocated to LTE for MBB services (specified by the **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** parameter) determines the proportion of RBs allocated for LTE downlink semi-persistent scheduling as follows:

Proportion of RBs allocated for LTE downlink semi-persistent scheduling = **(SpectrumCloud.DlSpsRestrictRatio × LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio) × 100%**

For details about LTE downlink semi-persistent scheduling, see *VoLTE* in *eRAN Feature Documentation*.

Simultaneous Use with WBB

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used with wireless broadband (WBB), it is recommended that the **WBB_MBB_CONTROL_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter be selected. When this option is selected, the eNodeB allocates resources to LTE WBB and MBB services based on the actually available LTE spectrum resources. The percentages of resources actually allocated to MBB services comply with the settings of the **CellWttxParaCfg.MbbUserDlPrbUpLimit** and **CellWttxParaCfg.MbbUserUlPrbUpLimit** parameters; the percentages of resources actually allocated to WBB services comply with the settings of the **CellWttxParaCfg.WbbUserDlPrbUpLimit** and **CellWttxParaCfg.WbbUserUlPrbUpLimit** parameters. For details about WBB, see *WBB* in *eRAN Feature Documentation*.

To ensure spectral efficiency, it is recommended that the following equations be met:

- **CellWttxParaCfg.MbbUserDlPrbUpLimit + CellWttxParaCfg.WbbUserDlPrbUpLimit = 100%**
- **CellWttxParaCfg.MbbUserUlPrbUpLimit + CellWttxParaCfg.WbbUserUlPrbUpLimit = 100%**

Simultaneous Use with LTE Uplink Voice Packet Loss Optimization

LTE uplink voice packet loss optimization includes two functions: uplink VoLTE enhancement (controlled by the **LTE_UPLINK_VOLTE_ENH_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter) and uplink VoLTE optimization (controlled by the **LTE_UPLINK_VOLTE_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter).

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, if uplink VoLTE enhancement is enabled in a cell serving LTE voice service UEs, uplink RBs can be reserved for these UEs. This prevents idle PUCCH RBs from being occupied during scheduling of these UEs in the uplink, reducing fragmentation for voice service UEs. In addition, when PUSCH resources are requested for uplink retransmission in a subframe during LTE voice services, the resource requirements for the initial transmission corresponding to this subframe are considered. The number of RBs reserved for LTE uplink voice service UEs is specified by the **SpectrumCloud.UlVoLTEGuaranteedRbNum** parameter. This parameter can be set to a non-zero value only when LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, uplink VoLTE optimization can be enabled. With this function, LTE always takes precedence in subframes with uplink retransmission of LTE uplink voice services, resource reservation for TTI bundling is optimized, and the MCS index is reduced in the uplink for voice service UEs at the cell center and at medium distances from the cell center. This function takes effect only when the function switch is turned on and either of the following conditions is met:

1. The **LNR_RES_ALLOC_ADAPT_SW** option of the **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** parameter is deselected.
2. Both the **LNR_RES_ALLOC_ADAPT_SW** option of the **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** parameter and the **LNR_UPLINK_VOLTE_GTEE_SW** option of the **SpectrumCloud.SpectrumCloudExtSwitch** parameter are selected.

You are advised to enable both uplink VoLTE enhancement and uplink VoLTE optimization in the following scenarios:

1. The uplink voice packet loss rate does not exceed 0.1% in LTE-only scenarios and the rate increases by more than 0.05 percentage points after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.
2. The uplink voice packet loss rate is higher than 0.1% in LTE-only scenarios and the rate increases by more than 20% after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.

Simultaneous Use with DRX

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with discontinuous reception (DRX), the following configurations are recommended for the involved LTE FDD cell that prevents the DRX On Duration period from being extended due to MBSFN subframes, NR CSI-RS, or NR TRS:

- Select the **DSS_GAP_POS_OPT_SW** option of the **CellDrxPara.DrxAlgoExtSwitch** parameter to optimize the position of gap-assisted measurements in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. This configuration mitigates the increase in the service drop rate caused by an overlap between the DRX On Duration period and gap-assisted measurements.
- Select the **DSS_VOICE_NSA_SA_OD_POS_OPT_SW** option of the **CellDrxPara.DrxAlgoExtSwitch** parameter to optimize the position of the DRX On Duration period for NSA- or SA-capable voice service UEs in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, and set the **CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod** parameter to **40MS**, **80MS**, **160MS**, or **320MS** to specify the interval at which aperiodic CQI reporting is triggered for these UEs in spectrum sharing scenarios. These configurations mitigate the increase in the service drop rate caused by an overlap between the DRX On Duration period and gap-assisted measurements, NR CSI-RS, NR TRS, or MBSFN subframes.

4.1.1.3.2 NR Functions

Simultaneous Use with Downlink CA

When downlink intra-FR inter-band CA or downlink intra-band CA is enabled after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the NR PUCCH

occupies more RBs, which leads to RB resource conflicts with the LTE PUCCH. In this case, the LTE PUCCH RB configuration needs to be modified using the parameters listed in [Table 4-12](#). Among these parameters, it is recommended that **CellRbReserve.RbRsvStartIndex** and **CellRbReserve.RbRsvEndIndex** be set to the indexes of symmetrical RBs adjacent to the NR PUCCH.

Table 4-12 Parameters for configuring the LTE PUCCH RBs

Parameter	Function
CellRbReserve.RbRsvMode	Specifies the RB reservation mode.
CellRbReserve.RbRsvType	Specifies the type of physical-layer reserved PRBs.
CellRbReserve.RbRsvStartIndex	Specifies the start index of the LTE PUCCH PRBs at the lower end of the frequency domain.
CellRbReserve.RbRsvEndIndex	Specifies the end index of the LTE PUCCH PRBs at the upper end of the frequency domain.

In addition, the number of RBs occupied by the NR PUCCH must be as follows:

- NSA networking: Number of RBs occupied by the NR PUCCH = $(\text{NRDUCellPucch.Format1RbNum} + \text{NRDUCellPucch.Format3RbNum} + \text{NRDUCellPucch.Format4RbNum} + \text{NRDUCellPucch.CsiDedicatedRbNum} + \text{NRDUCellPucch.Format4CsiDedicatedRbNum})/2$ – Number of RBs occupied by the guard band at either end of the NR frequency domain
- SA networking, and NSA and SA hybrid networking: Number of RBs occupied by the NR PUCCH = $(\text{NRDUCellPucch.Format1RbNum} + \text{NRDUCellPucch.Format3RbNum} + \text{NRDUCellPucch.Format4RbNum} + \text{NRDUCellPucch.CsiDedicatedRbNum} + \text{NRDUCellPucch.Format4CsiDedicatedRbNum})/2$ – Number of RBs occupied by the guard band at either end of the NR frequency domain + Number of RBs occupied by the NR common PUCCH at either end of the frequency domain

In NSA networking, the NR PUCCH does not include the common PUCCH. In SA networking or NSA and SA hybrid networking, the NR PUCCH always includes the common PUCCH, which occupies four RBs, with two RBs at each end of the frequency domain. For details about the common PUCCH, see descriptions of the PUCCH in *Channel Management in 5G RAN Feature Documentation*.

NOTICE

If the LTE or NR cell is not deactivated before the preceding configurations are performed:

- When the **FORBID_PARA_CFG_VERIFY_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter is deselected, a command to reconfigure **Cell.CrsPortNum**, **Cell.TxRxMode**, **CellRbReserve.RbRsvMode**, **CellRbReserve.RbRsvStartIndex**, or **CellRbReserve.RbRsvEndIndex** will fail to be executed.
- When the **FORBID_PARA_CFG_VERIFY_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter is selected, it is possible that downlink intra-FR inter-band CA or downlink intra-band CA fails to take effect or that the NR cell goes out of service.

Evaluate the preceding risks and determine whether to deactivate the LTE and NR cells before performing the preceding configurations.

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with downlink intra-band CA, the **LNR_PUCCH_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter must be selected.

For details about downlink intra-FR inter-band CA and downlink intra-band CA, see *Carrier Aggregation in 5G RAN Feature Documentation*.

The in-use modules, bandwidth, and RBs must be considered in the configuration of the start position of the LTE PUCCH at the lower end and the end position of the LTE PUCCH at the upper end in the frequency domain. If the RRU3832, RRU3838, or RRU3839 is used, sufficient RBs must be reserved for NR before LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled. For example, the value of **CellRbReserve.RbRsvStartIndex** must be greater than or equal to 4 in the case of the cell bandwidth being 10 MHz. Otherwise, the corresponding NR cell cannot be set up as a result of insufficient reserved RBs.

Simultaneous Use with Intra-Base-Station DL CoMP

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with intra-base-station DL CoMP, the serving cell and cooperating cells for intra-base-station DL CoMP must be configured as NR spectrum sharing cells and added to different spectrum sharing cell groups. The NR cells involved in intra-base-station DL CoMP must meet the configuration requirements described in [Table 4-13](#), and the LTE cells that share spectrum with the NR cells involved in intra-base-station DL CoMP must meet the configuration requirements described in [Table 4-14](#).

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with intra-base-station DL CoMP on the NR side, intra-base-station DL CoMP will be affected by the semi-persistent scheduling (SPS) function on the LTE side. When SPS takes effect, intra-base-station DL CoMP on the NR side cannot take effect.

Table 4-13 Configuration requirements for NR cells involved in intra-base-station DL CoMP

Parameter ID	Option	Configuration Requirement
gNBDULteNrSpctShrCg.FddLteNrSpctShrSwitch	LTE_CRS_PORT_1_RM_SW and LTE_CSI_RS_AVOID_SW	Settings must be consistent between the serving cell and cooperating cells.
NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	PWR_SPCT_DENSITY_ADAPT_SW	This option must be deselected for both the serving cell and cooperating cells.
NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	LTE_CRS_RATEMATCH_POLICY_SW, LTE_CRS_RM_ALL_SYM_4PORT_SW, LTE_CRS_RATEMATCH_ALL_SYM_SW, and LTE_SS_PBCH_RM_OPT_SW	Settings must be consistent between the serving cell and cooperating cells.
NRDUCellPdsch.RateMatchSwitch	PDCCH_RATEMATCH_SW	Settings must be consistent between the serving cell and cooperating cells.
NRDUCellDLSchRes.SpctShrPdschStartSymbol	None	Settings must be consistent between the serving cell and cooperating cells.

Table 4-14 Configuration requirements for the LTE cells that share spectrum with the NR cells involved in intra-base-station DL CoMP

Parameter ID	Option	Configuration Requirement
LteNrSpctShrCellGrp.LteNrSpctShrSwitch	LTE_CSI_RS_AVOID_POLICY_SW	The LTE cells must have consistent settings.
Cell.CrsPortNum	None	The LTE cells must have consistent settings.

Parameter ID	Option	Configuration Requirement
Cell.PhyCellId	None	When the Cell.CrsPortNum parameter is set to a value indicating one CRS port, the LTE cells must have the same PCI mod 6 value. When the Cell.CrsPortNum parameter is set to a value indicating two or more CRS ports, the LTE cells must have the same PCI mod 3 value.
ENodeBAlgoSwitch.LcsSwitch	OtdoaSwitch and OTDOAInterFreqRSTD-MeasSwitch	These two options must be deselected for all LTE cells.
CellEmtcAlgo.EmtcAlgoSwitch	EMTC_SWITCH	This option must be deselected for all LTE cells.
CellMbsfnSfEnhConfig.SubframeAllocationMode , CellMbsfnSfEnhConfig.RadioFrameAllocPrd , CellMbsfnSfEnhConfig.RadioFrameAllocOfs , CellMbsfnSfEnhConfig.OneFrameAllocation , CellMbsfnSfEnhConfig.FourFrameAllocation , and CellDlschAlgo.MbsfnSfCfg	None	The LTE cells must have consistent settings.
LteNrSptShrCellGrp.LteNrSptShrSwitch	LTE_BIG_EVENT_ASSURANCE_SW	This option must be deselected for all LTE cells.
Cell.CustomizedBandWidthCfgInd	None	The LTE cells must have consistent settings.
Cell.DLPuncturedRbNum	None	The LTE cells must have consistent settings.

Parameter ID	Option	Configuration Requirement
SpectrumCloud.SpectrumCloudEnhSwitch	POWER_POLARIZATION_SW and LTE_PDCCH_OVERHEAD_OPT_SW	The LTE cells must have consistent settings.
NsaDcMgmtConfig.NsaDcAlgoSwitch	TDM_SWITCH	This option must be deselected for all LTE cells.

For details about intra-base-station DL CoMP, see *CoMP* in *5G RAN Feature Documentation*.

4.1.1.3.3 Common Functions Supported by Both LTE and NR

Simultaneous Use with CPRI Compression

In LTE FDD and NR Flash Dynamic Spectrum Sharing, when the co-carrier co-CPRI data function and the CPRI compression function are both enabled, the CPRI compression ratios must be the same for the LTE and NR cells.

The CPRI compression type of an LTE cell is specified by the **Cell.CPRICompression** parameter, and the CPRI compression type of an NR cell is specified by the **NRDUCellTrp.CpriCompression** parameter. [Table 4-15](#) describes the configuration requirements. For details about CPRI compression, see *CPRI Compression* in *eRAN Feature Documentation* and *CPRI Compression* in *5G RAN Feature Documentation*.

Table 4-15 Configuration requirements on the CPRI compression types of the LTE and NR cells

Bandwidth h	TX/RX Mode	CPRI Compre ssion Ratio	CPRI Compression Type of the LTE Cell	CPRI Compression Type of the NR Cell
10 MHz	2T2R/2T4R/ 4T4R/8T8R	No CPRI compre ssion	NO_COMPRESSIO N	NO_COMPRESSIO N
	2T2R/2T4R/ 4T4R/8T8R	3:2 CPRI compre ssion	NORMAL_COMPR SSION	1DOT5_COMPRES SION
	16T16R	2:1 CPRI compre ssion ^a	ENHANCED_COM PRESSION	2_COMPRESSION

Bandwidth	TX/RX Mode	CPRI Compression Ratio	CPRI Compression Type of the LTE Cell	CPRI Compression Type of the NR Cell
	32T32R	3.2:1 CPRI compression ^a	3_2_COMPRESSION	3DOT2_COMPRESSION
		2.2:1 CPRI compression	2_2_COMPRESSION	2DOT2_COMPRESSION
		2:1 CPRI compression ^b	ENHANCED_COMPRESSION	2_COMPRESSION
		3.2:1 CPRI compression ^b	3_2_COMPRESSION	3DOT2_COMPRESSION
15 MHz/20 MHz	2T2R/2T4R/4T4R/8T8R	No CPRI compression	NO_COMPRESSION	NO_COMPRESSION
	2T2R/2T4R/4T4R/8T8R	2:1 CPRI compression	NORMAL_COMPRESSION	2_COMPRESSION
	16T16R	2:1 CPRI compression ^a	ENHANCED_COMPRESSION	2_COMPRESSION
		3.2:1 CPRI compression ^a	3_2_COMPRESSION	3DOT2_COMPRESSION
	32T32R	2.2:1 CPRI compression	2_2_COMPRESSION	2DOT2_COMPRESSION
		2:1 CPRI compression ^b	NORMAL_COMPRESSION	2_COMPRESSION
		3.2:1 CPRI compression ^b	3_2_COMPRESSION	3DOT2_COMPRESSION

Bandwidth	TX/RX Mode	CPRI Compression Ratio	CPRI Compression Type of the LTE Cell	CPRI Compression Type of the NR Cell
a: This compression ratio is supported only when the interface board is a UBBPh3a. b: This compression ratio is supported only when the interface board is a UBBPh2/UBBPh3/UBBPh3a/UBBPh3b and the versions of NR and LTE are V100R020C10 or later.				

Simultaneous Use with LTE and NR Flexible Frame Offset

In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, when the LTE and NR flexible frame offset function is enabled, the frame offset of the NR cell in a spectrum sharing group must be 3 ms later than that of the LTE cell in the same group. To achieve this, the "LTE and NR 3-ms frame offset difference" function must be enabled and the NR and LTE cell frame offsets must be configured accordingly.

The "LTE and NR 3-ms frame offset difference" function takes effect when the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the NR parameter **NRDUCellSpctShare.LnrSpectrumShareSwitch** parameter is selected and the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the LTE parameter **SpectrumCloud.SpectrumCloudExtSwitch** parameter is selected.

The "LTE and NR 3-ms frame offset difference" function can be enabled only in LTE FDD and NR spectrum sharing scenarios. When it is enabled, the **CellMbsfnSfEnhConfig.SubframeAllocationMode** parameter must be set to **NOT_CONFIG**.

The effective LTE and NR cell frame offsets are determined as follows:

- NR cell frame offset: If the **gNBFreqBandConfig.FrameOffset** parameter is set, the frame offset equals the value of this parameter. Otherwise, the frame offset equals the value of the **gNodeBParam.FrameOffset** parameter.
- LTE cell frame offset: If the **CellFrameOffset.FrameOffset** parameter is set, the frame offset equals the value of this parameter. Otherwise, the frame offset equals the value of the **ENodeBFrameOffset.FddFrameOffset** parameter.

For details about the LTE and NR flexible frame offset function, see *Cell Management* in *5G RAN Feature Documentation*.

4.1.2 LTE FDD and NR SDL Dynamic Spectrum Sharing (Single LTE Cell)

4.1.2.1 Principles

The principles of LTE FDD and NR Flash Dynamic Spectrum Sharing in the case of an NR SDL cell are similar to those in the case of an NR FDD cell described in [4.1.1.1 Principles](#).

The difference lies in the function switch on the NR side, as an NR SDL cell works only in the downlink while an NR FDD cell works in both the uplink and downlink. To enable this function in the case of an NR SDL cell, turn on function switches as follows:

- LTE: Set the **SpectrumCloud.SpectrumCloudSwitch** parameter to **LTE_NR_SPECTRUM_SHR**.
- NR: Select the **LTE_NR_SDL_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudSwitch** parameter.

TA offsets cannot be configured in LTE FDD and NR SDL dynamic spectrum sharing scenarios.

4.1.2.2 Key Technologies

As an NR SDL cell works only in the downlink, some key technologies differ from the case of an NR FDD cell. [Table 4-16](#) lists these differences.

Table 4-16 Differences in key technologies

Key Technology	Difference	Reference
Coordinated scheduling of physical channel resources	When the NR cell is an NR SDL cell, only the downlink (not uplink) physical channel resource scheduling needs to be properly coordinated.	4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources
Flexible spectrum priority mode	None	4.1.1.2.2 Flexible Spectrum Priority Mode
Flexible power allocation mode	None	4.1.1.2.3 Flexible Power Allocation Mode
Co-carrier co-CPRI data	None	4.1.1.2.4 Co-carrier Co-CPRI Data
LTE key event assurance	None	4.1.1.2.5 LTE Key Event Assurance

4.1.2.3 Simultaneous Use with Other Functions

As an NR SDL cell works only in the downlink, simultaneous use with other functions differs from the case of an NR FDD cell. [Table 4-17](#) lists these differences.

Table 4-17 Differences in simultaneous use with other functions

Scenario of Simultaneous Use with Other Functions	Difference	Reference
Simultaneous use with NB-IoT deployed in LTE guard band mode	When the NR cell is an NR SDL cell, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be used together with NB-IoT deployed in LTE guard band mode.	4.1.1.3.1 LTE Functions
Simultaneous use with LTE uplink interference suppression	When the NR cell is an NR SDL cell, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be used together with LTE uplink interference suppression.	4.1.1.3.1 LTE Functions
Simultaneous use with LTE downlink semi-persistent scheduling	None	4.1.1.3.1 LTE Functions
Simultaneous use with WBB	None	4.1.1.3.1 LTE Functions
Simultaneous use with LTE uplink voice packet loss optimization	When the NR cell is an NR SDL cell, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be used together with LTE uplink voice packet loss optimization.	4.1.1.3.1 LTE Functions
Simultaneous use with DRX	None	4.1.1.3.1 LTE Functions
Simultaneous use with downlink CA	None	4.1.1.3.2 NR Functions
Simultaneous use with intra-base-station DL CoMP	None	4.1.1.3.2 NR Functions
Simultaneous use with CPRI compression	None	4.1.1.3.3 Common Functions Supported by Both LTE and NR
Simultaneous use with LTE and NR flexible frame offset	None	4.1.1.3.3 Common Functions Supported by Both LTE and NR

4.1.3 LTE FDD and NR FDD Dynamic Spectrum Sharing (Multiple LTE Cells)

Static Multiple Beam is one of the LTE FDD multi-sector solutions. Static Multiple Beam generates at least two beams by means of beamforming, with each beam serving a sector split cell, thereby maximizing network capacity. LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) enables the two or more LTE FDD sector split cells served by the beams generated by Static Multiple Beam to dynamically share the same spectrum resources with an NR FDD cell, improving spectrum utilization.

Static Multiple Beam can be used in Smart 8T8R scenarios, RuralLink antenna solution, LumiLink solution, and massive MIMO scenarios. In Smart 8T8R scenarios, it generates two beams, which correspond to two LTE FDD sector split cells. In RuralLink antenna solution and LumiLink solution, it generates three beams, which correspond to three LTE FDD sector split cells. In 16T16R massive MIMO scenarios, it generates two to three beams, which correspond to two to three LTE FDD sector split cells. In 32T32R massive MIMO scenarios, it generates two to four beams, which correspond to two to four LTE FDD sector split cells. For details about Static Multiple Beam in Smart 8T8R scenarios, see *Smart 8T8R (FDD)* in *eRAN Feature Documentation*. For details about Static Multiple Beam in RuralLink antenna solution, see *RuralLink Solution*. For details about Static Multiple Beam in LumiLink solution, see *LumiLink Solution*. For details about Static Multiple Beam in 16T16R or 32T32R massive MIMO scenarios, see *Massive MIMO Basics (FDD)* in *eRAN Feature Documentation*.

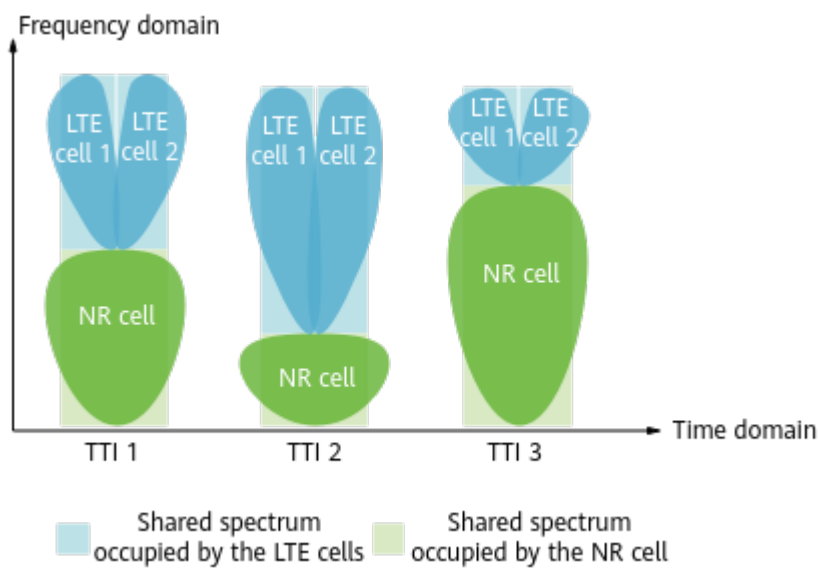
For ease of description, LTE and NR cells are used to refer to LTE FDD sector split cells served by the beams generated by Static Multiple Beam and NR FDD cells, respectively.

4.1.3.1 Principles

This function enables the multiple LTE FDD sector split cells served by the beams generated by Static Multiple Beam and an NR FDD cell to dynamically share time-frequency resources on a shared spectrum segment based on their traffic volume requirements. Figure 4-9 illustrates the working principles of this function. The LTE split cells and the NR cell must be configured with the same bandwidth, which can be 10 MHz, 15 MHz, or 20 MHz. The cells can share spectrum once they are added to associated spectrum sharing cell groups. Specifically:

- Time domain: Flash spectrum sharing is supported on a 1 ms basis, meaning that spectrum resources can be coordinated and scheduled every 1 ms.
- Frequency domain: Dynamic spectrum sharing is performed per RB. Spectrum resources are dynamically allocated to LTE and NR cells based on their traffic volume requirements. The same resources are allocated to the LTE split cells.

Figure 4-9 Working principles of flash dynamic spectrum sharing (multiple LTE cells)



This function is enabled by turning on function switches and configuring spectrum sharing cell groups. In addition, the LTE and NR cells sharing spectrum resources must have aligned radio frames and subframes. (The alignment is ensured by configuring frame offsets and TA offsets.) [Table 4-18](#) describes parameter settings for enabling this function.

Table 4-18 Activation of LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells)

Operation	LTE Parameter Setting	NR Parameter Setting
Turning on function switches on the LTE and NR sides	Set the SpectrumCloud.SpectrumCloudSwitch parameter to LTE_NR_SPECTRUM_SHR for each LTE split cell.	Select the LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter.

Operation	LTE Parameter Setting	NR Parameter Setting
Configuring spectrum sharing cell groups	Add all planned LTE split cells to an LTE spectrum sharing cell group by setting the SpectrumCloud.LteNrSpectrumShrCellGrpId and LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId parameters.	1. Configure an association between the LTE spectrum sharing cell group and an NR spectrum sharing cell group by setting the gNBDULteNrSpctShrCg.NrSpctShrCellGrpId and gNBDULteNrSpctShrCg.LteSpctShrCellGrpId parameters. 2. Add a planned NR cell to the NR spectrum sharing cell group by setting the NRDUCellSptCloud.NrDUCellId and NRDUCellSptCloud.NrSpctShrCellGrpId parameters.
Configuring frame offsets	Set the CellFrameOffset.FrameOffset or ENodeBFrameOffset.FddFrameOffset parameter. If both parameters are configured, the value of the CellFrameOffset.FrameOffset parameter is used. Ensure that the LTE split cells are configured with the same frame offset.	Set the gNodeBParam.FrameOffset or gNBFreqBandConfig.FrameOffset parameter. If both parameters are configured, the value of the gNBFreqBandConfig.FrameOffset parameter is used.
Configuring TA offsets	Set the CellFrameOffset.TaOffset parameter. Ensure that the LTE split cells are configured with the same TA offset.	Set the NRDUCell.TaOffset parameter.

You are advised to enable "cross-RAT configuration check in DSS scenarios" by selecting both the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter and the **DSS_CROSS_RAT_CFG_CHECK_SW** option of the **gNBOamParam.CfgInterceptPolicySw** parameter. This function checks whether the configurations of NR and LTE cells that are configured to be activated in the same spectrum sharing group meet all requirements.

4.1.3.2 Key Technologies

4.1.3.2.1 Coordinated Scheduling of Physical Channel Resources

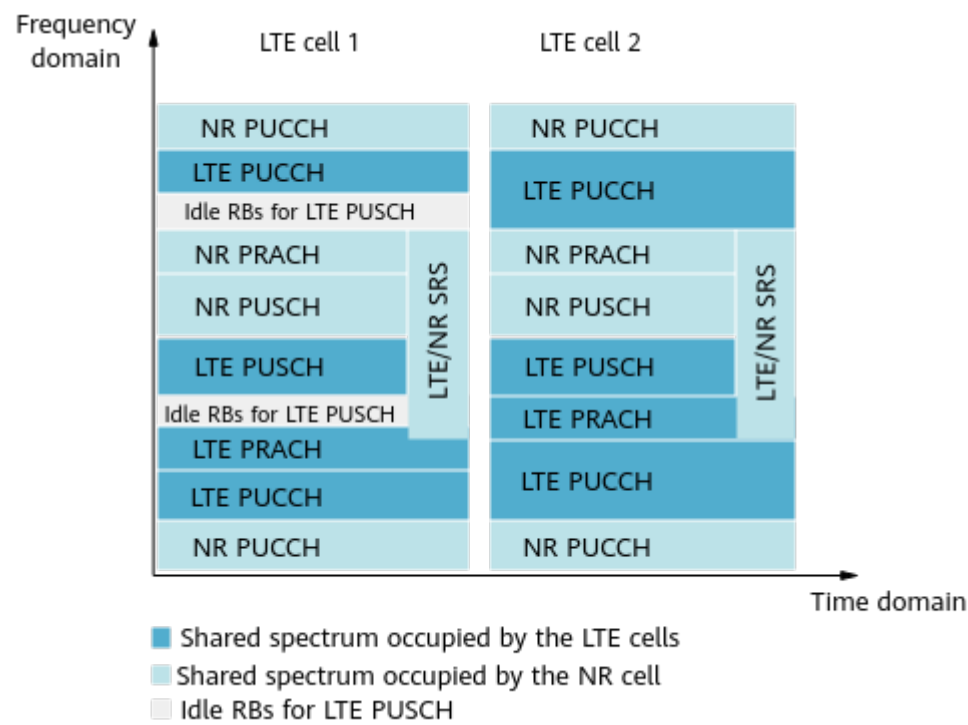
Some channels will conflict with each other during spectrum sharing between LTE and NR cells in a spectrum sharing cell group. Such conflicts can be avoided by

properly coordinating uplink and downlink physical channel resource scheduling, improving spectrum utilization.

Uplink Physical Channels

The LTE cells in this function use the same spectrum resources through spatial multiplexing. [Figure 4-10](#) shows the positions of uplink physical channels and signals on the spectrum shared by LTE and NR cells.

Figure 4-10 Positions of uplink physical channels and signals



The uplink physical channel resources of the LTE cells need to be merged before spectrum sharing with the NR cell. [Table 4-19](#) describes how the channel resources are merged.

Table 4-19 Uplink physical channel resource merging of the LTE cells

Channel/ Signal	Merging Mode
PUCCH	The combined PUCCH and PRACH resources of the LTE split cells are used as the total resources of these channels allocated to the LTE cells. If these resources are not fully occupied by the PUCCH and PRACH of the LTE cells, the idle resources can be used by the PUSCH of the LTE cells.
PRACH	
SRS	The largest amount of SRS time-frequency resources among the LTE split cells is considered as shared SRS resources allocated to the LTE and NR cells.

Channel/ Signal	Merging Mode
PUSCH	The largest amount of PUSCH resources among the LTE split cells is considered as the PUSCH resources allocated to these LTE cells.

Spectrum resources allocated to the NR cell are equal to the total shared spectrum resources minus the spectrum resources allocated to the LTE split cells. The spectrum resources allocated to the LTE split cells are the sum of the channel resources described in [Table 4-19](#).

For details about the coordinated scheduling of uplink physical channel resources, see [4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources](#). The LTE split cells must have consistent settings.

Downlink Physical Channels

The downlink physical channel resources of the LTE cells need to be merged before spectrum sharing with the NR cell. [Table 4-20](#) describes how the channel resources are merged.

Table 4-20 Downlink physical channel resource merging of the LTE cells

Channel/ Signal	Merging Mode
CRS	Choose the LTE performance preferred or NR performance preferred policy based on the network plan. • LTE performance preferred: The LTE cells must have different PCIs as well as different PCI mod 3 values. In this case, the LTE cells have two or three CRS patterns. The LTE_CRS_RATEMATCH_ALL_SYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter must be selected to enable CRS rate matching on all RBs in the symbols where LTE CRS appears, in order to eliminate the interference of the LTE cells on the NR cell. • NR performance preferred: The LTE cells must have different PCIs but the same PCI mod 3 value. In this case, the LTE cells have the same CRS pattern. The LTE_CRS_RATEMATCH_ALL_SYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter must be deselected to allow the gNodeB to perform CRS rate matching based on one LTE CRS pattern.
SS/PBCH	The SS/PBCH time-frequency domain positions of the LTE cells are the same.
PDCCH	The largest number of PDCCH symbols among the LTE split cells is used as the number of PDCCH symbols of these LTE cells.

Channel/ Signal	Merging Mode
PDSCH	The largest amount of PDSCH resources among the LTE split cells is used as the amount of PDSCH resources of these LTE cells.

For details about the coordinated scheduling of downlink physical channel resources, see [4.1.1.2.1 Coordinated Scheduling of Physical Channel Resources](#). The LTE split cells must have consistent settings.

4.1.3.2.2 Flexible Spectrum Priority Mode

The LTE split cells must be added to the same LTE spectrum sharing cell group, and must be configured with the same percentage of spectrum resources to be preferentially allocated and inter-RAT preferential guarantee policy for GBR services. For details about the flexible spectrum priority mode, see [4.1.1.2.2 Flexible Spectrum Priority Mode](#).

4.1.3.2.3 Flexible Power Allocation Mode

The LTE split cells must be configured with the same power allocation mode. For details about the power allocation mode, see [4.1.1.2.3 Flexible Power Allocation Mode](#).

4.1.3.2.4 Co-carrier Co-CPRI Data

The LTE split cells must be configured with the same co-carrier co-CPRI data policy. For details about co-carrier co-CPRI data, see [4.1.1.2.4 Co-carrier Co-CPRI Data](#).

4.1.3.2.5 LTE Key Event Assurance

The LTE key event assurance function preferentially ensures the capacity of LTE cells in NSA networking when the LTE cells are heavily loaded in scenarios such as concerts, sports events, and major holidays. This function is enabled by selecting the **LTE_BIG_EVENT_ASSURANCE_SW** option of the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrSwitch**, and LTE parameters **LteNrSpctShrCellGrp.LteBigEventUeNumThld** and **LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset** need to be configured to specify the LTE key event user number threshold and the threshold offset. The NR parameter **gNBDULteNrSpctShrCg.LteBigEventAssureUeNumThld** also needs to be configured, which specifies the user number threshold for LTE key event assurance for the NR cell. This function is applicable only to Option 3x architecture in NSA networking.

The LTE key event assurance function is triggered when the following conditions are met:

- Either of the LTE cells meets this condition: Number of LTE UEs in RRC connected mode > **LteNrSpctShrCellGrp.LteBigEventUeNumThld** + **LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset**

- Number of NR UEs in RRC connected mode $\leq$ **gNBDULteNrSpctShrCg.LteBigEventAssureUeNumThld**

When the LTE key event assurance function is triggered, the uplink and downlink data split modes for the NR cell change to MCG only. The NR cell enters the LTE key event assurance state, and all the shared spectrum resources are allocated to the LTE cells to ensure their capacity.

When the number of LTE UEs in RRC connected mode in each LTE cell becomes less than or equal to **LteNrSpctShrCellGrp.LteBigEventUeNumThld** minus **LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset**, the NR cell exits the LTE key event assurance state and enters the non-LTE-key-event-assurance state.

To protect the NR cell from ping-pong switchovers between these two states, a 5-minute protection timer is introduced to the NR cell.

4.1.3.3 Simultaneous Use with Other Functions

4.1.3.3.1 LTE Functions

Simultaneous Use with LTE Downlink Semi-persistent Scheduling

When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is used together with LTE downlink semi-persistent scheduling, the upper limit on the proportion of RBs that can be allocated for LTE downlink semi-persistent scheduling is configurable (through the **SpectrumCloud.DlSpsRestrictRatio** parameter). It is recommended that this parameter be set to the same value, and that the **SpsSchSwitch** option of the **CellAlgoSwitch.DlSchSwitch** parameter be set consistently for the LTE cells. This proportion together with the percentage of spectrum resources to be preferentially allocated to LTE for MBB services (specified by the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio**) determines the proportion of RBs allocated for LTE downlink semi-persistent scheduling as follows: The proportion of RBs allocated for LTE downlink semi-persistent scheduling = $(\text{SpectrumCloud.DlSpsRestrictRatio} \times \text{LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio}/100)\%$.

Simultaneous Use with LTE PRACH Frequency-Domain Position Adaptation

With LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) enabled, more resources are allocated to LTE cells and fewer resources are allocated to the NR cell if PRACH frequency-domain position adaptation is enabled for only some LTE cells and the PRACH frequency-domain positions in LTE cells are different. Therefore, it is recommended that the **PrachFreqAdjSwitch** option of the **CellAlgoSwitch.RachAlgoSwitch** parameter be set consistently for the LTE cells.

Simultaneous Use with LTE Uplink Semi-persistent Scheduling and LTE TTI Bundling

When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled, uplink semi-persistent scheduling is enabled in only some LTE cells and takes effect, and TTI bundling is enabled only in the rest LTE cells and takes effect, the resources for semi-persistent scheduling and those for TTI bundling are

respectively merged for these LTE cells, and consequently the resources for the NR cell decrease. Therefore, it is recommended that the LTE cells use the same settings of the **TtiBundlingSwitch** and **SpsSchSwitch** options of the **CellAlgoSwitch.UlSchSwitch** parameter.

Simultaneous Use with WBB

When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is used together with WBB, it is recommended that the **WBB_MBB_CONTROL_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter be selected for each LTE cell. When this option is selected, the eNodeB allocates resources to LTE WBB and MBB services based on the actually available LTE spectrum resources. The percentages of resources actually allocated to MBB services comply with the settings of the **CellWttxCfg.MbbUserDlPrbUpLimit** and **CellWttxCfg.MbbUserUlPrbUpLimit** parameters; the percentages of resources actually allocated to WBB services comply with the settings of the **CellWttxCfg.WbbUserDlPrbUpLimit** and **CellWttxCfg.WbbUserUlPrbUpLimit** parameters. For details about WBB, see *WBB in eRAN Feature Documentation*.

To ensure spectral efficiency, it is recommended that the following equations be met: **CellWttxCfg.MbbUserDlPrbUpLimit** + **CellWttxCfg.WbbUserDlPrbUpLimit** = 100%, and **CellWttxCfg.MbbUserUlPrbUpLimit** + **CellWttxCfg.WbbUserUlPrbUpLimit** = 100%.

Simultaneous Use with LTE Uplink Voice Packet Loss Optimization

For details about this scenario, see [4.1.1.3.1 LTE Functions](#).

Simultaneous Use with DRX

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with DRX, the following configurations are recommended for the involved LTE FDD cells that prevent the DRX On Duration period from being extended due to MBSFN subframes, NR CSI-RS, or NR TRS:

- Select the **DSS_GAP_POS_OPT_SW** option of the **CellDrxCfg.DrxAlgoExtSwitch** parameter for each cell to optimize the position of gap-assisted measurements in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. This configuration mitigates the increase in the service drop rate caused by an overlap between the DRX On Duration period and gap-assisted measurements.
- Select the **DSS_VOICE_NSA_SA_OD_POS_OPT_SW** option of the **CellDrxCfg.DrxAlgoExtSwitch** parameter for each cell to optimize the position of the DRX On Duration period for NSA- or SA-capable voice service UEs in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, and set the **CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod** parameter to the same value in the range of [40MS, 80MS, 160MS, 320MS] for all the involved cells to specify the interval at which aperiodic CQI reporting is triggered for these UEs in spectrum sharing scenarios. These configurations mitigate the increase in the service drop rate caused by an overlap between the DRX On Duration

period and gap-assisted measurements, NR CSI-RS, NR TRS, or MBSFN subframes.

4.1.3.3.2 NR Functions

Simultaneous Use with Downlink CA

When downlink intra-FR inter-band CA or downlink intra-band CA is enabled after LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled, the NR PUCCH occupies more RBs, which leads to RB resource conflicts with the LTE PUCCH. In this case, the LTE PUCCH RB configuration needs to be modified using the parameters listed in Table 4-21. Among these parameters, it is recommended that **CellRbReserve.RbRsvStartIndex** and **CellRbReserve.RbRsvEndIndex** be set to the indexes of symmetrical RBs adjacent to the NR PUCCH. The LTE cells must have the same PUCCH RB configuration.

Table 4-21 Parameters for configuring the LTE PUCCH RBs

Parameter	Function
CellRbReserve.RbRsvMode	Specifies the RB reservation mode.
CellRbReserve.RbRsvType	Specifies the type of physical-layer reserved PRBs.
CellRbReserve.RbRsvStartIndex	Specifies the start index of the LTE PUCCH PRBs at the lower end of the frequency domain.
CellRbReserve.RbRsvEndIndex	Specifies the end index of the LTE PUCCH PRBs at the upper end of the frequency domain.

NOTICE

If the LTE or NR cell is not deactivated before the preceding configurations are performed:

- When the **FORBID_PARA_CFG_VERIFY_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter is deselected, a command to reconfigure **Cell.CrsPortNum**, **Cell.TxRxMode**, **CellRbReserve.RbRsvMode**, **CellRbReserve.RbRsvStartIndex**, or **CellRbReserve.RbRsvEndIndex** will fail to be executed.
- When the **FORBID_PARA_CFG_VERIFY_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter is selected, downlink intra-FR inter-band CA may fail to take effect, or the NR cell may become out of service.

Evaluate the preceding risks and determine whether to deactivate the LTE and NR cells before performing the preceding configurations.

When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with downlink intra-band CA, the **LNR_PUCCH_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter must be selected.

Simultaneous Use with Intra-Base-Station DL CoMP

For details about the configuration requirements in this scenario, see [4.1.1.3.2 NR Functions](#).

4.1.3.3.3 Common Functions Supported by Both LTE and NR

Simultaneous Use with CPRI Compression

The LTE and NR cells with both LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) and CPRI compression enabled must have the same CPRI compression ratio. For details about the configuration requirements, see [4.1.1.3.3 Common Functions Supported by Both LTE and NR](#).

Simultaneous Use with LTE and NR Flexible Frame Offset

In LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) scenarios, when the LTE and NR flexible frame offset function is enabled, the frame offset of the NR cell must be 3 ms later than that of each LTE cell. For details about the configuration requirements, see [4.1.1.3.3 Common Functions Supported by Both LTE and NR](#).

4.2 Network Analysis

4.2.1 Benefits

This function increases the downlink cell throughput. The benefits provided by this function vary depending on the networking type and the number of LTE cells. [Table 4-22](#) and [Table 4-23](#) describe the benefits provided by LTE FDD and NR Flash Dynamic Spectrum Sharing (single LTE cell) when compared with a typical static refarming solution. [Table 4-24](#) and [Table 4-25](#) describe the benefits provided by LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) when compared with a typical static refarming solution.

Table 4-22 Downlink cell peak throughput gain in NSA networking (single LTE cell)

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
2T	10 MHz	5 MHz	5 MHz	174%–195%	166%–174%
	15 MHz	10 MHz	5 MHz	128%–143%	259%–302%
	20 MHz	10 MHz	10 MHz	172%–192%	161%–189%
4T	10 MHz	5 MHz	5 MHz	174%–195%	155%–167%
	15 MHz	10 MHz	5 MHz	128%–143%	241%–257%

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
	20 MHz	10 MHz	10 MHz	172%–192%	150%–163%
8T	10 MHz	5 MHz	5 MHz	170%–190%	150%–167%
	15 MHz	10 MHz	5 MHz	125%–140%	236%–257%
	20 MHz	10 MHz	10 MHz	170%–190%	145%–163%
16T	10 MHz	5 MHz	5 MHz	165%–185%	140%–167%
	15 MHz	10 MHz	5 MHz	122%–137%	216%–257%
	20 MHz	10 MHz	10 MHz	165%–185%	138%–163%
32T	10 MHz	5 MHz	5 MHz	160%–180%	130%–167%
	15 MHz	10 MHz	5 MHz	118%–133%	195%–257%
	20 MHz	10 MHz	10 MHz	160%–180%	130%–163%
a: LTE bandwidth and NR bandwidth in a static refarming solution when this function is not enabled b: LTE cell gain and NR cell gain provided by this function compared with a static refarming solution. If LTE and NR share spectrum power while sharing spectrum, the NR cell gain decreases by 10% to 20%, and the LTE cell gain is not affected. If LTE and NR use separate spectrum power while sharing spectrum, the LTE and NR cell gains are not affected.					

Table 4-23 Downlink cell peak throughput gain in SA networking and NSA and SA hybrid networking (single LTE cell)

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
2T	10 MHz	5 MHz	5 MHz	170%–195%	166%–174%
	15 MHz	10 MHz	5 MHz	125%–143%	259%–302%
	20 MHz	10 MHz	10 MHz	168%–192%	161%–189%
4T	10 MHz	5 MHz	5 MHz	170%–195%	155%–167%
	15 MHz	10 MHz	5 MHz	125%–143%	241%–257%
	20 MHz	10 MHz	10 MHz	168%–192%	150%–163%
8T	10 MHz	5 MHz	5 MHz	170%–190%	150%–167%
	15 MHz	10 MHz	5 MHz	125%–140%	236%–257%

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
	20 MHz	10 MHz	10 MHz	165%–190%	145%–163%
16T	10 MHz	5 MHz	5 MHz	160%–185%	140%–167%
	15 MHz	10 MHz	5 MHz	117%–137%	216%–257%
	20 MHz	10 MHz	10 MHz	156%–185%	138%–163%
32T	10 MHz	5 MHz	5 MHz	150%–180%	130%–167%
	15 MHz	10 MHz	5 MHz	108%–133%	195%–257%
	20 MHz	10 MHz	10 MHz	147%–180%	130%–163%
a: LTE bandwidth and NR bandwidth in a static refarming solution when this function is not enabled b: LTE cell gain and NR cell gain provided by this function compared with a static refarming solution. If LTE and NR share spectrum power while sharing spectrum, the NR cell gain decreases by 10% to 20%, and the LTE cell gain is not affected. If LTE and NR use separate spectrum power while sharing spectrum, the LTE and NR cell gains are not affected.					

Table 4-24 Downlink cell peak throughput gain in NSA networking (multiple LTE cells)

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
8T LTE performance preferred	40 MHz	20 MHz	20 MHz	92%–100%	165%–190%
	10 MHz	5 MHz	5 MHz	170%–190%	90%–100%
	15 MHz	10 MHz	5 MHz	125%–140%	141%–154%
	20 MHz	10 MHz	10 MHz	170%–190%	90%–100%
8T NR performance preferred	10 MHz	5 MHz	5 MHz	136%–152%	150%–167%
	15 MHz	10 MHz	5 MHz	100%–112%	236%–257%
	20 MHz	10 MHz	10 MHz	136%–152%	145%–163%
16T LTE performance preferred	10 MHz	5 MHz	5 MHz	160%–185%	88%–100%
	15 MHz	10 MHz	5 MHz	117%–137%	134%–154%
	20 MHz	10 MHz	10 MHz	160%–185%	88%–100%

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
16T NR performance preferred	10 MHz	5 MHz	5 MHz	129%–144%	140%–167%
	15 MHz	10 MHz	5 MHz	95%–106%	216%–257%
	20 MHz	10 MHz	10 MHz	129%–144%	138%–163%
32T LTE performance preferred	10 MHz	5 MHz	5 MHz	150%–180%	85%–100%
	15 MHz	10 MHz	5 MHz	108%–133%	127%–154%
	20 MHz	10 MHz	10 MHz	150%–180%	85%–100%
32T NR performance preferred	10 MHz	5 MHz	5 MHz	122%–136%	130%–167%
	15 MHz	10 MHz	5 MHz	90%–100%	195%–257%
	20 MHz	10 MHz	10 MHz	122%–136%	130%–163%
a: LTE bandwidth and NR bandwidth in a static refarming solution when this function is not enabled b: LTE cell gain and NR cell gain provided by this function compared with a static refarming solution. If LTE and NR share spectrum power while sharing spectrum, the NR cell gain decreases by 10% to 20%, and the LTE cell gain is not affected. If LTE and NR use separate spectrum power while sharing spectrum, the LTE and NR cell gains are not affected.					

Table 4-25 Downlink cell peak throughput gain in SA networking and NSA and SA hybrid networking (multiple LTE cells)

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
8T LTE performance preferred	40 MHz	20 MHz	20 MHz	92%–100%	165%–190%
	10 MHz	5 MHz	5 MHz	170%–190%	90%–100%
	15 MHz	10 MHz	5 MHz	125%–140%	141%–154%
	20 MHz	10 MHz	10 MHz	170%–190%	90%–100%
8T NR performance preferred	10 MHz	5 MHz	5 MHz	136%–152%	150%–167%
	15 MHz	10 MHz	5 MHz	100%–112%	236%–257%
	20 MHz	10 MHz	10 MHz	136%–152%	145%–163%

Cell TX Mode	Total Bandwidth	LTE Bandwidth ^a	NR Bandwidth ^a	LTE Cell Downlink Peak Throughput Gain ^b	NR Cell Downlink Peak Throughput Gain ^b
16T LTE performance preferred	10 MHz	5 MHz	5 MHz	160%–185%	88%–100%
	15 MHz	10 MHz	5 MHz	117%–137%	134%–154%
	20 MHz	10 MHz	10 MHz	160%–185%	88%–100%
16T NR performance preferred	10 MHz	5 MHz	5 MHz	129%–144%	140%–167%
	15 MHz	10 MHz	5 MHz	95%–106%	216%–257%
	20 MHz	10 MHz	10 MHz	129%–144%	138%–163%
32T LTE performance preferred	10 MHz	5 MHz	5 MHz	150%–180%	85%–100%
	15 MHz	10 MHz	5 MHz	108%–133%	127%–154%
	20 MHz	10 MHz	10 MHz	150%–180%	85%–100%
32T NR performance preferred	10 MHz	5 MHz	5 MHz	122%–136%	130%–167%
	15 MHz	10 MHz	5 MHz	90%–100%	195%–257%
	20 MHz	10 MHz	10 MHz	122%–136%	130%–163%
a: LTE bandwidth and NR bandwidth in a static refarming solution when this function is not enabled b: LTE cell gain and NR cell gain provided by this function compared with a static refarming solution. If LTE and NR share spectrum power while sharing spectrum, the NR cell gain decreases by 10% to 20%, and the LTE cell gain is not affected. If LTE and NR use separate spectrum power while sharing spectrum, the LTE and NR cell gains are not affected.					

Specific gains can only be achieved when certain conditions are met.

- The LTE cell gain can only be achieved when all of the following conditions are met:
 - The LTE cell is fully loaded and no online UEs exist in the NR cell.
 - The SSB period for the NR cell (specified by the NR parameter **NRDUCell.SsbPeriod**) is set to **MS20(20)** or a larger value.
 - The NR cell SIB1 period (specified by the NR parameter **NRDUCell.Sib1Period**) is set to **MS40(40)**.
 - The NR cell uses a single SSB beam.
 - The LTE cell works in TM4.
- The NR cell gain can only be achieved when all of the following conditions are met:
 - The NR cell is fully loaded and no online UEs exist in the LTE cell.

- The SIB1 period for the NR cell is set to **MS40(40)** or a larger value. In contrast, the SIB1 period is set to **MS20(20)** when the static refarming solution is used.
- The TRS period for the NR cell is set to **MS80(80)**. In contrast, the TRS period is set to **MS20(20)** when the static refarming solution is used.
- The NR cell uses a single SSB beam.

If one of the preceding conditions is not met, actual gains achieved will be lower than those listed in the tables above, but no negative impacts will occur.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- LTE-to-NR spectrum sharing has the following impacts on LTE networks:
 - NR synchronization signals, the PBCH, reference channels, and system information interfere with LTE. The interference leads to LTE wideband quality-related indicators being inaccurately measured. The indicators include the signal to interference plus noise ratio (SINR), reference signal received quality (RSRQ), wideband channel quality indicator (CQI), precoding matrix indication (PMI), and rank indication (RI). Consequently:
 - The average cell uplink and downlink initial block error rate (IBLER) and residual block error rate (RBLER) increase.
- Average cell uplink IBLER = $(L.Traffic.UL.SCH.QPSK.ErrTB.Ibler + L.Traffic.UL.SCH.16QAM.ErrTB.Ibler + L.Traffic.UL.SCH.64QAM.ErrTB.Ibler + L.Traffic.UL.SCH.256QAM.ErrTB.Ibler) / (L.Traffic.UL.SCH.QPSK.TB + L.Traffic.UL.SCH.16QAM.TB + L.Traffic.UL.SCH.64QAM.TB + L.Traffic.UL.SCH.256QAM.TB)$. [Table 4-26](#) lists the related counters.
- Average cell downlink IBLER = $(L.Traffic.DL.SCH.QPSK.ErrTB.Ibler + L.Traffic.DL.SCH.16QAM.ErrTB.Ibler + L.Traffic.DL.SCH.64QAM.ErrTB.Ibler + L.Traffic.DL.SCH.256QAM.ErrTB.Ibler) / (L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB)$. [Table 4-27](#) lists the related counters.
- Average cell uplink RBLER = $(L.Traffic.UL.SCH.QPSK.ErrTB.Rbler + L.Traffic.UL.SCH.16QAM.ErrTB.Rbler + L.Traffic.UL.SCH.64QAM.ErrTB.Rbler + L.Traffic.UL.SCH.256QAM.ErrTB.Rbler) / (L.Traffic.UL.SCH.QPSK.TB + L.Traffic.UL.SCH.16QAM.TB + L.Traffic.UL.SCH.64QAM.TB + L.Traffic.UL.SCH.256QAM.TB)$. [Table 4-28](#) lists the related counters.
- Average cell downlink RBLER = $(L.Traffic.DL.SCH.QPSK.ErrTB.Rbler + L.Traffic.DL.SCH.16QAM.ErrTB.Rbler + L.Traffic.DL.SCH.64QAM.ErrTB.Rbler + L.Traffic.DL.SCH.QPSK.TB) / (L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB)$. [Table 4-29](#) lists the related counters.

Table 4-26 Counters used for measuring the average cell uplink IBLER on the LTE side

Counter ID	Counter Name
1526728186	L.Traffic.UL.SCH.QPSK.ErrTB.Ibler
1526728188	L.Traffic.UL.SCH.16QAM.ErrTB.Ibler
1526728190	L.Traffic.UL.SCH.64QAM.ErrTB.Ibler
1526749528	L.Traffic.UL.SCH.256QAM.ErrTB.Ibler
1526727366	L.Traffic.UL.SCH.QPSK.TB
1526727367	L.Traffic.UL.SCH.16QAM.TB
1526727368	L.Traffic.UL.SCH.64QAM.TB
1526749524	L.Traffic.UL.SCH.256QAM.TB

Table 4-27 Counters used for measuring the average cell downlink IBLER on the LTE side

Counter ID	Counter Name
1526728180	L.Traffic.DL.SCH.QPSK.ErrTB.Ibler
1526728182	L.Traffic.DL.SCH.16QAM.ErrTB.Ibler
1526728184	L.Traffic.DL.SCH.64QAM.ErrTB.Ibler
1526739660	L.Traffic.DL.SCH.256QAM.ErrTB.Ibler
1526727354	L.Traffic.DL.SCH.QPSK.TB
1526727355	L.Traffic.DL.SCH.16QAM.TB
1526727356	L.Traffic.DL.SCH.64QAM.TB
1526739656	L.Traffic.DL.SCH.256QAM.TB

Table 4-28 Counters used for measuring the average cell uplink RBLER on the LTE side

Counter ID	Counter Name
1526728187	L.Traffic.UL.SCH.QPSK.ErrTB.Rbler
1526728189	L.Traffic.UL.SCH.16QAM.ErrTB.Rbler
1526728191	L.Traffic.UL.SCH.64QAM.ErrTB.Rbler
1526749530	L.Traffic.UL.SCH.256QAM.ErrTB.Rbler

Counter ID	Counter Name
1526727366	L.Traffic.UL.SCH.QPSK.TB
1526727367	L.Traffic.UL.SCH.16QAM.TB
1526727368	L.Traffic.UL.SCH.64QAM.TB
1526749524	L.Traffic.UL.SCH.256QAM.TB

Table 4-29 Counters used for measuring the average cell downlink RBLER on the LTE side

Counter ID	Counter Name
1526728181	L.Traffic.DL.SCH.QPSK.ErrTB.Rbler
1526728183	L.Traffic.DL.SCH.16QAM.ErrTB.Rbler
1526728185	L.Traffic.DL.SCH.64QAM.ErrTB.Rbler
1526739661	L.Traffic.DL.SCH.256QAM.ErrTB.Rbler
1526727354	L.Traffic.DL.SCH.QPSK.TB
1526727355	L.Traffic.DL.SCH.16QAM.TB
1526727356	L.Traffic.DL.SCH.64QAM.TB
1526739656	L.Traffic.DL.SCH.256QAM.TB

- The access success rate, handover success rate, and RRC connection reestablishment success rate decrease.

Access success rate = **RRC Setup Success Rate**

Handover success rate = $(L.HHO.IntraeNB.IntraFreq.ExecSuccOut + L.HHO.IntraeNB.InterFreq.ExecSuccOut + L.HHO.IntereNB.IntraFreq.ExecSuccOut + L.HHO.IntereNB.InterFreq.ExecSuccOut) / (L.HHO.IntraeNB.IntraFreq.ExecAttOut + L.HHO.IntraeNB.InterFreq.ExecAttOut + L.HHO.IntereNB.IntraFreq.ExecAttOut + L.HHO.IntereNB.InterFreq.ExecAttOut)$. [Table 4-30](#) lists the related counters.

RRC connection reestablishment success rate = $L.RRC.ReEst.Succ / L.RRC.ReEst.Att$. [Table 4-31](#) lists the related counters. Selecting the **SSB_ADJ_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** parameter will further decrease the values of these counters.

Table 4-30 Counters used for measuring the handover success rate

Counter ID	Counter Name
1526726997	L.HHO.IntraeNB.IntraFreq.ExecSuccOut
1526727000	L.HHO.IntraeNB.InterFreq.ExecSuccOut
1526727003	L.HHO.IntereNB.IntraFreq.ExecSuccOut
1526727006	L.HHO.IntereNB.InterFreq.ExecSuccOut
1526726996	L.HHO.IntraeNB.IntraFreq.ExecAttOut
1526726999	L.HHO.IntraeNB.InterFreq.ExecAttOut
1526727002	L.HHO.IntereNB.IntraFreq.ExecAttOut
1526727005	L.HHO.IntereNB.InterFreq.ExecAttOut

Table 4-31 Counters used for measuring the RRC connection reestablishment success rate

Counter ID	Counter Name
1526727086	L.RRC.ReEst.Succ
1526727085	L.RRC.ReEst.Att

- The bearer service drop rate increases.
Bearer service drop rate = **Service Drop Rate**
- The PDCCH DTX proportion increases.

$$\text{PDCCH DTX proportion} = \frac{(\text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl1} + \text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl2} + \text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl4} + \text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl8})}{(\text{L.ChMeas.PDCCH.AggLvl1Num} + \text{L.ChMeas.PDCCH.AggLvl2Num} + \text{L.ChMeas.PDCCH.AggLvl4Num} + \text{L.ChMeas.PDCCH.AggLvl8Num})}$$

Table 4-32 lists the related counters.

Table 4-32 Counters related to PDCCH DTX

Counter ID	Counter Name
526728664	L.ChMeas.PDCCH.DL.DTXNum.AggLvl1

Counter ID	Counter Name
1526728665	L.ChMeas.PDCCH.DL.DTXNum.AggLvl 2
1526728666	L.ChMeas.PDCCH.DL.DTXNum.AggLvl 4
1526728667	L.ChMeas.PDCCH.DL.DTXNum.AggLvl 8
1526728668	L.ChMeas.PDCCH.AggLvl1Num
1526728669	L.ChMeas.PDCCH.AggLvl2Num
1526728670	L.ChMeas.PDCCH.AggLvl4Num
1526728671	L.ChMeas.PDCCH.AggLvl8Num

- The spectrum resources available for LTE are reduced, resulting in a fluctuation in the average uplink and downlink throughput of the LTE cell, a decrease in the average uplink and downlink throughput of UEs, and an increase in the uplink and downlink voice packet loss rates in the cell.
 - Average uplink cell throughput = **Cell Uplink Average Throughput**
 - Average downlink cell throughput = **Cell Downlink Average Throughput**
 - Average uplink UE throughput = **User Uplink Average Throughput**
 - Average downlink UE throughput = **User Downlink Average Throughput**
 - Uplink voice packet loss rate in a cell = **Uplink Packet Loss Rate (VoIP)**
 - Downlink voice packet loss rate in a cell = **Downlink Packet Loss Rate (VoIP)**
- NR-to-LTE spectrum sharing has the following impacts on NR networks:

The spectrum resources available for NR are reduced, resulting in a fluctuation in the average uplink and downlink throughput of the NR cell, and a decrease in the average uplink and downlink throughput of UEs.

 - Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
 - Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**
 - Average uplink UE throughput = **User Uplink Average Throughput (DU)**
 - Average downlink UE throughput = **User Downlink Average Throughput (DU)**

 NOTE

LTE synchronization signals, the PBCH, reference channels, and system information do not interfere with NR. As a result, NR wideband quality-related indicators are not affected.

A change in the CRS bandwidth or power in a neighboring LTE cell will affect performance counters of the NR cell in both the uplink and downlink, such as the IBLER, RBLER, average MCS index, average cell throughput, and average UE throughput.

- Measurement for counters related to uplink interference is affected because available RBs are allocated to LTE and NR in real time when the two RATs share spectrum dynamically. As a result, the values of the following counters may change:
 - **L.UL.Interference.Avg**
 - **L.UL.Interference.Max**
 - **L.UL.Interference.Avg.PRBO** to **L.UL.Interference.Avg.PRBO99**
 - **N.UL.NI.Avg**
 - **N.UL.NI.Max**
 - **N.UL.NI.Avg.PRBO** to **N.UL.NI.Avg.PRBO105**
- This function involves the estimation and allocation of spectrum resources on both the LTE and NR sides, and therefore has the following impacts on the two sides:
 - Increased board user-plane CPU usage
 - Average user-plane CPU usage of an eNodeB board:
L.Traffic.Board.UPlane.CPULoad.AVG
 - Average user-plane CPU usage of a gNodeB board:
VS.NRBoard.UPlane.CPULoad.Avg
 - Increased ping delay
 - More messages exchanged between the LTE and NR sides. When the number of cells or UEs served by a board is excessively large, LTE and NR cells enter the congestion state. In this situation, resources are statically allocated to the LTE and NR cells, improving feature reliability.
- After this function is enabled on the LTE and NR sides, if MBSFN subframes are configured on TM9-dedicated carriers, the control format indicator (CFI) value and control channel element (CCE) usage change as the CFI range differs between MBSFN subframes and common subframes. The CCE usage can be calculated using the following formula: CCE usage = $(L.ChMeas.CCE.CommUsed + L.ChMeas.CCE.ULUsed + L.ChMeas.CCE.DLUsed) / L.ChMeas.CCE.Avail$.
- On the NR side, if this function is enabled, fewer resources are available for the PDCCH due to avoidance of conflicts between the PDCCH and SSB. As a result, the CCE allocation success rate and the user-perceived throughput decrease in both the uplink and downlink.
- If the LTE key event assurance function is enabled and the NR cell enters the LTE heavy traffic state, the impacts are as follows:
 - The values of the **N.PDCP.Vol.UL.X2U.TrfPDU.Rx** and **N.PDCP.Vol.DL.X2U.TrfPDU.Tx** counters increase.

- The values of the **L.Thrp.bits.UL.RLC.Send** and **L.Thrp.bits.DL.RLC.Send** counters increase.
- If the co-carrier co-CPRI data function is enabled and the UE is located near the cell center, the gNodeB implements time synchronization between the LTE and NR networks. This has the following impacts:
 - On the LTE side: The values of counters **L.TA.UE.Index0** to **L.TA.UE.Index6** slightly increase, and the values of counters **L.RA.TA.UE.Index0** to **L.RA.TA.UE.Index11** slightly change.
 - On the NR side:
 - The values of counters **N.RA.TA.UE.Index0** to **N.RA.TA.UE.Index12** slightly increase.
 - If CA is also enabled, the average uplink throughput (**Cell Uplink Average Throughput (DU)**) of both the PCell and SCells fluctuates.
 - If super uplink is also enabled, the average uplink throughput (**Cell Uplink Average Throughput (DU)**) of the SUL cell fluctuates.
- If the LTE TA offset (specified by the **CellFrameOffset.TaOffset** parameter) is changed from **0TS** to another value, the values of counters **L.TA.UE.Index0** to **L.TA.UE.Index15** and the values of counters **L.RA.TA.UE.Index0** to **L.RA.TA.UE.Index11** slightly fluctuate.
- If the paging time sequence adjustment function is disabled and NR paging and LTE paging collide in timing, the paging success rate will decrease. If the function is enabled, NR paging will give way to LTE paging, and the paging success rate will increase.
- If the four-beam function is enabled, the impacts are as follows:
 - The measured values of wideband LTE performance indicators related to quality decrease, for example, the SINR, RSRQ, wideband CQI, PMI, and RI. The BLER increases. As a result, the **Service Drop Rate** increases, and the values of the following indicators decrease:
 - RRC connection setup success rate = **RRC Setup Success Rate**
 - Outgoing intra-frequency handover success rate = **Intra-Frequency Handover Out Success Rate**
 - Outgoing inter-frequency handover success rate = **Inter-Frequency Handover Out Success Rate**
 - Random access success rate = **L.RA.Dedicate.Msg3Rcv/L.RA.Dedicate.Att**
 - Average downlink UE throughput = **User Downlink Average Throughput**
 - The values of the following NR indicators decrease:
 - RRC connection setup success rate = **RRC Setup Success Rate (CU)**
 - QoS flow setup success rate of all services = **QoS Flow Setup Success Rate (CU)**
 - Handover success rate = **(L.HHO.IntraeNB.IntraFreq.ExecSuccOut + L.HHO.IntraeNB.InterFreq.ExecSuccOut +**

$$\frac{\text{L.HHO.IntereNB.IntraFreq.ExecSuccOut} + \text{L.HHO.IntereNB.InterFreq.ExecSuccOut}}{(\text{L.HHO.IntraeNB.IntraFreq.ExecAttOut} + \text{L.HHO.IntraeNB.InterFreq.ExecAttOut} + \text{L.HHO.IntereNB.IntraFreq.ExecAttOut} + \text{L.HHO.IntereNB.InterFreq.ExecAttOut})}$$

- Random access success rate = $\frac{\text{N.RA.Contention.Resolution.Succ}}{\text{N.RA.Contention.Att}}$
- Average downlink UE throughput = **User Downlink Average Throughput (DU)**

The value of the following NR indicator increases:

$$\text{CCE allocation failure rate} = 100 \times (1 - (\frac{\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num}}{\text{N.CCE.DL.AllocReq.Num}}))$$

- When LTE uplink voice packet loss optimization is enabled and the conditions for this function to take effect are met in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the resources allocated to LTE increase, which may change the uplink user-perceived rate on the LTE or NR side; in addition, the values of the following indicators decrease:
 - VoLTE uplink packet loss rate = **Uplink Packet Loss Rate (VoIP)**
 - Uplink IBLER of uplink voice services = $\frac{(\text{L.Traffic.UL.SCH.QPSK.ErrTB.Ibler.QCI.1} + \text{L.Traffic.UL.SCH.16QAM.ErrTB.Ibler.QCI.1} + \text{L.Traffic.UL.SCH.64QAM.ErrTB.Ibler.QCI.1})}{(\text{L.Traffic.UL.SCH.QPSK.TB.QCI.1} + \text{L.Traffic.UL.SCH.16QAM.TB.QCI.1} + \text{L.Traffic.UL.SCH.64QAM.TB.QCI.1})} \times 100\%$
- The uplink IBLER of uplink voice services slightly decreases.
 - **User Uplink Average Throughput** = $\frac{(\text{L.Thrp.bits.UL} - \text{L.Thrp.bits.UE.UL.SmallPkt})}{\text{L.Thrp.Time.UE.UL.RmvSmallPkt}}$
 - Average uplink UE throughput = **User Uplink Average Throughput (DU)**
- If LTE and NR SRS resource allocation optimization is enabled, there may be a short period of time when NR UEs do not periodically report SRS or have SRS subframes migrated. During this period, the uplink and downlink user-perceived rates decrease.
- When this function is used together with DRX, the impacts vary as follows:
 - If the **DSS_GAP_POS_OPT_SW** option of the **CellDrxPara.DrxAlgoExtSwitch** parameter is selected, there will be fewer service drops caused by an overlap between gap-assisted measurements and the DRX On Duration period in spectrum sharing scenarios.
 - If the **DSS_VOICE_NSA_SA_OD_POS_OPT_SW** option of the **CellDrxPara.DrxAlgoExtSwitch** parameter is selected, there will be fewer service drops caused by subframe unavailability during the DRX On Duration period in spectrum sharing scenarios and the downlink rates of voice service UEs may slightly decrease.

In this case, for NSA- or SA-capable voice service UEs in spectrum sharing scenarios, a smaller effective value of the **CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod** parameter leads to less

frequent aperiodic CQI reporting and less uplink overhead for aperiodic CQI reporting; a larger effective value of this parameter leads to the opposite effects.

- If the "LTE and NR 3-ms frame offset difference" function is enabled, the arrangement of subframes prioritized for one RAT changes and MBSFN subframes cannot be configured. As a result, the scheduling behavior in LTE and NR cells engaged in LTE FDD and NR Flash Dynamic Spectrum Sharing may change, leading to changes in the average UE and cell throughputs in the uplink and downlink for both LTE and NR. In addition, this function affects the benefits brought by functions that depend on MBSFN subframes.

4.3 Requirements

4.3.1 Licenses

LTE FDD and NR Flash Dynamic Spectrum Sharing requires both feature and capacity licenses. [Table 4-33](#) and [Table 4-34](#) list the required feature licenses. [Table 4-35](#) and [Table 4-36](#) list the required capacity licenses.

Table 4-33 Feature licenses (for 3900 and 5900 series base stations)

RAT	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
LTE FDD	MRFD-16 0222	LTE FDD and NR Flash Dynamic Spectrum Sharing (LTE FDD)	LT1S0LF NSS00	LTE FDD and NR Flash Dynamic Spectrum Sharing(LTE FDD)	eNod eB	per Cell
NR	MRFD-16 0262	LTE FDD and NR Flash Dynamic Spectrum Sharing (NR)	NR0S00F NSS00	LTE FDD and NR Flash Dynamic Spectrum Sharing (NR)	gNod eB	per Cell

RAT	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
For the license control item LT1S0LFNSS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the SpectrumCloud.SpectrumCloudSwitch parameter cannot be set to LTE_NR_SPECTRUM_SHR due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. For the license control item NR0S00FNSS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the LTE_NR_FDD_SPCT_SHR_SW and LTE_NR_SDL_SPCT_SHR_SW options of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter cannot be selected due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. On the LTE side, the license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the eNBLicenseAlmThd.LicenseCheckCtrlSw parameter. On the NR side, the license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.						

Table 4-34 Feature licenses (for DBS3900 LampSite and DBS5900 LampSite)

RAT	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
LTE FDD	MRFD-16 0222	LTE FDD and NR Flash Dynamic Spectrum Sharing (LTE FDD)	LT1SDYN SHR00	LTE FDD and NR Flash Dynamic Spectrum Sharing (LampSite FDD)	eNo deB	per Carrier per pRRU
NR	MRFD-16 0262	LTE FDD and NR Flash Dynamic Spectrum Sharing (NR)	NR0SDY NSHR00	LTE FDD and NR Flash Dynamic Spectrum Sharing (LampSite NR)	gNo deB	per Carrier per pRRU
For the license control item NR0SDYNNSHR00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified. For the license control item LT1SDYNNSHR00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the SpectrumCloud.SpectrumCloudSwitch parameter cannot be set to LTE_NR_SPECTRUM_SHR due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the eNBLicenseAlmThd.LicenseCheckCtrlSw parameter.						

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

Table 4-35 Capacity licenses (for 3900 and 5900 series base stations)

RAT	Model	License Control Item	NE	Sales Unit
LTE FDD	LT1SRFSPCS00	RF Spectrum Sharing License(FDD)	eNodeB	per Band per RU
LTE FDD	LT1S5000RFSS	Spectrum Sharing License for 5000 Series RF Module(FDD)	eNodeB	per Band per RU
NR	NR0S0RFSSL00	RF Spectrum Sharing License (NR)	gNodeB	per Band per RU
NR	NR0SSSLRFM00	Spectrum Sharing License for 5000 Series RF Module (NR)	gNodeB	per Band per RU

Table 4-36 Capacity licenses (for DBS3900 LampSite and DBS5900 LampSite)

RAT	Model	License Control Item	NE	Sales Unit
LTE FDD	LT1SFDSPCS30	Spectrum Sharing License for LampSite pRRU (LTE FDD)	eNodeB	per Band per pRRU
LTE FDD	LT1SFDSPCS50	Spectrum Sharing License for 5000 Series LampSite pRRU(LTE FDD)	eNodeB	per Band per pRRU
NR	NR0SSPECTT00	Spectrum Sharing License for LampSite pRRU (NR)	gNodeB	per Band per pRRU
NR	NR0SSPECTT50	Spectrum Sharing License for 5000 Series LampSite pRRU (NR)	gNodeB	per Band per pRRU

The LTE and NR flexible frame offset function requires a capacity license, as listed in [Table 4-37](#).

Table 4-37 Capacity license for the LTE and NR flexible frame offset function

RAT	Model	License Control Item	NE	Sales Unit
NR	NR0SFLCFOS00	Flexible Cell Frame Offset (NR FDD)	gNodeB	per RU

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

When 5000 series RF modules are used, customers must purchase the Spectrum Sharing License for 5000 Series RF Module. When other RF modules are used, customers must purchase the RF Spectrum Sharing License. When 5000 series RF modules and other RF modules are both used, customers must purchase both capacity licenses.

Customers must purchase the Spectrum Sharing License for 5000 Series LampSite pRRU when 5000 series LampSite pRRUs are used. When non-5000 series LampSite pRRUs are used, customers must purchase the Spectrum Sharing License for LampSite pRRU. When 5000 series LampSite pRRUs and non-5000 series LampSite pRRUs are both used, customers must purchase both capacity licenses.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- Prerequisite functions on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Normal CP	Cell.UICyclicPrefix	<i>Extended CP</i>	Set this parameter to NORMAL_CP .

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentatio n)	Description
LTE FD D	Work mode	Cell.WorkMode	<i>Cell Management</i>	When the NR cell is an NR SDL cell, set this parameter to DL_ONLY . In other cases, set this parameter to UL_DL .
LTE FD D	PRACH Frequency Offset Strategy	RACHCfg.PrachFreqOffsetStrategy	<i>Random Access Control</i>	Set this parameter to FULLY_AUTOMATIC .
LTE FD D	CSI-RS Period	CellCsiRsParaCfg.CsiRsPeriod	<i>Dedicated Carrier for TM9</i>	When the CellCsiRsParaCfg.CsiRsSwitch parameter is set to FIXED_CFG or ADAPTIVE_CFG, the following requirements must be met: • If the NR SSB period (specified by the NR parameter NRDUCell.SsbPeriod) is greater than or equal to MS20(20), CellCsiRsParaCfg.CsiRsPeriod must be set to ms10 or a larger value. • If the NR SSB period is less than or equal to MS10(10), CellCsiRsParaCfg.CsiRsPeriod must be set to ms20 or a larger value.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentatio n)	Description
LTE FD D	PHICH resource	PHICHCfg.Phi chResource	<i>Physical Channel Resource Management</i>	The dependency relationship exists only when the cell bandwidth is 10 MHz and requires this parameter to be set to ONE_SIXTH, HALF, or ONE .
LTE FD D	SRS Configurati on Indicator	SRSCfg.SrsCf gInd	<i>Physical Channel Resource Management</i>	Set this parameter to BOOLEAN_TRUE .
LTE FD D	SRS subframe configuratio n	SRSCfg.SrsSub frameCfg	<i>Physical Channel Resource Management</i>	When SRS automatic neighboring cell measurement is enabled (the NCellSrsMeasPara .SrsAutoNCellMea sSwitch parameter set to ON), the SRSCfg.SrsSubfra meCfg parameter must be set to SC2 . When SRS automatic neighboring cell measurement is disabled, the SRSCfg.SrsSubfra meCfg parameter must be set to SC3, SC7, SC14, or SC0 . When the "LTE and NR 3-ms frame offset difference" function is enabled, the SRSCfg.SrsSubfra meCfg parameter must be set to SC6, SC13, SC14, or SC0 .

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentatio n)	Description
LTE FD D	Dynamic adjustment to cell-specific SRS subframe configuration	SrsSubframeRecfSwitch option of the CellAlgoSwitch.SrsAlgoSwitch parameter	<i>Physical Channel Resource Management</i>	Select this option of this parameter only when the LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter is deselected.
LTE FD D	Uplink adaptive HARQ	CellUlschAlgo.AdaptHarqSwitch	<i>Uplink Scheduling</i>	Set this parameter to ADAPTIVE_HARQ_SW_ON or ADAPTIVE_HARQ_SW_SEMI_ON .

- Prerequisite functions on the NR side

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentatio n)	Description
NR FD D	Basic functions of downlink scheduling	None	<i>Downlink Scheduling</i>	The NRDUCellPdsch.DlDmrsMaxLength parameter must be set to 1SYMBOL .
NR FD D	Search Space Zero	NRDUCellPdcch.SearchSpaceZero	<i>Channel Management</i>	The NRDUCellPdcch.SearchSpaceZero parameter must be set to DEFAULT .

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentatio n)	Description
NR FD D	SSB Frequency Position	NRDUCell.SsbF reqPos	None	When the SSB_ADJ_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter is selected, the spacing between the SSB center frequency and NR cell center frequency must be greater than or equal to 17 RBs.
NR FD D	SSB beam adjustment	SSB_ADJ_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	None	SSB beam adjustment must be enabled before LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled for NR LumiLink cells.
NR FD D	LumiLink solution (NR)	EASY_OMNI_S ITE_MODE_SW option of the NRDUFddCellT rpBeam.Scenar ioModeSw parameter	<i>LumiLink Solution</i>	LTE FDD and NR Flash Dynamic Spectrum Sharing can take effect in NR LumiLink cells only if the NRDUCell.DlBandwidth parameter is set to CELL_BW_20M for the cells.

4.3.2.2 Mutually Exclusive Functions

- Mutually exclusive functions on the LTE side
 - CIoT functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
NB - IoT	LTE in-band deployment	CellRbReserve.RbRsvMode set to NB_RESERVED or NB_DEPLOYMENT	<i>NB-IoT Basics (FDD)</i>
NB - IoT	NR guard band deployment	Prb.DeployMode set to GUARD_BAND and Prb.NrCellGbCfgInd set to CFG	<i>NB-IoT Basics (FDD)</i>
NB - IoT	eMTC introduction ^a	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC</i>
a: The "LTE and NR 3-ms frame offset difference" function (controlled by the LNR_3MS_FRAME_OFFSET_DIFF_SW option of the SpectrumCloud.SpectrumCloudExtSwitch parameter) cannot work with eMTC introduction.			

– Inter-RAT coordination functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>
LT E FD D	UMTS and LTE Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>
LT E FD D	UMTS and LTE Spectrum Sharing Based on DC-HSDPA	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>
LT E FD D	UMTS and LTE Dynamic Power Sharing	UMTS_LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>UMTS and LTE Dynamic Power Sharing</i>

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>
LT E FD D	Extreme power sharing	LTE_EXTREME_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>
LT E FD D	GSM and LTE Dynamic Power Sharing	CellAlgoSwitch.GLPwrShare	<i>GSM and LTE Dynamic Power Sharing</i>
LT E FD D	LTE and NR SRS resource allocation optimization ^a	LTE_NR_SRS_ALLOC_OPT_SW option of the LteNrSpctShrCellGrp.LteNrSpctShrSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>
LT E FD D	Absence of SRS configuration for LTE UEs ^a	LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>
a: This function cannot work with the "LTE and NR 3-ms frame offset difference" function (controlled by the LNR_3MS_FRAME_OFFSET_DIFF_SW option of the SpectrumCloud.SpectrumCloudExtSwitch parameter).			

– Energy saving functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Multi-RAT Carrier Joint Intelligent Shutdown (UL/GUL)	InterRatCellShutdown.ForceShutdownSwitch	<i>Multi-RAT Coordinated Energy Saving</i>
LT E FD D	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction</i>

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	RF channel dynamic muting ^a	RF_CHN_DYN_MUTING_ SW option of the CellRfChnDynMut- ing.RfChnDynMutingAl- goSwitch parameter	<i>Energy Conservation and Emission Reduction</i>
a: In 2T, 4T, and 8T scenarios where the SymbolPwrSaving.MultiRatSymbolShutdownSw parameter is set to ON , LTE FDD and NR Flash Dynamic Spectrum Sharing can be enabled together with RF channel dynamic muting. In other scenarios, the two functions cannot work together.			

– MIMO functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Virtual 4T4R	Virtual4T4RSwitch option of the CellAlgoSwitch.EmimoS witch parameter	<i>Virtual 4T4R (FDD)</i>
LT E FD D	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitC ellGroupOptSwitch parameter	<i>Smart 8T8R (FDD)</i>
LT E FD D	Unequal downlink port power ^a	PDSCHCfg.TxPowerOffs etAnt2 PDSCHCfg.TxPowerOffs etAnt3	<i>Smart 8T8R (FDD)</i>
LT E FD D	Uplink SRS-based frequency selective scheduling ^a	CellUlschAlgo.UlSrsFreq SelSchSinrThld set to a value other than 255	<i>Smart 8T8R (FDD)</i>
LT E FD D	Enhanced extreme power sharing	LTE_ENH_EXT_POWER_S HARING_SW option of the CellDynPowerShar- ing.DynamicPowerShar- ingSwitch parameter	<i>Smart 8T8R (FDD)</i>
LT E FD D	Centralized power sharing	LTE_CENTRAL_POWER_ SHARING_SW option of the CellDynPowerShar- ing.DynamicPowerShar- ingSwitch parameter	<i>Smart 8T8R (FDD)</i>

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Power optimization for Static Multiple Beam	8T_SMB_PWR_OPT_SW option of the PDSCHCfg.TxChnPowerCfgSw parameter	<i>Smart 8T8R (FDD)</i>
LT E FD D	Fast power sharing ^b	SectorSplitGroup.PwrSharingHighLoadThld set to a value other than 100	<i>Smart 8T8R (FDD)</i>
LT E FD D	Downlink turbo pilot	SectorSplitGroup.TurboPilotPowerOffset set to a non-zero value, and Cell.TxRxMode set to 8T8R	<i>Massive MIMO Enhancements (FDD)</i>
LT E FD D	Intelligent beam weight optimization ^c	INTELLIGENT_BEAM_WEIGHT_OPT_SW option of the SectorSplitGroup.SectorSplitSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i>
a: This mutually exclusive relationship exists only when LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled. b: This mutually exclusive relationship exists only when hybrid power sharing is enabled. c: In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, an activated LTE cell cannot have intelligent beam weight optimization enabled together with "absence of SRS configuration for LTE UEs" (controlled by the LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter).			

– Multi-RAT coordination functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	GSM and LTE Buffer Zone Optimization	AvoidGeranInterfSwitch option of the CellAlgoSwitch.AvoidInterfSwitch parameter	<i>GSM and LTE Buffer Zone Optimization</i>
LT E FD D	UL Refarming Zero Bufferzone	AvoidUtranInterfSwitch option of the CellAlgoSwitch.AvoidInterfSwitch parameter	<i>UL Refarming Zero Bufferzone</i>

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	UMTS and LTE Zero Bufferzone	UMTS_LTE_ZERO_BUFFER_ZONE_SW option of the ULZeroBufferZone.ZeroBufZoneSwitch parameter	<i>UMTS and LTE Zero Bufferzone</i>
LT E FD D	CDMA and LTE Zero Bufferzone	AvoidCDMAInterfSwitch option of the CellAlgoSwitch.AvoidInterfSwitch parameter	<i>CDMA and LTE Zero Bufferzone</i>

– Coverage improvement functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Cell radius greater than 100 km	Cell.CellRadius	<i>Extended Cell Range</i>
LT E FD D	Superior Uplink Coverage	CellAlgoExtSwitch.UlCoverageEnhancementSw	<i>Superior Uplink Coverage (FDD)</i>

– Interference mitigation functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Super combined cell	SuperCombCellSwitch option of the CellAlgoSwitch.SfnAlgoSwitch parameter	<i>Super Combined Cell (FDD)</i>
LT E FD D	Dynamic TDM eICIC	CellAlgoSwitch.EicicSwitch	<i>TDM eICIC (FDD)</i>
LT E FD D	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUT_DOWN_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Intelligent PIM interference avoidance ^a	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>
LT E FD D	Adaptive SFN/SDMA (FDD) ^b	CellAlgoSwitch.SfnUlSchSwitch set to ADAPTIVE or CellAlgoSwitch.SfnDlSchSwitch set to ADAPTIVE	<i>SFN</i>
LT E FD D	SFN algorithm	CellAlgoSwitch.SfnAlgoSwitch	None
a: Intelligent PIM interference avoidance cannot be enabled when the EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter is selected and the SpectrumCloud.SpectrumCloudSwitch parameter is set to LTE_NR_SPECTRUM_SHR. b: When RRUs are combined or RRUs and pRRUs are combined to serve an SFN cell, adaptive SFN/SDMA (FDD) cannot work with LTE FDD and NR Flash Dynamic Spectrum Sharing.			

– Power management functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Coordinated scheduling-based power control	CellCspcPara.CellCspcSwitch	<i>CSPC</i>
LT E FD D	Cell Power Limit ^a	PDSCHCfg.EmfPowerLimitSwitch set to ON , and PDSCHCfg.CellPowerLimit set to a value other than 0	<i>On-Demand TX Power Allocation Under EME</i>
a: This mutually exclusive relationship exists only when LTE FDD and NR share spectrum power during spectrum sharing.			

– Basic functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	UL CRA	UL_COORD_RES_ALLOC_SWITCH option of the ULCsAlgoPara.UlCsSw parameter	<i>Uplink Coordinated Scheduling</i>
LT E FD D	UL CPC	UL_COORD_PC_SWITCH option of the ULCsAlgoPara.UlCsSw parameter	<i>Uplink Coordinated Scheduling</i>
LT E FD D	Manual RB blocking	CellRbReserve.RbRsvMode set to RB_MASKING or RB_MASKING_WITH_SR S	<i>Downlink Scheduling</i>
LT E FD D	Flexible PUCCH configuration ^a	PucchFlexCfgSwitch option of the CellAlgoSwitch.PucchAlgoSwitch parameter	<i>Physical Channel Resource Management</i>
LT E FD D	ePDCCH	CellPdcchAlgo.EpdccAlgoSwitch	<i>Physical Channel Resource Management</i>
LT E FD D	Extended PHICH	PHICHCfg.PhichDuration	<i>Physical Channel Resource Management</i>
a: When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the settings of the CellRbReserve.RbRsvMode , CellRbReserve.RbRsvType , CellRbReserve.RbRsvStartIndex , and CellRbReserve.RbRsvEndIndex parameters can be used as an alternative to the flexible PUCCH configuration function.			

– Mobility management functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	High speed mobility ^a	Cell.HighSpeedFlag	<i>High Speed Mobility</i>

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	Ultra high speed mobility ^a	Cell.HighSpeedFlag	<i>High Speed Mobility</i>
a: This mutually exclusive relationship exists only when the LTE_NR_SRS_ALLOC_OPT_SW option of the LteNrSpctShrCellGrp.LteNrSpctShrSwitch parameter is selected.			

– Functions related to RAN services

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	In-band relay	InBandRelayDeNbS- witch, InBandRelayReNbS- witch, and InBandRelayDistantDe- ploySw options of the CellAlgoSwitch.RelaySw itch parameter	<i>Relay</i>
LT E FD D	Uplink Relay Subframe Ratio ^a	CellRelayAlgoPara.UIRe laySubframeRatio	<i>Relay</i>
LT E FD D	Downlink Relay Subframe Ratio ^a	CellRelayAlgoPara.DIRe laySubframeRatio	<i>Relay</i>
LT E FD D	eMBMS	CellMBMSCfg.MBMSSw itch	<i>eMBMS</i>
a: When LTE FDD and NR Flash Dynamic Spectrum Sharing and out-of-band relay are both enabled, the values of both CellRelayAlgoPara.UIRelaySubframeRatio and CellRelayAlgoPara.DIRelaySubframeRatio must be less than the value of LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio .			

– Other functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	LTE flexible spectrum	Cell.CustomizedBandWidthCfgrInd set to FLEX_CFG	<i>LTE Flexible Spectrum (FDD)</i>
LT E FD D	Flexible bandwidth based on overlapping carriers	DdCellGroup.DdBandwidth	<i>Flexible Bandwidth based on Overlap Carriers (FDD)</i>
LT E FD D	PUSCH FH	CellUlschAlgo.UlHoppingType	None
LT E FD D	LTE flexible bandwidth	CellLteFlexBw.LteFlexBwSwitch	None
LT E FD D	Short TTI	SHORT_TTI_SW option of the CellShortTtiAlgo.SttiAlgorithmSwitch parameter	<i>Short TTI (FDD)</i>
LT E FD D	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARR_SW option of the ContigIntraBand-Carr.ContigIntraBand-CarrSw parameter	<i>Seamless Intra-Band Carrier Joining (FDD)</i>
LT E FD D	TM4 and TM6 Adaptation	CL_TM_ADAPTATION_OPTION_SW option of the CellMimoParaCfg.MimoSwitch parameter	<i>Low-Band Booster</i>
LT E FD D	Uplink data transmission path selection ^a	NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>
LT E FD D	Uplink PUCCH RB occupation by PUSCH ^b	PuschUsePucchRbSwitch option of the CellAlgoSwitch.UlschSwitch parameter	None
LT E FD D	CSI-RS Period ^c	CellCsiRsParaCfg.CsiRsPeriod	None

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LT E FD D	CSI-RS subframe selection ^d	CSIRS_SUBFRAME_SELECT_SW option of the SpectrumCloud.LnrSpectrumShrOptSwitch parameter	None
a: This mutually exclusive relationship exists only when the LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter is selected. b: This mutually exclusive relationship exists only when the LTE_UPLINK_VOLTE_ENH_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter is selected. c: When the "LTE and NR 3-ms frame offset difference" function (controlled by the LNR_3MS_FRAME_OFFSET_DIFF_SW option of the NRDUCellSptShare.LnrSpectrumShareSwitch parameter) is enabled, the CellCsiRsParaCfg.CsiRsPeriod parameter cannot be set to ms5 or ms10. d: The "LTE and NR 3-ms frame offset difference" function cannot work with this function.			

- Mutually exclusive functions on the NR side
 - MIMO functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	PUSCH coordinated power control	PUSCH_COORD_PWR_CTRL_SW option of the NRDUCellULPcConfig.UlPwrCtrlAlgoExtSwitch parameter	<i>Smart Coverage (FDD)</i> <i>Smart 8T8R (FDD)</i> <i>Smart Massive MIMO (FDD)</i>
NR FD D	Coordinated scheduling	DL_COORDINATED_SWITCH option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Smart Coverage (FDD)</i> <i>Smart 8T8R (FDD)</i> <i>Smart Massive MIMO (FDD)</i>
NR FD D	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Inter-gNodeB Distributed MIMO	INTER_GNB_DL_D_MIMO_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>
NR FD D	MU-MIMO measurement enhancement	MU_MIMO_MEAS_ENH_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementsw parameter	<i>Smart 8T8R (FDD)</i>
NR FD D	Frequency selective scheduling	FREQ_SEL_SCH_SW option of the NRDUCellDlSch.DlSchAlgoSwitch parameter	<i>Smart Coverage (FDD)</i> <i>Smart 8T8R (FDD)</i>
NR FD D	UE-specific PDCCH symbol number fast adaptation	UE_PDCCH_SYM_NUM_FAST_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Smart Coverage (FDD)</i> <i>Smart 8T8R (FDD)</i> <i>Smart Massive MIMO (FDD)</i>

– NSA networking functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Uplink single-side transmission ^a	SINGLE_UL_NSA_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>
NR FD D	NSA DC enhanced uplink power control ^a	NSA_DC_ENH_UL_POWER_CONTROL_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	None
a: This mutually exclusive relationship exists only when the cell TX/RX mode is 16T16R or 32T32R.			

– UCN functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Cell Combination	NRDUCell.NrDuCellNet workingMode set to HYPER_CELL_COMBINE_ MODE and NRDUCellMultiTrp.Data TransMode set to BASIC_MODE	<i>Cell Combination</i>
NR FD D	Joint Transmission Cell Combination	NRDUCell.NrDuCellNet workingMode set to HYPER_CELL_COMBINE_ MODE and NRDUCellMultiTrp.Data TransMode set to JOINT_MODE	<i>Cell Combination</i>
NR FD D	Hyper Cell	NRDUCell.NrDuCellNet workingMode set to HYPER_CELL	<i>Hyper Cell</i>
NR FD D	Inter-TRP handover in CA SCells ^a	SCell_INTER_TRP_HO_S W option of the NRDUCellCarrMgmt.Ca EnhancedAlgoSwitch parameter	<i>Cell Combination</i>
NR FD D	Inter-TRP SSB staggering ^a	INTER_TRP_SSB_STAGG ER_SWITCH option of the NRDUCellMultiTrp.Mult iTrpAlgoSwitch parameter	<i>Cell Combination</i>
NR FD D	Inter-TRP CSI-RS staggering based on TRP ID mod 2 values ^a	NRDUCellCsirs.InterTrp CsirsStaggerMode set to MOD2_STAGGER	<i>Cell Combination</i>
a: This mutually exclusive relationship exists only when NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode is set to DPS_MODE .			

- Coverage improvement functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	SSB power aggregation	SSB_POWER_AGG_OPT_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter	<i>Turbo Coverage (FDD)</i>
NR FD D	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>
Lo w- fre qu en cy NR TD D	UL and DL Decoupling ^a	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>
a: When UL and DL Decoupling is enabled in SUL and NR FDD co-deployment mode, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be enabled.			

– Interference mitigation functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Intra-frequency paging interference avoidance	INTRA_FREQ_PAGING_AVOID_SW option of the NRDUCellPagingConfig.PagingSchAlgoSwitch parameter	<i>Interference Avoidance</i>
NR FD D	Common channel interference avoidance ^a	SSB_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommonChnIntrfAvoidSwitch parameter	<i>Interference Avoidance</i>
NR FD D	Manual configuration of interference ranges for downlink interference identification ^a	NRDUCellIntrfIdent.IntrfIdentificationMethod set to DL_MANUAL	<i>Interference Avoidance</i>

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Channel power reduction for PIM interference avoidance	PIM_INTERF_CHN_PWR_REDUCE_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance</i>
NR FD D	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUT_DOWN_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance</i>
a: This function does not work with PSD adaptation (controlled by the PWR_SPCT_DENSITY_ADAPT_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter).			

– Basic functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Enhanced paging capacity mode 1	NRDUCellPagingConfig.PagingCapacityMode set to ENHANCE_MODE_1	<i>5G Networking and Signaling</i>
NR FD D	Basic functions of downlink scheduling ^a	None	<i>Downlink Scheduling</i>
NR FD D	Unbalanced transmit power configurations between channels ^b	NRDUCellCoverage.TransmitPowerOffsetChn0 , NRDUCellCoverage.TransmitPowerOffsetChn1 , NRDUCellCoverage.TransmitPowerOffsetChn2 , NRDUCellCoverage.TransmitPowerOffsetChn3 , NRDUCellCoverage.TransmitPowerOffsetChn4 , NRDUCellCoverage.TransmitPowerOffsetChn5 , NRDUCellCoverage.TransmitPowerOffsetChn6 , or NRDUCellCoverage.TransmitPowerOffsetChn7 set to a non-zero value	<i>Cell Management</i>

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>
NR FD D	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>
NR FD D	SSB beam switching	NRDUCellCsirs.SsbBeamSwitchingMode set to MAC_CE_MODE	<i>Channel Management</i>
NR FD D	SRS transmission using common PUCCH resources ^c	COMMON_PUCCH_AVOID_SRS_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>
NR FD D	Configuration of the maximum aggregation level for the common PDCCH	NRDUCellPdcch.MaxCommonPdcchAggLvl set to AGGLVL16	<i>Channel Management</i>
NR FD D	SSB beam sweeping period ^d	NRDUCell.SsbPeriod set to MS5	<i>Channel Management</i>

a: LTE FDD and NR Flash Dynamic Spectrum Sharing is mutually exclusive with the **NRDUCellPdsch.DlAdditionalDmrsPos** parameter (set to **POS2**) in the basic functions of downlink scheduling.

b: This mutually exclusive relationship exists only when LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled.

c: This mutually exclusive relationship exists only when the **NRDUCellBwp.UlMinCarrierBw** parameter is set to **NOT_CONFIG**.

d: When the "LTE and NR 3-ms frame offset difference" function (controlled by the **LNR_3MS_FRAME_OFFSET_DIFF_SW** option of the **NRDUCellSptShare.LnrSpectrumShareSwitch** parameter) is enabled, the **NRDUCell.SsbPeriod** parameter cannot be set to **MS5**.

- Access control functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Service admission ^a	QOS_FLOW_ADMISSION_SW option of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>
NR FD D	Service preemption ^a	QOS_FLOW_PREEMPTION_SW option of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>
a: This mutually exclusive relationship exists only when LTE FDD and NR SDL cells share spectrum.			

– Energy saving functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Energy saving based on flexible frequency-domain scheduling ^a	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>
NR FD D	Common-channel dynamic shutdown in TTI-level channel shutdown	NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter and the TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>
NR FD D	Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>
NR FD D	Dynamic RF channel shutdown ^b	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>
a: This mutually exclusive relationship exists only when PSD adaptation is enabled. b: This mutually exclusive relationship exists if the NR cell engaged in LTE FDD and NR Flash Dynamic Spectrum Sharing is an SDL cell. If the NR cell is an FDD cell, among the sub-functions of dynamic RF channel shutdown, only data-channel dynamic shutdown can be enabled (by setting NRDUCellPowerSaving.ComChnDynShutdownSwitch to OFF) together with LTE FDD and NR Flash Dynamic Spectrum Sharing, but only in 4T4R or 8T8R scenarios.			

– Spectrum management functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Flexible bandwidth	FLEXIBLE_BANDWIDTH_SW option of the NRDUCellFlexBw.FlexBwAlgoSwitch parameter	<i>Flexible Bandwidth (FDD)</i>

– Functions related to RAN services

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	RedCap SSSG switching	REDCAP_SSSG_SWITCHING_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter REDCAP_SSSG_SWITCH_EXT_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>RedCap</i>

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Adaptive TBoMS	TBOMS_SW option of the NRDUCellRedCap.RedCapCoverageAlgoSw parameter	<i>RedCap</i>
NR FD D	PEI ^a	PEI_SW option of the NRDUCellUePwrSaving.IdleUePwrSavingSwitch parameter	<i>RedCap</i>
a: The paging early indication (PEI) function cannot be enabled if LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled and the cell bandwidth is less than or equal to 10 MHz.			

– Other functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	Dynamic voltage adjustment	DYN_VOLT_ADJ_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None
NR TD D	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None
NR FD D	R16 Type2 ^a	R16_TYPE2_SW option of the NRDUCellPdschPre-code.DlCodebookTypeSwitch parameter	None
NR FD D	DSS inter-RAT PDCCH spatial multiplexing ^b	DSS_INTER_RAT_PDCCH_SPC_MUX_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
	a: The R16_TYPE2_SW option of the NRDUCellPdschPre-code.DlCodebookTypeSwitch parameter must be deselected for a cell that meets either of the following conditions: (1) The LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is selected and the TX/RX mode of the cell is not 4T4R, 8T8R, 16T16R, or 32T32R. (2) The LTE_NR_SDL_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is selected and the TX/RX mode of the cell is not 4T. b: This mutually exclusive relationship exists only when NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode is set to DPS_MODE.		

4.3.2.3 Function Impacts

- Function impacts on the LTE side
 - CloT functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
NB - IoT	LTE guard band deploymen t	Prb.DeployMo de	<i>NB-IoT Basics (FDD)</i>	• For 3900 and 5900 series base stations, when LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with NB-IoT deployed in LTE guard band mode, uplink and downlink RBs are reserved to reduce the interference between NB-IoT and NR. For details, see 4.1.1.3.1 LTE Functions. The RB reservation decreases the spectrum resources available for the NR cell, reducing the NR cell throughput. • For DBS3900 LampSite and DBS5900 LampSite, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be used together with NB-IoT deployed in LTE guard band mode.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				When the NR cell is an NR SDL cell, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be used together with NB-IoT deployed in LTE guard band mode.
LT E FD D	eMTC introductio n	EMTC_SWITC H option of the CellEmtcAlgo. EmtcAlgoSwit ch parameter	<i>eMTC</i>	eMTC introduction does not take effect in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios.

– Inter-RAT coordination functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	LTE spectrum coordinatio n	SpectrumCoo rdinationSwit ch option of the CaMgtCfg.Cel lCaAlgoSwitc h parameter	<i>LTE Spectrum Coordination</i>	The number of uplink RBs available for LTE decreases, which reduces the proportion of UEs for which LTE spectrum coordination takes effect.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE FDD and NR Flash Dynamic Power Sharing is enabled, the power sharing capability of the LTE spectrum sharing cell is lower than that of common cells. As a result, the power sharing benefits decrease.
LT E FD D	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_DIM_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	The benefits of LTE-NR multi-dimensional scheduling optimization decrease in cells with MBSFN subframes configured.
LT E FD D	LNR CRS muting	LNR_CRSMUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, LNR CRS muting takes effect in a lower proportion of cases and consequently offers fewer benefits than in NR-only scenarios.

– CoMP functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	UL CoMP	CellAlgoSwitch.UplinkCompSwitch	<i>UL CoMP</i>	LTE UEs perform measurement on the entire bandwidth. When the LTE CRS transmission avoids the NR SSB, LTE RSRP measurement results become inaccurate. Consequently, the number of cell edge users (CEUs) in the LTE cell decreases, that is, the number of UEs for which UL CoMP takes effect decreases.
LT E FD D	DL CoMP with TM9	Tm9JtSwitch option of the CellAlgoSwitch.DLCompSwitch parameter	<i>DL CoMP (FDD)</i>	DL CoMP with TM9 does not take effect when used together with LTE FDD and NR Flash Dynamic Spectrum Sharing, as the conditions for DL CoMP with TM9 to take effect cannot be met.

- MIMO functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Uplink MU-MIMO	ULVmimoSwitch option of the CellAlgoSwitch . ULSchSwitch parameter	<i>MIMO</i>	The number of uplink RBs available for LTE decreases, which increases the pairing success rate of uplink MU-MIMO. Consequently, the BLER increases.
LT E FD D	TM9 adaptation	4T_TM9_ADAPTATION_SW option of the CellMimoParaCfg . MimoSwitch parameter	<i>MIMO</i>	When an LTE cell is configured with MBSFN subframes, TM9 adaptation does not take effect.
LT E FD D	Downlink MU-MIMO	EmimoMuMimoSwitch and 4TxTM9MuMimoSwitch options of the CellAlgoSwitch . EmimoSwitch parameter	<i>eMIMO (FDD)</i>	The number of downlink RBs available for LTE decreases, which increases the pairing success rate of downlink MU-MIMO. Consequently, the BLER increases.
LT E FD D	Inter- carrier power sharing	INTER_CARRIER_PWR_SHR_SWITCH option of the SectorSplitGroup . IntelligentBeamSchSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the power that can be shared between carriers decreases.
LT E FD D	Dynamic power sharing for TM4 UEs	DYNAMIC_TM4_PWR_SHR_SW option of the SectorSplitGroup . IntelligentBeamSchSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i>	After LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, dynamic power sharing for TM4 UEs cannot take effect.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Dynamic Massive Beam	None	<i>Smart 8T8R (FDD)</i>	For a cell not served by an integrated 8T8R RRU, Dynamic Massive Beam does not take effect if it is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing.
LT E FD D	Hybrid power sharing	LTE_HYBRID_POWER_SHARING_SW option of the CellDynPowerSharing.Dyn amicPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, hybrid power sharing does not take effect for UEs using CRS-based demodulation and QAM modulation in the downlink.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Cell Beam Azimuth Offset	eUCellSectorEqm.CellBeamAzimuthOffset set to a value other than 0	<i>Smart 8T8R (FDD)</i>	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, if the azimuth offset of beams in an LTE cell is adjusted (by setting the eUCellSectorEqm.CellBeamAzimuthOffset parameter to a value other than 0), LTE CRS full-symbol rate matching can be used to mitigate the impact of the change in the LTE cell coverage on the NR cell as a result of the azimuth offset adjustment. (LTE CRS full-symbol rate matching is controlled by the LTE_CRS_RATEMATCH_ALL_SYM_SW option of the NRDUCellAlgorithmSwitch.SpectrumC loudEnhSwitch parameter).

- Interference mitigation functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	eNodeB- based interferenc e randomizat ion	CellAlgoSwitch.<i>InterfRandSwitch</i>	<i>ICIC</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the amount of spectrum resources available for LTE decreases. As a result, the benefits provided by eNodeB-based interference randomization decrease.
LT E FD D	Uplink interferenc e randomizat ion	CellUlschAlgo.<i>ULRbAllocationStrategy</i> set to FS_INRANDOM_ADAPTIVE	<i>ICIC</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the amount of spectrum resources available for LTE decreases. As a result, the benefits provided by uplink interference randomization decrease.

– Basic functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Accurate uplink RB scheduling	UL_RB_PRECI SE_SCH_SW option of the CellAlgoSwitc h.UlSchExtSw itch parameter	<i>Uplink Scheduling</i>	Accurate uplink RB scheduling may produce fewer benefits when it is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing.
LT E FD D	UL Multi- Cluster	ULMultiCluste rSwitch option of the CellAlgoSwitc h.UlSchExtSw itch parameter	<i>Uplink Scheduling</i>	The number of uplink RBs available for LTE decreases, which reduces the benefits provided by UL Multi- Cluster.
LT E FD D	Uplink frequency selective scheduling	None	<i>Uplink Scheduling</i>	The number of uplink RBs available for LTE decreases, which reduces the benefits provided by uplink frequency selective scheduling.
LT E FD D	Uplink short- interval SPS	CellUlSchAlgo .IntvlOfULSps WithSkipping	<i>Uplink Scheduling</i>	The number of uplink RBs available for LTE decreases, which reduces the benefits provided by uplink short- interval SPS to UEs.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	SRS Configurati on Indicator	SRSCfg.SrsCfg Ind	<i>Physical Channel Resource Management</i>	As LTE must share SRS resources with NR, the LTE cell triggers the expansion of the cell-specific SRS subframes in advance, which prolongs the SRS period of LTE UEs.

– Energy saving functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	LNR coordinate d channel shutdown	CellRfShutdo wn.MultiRat/ ointChnShutd ownSw	<i>Multi-RAT Coordinated Energy Saving</i>	For FDD massive MIMO modules, LNR coordinated channel shutdown does not take effect after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	pRRU deep dormancy in intelligent mode	E nodeb M pru E s.MpruDorma ncyDIEarfcn	<i>Multi-RAT Coordinated Energy Saving</i>	For DBS3900 LampSite and DBS5900 LampSite, when LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with pRRU deep dormancy in intelligent mode, the demodulation performance of the common PUCCH in the NR cell deteriorates.
LT E FD D	DRX	C ellDrxPara .D rxAlgSwitch	<i>DRX and Signaling Control</i>	The number of RBs available for LTE decreases, leading to a decrease in the average number of scheduling times. As a result, the average number of UEs that enter DRX mode, as well as the frequency of UEs entering and exiting DRX mode increase.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Dynamic DRX	CellAlgoSwitch.DynDrxSwitch	<i>DRX and Signaling Control</i>	The number of RBs available for LTE decreases, leading to a decrease in the average number of scheduling times. As a result, the average number of UEs that enter DRX mode, as well as the frequency of UEs entering and exiting DRX mode increase.
LT E FD D	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltSwitch	<i>Energy Conservation and Emission Reduction</i>	Dynamic voltage adjustment does not take effect when enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, as its triggering conditions cannot be met.
LT E FD D	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After the switching to one beam through 8T beam mode shift for energy saving, the LTE side will experience a higher PRB usage and start to preempt NR resources. This may affect NR user experience.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	RF channel intelligent shutdown	CellRfShutdo wn.RfShutdo wnSwitch	<i>Energy Conservation and Emission Reduction</i>	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, if the CellDlpcPdschPa.PaPcOff parameter is set to DB_3_P_A or DB_1DOT77_P_A , NR power may be insufficient, causing NR access failures. In this case, selecting the RF_SHUTDOWN_DSS_NR_PWR_OPT_SW option of the CellRfShutdown.RfShutdownAlgoSwitch parameter and setting the CellPwrSavingAlgo.DssNrPwrInsuffTtiThld parameter to 20 may trigger exit from RF channel intelligent shutdown, thereby reducing the risk of NR access failures.
LT E FD D	Port sparseness	PORT_SPARSE NESS_SW option of the CellPwrSavin gAlgo.PwrSav ingAlgoSwitc h parameter	<i>Energy Conservation and Emission Reduction</i>	Port sparseness may provide fewer benefits when it is in effect together with LTE FDD and NR Flash Dynamic Spectrum Sharing.

- Functions related to RAN services

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FDD	Out-of- band relay	OutOfBandRelaySwitch option of the CellAlgoSwitch . RelaySwitch parameter	<i>Relay</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing and out-of-band relay are both enabled, the following impacts will occur: • NR occupies some scheduling resources when LTE and NR dynamically share resources. This means that fewer resources are available for LTE in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios than in LTE-only scenarios. As a result, a donor eNodeB (DeNB) has fewer resources, which further affects the performance of the corresponding relay BTSs (ReBTSs). The following methods can be used to

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				mitigate the impacts: - Increasing the percentage of resources preferentially allocated to LTE (using the LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio parameter) to ensure the performance of relay UEs to a certain extent - Adjusting the uplink and downlink relay subframe ratios as well as the percentage of resources preferentially allocated to LTE to ensure available resources for relay services on the LTE side • In LTE FDD and NR Flash Dynamic

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				Spectrum Sharing scenarios, common LTE UEs and common NR UEs have fewer available resources and their performance is affected by LTE relay UEs with a higher priority. The impact severity depends on the uplink and downlink relay subframe ratios as well as the percentage of resources preferentially allocated to LTE. • In extreme scenarios (for example, signaling storms and concurrent access of excessive UEs on the NR side) with a large proportion of system information, control information, or signaling, fewer

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				resources are available for relay UEs. As a result, the performance of relay UEs and delay-related indicators deteriorate. • When services with QoS guarantee requirements are running on the NR side, there may be no available NR resources in LTE relay subframes, affecting NR performance. To ensure NR performance, the uplink and downlink relay subframe ratios can be reduced. • LTE FDD and NR Flash Dynamic Spectrum Sharing increases the number of messages exchanged between the LTE and NR sides. When the number of cells or UEs served by a

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				board is too large, LTE and NR cells will be congested. In this case, resources are allocated to LTE and NR based on static configurations, which has a great impact on the performance of DeNBs and ReBTSS. The access of relay UEs and common LTE UEs can be guaranteed, but their user experience cannot be guaranteed.
LT E FD D	Uplink semi- persistent scheduling	SpsSchSwitch option of the CellAlgoSwitch .ULSchSwitch h parameter	<i>VoLTE</i>	The number of uplink RBs available for LTE decreases, which reduces the benefits provided by uplink semi-persistent scheduling.
LT E FD D	Downlink semi- persistent scheduling	SpsSchSwitch option of the CellAlgoSwitch .DLSchSwitch h parameter	<i>VoLTE</i>	The number of downlink RBs available for LTE decreases, which reduces the benefits provided by downlink semi-persistent scheduling.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	TTI bundling	TtiBundlingSwitch option of the CellAlgoSwitch.UlSchSwitch parameter	<i>VoLTE</i>	The number of uplink RBs available for LTE decreases. Consequently, the maximum number of UEs that can enter the TTI bundling state decreases.
LT E FD D	Uplink RB reservation	UlVoipRbRsvSwitch option of the CellAlgoSwitch.UlSchExtSwitch parameter	<i>VoLTE</i>	It is not recommended that uplink RB reservation be enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing. This is because interference will be generated on voice service UEs that occupy the reserved RBs in neighboring LTE cells in cases where reserved RBs are configured in the shared spectrum and some of the shared spectrum is occupied by NR.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LT E FDD	LCS	ENodeBAlgoSwitch.LcsSwitch	<i>LCS</i>	The positioning reference signal (PRS) sent for LCS causes interference to NR, increasing the bit error rate (BER) on the NR side. As a result, the performance of the NR cell deteriorates. Therefore, it is not recommended that LCS be enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing.

– Mobility management functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LT E FDD	Intra-RAT mobility load balancing	CellAlgoSwitch.MlbAlgoSwitch	<i>Intra-RAT Mobility Load Balancing</i>	If the PRB usage of LTE changes or the number of RBs available for LTE changes, the intra-RAT MLB policy changes.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	High speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i>	SFN is required for downlink AFC of high speed mobility to deliver optimal benefits in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios.
LT E FD D	Ultra high speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i>	SFN is required for downlink AFC of ultra high speed mobility to deliver optimal benefits in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios.

– Carrier management functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Uplink CA	CaUL2CCSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The number of uplink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD uplink carrier aggregation (CA) takes effect.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Downlink CA	None	<i>Carrier Aggregation</i>	The number of downlink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD downlink CA takes effect.
LT E FD D	Downlink 3CC aggregatio n	CaDl3CCSwitch option of the CaMgtCfg.Cel lCaAlgoSwitc h parameter	<i>Carrier Aggregation</i>	When downlink 3CC aggregation is used together with LTE FDD and NR Flash Dynamic Spectrum Sharing, and the percentage of resources preferentially allocated to LTE is set to 0 , LTE UEs cannot access the network. It is recommended that the PUCCH configuration be optimized.
LT E FD D	Flexible CA	MultiCarrierF lexCaSwitch option of the CaMgtCfg.Cel lCaAlgoSwitc h parameter	<i>Carrier Aggregation</i>	The number of uplink and downlink RBs available for LTE decreases, and the selected serving cell combination may change as a result.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Downlink massive CA	DLMassiveCaSwitch option of the CaMgtCfg.Cel lCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	LTE cells with LTE FDD and NR Flash Dynamic Spectrum Sharing enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured, which decreases the LTE network throughput.

– Other functions

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	RAN sharing with common carrier	ENodeBSharingMode.EnodeBSharingMode set to SHARED_FREQ or HYBRID_SHARED	<i>RAN Sharing</i>	When RAN sharing with common carrier is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the number of available RBs for an LTE operator is calculated as follows: (Total number of RBs – Number of RBs shared with NR) x RB allocation proportion configured for the operator.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Compact bandwidth	Cell.Customiz edBandWidth Cfglnd	<i>Compact Bandwidth (FDD)</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the compact bandwidth function supports only the following bandwidths: • 9.6 MHz • 9.8 MHz • 14.6 MHz • 14.8 MHz • 17.5 MHz • 19.6 MHz • 19.8 MHz When a compact bandwidth of 9.6 MHz or 9.8 MHz is used, the NR SSB cannot be configured due to the few number of available RBs. Therefore, it is recommended that the number of LTE CRS ports (specified by the Cell.CrsPortNum parameter) be set to 1 or 2 for the LTE cell. This configuration suggestion applies only to SA networking. This impact relationship exists only when 3900 or 5900 series base stations (macro base

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				stations) are used. When the Cell.CustomizedBandwidthCfgInd parameter is set to CFG and the Cell.TxRxMode parameter is set to 16T16R or 32T32R , LTE FDD and NR Flash Dynamic Spectrum Sharing cannot work with compact bandwidth.
LT E FD D	Uplink data transmission path selection	NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If LTE key event assurance is enabled, uplink data transmission path selection does not take effect.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Physical Resource Slicing	PHY_RES_SLIC ING_SWITCH option of the CellAlgoExtS witch.Operat orFunSwitch parameter	<i>Physical Resource Slicing (FDD)</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with Physical Resource Slicing, the number of available PRBs in each LTE PRB group varies as follows: • If resources are reserved based only on SPIDs, the number of available PRBs in an LTE PRB group equals (Total spectrum resources – Number of PRBs shared with the non-LTE side) x Proportion of reserved uplink or downlink PRBs for the PRB group. • If resources are reserved based on both operators and SPIDs, the number of available PRBs in an LTE PRB group equals (Total spectrum resources – Number of

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
				PRBs shared with the non-LTE side) x Proportion of reserved uplink or downlink PRBs for the corresponding operator x Proportion of reserved uplink or downlink PRBs for the PRB group.
LT E FD D	Downlink maximum cell transmit power increase	EDIMaxTXPwrSwitch option of the CellAlgoSwitch . DLPcAlgoSwitch parameter	None	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, if the function of downlink maximum cell transmit power increase is enabled, the PSDs of LTE and NR will exceed the required values. As a result, cells may fail to be activated.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Load simulation	CellSimuLoad.SimuLoadMode	None	Only global and subframe-level load simulation can be used in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. Global load simulation is enabled when the CellSimuLoad.SimuLoadMode parameter is set to GLOBAL_SIMU_LOAD. Subframe-level load simulation is enabled when the CellSimuLoad.SimuLoadMode parameter is set to SUBFRAME_SIMU_LOAD. If different load simulation modes are used in multiple LTE split cells that share spectrum with an NR cell, load simulation will not achieve expected results.

- Function impacts on the NR side
 - Inter-RAT coordination functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the power that can be shared between NR carriers decreases.
NR FD D	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE FDD and NR Flash Dynamic Power Sharing is enabled, the power sharing capability of the NR spectrum sharing cell is lower than that of common cells. As a result, the power sharing benefits decrease.

- MIMO functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Uplink MU grouping and pairing	UL_MU_GRP_ PAIR_SW option of the NRDUCellUIM imo.UlMuMi moAlgoSwitc h parameter	<i>Smart 8T8R</i> <i>(FDD)</i> <i>Smart Massive</i> <i>MIMO (FDD)</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the NR PDCCH and LTE PDCCH/PCFICH/ PHICH need to avoid conflicts with each other, and the NRDUCellPdcch. OccupiedSymbol Num parameter must be set to 1SYM or 2SYM . As a result, if CCE resources are insufficient, the gains of uplink MU grouping and pairing are affected, but there are no negative impacts.
NR FD D	Frequency- domain high- resolution power control	FREQ_HIGH_R ESOLUT_PWR _CTRL_SW option of the NRDUCellPds ch.DlPdschAlg oSwitch parameter	<i>Smart</i> <i>Coverage</i> <i>(FDD)</i> <i>Smart 8T8R</i> <i>(FDD)</i> <i>Smart Massive</i> <i>MIMO (FDD)</i>	Frequency- domain high- resolution power control is not supported in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	PDCCH CCE ratio adjustment in common slots	PDCCH_COM M_SLOT_RATI O_ADJ_SW option of the NRDUCellPdc ch.PdcchAlgo EnhSwitch parameter	<i>Smart Coverage (FDD)</i> <i>Smart 8T8R (FDD)</i> <i>Smart Massive MIMO (FDD)</i>	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, PDCCH CCE ratio adjustment in common slots can take effect only if the cell bandwidth is greater than or equal to 15 MHz.
NR FD D	Downlink control channel power aggregatio n optimizatio n	The following parameter settings are present: • NrDuCellC hnPwr.Max PdcchAggP wrOffset set to 30 • NrDuCellC hnPwr.Pwr AggLevel set to LEVEL_1 • NRDUCellP dcchAlgo.C ce4PwrAgg SinrOffset set to 20 • NRDUCellP dcchAlgo.P dcchEdgeP wrCtrlMax Offset set to 0	<i>Smart Coverage (FDD)</i>	Downlink control channel power aggregation optimization is not supported in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	PDCCH spare power allocation	PDCCH_SPAR E_POWER_AL LOC_SW option of the NRDUCellPdc ch.PdcchAlgo EnhSwitch parameter	<i>Smart Coverage (FDD)</i>	PDCCH spare power allocation is not supported in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	3D coverage pattern	NRDUFddCell TrpBeam.Cov erageScenario set to SCENARIO_20 2, SCENARIO_20 5, SCENARIO_20 7, SCENARIO_20 8, SCENARIO_24 0, SCENARIO_24 1, SCENARIO_24 2, SCENARIO_20 1, SCENARIO_22 0, or SCENARIO_20 4	<i>Beam Management</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled: Four SSB beams can be used only if all the following conditions are met: • The NR cell bandwidth is 20 MHz. • The NRDUFddCell TrpBeam.Cove rageScenario parameter is set to SCENARIO_20 2, SCENARIO_20 5, SCENARIO_20 8, SCENARIO_24 0, SCENARIO_24 1, or SCENARIO_24 2. • The SSB_ADJ_SW option of the NRDUCellAlgo Switch.Spectr umCloudEnhS witch parameter is selected. • When spectrum power sharing mode is used,

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				the values in dB corresponding to the configured values of the CellChPwrCfg.PbchPwr and CellChPwrCfg.SchPwr parameters do not exceed the PA to be used when the even power distribution is used for the PDSCH (specified by the CellDlpcPdschPa.PaPcOff parameter). • The EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter is deselected. Two SSB beams can be used only if all the following conditions are met: • The NRDUfddCellTrpBeam.CoverageScenario parameter is set to SCENARIO_201,

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				SCENARIO_204, SCENARIO_207, or SCENARIO_220. • The SSB_ADJ_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter is selected. • The EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter is deselected. In other situations, only one SSB beam can be used, and the coverage scope is the same as that in the default coverage scenario (indicated by the value DEFAULT).

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Intra- gNodeB Distributed MIMO	INTRA_GNB_DL_D_MIMO_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>	After LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, intra-gNodeB coherent joint transmission and robust weight for joint transmission in Intra-gNodeB Distributed MIMO do not take effect.

– NSA networking functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Uplink fallback to LTE	UL_FALLBACK_TO_LTE_SWITCH option of the NRCeIlNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	If LTE key event assurance is enabled, uplink fallback to LTE does not take effect.
NR FD D	Uplink data transmissio n path selection	NSA_DC_UL_PATH_SELECTION_SW option of the NRCeIlNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If LTE key event assurance is enabled, uplink data transmission path selection does not take effect.

– Coverage improvement functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	DMRS interferenc e reduction	DMRS_RESOU RCE_ADJ_SW option of the NRDUCellChn CovAlgo.UlCo verageAlgoS witch parameter	<i>Turbo Coverage (FDD)</i>	The benefits provided by DMRS interference reduction reduce when it is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
Lo w- fre qu en cy NR TD D	Super uplink with SUL and NR FDD co- deploymen t	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with super uplink with SUL and NR FDD co-deployment, the amount of uplink spectrum resources that can be allocated to implement super uplink decreases if the LTE FDD or NR traffic volume increases. As a result, the uplink throughput of the SUL carrier decreases. If uplink semi-persistent scheduling, eMTC, or compact bandwidth is also enabled in the LTE cell, super uplink with SUL and NR FDD co-deployment cannot take effect as these functions periodically or continuously occupy spectrum resources. When the NR TDD and NR FDD cells enabled with super uplink with SUL and NR FDD co-deployment are configured on the same

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				baseband processing unit, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be enabled together with super uplink with SUL and NR FDD co-deployment.
NR FD D	Extended cell range (FDD)	NRDUCell.Cel lRadius set to a value greater than 14500 and less than or equal to 100000	<i>Extended Cell Range</i>	The LNR_PRACH_RES _ALLOC_OPT_SW option of the NRDUCellAlgoS witch.SpectrumC loudEnhSwitch parameter must be selected when both LTE FDD and NR Flash Dynamic Spectrum Sharing and extended cell range (FDD) are in effect in a cell with a heavy uplink load. This is because the access success rate of NR UEs decreases and the probability of false alarms increases in this scenario.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	RuralLink antenna solution (NR)	EASY_OMNI_S ITE_MODE_S W option of the NRDUFddCell TrpBeam.Scen arioModeSw parameter	<i>RuralLink Solution</i>	The RuralLink antenna solution can take effect together with LTE FDD and NR Flash Dynamic Spectrum Sharing only if each of the following conditions is met: • The NR cell bandwidth is 15 MHz or 20 MHz. • The SSB_ADJ_SW option of the NRDUCellAlgo Switch.Spectr umCloudEnhS witch parameter is selected.

- Energy saving functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Energy saving based on flexible frequency- domain scheduling	FLEX_FREQ_S CH_ENERGY_S AVING_SW option of the NRDUCellAlg oSwitch.Power rSavingSwitc h parameter	<i>Energy Conservation and Emission Reduction</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits provided by energy saving based on flexible frequency-domain scheduling decrease, and the capacity on the LTE side decreases.
NR FD D	Power saving scheduling	POWER_SAVI NG_SCHEDUL E_SW option of the NRDUCellAlg oSwitch.Power rSavingSwitc h parameter	<i>Energy Conservation and Emission Reduction</i>	When both LTE FDD and NR Flash Dynamic Spectrum Sharing and power saving scheduling take effect, the benefits provided by power saving scheduling decrease, and the LTE capacity may decrease. If the gNBPDcpParamG roup.DlDataPdcp SplitMode parameter is set to SCG_AND_MCG in this scenario, the downlink throughput of NR cells decreases.

– Basic functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FDD	Downlink power control	None	<i>Power Control</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the downlink power cannot be increased.
NR FDD	Downlink adaptive selection of resource allocation type 0 or 1	TYPE0_TYPE1_ADAPTIVE_SCH_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	Before enabling downlink adaptive selection of resource allocation type 0 or 1 in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, you are advised to select the DL_RB_ESTIMATE_OPT_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter to prevent UE data packets from being truncated.
NR FDD	DC component rate matching	DC_COMPONENT_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, configurations for DC component rate matching will become invalid.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	PDCCH rate matching	PDCCH_RATE MATCH_SW option of the NRDUCellPds ch.RateMatch Switch parameter	<i>Downlink Scheduling</i>	When both PDCCH rate matching and LTE FDD and NR Flash Dynamic Spectrum Sharing are enabled, fewer RBs are available for LTE in the downlink and the MCS index decreases. As a result, the downlink UE throughput and cell capacity may decrease.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Load- based adaptive downlink scheduling	NRDUCellDIS ch.DISchPolic y set to ADAPTIVE	<i>Downlink Scheduling</i>	When basic functions of LTE FDD and NR Flash Dynamic Spectrum Sharing take effect, the number of available downlink RBs in an NR cell decreases. As a result, the downlink PRB usage may be lower than the downlink scheduling heavy- load PRB usage threshold for triggering the load-based adaptive downlink scheduling function, reducing the scenarios where the load- based adaptive downlink scheduling function can provide benefits.
NR FD D	RB allocation for downlink experience optimizatio n	NRDUCellDIS ch.Res.DIExpO ptRblncrease Pct set to a non-zero value	<i>Downlink Scheduling</i>	When RB allocation for downlink experience optimization takes effect together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the capacity on the LTE side decreases.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Unbalance d transmit power configurati ons between channels	NRDUCellCov erage.Transmi tPowerOffset Chn0 , NRDUCellCov erage.Transmi tPowerOffset Chn1 , NRDUCellCov erage.Transmi tPowerOffset Chn2 , or NRDUCellCov erage.Transmi tPowerOffset Chn3 set to a non-zero value	<i>Cell Management</i>	When the function of unbalanced transmit power configurations between channels is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, and the spectrum power sharing mode is used, the number of downlink available RBs in the NR cell decreases.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	LTE and NR flexible frame offset	LNR_FRAME_ OFFSET_SW option of the RRUEXTINFO. RruOptlFeats w parameter	<i>Cell Management</i>	Assume that the LTE and NR flexible frame offset function is enabled. LTE FDD and NR Flash Dynamic Spectrum Sharing can take effect only when the "LTE and NR 3-ms frame offset difference" function is enabled. (The "LTE and NR 3-ms frame offset" function is controlled by the LNR_3MS_FRAM E_OFFSET_DIFF_S W option of the NR parameter NRDUCellSpctSh are.LnrSpectrum ShareSwitch and the LNR_3MS_FRAM E_OFFSET_DIFF_S W option of the LTE parameter SpectrumCloud.S pectrumCloudExt Switch).

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	SSB Time Position	NRDUCell.Ssb TimePos	<i>Channel Management</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled: • The NRDUCell.Ssb TimePos parameter can be set to a value other than DEFAULT only if the NRDUFddCell TrpBeam.CoverageScenario parameter is set to DEFAULT or SCENARIO_203. • If the LTE_NR_FDD_SPCT_SHR_AS YM_SW option of the NRDUCellAlgo Switch.SpectrumCloudEnhSwitch parameter is deselected, the following conditions must be met: – If the SSB_ADJ_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				is selected, the NRDUCell.SsbTimePos parameter must be set to DEFAULT. - If the SSB_ADJ_SW option of the NRDUCellAlogoSwitch.SpectrumCloudEnhSwitch parameter is deselected, the NRDUCell.SsbTimePos parameter must be set to DEFAULT or SSB23_PCI_MOD2_STAGGER.

- Spectrum management functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Compact bandwidth	NRDUCell.CustomizedBwConfglnd set to CONFIG	<i>Scalable Bandwidth</i>	When compact bandwidth is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing and a compact bandwidth of 9.6 MHz or 9.8 MHz is used, the NR SSB cannot be configured due to the few number of available RBs. Therefore, it is recommended that the number of CRS ports (specified by the Cell.CrsPortNum parameter) be set to 1 or 2 for the LTE cell. This configuration suggestion applies only to SA networking. This impact relationship exists only when 3900 or 5900 series base stations are used. When the Cell.CustomizedBandwidthCfglnd parameter is set to CFG and the Cell.TxRxMode parameter is set to 16T16R or 32T32R, LTE FDD and NR Flash Dynamic Spectrum Sharing

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				cannot work with compact bandwidth. When the NR cell is an NR SDL cell, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be used together with compact bandwidth.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	UE bandwidth adaptation	UE_BW_ADAP TIVE_SW option of the NRDUCellBw p.BwpConfigS witch parameter	<i>Scalable Bandwidth</i>	When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled with UE bandwidth adaptation: • The frequency-domain positions of the PRACH and PUCCH are confined to the bandwidth range specified by the NRDUCellBwp.UlMinCarrierBw parameter, and the frequency-domain position of PDCCH CORESET 0 is confined to the bandwidth range specified by the NRDUCellBwp.DlMinCarrierBw parameter. • If the NRDUCell.CustomizedUlBandwidth parameter is set to CELL_BW_17DOT5M, the NRDUCellBwp.UlMinCarrierBw parameter must be set to 10M.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				• If the NRDUCell.Cus tomizedDlBan dwidth parameter is set to CELL_BW_17D OT5M, the NRDUCellBwp .DlMinCarrier Bw parameter must be set to 10M. • If the NRDUCell.Cus tomizedDlBan dwidth parameter is set to CELL_BW_17D OT5M and "LTE and NR PUCCH resource optimization" (controlled by the LNR_PUCCH_R ES_OPT_SW option of the NRDUCellPucc h.PucchAlgoEx tSwitch parameter) is enabled, setting the CellRbReserve .RbRsvStartIn dex parameter to an excessively large value may cause the uplink NR and LTE channels to overlap and

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				the NR cell to fail to be activated. Considering this risk, the recommended setting of the CellRbReserve.RbRsvStartIndex parameter is as follows: In high-frequency compact bandwidth scenarios, it is recommended that the CellRbReserve.RbRsvStartIndex parameter be set to (1) 12 or less if PUCCHCfg.DeltaShift is set to DS1_DELTA_SHIFT; (2) 11 or less if PUCCHCfg.DeltaShift is set to DS2_DELTA_SHIFT; (3) 8 or less if PUCCHCfg.DeltaShift is set to DS3_DELTA_SHIFT. In low-frequency compact bandwidth scenarios, it is recommended

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				that the CellRbReserve.RbRsvStartIndex parameter be set to 23 or less. • For a cell with the NRDUCell.CustomizedDLBandwidth parameter set to CELL_BW_17DOT5M, downlink intra-FR inter-band CA or downlink intra-band CA can take effect only when the LNR_PUCCH_RESOURCES_OPT_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter is selected.

- Network sharing functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	RAN sharing with common carrier	SHARED_FRE Q option of the gNBSharingM ode.gNBMulti OpSharingMo de parameter	<i>Multi- Operator Sharing</i>	When RAN sharing with common carrier is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the number of available RBs for an NR operator is calculated as follows: (Total number of RBs – Number of RBs shared with LTE) x RB allocation proportion configured for the operator.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Operator-specific RB management	RB_DYNAMIC_SHARING_SW and MAX_RB_LIMIT_SW options of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When both the function of operator-specific limitation on the maximum number of available RBs and LTE FDD and NR Flash Dynamic Spectrum Sharing are enabled, the maximum number of RBs available for an operator is calculated using the following formula: Maximum number of RBs available for an operator = Available bandwidth of the NR cell (not the bandwidth configured for the NR cell) x Maximum percentage of downlink RBs.
NR FD D	Percentage-based dynamic RB sharing among operators	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When both percentage-based dynamic RB sharing among operators and PSD adaptation are enabled, the downlink cell throughput may decrease.

- Network slicing functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	NR DU cell resource manageme nt between network slices	RB_DYNAMIC _CONTROL_S W option of the NRDUCellAlg oSwitch.Netw orkSliceAlgoS witch parameter	<i>Network Slicing</i>	When both the NR DU cell resource management between network slices function and LTE FDD and NR Flash Dynamic Spectrum Sharing are enabled, the number of RBs available for a network slice group is calculated using the following formula: Number of RBs available for a network slice group = Available bandwidth of the NR cell (not the bandwidth configured for the NR cell) x Percentage of RBs configured for the network slice group. When both the NR DU cell resource management between network slices function and PSD adaptation are enabled, the downlink cell throughput may decrease.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORKSLICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgoExtSwitch parameter	<i>Network Slicing</i>	When both "user-rate-based dynamic network slice resource management" and LTE FDD and NR Flash Dynamic Spectrum Sharing take effect, dynamic resource adjustment is inaccurate, resulting in decreased user-perceived rates of UEs in a network slice group.

- Resource control functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	RFSP- group- based RB resource control	RB_DYNAMIC _CONTROL_S W option of the NRDUCellCo mmonSw.Rfs pAlgoSwitch parameter	<i>Resource Control in NSA Networking</i>	When RFSP- group-based RB resource control is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the number of RBs available for an RFSP group equals the number of RBs on the available bandwidth (not configured bandwidth) of the NR cell multiplied by the RB proportion configured for the RFSP group.
NR FD D	QCI-based resource control	RB_DYNAMIC _CONTROL_S W option of the NRDUCellFea tureSw.QciBa sedDiffServEx pSw parameter	<i>QCI-based Resource Control</i>	When QCI-based resource control is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the number of RBs available for a QCI group equals the number of RBs on the available bandwidth (not configured bandwidth) of the NR cell multiplied by the RB proportion configured for the QCI group.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When both "user-rate-based dynamic QCI resource management" and LTE FDD and NR Flash Dynamic Spectrum Sharing take effect, dynamic resource adjustment is inaccurate, resulting in decreased user-perceived rates of UEs in a QCI group.

– Carrier management functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	The single downlink control information (DCI) function does not take effect for a UE if LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled in the PCell of the UE.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Downlink fast intra- gNodeB CA	CA_ULTRA_LO W_LATENCY_ SPLIT_SW option of the NRDUCellCarr Mgmt.CaEnha ncedAlgoExtS witch parameter	<i>CA Experience Improvement</i>	Downlink fast intra-gNodeB CA does not take effect on a secondary component carrier (SCC) that participates in dynamic spectrum sharing.
NR FD D	Downlink intra-FR inter-band CA	INTRA_FR_IN TER_BAND_C A_SW option of the NRDUCellAlg oSwitch.CaAl goSwitch parameter	<i>Carrier Aggregation</i>	If downlink intra- FR inter-band CA is enabled after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the related configuration requirements described in 4.1.1.3.2 NR Functions must be met. Otherwise, downlink intra-FR inter-band CA does not take effect, or the NR cell will go out of service. When LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled for a cell, the cell is not configured as an SCell for a UE if the UE does not support LTE CRS rate matching in the operating band of the cell.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Downlink intra-band CA	INTRA_BAND _CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If downlink intra-band CA is enabled after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the related configuration requirements described in 4.1.1.3.2 NR Functions must be met. Otherwise, downlink intra-band CA does not take effect, or the NR cell will go out of service.

– Functions related to RAN services

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Cloud X services	CLOUD_VR_SERVICE_SWITCH option of the NRCellAlgoSwitch.CloudVrSwitch parameter	<i>Ultimate Cloud X Experience</i>	In a spectrum sharing cell in the LTE key event assurance state, the downlink data of Cloud X bearers is transmitted in the master cell group (MCG).

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Uplink RB reservation	UL_RB_RSV_SW option of the NRCellAlgoSwitch . <i>VonrSwitch</i> parameter	<i>VoNR</i>	It is not recommended that uplink RB reservation be enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing. This is because if the reserved RBs are configured in the shared spectrum and some of the shared spectrum is occupied by LTE, interference will be generated on voice service UEs that occupy the reserved RBs in neighboring NR cells.

– Other functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Intra-base- station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch . <i>CompSwitch</i> parameter	<i>CoMP</i>	After intra-base-station DL CoMP is enabled, available LTE resources decrease and LTE user experience deteriorates.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Interferenc e avoidance for indoor cells	INDOOR_INT RF_AVOID_S W option of the NRDUCellAlg oSwitch.Indoo rOutdoorCoo perateSw parameter	<i>3D Networking Experience Improvement</i>	After LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, the benefits provided by the function of interference avoidance for indoor cells decrease because the start position for PDSCH scheduling is no longer fixed and some of the positions for PDSCH scheduling in indoor and outdoor cells are already different on the NR side. If the positions for PDSCH scheduling are completely different between indoor and outdoor cells on the NR side after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled, enabling the function of interference avoidance for indoor cells leads to occasional decreases in the throughputs of these cells but improves the downlink spectral

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
				efficiency of the entire network.
NR FD D	High-speed Railway Superior Experience	NRDUCell.Hig hSpeedFlag	<i>High Speed Mobility</i>	When High-speed Railway Superior Experience and LTE FDD and NR Flash Dynamic Spectrum Sharing are both enabled, the high moving speeds of UEs in high-speed cells adversely affect the performance of flash dynamic spectrum sharing UEs. In addition, flash dynamic spectrum sharing UEs do not support two downlink additional DMRS positions, affecting the PDSCH demodulation performance of UEs in high-speed cells. Therefore, it is recommended that these two functions not be used together.
Lo w- fre qu en cy NR TD D	Flexible spectrum access	FLEX_SPCT_A CCESS_SW option of the NRDUCellCarr Mgmt.MultiB andFlexSchS witch parameter	<i>Uplink Flexible Spectrum Scheduling</i>	Flexible spectrum access (FSA) does not take effect in 16T16R or 32T32R NR FDD cells with LTE FDD and NR Flash Dynamic Spectrum Sharing enabled.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	Cell load simulation	NRDUCellSim ulateLoad. <i>TestMode</i>	None	Only FDD and subframe-level load simulation can be used in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios. FDD load simulation is enabled when the NRDUCellSimulateLoad. <i>TestMode</i> parameter is set to FDD_SIMULATED_LOAD_MODE . Subframe-level load simulation is enabled when the NRDUCellSimulateLoad. <i>TestMode</i> parameter is set to SUBFRM_SIMULATED_LOAD_MODE .

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Huawei devices (including the eNodeB and gNodeB) must be used on the radio access network.

Base Station Models

- LTE
 - 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, BBU5900A, BookBBU5901a, or BookBBU5901.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910, and DBS5900 LampSite must be configured with the BBU5900 or BBU5910. When applied to DBS3900

LampSite and DBS5900 LampSite, LTE FDD and NR Flash Dynamic Spectrum Sharing is a trial feature and supports sharing with only one LTE cell.

- NR
 - 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, BBU5900A, BookBBU5901a, or BookBBU5901.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910, and DBS5900 LampSite must be configured with the BBU5900 or BBU5910. When applied to DBS3900 LampSite and DBS5900 LampSite, LTE FDD and NR Flash Dynamic Spectrum Sharing is a trial feature and supports sharing with only one LTE cell.

When the NR cell is an NR SDL cell, only 3900 and 5900 series base stations support flash dynamic spectrum sharing between LTE FDD and NR SDL cells.

Boards

- LTE

When the cell TX/RX mode is 2T2R, 2T4R, or 4T4R, only a UBBPd/UBBPe/UBBPg/UBBPh supports this function. [Table 4-38](#) lists the cell number specifications of different baseband processing units after this function is enabled. When the cell bandwidth is 17.5 MHz, only a UBBPe/UBBPg/UBBPh supports this function. Among main control boards, only a UMPTb/UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga/UMPTga supports this function.

When the cell TX/RX mode is 8T8R, only a UBBPe4/UBBP6/UBBP18/UBBPg/UBBPh supports this function. When LTE FDD and NR Flash Dynamic Spectrum Sharing (single LTE cell) is enabled, the number of cells supported by each baseband processing unit is the same as that before this function is enabled. When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled, the cell number specifications of different baseband processing units are listed in [Table 4-39](#). Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga supports this function.

When the cell TX/RX mode is 16T16R, FDD baseband processing units compatible with 16T16R massive MIMO support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *LTE Technical Specifications of Baseband Processing Units in 3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga supports this function.

When the cell TX/RX mode is 32T32R, FDD massive-MIMO-capable baseband processing units (except the UBBPex2) support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *LTE Technical Specifications of Baseband Processing Units in 3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTga/UMPTga/UMPTga supports this function.
- NR

When the cell TX/RX mode is 2T2R, 2T4R, or 4T4R, only a UBBPg1/UBBPg1a/UBBPg2/UBBPg2a/UBBPg3/UBBPg3b/UBBPh supports this function. The cell

number specifications of these baseband processing units are the same as those before this function is enabled. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTHa/UMPTHb supports this function.

When the cell TX/RX mode is 8T8R, only a UBBPg/UBBPh supports this function. The cell number specifications of the baseband processing unit are the same as those before this function is enabled. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTHa/UMPTHb supports this function.

When the cell TX/RX mode is 16T16R, FDD baseband processing units compatible with 16T16R massive MIMO support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *NR Technical Specifications of Baseband Processing Units in 3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTHa/UMPTHb supports this function.

When the cell TX/RX mode is 32T32R, FDD massive-MIMO-capable baseband processing units support this function. The number of cells supported by each baseband processing unit is the same as that before this function is enabled. For details, see *NR Technical Specifications of Baseband Processing Units in 3900 & 5900 Series Base Station Product Documentation*. Among main control boards, only a UMPTe/UMPTg/UMPTga/UMPTHa/UMPTHb supports this function.

When the NR cell is an NR SDL cell, the cell TX mode must be 2T or 4T.

Table 4-38 Cell number specifications of LTE baseband processing units (single LTE cell)

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD or LM ^a	UBBPd5	Decrease or remain unchanged The cell number specifications decrease from six (2T2R) cells to three cells, or remain three (2T4R or 4T4R) cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing ^b .
LTE FDD or LM	UBBPd6	Decrease The cell number specifications decrease from six (2T2R, 2T4R, or 4T4R) cells to three cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing ^b .

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD or LM	UBBPe1	Remain unchanged The cell number specifications remain unchanged and are three (2T2R) cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing.
LTE FDD or LM	UBBPe2	Remain unchanged The cell number specifications remain unchanged and are three (2T2R, 2T4R, or 4T4R) cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing.
LTE FDD or LM	UBBPe3	Remain unchanged The cell number specifications remain unchanged and are six (2T2R) cells or three (2T4R or 4T4R) cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells.
LTE FDD or LM	UBBPe4	Remain unchanged The cell number specifications remain unchanged and are six (2T2R, 2T4R, or 4T4R) cells or three (8T8R) cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells.
LTE FDD or LM	UBBPe18	Remain unchanged The cell number specifications remain unchanged and are six (2T2R, 2T4R, or 4T4R) cells or three (8T8R) cells.	A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing.

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD, LM, or other working modes involving LTE FDD	UBBPe5	Remain unchanged • LTE FDD or LM The cell number specifications remain unchanged and are nine (2T2R, 2T4R, or 4T4R) cells. • Working mode combinations other than LM: The cell number specifications remain unchanged and are six (2T2R, 2T4R, or 4T4R) cells.	• LTE FDD or LM: A maximum of six cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells. • Working mode combinations other than LM: A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells.

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD, LM, or other working modes involving LTE FDD	UBBPe6	Remain unchanged • LTE FDD: The cell number specifications remain unchanged and are 12 (2T2R, 2T4R, or 4T4R) cells. • LM (with BBP.SRT set to DEFAULT): The cell number specifications remain unchanged and are 12 (2T2R, 2T4R, or 4T4R) cells. • LM (with BBP.SRT set to NBIOT_ENHANCE): The cell number specifications remain unchanged and are nine (2T2R, 2T4R, or 4T4R) cells. • Working mode combinations other than LM: The cell number specifications remain unchanged and are nine (2T2R, 2T4R, or 4T4R) cells. • LTE FDD: The cell number specifications remain unchanged and are six (8T8R) cells.	• LTE FDD: A maximum of six cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells. • LM (with BBP.SRT set to DEFAULT): A maximum of six cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells. • LM (with BBP.SRT set to NBIOT_ENHANCE): A maximum of six cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells. • Working mode combinations other than LM: A maximum of three cells can be enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, while the rest are common cells.

Working Mode	Baseband Processing Unit Model	Supported Cell Number Specifications (Compared with Those Before This Function Is Enabled)	Supported Spectrum Sharing Cell Number Specifications
LTE FDD, LM, LN, or other working modes involving LTE FDD	UBBPg/UBBPh	Remain unchanged	Same as the cell number specifications before this function is enabled
a: LM refers to LTE FDD and NB-IoT. b: The UBBPe1 is used as an example for illustrating the cell number specifications. If this board also serves cells enabled with LTE FDD and NR Uplink Spectrum Sharing, this board supports a maximum of three cells enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing and with LTE FDD and NR Uplink Spectrum Sharing.			

Table 4-39 Cell number specifications of LTE baseband processing units (multiple LTE cells)

Working Mode	Baseband Processing Unit Model	Supported LTE Cell Number Specifications
LTE FDD	UBBPe4/UBBPe6/UBBPe18	Three LTE sector split groups are supported, with two LTE cells in each group. A maximum of three LTE sector split groups support this function.
LTE FDD	UBBPg/UBBPh	The supported LTE sector split group specifications are the same as those before this function is enabled.

NOTICE

The number of cells configured on a baseband processing unit cannot exceed the specifications listed in [Table 4-38](#) or [Table 4-39](#). If this condition is not met, NR indicators will be negatively affected, including the RRC connection setup success rate, QoS flow setup success rate, and user-perceived rate.

To prevent LTE cell activation failures caused by the cell activation sequence, cells with this function enabled must be bound to baseband processing units supporting this function. If cells are not bound to such baseband processing units, cell activation may fail when the baseband processing units supporting this

function are fully occupied while other baseband processing units do not support this function.

When a UBBPh2/UBBPh3/UBBPh3a/UBBPh3b is used as the interface board to serve 32T32R cells using 2:1 or 3.2:1 CPRI compression and the co-carrier co-CPRI data function is enabled, cell number specifications are represented by an indicator calculated using the following formula: Number of LTE cells in a spectrum sharing cell group x Number of antennas. The sum of the values of this indicator for all spectrum sharing cell groups must be less than or equal to 192. If the number of cells has reached the corresponding specification, subsequent cell setups will fail. If the networking requirements are not met or the UBBPh capacity in terms of the number of cells in baseband combination mode is insufficient, cell setups will fail. For LTE cells, ALM-29240 Cell Unavailable and ALM-29248 RF Out of Service will be reported with "The BBP capability is insufficient" as the cause value. For NR cells, ALM-29870 NR DU Cell TRP Unavailable and ALM-29871 NR DU Cell TRP Capability Degraded will be reported with "The baseband capability is insufficient" as the cause value.

When a UBBPh3a is used as the interface board to serve 16T16R cells using 2:1 or 3.2:1 CPRI compression and the co-carrier co-CPRI data function is enabled, cell number specifications are represented by an indicator calculated using the following formula: Number of LTE cells in a spectrum sharing cell group x Number of antennas. The sum of the values of this indicator for all spectrum sharing cell groups must be less than or equal to 192. If the number of cells has reached the corresponding specification, subsequent cell setups will fail. If the networking requirements are not met or the UBBPh capacity in terms of the number of cells in baseband combination mode is insufficient, cell setups will fail. For LTE cells, ALM-29240 Cell Unavailable and ALM-29248 RF Out of Service will be reported with "The BBP capability is insufficient" as the cause value. For NR cells, ALM-29870 NR DU Cell TRP Unavailable and ALM-29871 NR DU Cell TRP Capability Degraded will be reported with "The baseband capability is insufficient" as the cause value.

Only a UBBPe or a later series baseband processing unit supports the "LTE and NR 3-ms frame offset difference" function.

RF Modules

- 3900 and 5900 series base stations: All 3000 and 5000 series RF modules that meet the following conditions support this function:
 - The RF module is shared by LTE and NR. When the NR cell is an NR SDL cell, the RF module must be configured with the NR SDL mode.
 - The 10 MHz, 15 MHz, or 20 MHz bandwidth is supported and can be configured.
 - The following frequency bands are supported and can be configured: 700 MHz, 800 MHz, 850 MHz, 900 MHz, 1800 MHz, 2100 MHz, 2600 MHz, AWS, and PCS. NR SDL cells can work only in the 1400 MHz frequency band.
 - The following TX/RX modes are supported and can be configured: 2T2R, 2T4R, 4T4R, 8T8R, 16T16R, and 32T32R. NR SDL cells support only 2T or 4T.
Of all 8T8R RF modules, the AAU5972 (active module), integrated 8T8R RRUs, and 8T8R RRUs formed by combining 4T4R RRUs support this function.

Of all 16T16R RF modules, those supporting both LTE FDD and NR FDD support this function.

Of all 32T32R RF modules, those supporting both LTE FDD and NR FDD support this function.

32T32R RF modules support only one LTE and NR spectrum sharing cell group. [Table 4-40](#) lists the carrier number specifications before and after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled in a frequency band.

Table 4-40 Carrier number specifications of 32T32R RF modules

RF Module Type	Original Carrier Number Specifications	Carrier Number Specifications After LTE FDD and NR Flash Dynamic Spectrum Sharing Is Enabled
Single-band module	Two carriers in one band	Three carriers in one band
Dual-band module (AAU5768 and AAU5765 excluded)	Two carriers in one band, four carriers in two bands	Three carriers in one band, five carriers in two bands
Dual-band module (AAU5768 and AAU5765)	Three carriers in one band, four carriers in two bands	Four carriers in one band, five carriers in two bands

When NB-IoT needs to be deployed in a guard band of an LTE cell with LTE FDD and NR Flash Dynamic Spectrum Sharing enabled, only the following RF modules can be used: RRU3262, RRU3965, RRU5309, RRU5508 (700 MHz), RRU5512 (700 MHz), RRU5512t (800 MHz), RRU5519et (800 MHz), RRU5909, RRU5509t, RRU5308, RRU5301, RRU5905, and RRU5910.

The following RF modules support the "LTE and NR 3-ms frame offset difference" function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios:

- AAU578xAB, RRU592xAB, RRU552xAB, and RRU582xAB, where *x* is a digit and *A* and *B* stand for different letters. (*A* and *B* can each be omitted.)
- RRU5508 (02312XBW), RRU5901 (02312WPR), RRU5516 (02313XTQ), RRU5502 (02312BSJ), RRU5904 (02311UWT), RRU5502N (02313EYK), and RRU5509t (02312GEB)
- DBS3900 LampSite and DBS5900 LampSite: All 5000 series pRRUs that meet the following conditions support this function:
 - The RF module is shared by LTE and NR.
 - The pRRUs involved in an RF combination must use the same channels. Specifically, they must all use channels A and B, or use channels C and D.
 - The following frequency bands are supported and can be configured: 700 MHz, 800 MHz, 850 MHz, 900 MHz, 1800 MHz, 2100 MHz, 2600 MHz, AWS, and PCS.

- 2T2R is supported and can be configured.

Cells

NR SDL cells support LTE FDD and NR Flash Dynamic Spectrum Sharing with only a single LTE cell.

The **Cell.WorkMode** parameter must be set to **DL_ONLY** when the corresponding NR cell is an SDL cell and to **UL_DL** when the NR cell is of another type.

Assume that the **LTE_NR_FDD_SPCT_SHR_SW** option of **NRDUCellAlgoSwitch.SpectrumCloudSwitch** is selected, the **LTE_NR_FDD_SPCT_SHR_ASYM_SW** option of **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** is deselected, **NRDUCell.CustomizedBwConfigInd** is set to **CONFIG_LEFT_ALIGN**, **NRDUCell.DlBandwidth** is set to **CELL_BW_20M**, **NRDUCell.CustomizedDlBandwidth** is set to **CELL_BW_17DOT5M**, and **NRDUCellPdcch.OccupiedSymbolNum** is set to **1SYM**: The result of $\text{Rounddown}[(\text{Frequency corresponding to } \text{NRDUCell.SsbFreqPos} - \text{Frequency corresponding to } \text{NRDUCell.DlNarfcn} + 6 \times \text{NRB} \times \text{NRDUCell.SubcarrierSpacing}) / (12 \times \text{NRDUCell.SubcarrierSpacing})]$ must be greater than or equal to 22 and less than or equal to NRB minus 40. NRB indicates the number of RBs corresponding to the cell bandwidth specified in Table 5.3.2-1 in 3GPP TS 38.104.

LTE and NR cells must meet all of the following requirements:

- Bandwidth

The LTE and NR cell bandwidths are both 10 MHz, 15 MHz, or 20 MHz, and LTE and NR cells must have the same uplink and downlink bandwidths.

In addition to the preceding standard cell bandwidths, this function can be used with the compact bandwidth function. For details about the non-standard bandwidths supported by this function, see descriptions of compact bandwidth in [4.3.2.3 Function Impacts](#).

NOTICE

When a non-standard bandwidth is used, in addition to the hardware requirements described in this section, the hardware requirements of compact bandwidth must also be met. For details about the hardware requirements of compact bandwidth, see related descriptions in *Compact Bandwidth (FDD)* in *eRAN Feature Documentation* and in *Scalable Bandwidth* in *5G RAN Feature Documentation*.

- TX/RX mode
 - 3900 and 5900 series base stations:

The TX/RX modes of LTE and NR cells must be the same and must be one of 2T2R, 2T4R, 4T4R, 8T8R, 16T16R, or 32T32R.
 - DBS3900 LampSite and DBS5900 LampSite: 2T2R
- LTE and NR cells must have the same operating frequency band and center frequency.

- The PSD of the LTE and NR cells must be the same. If the total power of the LTE and NR cells exceeds the maximum transmit power of the RF module, the power of the cells needs to be reduced or the RF module needs to be replaced.
- The PA value must meet the following requirements when even power distribution is used for the PDSCH:
 - In the case of 2T2R, 2T4R, or 4T4R cell TX/RX mode: When the cell bandwidth is 10 MHz and the spectrum power sharing mode is used, the **CellDlpcPdschPa.PaPcOff** parameter cannot be set to **DB_6_P_A** or **DB_4DOT77_P_A**.
 - In the case of 8T8R cell TX/RX mode
 - When LTE FDD and NR Flash Dynamic Spectrum Sharing (single LTE cell) is enabled: Assume that the cell bandwidth is 10 MHz and the spectrum power sharing mode is used. If the **CellDlpcPdsch.RemainingPwrUseStrategy** parameter for the LTE cell is not set to **PORT_PWR_BALANCING**, the **CellDlpcPdschPa.PaPcOff** parameter cannot be set to **DB_6_P_A** or **DB_4DOT77_P_A**.
 - When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled: Assume that the cell bandwidth is 10 MHz and the spectrum power sharing mode is used. If the **PDSCHCfg.LogicalPortSwapSwitch** parameter for the two LTE cells is set to the same value, the **CellDlpcPdschPa.PaPcOff** parameter cannot be set to **DB_6_P_A** or **DB_4DOT77_P_A**.
 - In the case of 32T32R cell TX/RX mode: Assume that the cell bandwidth is 10 MHz and the spectrum power sharing mode is used. If one or three LTE cells are configured or the **SectorSplitGroup.TM4WeightScheme** parameter is set to **DEFAULT_MODE**, the **CellDlpcPdschPa.PaPcOff** parameter cannot be set to **DB_6_P_A** or **DB_4DOT77_P_A**.
 - In the case of 32T32R cell TX/RX mode: Assume that the cell bandwidth is 10 MHz, the spectrum power sharing mode is used, and the **BEAM_NUM_ADJ_SWITCH** option of the **SectorSplitGroup.SectorSplitSwitch** parameter is selected. If a **SectorSplitCellExt** MO is configured and one or three LTE cells are included in the sector split group indicated by the **SectorSplitCellExt.SectorSplitGroupId** parameter, or if the **SectorSplitGroupExt.TM4WeightScheme** parameter is set to **DEFAULT_MODE**, the **SectorSplitCellExt.PaPcOff** parameter cannot be set to **DB_6_P_A** or **DB_4DOT77_P_A**.
- LTE and NR cells must be deployed in 1:1 co-coverage mode, and must be served by the same transmit channels of an RF module.
- When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is required, the two LTE cells must have different PCIs as well as different PCI mod 3 values to preferentially ensure LTE performance. If NR performance needs to be preferentially ensured, the two LTE cells must have different PCIs but the same PCI mod 3 value.
- To enable both LTE FDD and NR Flash Dynamic Spectrum Sharing and dynamic point selection (DPS) transmission mode, the following conditions must be met:

- The TX/RX mode is not 8T8R, 16T16R, or 32T32R.
- LampSite base stations are not used.
- The transmit power configurations of activated cells are the same.

4.3.4 Networking

3900 and 5900 Series Base Stations

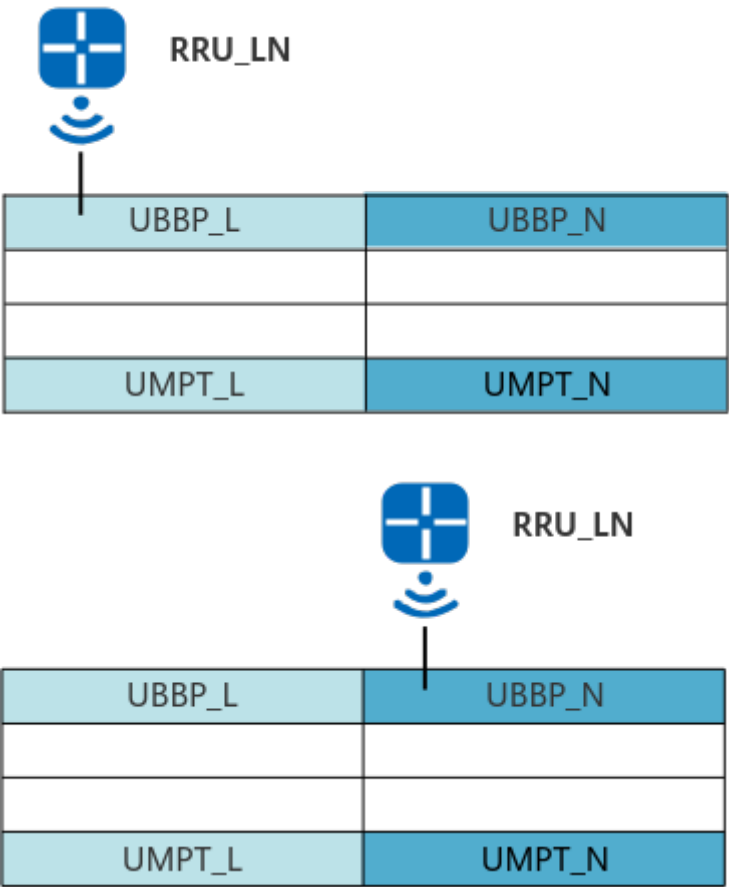
When the LTE and NR co-carrier co-CPRI data function is not enabled (the LTE parameter **LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw** is set to **OFF**):

- If the independent power configuration mode is used, LTE and NR must share a BBU.
- If the spectrum power sharing mode is used, LTE and NR must share a BBU. In addition, the **LNR_PWR_WITH_SPCT_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter must be selected to solve the power allocation conflicts caused by inconsistent LTE and NR latency when each uses their respective optical fibers over the CPRI port.

When the co-carrier co-CPRI data function is enabled (the LTE parameter **LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw** set to **ON**), only the following topologies are supported: the intra-BBU single-fiber CPRI MUX topology in separate-MPT and co-MPT LTE/NR scenarios, and the intra-board cold backup ring topology in co-MPT LTE/NR scenarios. These topologies do not pose special requirements on board installation positions. In an intra-BBU single-fiber CPRI MUX topology and an intra-board cold backup ring topology, LTE and NR cells can be established on the same or different baseband processing units. The following example illustrates the networking requirements using LTE and NR cells established on different baseband processing units.

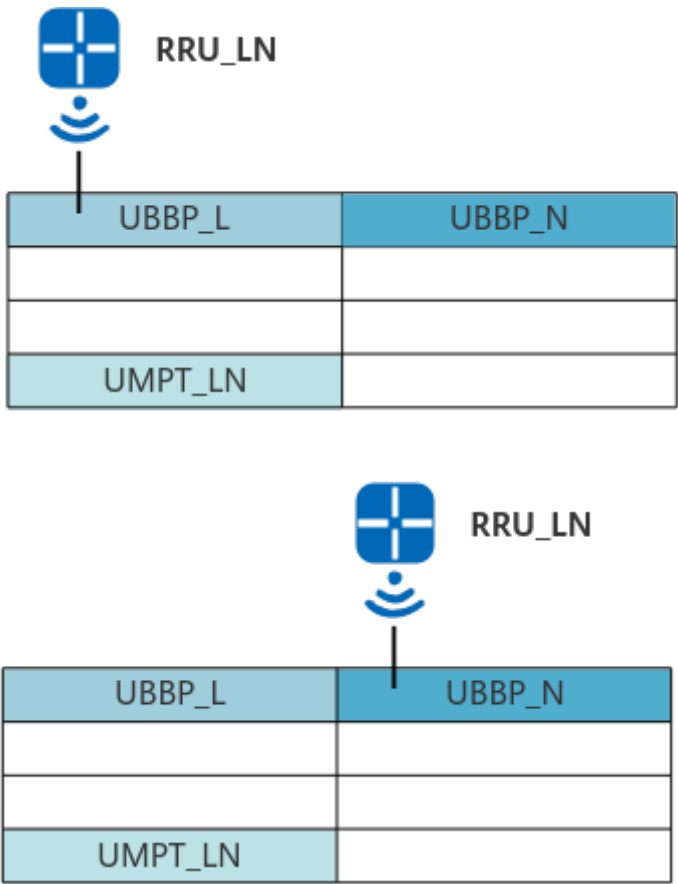
- Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RRU through an optical fiber, as shown in [Figure 4-11](#).

Figure 4-11 Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios



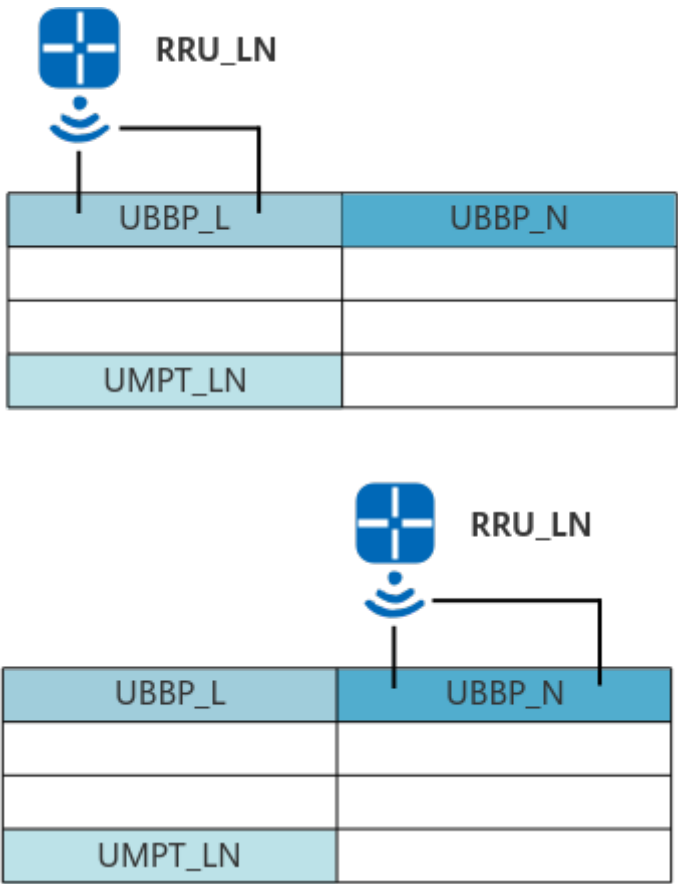
- Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RRU through an optical fiber, as shown in [Figure 4-12](#).

Figure 4-12 Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios



- Intra-board cold backup ring topology in co-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RRU through an optical fiber to form a ring topology, as shown in [Figure 4-13](#).

Figure 4-13 Intra-board cold backup ring topology in co-MPT LTE/NR scenarios



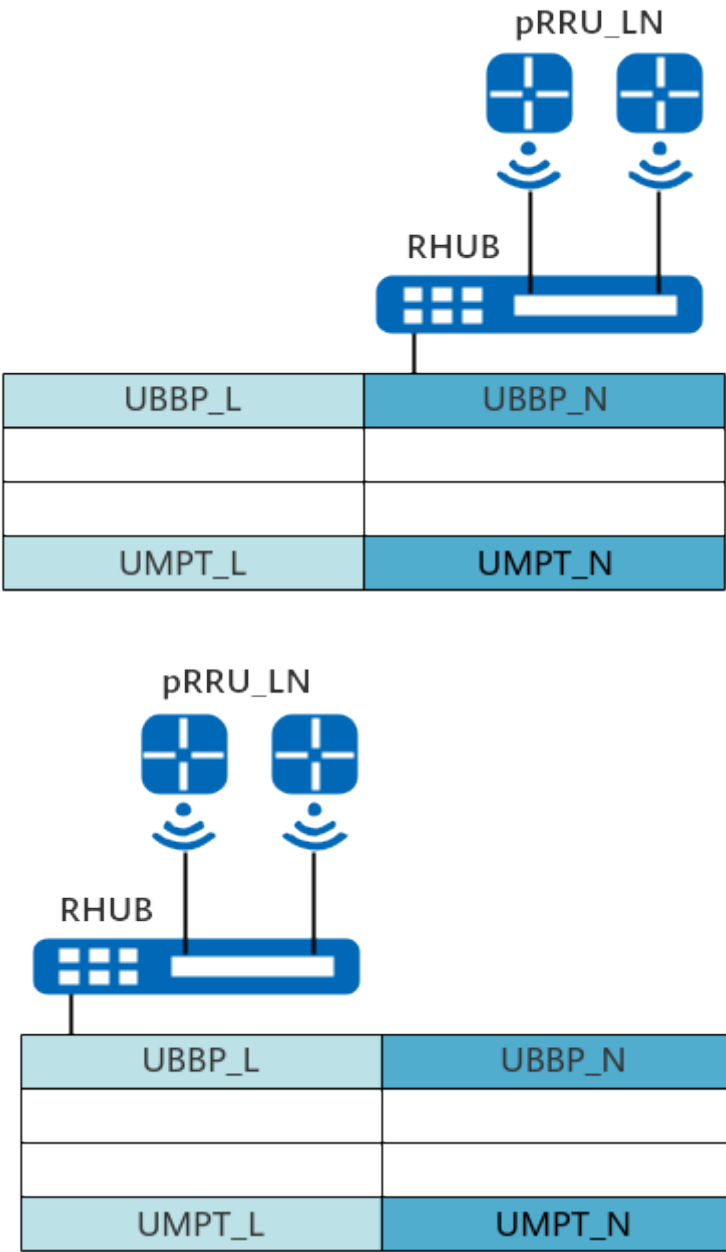
DBS3900 LampSite and DBS5900 LampSite

The co-carrier co-CPRI data function must be enabled by setting the LTE parameter **LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw** to **ON**.

When the preceding requirement is met, the intra-BBU CPRI MUX topology in separate-MPT and co-MPT scenarios can be used. No special requirements are imposed on board installation positions. LTE and NR cells can be established on the same or different baseband processing units. The following example illustrates the networking requirements using LTE and NR cells established on different baseband processing units.

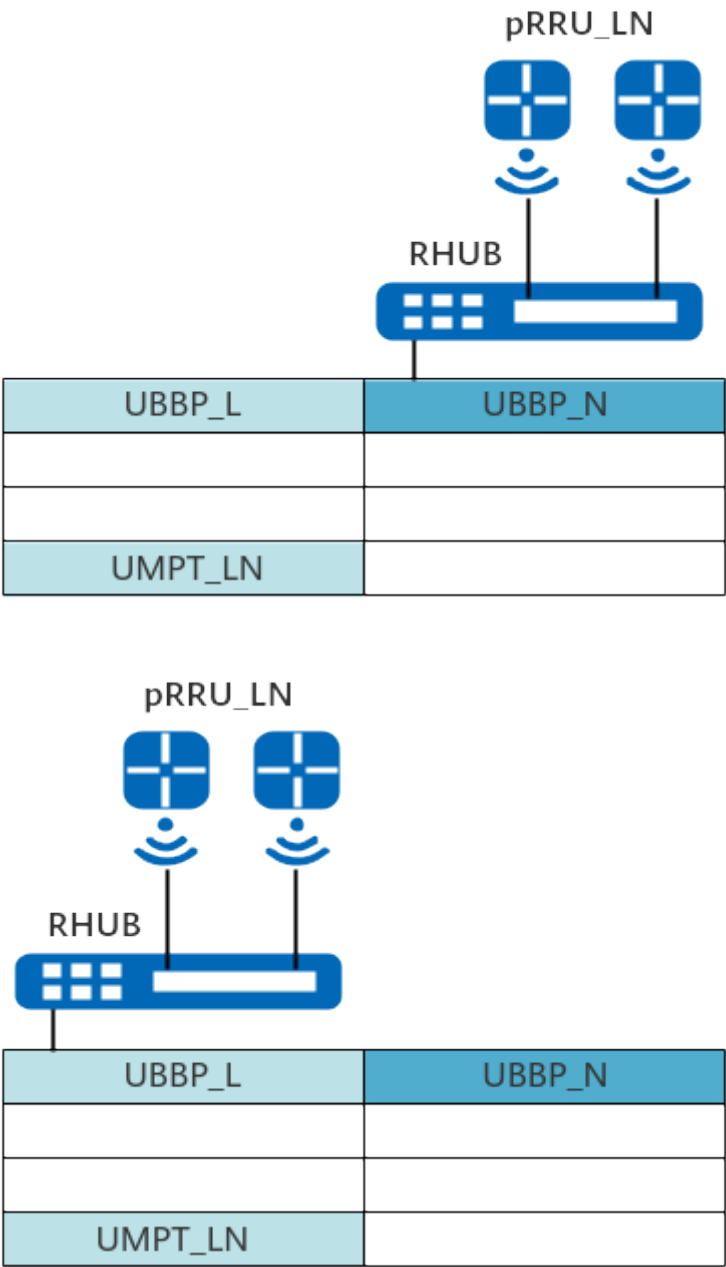
- Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RHUB through an optical fiber, as shown in [Figure 4-14](#).

Figure 4-14 Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios



- Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios: LTE and NR share a BBU, and the LTE or NR baseband processing unit is connected directly to the RHUB through an optical fiber, as shown in [Figure 4-15](#).

Figure 4-15 Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios



For the DBS3900 LampSite and DBS5900 LampSite, LTE and NR cells support only one sector equipment group. The number of pRRUs in a sector equipment group cannot exceed 16.

NOTICE

In separate-MPT scenarios, the versions of NR and LTE must be V100R016C00 or later for LTE FDD and NR Flash Dynamic Spectrum Sharing to take effect. In addition, if NR runs version N , LTE must run a version in the range of $[N - 2, N + 2]$. For example, if NR runs V100R019C10, the earliest version that LTE can run is V100R017C10.

4.3.5 Others

- MRFD-160222 LTE FDD and NR Flash Dynamic Spectrum Sharing (LTE FDD) and MRFD-160262 LTE FDD and NR Flash Dynamic Spectrum Sharing (NR) must be used simultaneously.
- It is recommended that this function be enabled in all cells in the planned deployment area to reduce the interference caused by the near-far effect.
- This function requires LTE FDD and NR FDD networks to be time-synchronized. For details about how to configure time synchronization, see *Synchronization* in *eRAN Feature Documentation* and *Synchronization* in *5G RAN Feature Documentation*.
- NR UEs must support the CRS rate matching function. This function allows the gNodeB to indicate the RE position of the LTE CRS, so that NR UEs can avoid the LTE CRS. UEs support the CRS rate matching function when the *rateMatchingLTE-CRS* IE is included in the *BandNR* IE and the value of the *rateMatchingLTE-CRS* IE is "supported". For details, see section 5.1.4.2 "PDSCH resource mapping with RE level granularity" in 3GPP TS 38.214 V15.5.0. Only the NR UEs that support CRS rate matching on the live network support LTE FDD and NR Flash Dynamic Spectrum Sharing. NR UEs that do not support CRS rate matching cannot access NR FDD cells enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing. They access LTE cells enabled with this function or access NR FDD cells not enabled with this function.

In NSA networking, when NR UEs that do not support CRS rate matching are handed over from a neighboring NR FDD cell not enabled with spectrum sharing to an NR FDD spectrum sharing cell and the handover is rejected, the following KPIs decrease:

- **SgNB Addition Success Rate**
- **SgNB Addition Success Rate (CU)**
- **Inter-SgNB PSCell Change Success Rate (CU)**
- **Inter-SgNB IntraFreq PSCell Change Success Rate (CU)**

In SA networking, when NR UEs that do not support CRS rate matching are handed over from a neighboring NR FDD cell not enabled with spectrum sharing to an NR FDD spectrum sharing cell and the handover is rejected:

- The values of the following indicators decrease:
 - QoS flow setup success rate = **QoS Flow Setup Success Rate (CU)**
 - UE context setup success rate = $(N.UECntx.Est.Succ / N.UECntx.Est.Att) \times 100\%$
 - PDU session establishment success rate = $(N.PDUSession.Est.Succ / N.PDUSession.Est.Att) \times 100\%$
 - E-UTRAN-to-NR incoming handover preparation success rate = $(N.HO.InterRAT.E2N.PrepSuccIn / N.HO.InterRAT.E2N.PrepAttIn) \times 100\%$
- The values of the following counters increase:
 - **N.QosFlow.FailEst.UeCapability**

- **N.UECntx.FailEst.UeCapability**
- **N.PDUSESSION.FailEst.UeCapability**
- **N.RRCRedirection.N2E.UeCapability**
- **L.IRATHO.E2N.Prep.FailOut.PrepFailure**
- If certain parameters are modified for an NR cell engaged in LTE FDD and NR Flash Dynamic Spectrum Sharing, the LTE parameters **CellRbReserve.RbRsvStartIndex** and **CellRbReserve.RbRsvEndIndex** must be modified accordingly for the LTE cell in the same spectrum sharing group as the NR cell. The new settings must ensure that the negotiation between the eNodeB and gNodeB will succeed. Otherwise, the NR cell will fail to be set up. These parameters include the following:
 - **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter, which controls common PUCCH RB resource optimization
 - **PUCCH_COMPRESSION_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter, which controls ultimate PUCCH compression
 - **NRDUCellPucch.CommonPucchResComprDegree** parameter, which specifies the compression degree of common PUCCH RBs

4.4 Operation and Maintenance

4.4.1 Data Configuration



If the activation of this function is to be verified through signaling message tracing, you need to start signaling tracing on the MAE-Access before activating this function in LTE and NR cells. Otherwise, messages regarding the activation of this function cannot be observed after this function is activated.

4.4.1.1 Data Preparation

[Table 4-41](#) and [Table 4-45](#) describe the parameters used for function activation. [Table 4-44](#) and [Table 4-46](#) describe the parameters used for function optimization. This section does not describe parameters related to cell establishment.

The configuration of some parameters can result in automatic cell restarts during function activation and deactivation, and modifying the parameter settings of one RAT will cause both LTE and NR cells to automatically restart. As such, it is recommended that you deactivate cells prior to parameter configuration and reactivate them after parameter configuration is complete. This will prevent the cells from being repeatedly restarted during parameter configuration.

To prevent LTE cell activation failures due to insufficient baseband processing resources, cells enabled with this function must be bound to baseband processing

units supporting this function. If cells are not bound to such baseband processing units, cell activation may fail when the baseband processing units supporting this function are fully occupied while other baseband processing units do not support this function.

Table 4-41 LTE parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Switch	SpectrumCloud.SpectrumCloudSwitch	Set this parameter to LTE_NR_SPECTRUM_SHR .

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	Set this parameter based on the network plan. - When SRS resource allocation to LTE is not required, select the LTE_UE_SRS_NOT_CONFIG_SW option of this parameter. Otherwise, deselect this option. The LTE_UE_SRS_NOT_CONFIG_SW option can only be selected when all of the following conditions are met: - The SRSCfg.SrsSubframeCfg parameter is set to SC3. - The SrsSubframeRecfsSwitch option of the CellAlgoSwitch.SrsAlgoSwitch parameter is deselected. - When the spectrum power sharing mode is used, and the LTE and NR baseband processing units need to be connected to the RRU through their respective optical fibers, select the LNR_PWR_WITH_SPECT_OPT_SW option. When the independent power configuration mode is used, LTE and NR can use their respective optical fibers without needing to select this option.

Parameter Name	Parameter ID	Setting Notes
		Select the WBB_MBB_CONTROL_OPT_SW option of this parameter when the simultaneous use with WBB is required.
LNR Spectrum Sharing Switch	LteNrSpctShrCellGrp.LteNrSpctShrSwitch	When the NR PDSCH needs to avoid the LTE CSI-RS at the RE level according to the network plan, select the LTE_CSI_RS_AVOID_POLICY_SW option of this parameter. This option cannot be selected when the LTE cell TX/RX mode is 8T8R.
LTE and NR Spectrum Sharing Cell Group ID ^a	SpectrumCloud.LteNrSpectrumShrCellGrpId	Set this parameter based on the network plan.
Spectrum Sharing Mode	SpectrumCloud.SpctShrMode	Set this parameter to LTE_NR_PWR_DYN_SHR_WITH_SPCT or LTE_NR_PWR_INDEPENDENT based on the network plan.
LTE and NR Spectrum Sharing Cell Group ID ^a	LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId	Set this parameter based on the network plan. The value of the LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId parameter must be the same as that of the gNBDULteNrSpctShrCg.LteSpctShrCellGrpId parameter.

Parameter Name	Parameter ID	Setting Notes
LTE Pri Res Ratio in LTE and NR Spct Shr	LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio	Set this parameter based on the network plan. The value of this parameter varies with the spectrum allocation policy, and the following lists the configuration suggestions for ensuring network performance when different spectrum allocation policies are adopted: • If spectrum resources need to be preferentially allocated to LTE, it is recommended that this parameter be set to a value within the range of 51 to 80. • If spectrum resources need to be preferentially allocated to NR, it is recommended that this parameter be set to a value within the range of 20 to 49. • If spectrum resources need to be evenly allocated to LTE and NR, it is recommended that this parameter be set to 50. If this parameter is set to a value in the range of 0 to 20 and the value of the SRSCfg.FddSrsCfgMode parameter is changed from DEFAULTMODE to ADAPTIVEMODE, the RRC connection setup success rate may decrease. If this parameter is set to a value less than 20 or greater than 80, the

Parameter Name	Parameter ID	Setting Notes
		access success rate may decrease or the service drop rate may increase in high load scenarios. If this parameter is set to 0, LTE UEs cannot access the network. Therefore, the value 0 is not recommended. If this parameter is set to 100, NR UEs cannot access the network. Therefore, the value 100 is not recommended.
DL SPS Restrict Ratio	SpectrumCloud.DISpsRestrictRatio	Set this parameter based on the network plan. The proportion of RBs allocated for LTE downlink semi-persistent scheduling = $(\text{SpectrumCloud.DISpsRestrictRatio} \times \text{LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio}) / 100\%$.
LTE and NR Co Carrier Co CPRI Data Switch	LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw	Set this parameter based on the network plan. This parameter must be set to ON only when LTE and NR share the same CPRI optical fibers and the CPRI bandwidth is insufficient. If the cell TX/RX mode is 32T32R, this parameter can be set to ON only when the CPRI compression ratio is 2.2:1, 2:1, or 3.2:1. If the cell TX/RX mode is 16T16R, this parameter can be set to ON only when the CPRI compression ratio is 2:1 or 3.2:1.
RB Reserve Mode ^b	CellRbReserve.RbRsvMode	Set this parameter to LNR_SPECTRUM_SHR_RB_RESERVED.

Parameter Name	Parameter ID	Setting Notes
RB Reserve Type ^b	CellRbReserve.RbRsvType	Set this parameter to UPLINK_MODE .
RB Reserve Start Index ^b	CellRbReserve.RbRsvStartIndex	Set this parameter based on the network plan.
RB Reserve End Index ^b	CellRbReserve.RbRsvEndIndex	Set this parameter based on the network plan.
Parameter Configuration Protection Switch	eNodeBResModeAlgo.ParamCfgProtectSw	Deselect the FORBID_PARA_CFG_VERIFY_SW option of this parameter.
Subframe Allocation Mode	CellMbsfnSfEnhConfig.SubframeAllocationMode	Set this parameter to FOUR_FRAME to avoid conflicts with NR SSB and SIB1.
Four Frame Allocation	CellMbsfnSfEnhConfig.FourFrameAllocation	The configuration for this parameter varies depending on the SSB period and the number of SSB beams (1, 2, or 4). - In NSA networking, set this parameter as described in Table 4-42. - In SA networking, set this parameter as described in Table 4-43. The subframes configured for MBSFN must include the subframes for NR SSB.
Radio Frame Allocation Period	CellMbsfnSfEnhConfig.RadioFrameAllocPrd	Set this parameter to N4 to match the setting of the CellMbsfnSfEnhConfig.SubframeAllocationMode parameter.
Radio Frame Allocation Offset	CellMbsfnSfEnhConfig.RadioFrameAllocOfs	Set this parameter to 0 .
Frame Offset ^c	CellFrameOffset.FrameOffset	Set this parameter to a value confirmed by Huawei engineers.

Parameter Name	Parameter ID	Setting Notes
FDD Frame Offset ^c	ENodeBFrameOffset.FddFrameOffset	Set this parameter to a value confirmed by Huawei engineers.
TA Offset	CellFrameOffset.TaOffset	Ensure that the TA offset specified by this parameter on the LTE side is the same as the TA offset (specified by the NRDUCell.TaOffset parameter) on the NR side. Otherwise, the actual TA offset on the LTE or NR side is inconsistent with the configured value. That is, the following configuration requirements must be met: • When the CellFrameOffset.TaOffset parameter is set to 0Ts, the NRDUCell.TaOffset parameter must be set to 0Tc. • When the CellFrameOffset.TaOffset parameter is set to 400Ts, the NRDUCell.TaOffset parameter must be set to 25600Tc. • When the CellFrameOffset.TaOffset parameter is set to 624Ts, the NRDUCell.TaOffset parameter must be set to 39936Tc. Before changing the value of this parameter, deactivate both the LTE and NR cells.

Parameter Name	Parameter ID	Setting Notes
a: In scenarios of a single LTE cell, an LTE spectrum sharing cell group can contain only one LTE cell. In scenarios of multiple LTE cells, an LTE spectrum sharing cell group can contain only the sector split cells of the same LTE FDD sector split group. b: In upgrade scenarios, these four parameters inherit the configurations of the source version and do not need to be configured. In new deployment scenarios, the configurations of these parameters follow the instructions on how to configure the LTE PUCCH in 4.1.1.2 Key Technologies. c: If both the CellFrameOffset.FrameOffset and ENodeBFrameOffset.FddFrameOffset parameters are configured, the frame offset specified by the CellFrameOffset.FrameOffset parameter takes effect for the LTE cell.		

Table 4-42 Configurations for **Four Frame Allocation** in NSA networking

Scenario	Configuration for Four Frame Allocation
One or two SSB beams are used.	- When the SSB period is set to MS5(5), set this parameter to 9586980. - When the SSB period is set to MS10(10), set this parameter to 8521760. - When the SSB period is set to MS20(20) or a larger value, set this parameter to 8390656.
Four SSB beams are used.	- When the SSB period is set to MS10(10), set this parameter to 8521760. - When the SSB period is set to MS20(20) or a larger value, set this parameter to 8390656.
Note: When the NRDUCell.SsbPeriod parameter, which specifies the SSB period, is set to MS5(5) on the NR side, the SubFrame1 and SubFrame6 options of the MBSFN Subframe Configuration parameter must be selected. When the SSB period is set to MS10(10), the SubFrame1 option must be selected.	

Table 4-43 Configurations for **Four Frame Allocation** in SA networking

Scenario	Configuration for Four Frame Allocation
One SSB beam is used.	- When the SSB period is set to MS5(5), set this parameter to 11684132. - When the SSB period is set to MS10(10), set this parameter to 10618912. - When the SSB period is set to MS20(20) or a larger value, set this parameter to 10487808.

Scenario	Configuration for Four Frame Allocation
Two SSB beams are used.	- When the SSB period is set to MS5(5), set this parameter to 15878436. - When the SSB period is set to MS10(10), set this parameter to 14813216. - When the SSB period is set to MS20(20) or a larger value, set this parameter to 14682112.
Four SSB beams are used.	- When the SSB period is set to MS10(10), set this parameter to 14813216. - When the SSB period is set to MS20(20) or a larger value, set this parameter to 14682112.
Note: When the NRDUCell.SsbPeriod parameter, which specifies the SSB period, is set to MS5(5) on the NR side, the SubFrame1 and SubFrame6 options of the MBSFN Subframe Configuration parameter must be selected. When the SSB period is set to MS10(10) , the SubFrame1 option must be selected.	

Table 4-44 LTE parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrNFreq.AggregationAttribute	Select the DSS_FREQ_INDICATION option of this parameter based on the network plan. This option can be selected only in SA networking or NSA and SA hybrid networking. It cannot be selected in NSA networking.

Parameter Name	Parameter ID	Setting Notes
LNR Spectrum Sharing Switch	LteNrSpctShrCellGrp.LteNrSpctShrSwitch	The LNR_RES_ALLOC_ADAPT_SW option of this parameter together with the LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio parameter determines the spectrum resource allocation mode for LTE and NR cells. For details, see 4.1.1.2.2 Flexible Spectrum Priority Mode. When the function of inter-RAT preferential guarantee for GBR services is required, select the GBR_PRI_ALLOC_SW option of this parameter.
LNR Spectrum Sharing Switch	LteNrSpctShrCellGrp.LteNrSpctShrSwitch	When the LTE key event assurance function is required according to the network plan, select the LTE_BIG_EVENT_ASSURANCE_SW option of this parameter.

Parameter Name	Parameter ID	Setting Notes
LNR Spectrum Sharing Switch	LteNrSpctShrCellGrp.LteNrSpctShrSwitch	Select the LTE_NR_SRS_ALLOC_OPT_SW option of this parameter when LTE and NR SRS resource allocation optimization is required based on the network plan. The preceding option can be selected only when the following conditions are all met: • The cell TX/RX mode is not 16T16R or 32T32R. • The Cell.HighSpeedFlag parameter must be set to LOW_SPEED. • The CellUlschAlgo.UlSrsFreqSelSchSinrThld parameter must be set to 255. • The NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter is deselected. • The NCellSrsMeasPara.SrsAutoNCellMeasSwitch parameter is set to OFF. • The LTE_UE_SRS_NOT_CONFIG_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter is deselected.

Parameter Name	Parameter ID	Setting Notes
LNR Spectrum Sharing Switch	LteNrSpctShrCellGrp.LteNrSpctShrSwitch	It is recommended that the LNR_RES_ALLOC_DIRECTION_OPT_SW option of this parameter be selected to enable LTE and NR resource allocation direction optimization.
LTE Big Event UE Number Threshold	LteNrSpctShrCellGrp.LteBigEventUeNumThld	Retain the default value.
LTE Big Event UE Number Threshold Offset	LteNrSpctShrCellGrp.LteBigEventUeNumThldOffset	Retain the default value.
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	It is recommended that the LTE_UPLINK_VOLTE_ENH_SW and LTE_UPLINK_VOLTE_OPT_SW options of this parameter be selected when LTE uplink voice packet loss optimization is required based on the network plan.
Spectrum Cloud Extend Switch	SpectrumCloud.SpectrumCloudExtSwitch	Select the LNR_UPLINK_VOLTE_GTEE_SW option of this parameter when the LNR_RES_ALLOC_ADAPT_SW option of the LteNrSpctShrCellGrp.LteNrSpctShrSwitch parameter is selected and uplink VoLTE optimization is required based on the network plan.
Uplink VoLTE Guaranteed RB Num	SpectrumCloud.UlVoLTEGuaranteedRbNum	The value 3 is recommended.

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Extend Switch	SpectrumCloud.SpectrumCloudExtSwitch	Select the LNR_3MS_FRAME_OFFSET_DIFF_SW option of this parameter when LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with the LTE and NR flexible frame offset function based on the network plan.
DRX Algorithm Extend Switch	CellDrxPara.DrxAlgoExtSwitch	When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with DRX based on the network plan, it is recommended that the DSS_GAP_POS_OPT_SW and DSS_VOICE_NSA_SA_OD_POS_OPT_SW options of this parameter be selected for a cell that prevents the DRX On Duration period from being extended due to MBSFN subframes, NR CSI-RS, or NR TRS.

Parameter Name	Parameter ID	Setting Notes
DSS NSA SA Voice UE Aperiodic CQI Trig Period	CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod	When LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with DRX based on the network plan, it is recommended that this parameter be set to 40MS , 80MS , 160MS , or 320MS for a cell that prevents the DRX On Duration period from being extended due to MBSFN subframes, NR CSI-RS, or NR TRS. The CellCqiAdaptiveCfg.DssNsaSaVoiceApCqiPeriod parameter takes effect only when the DSS_VOICE_NSA_SA_OD_POS_OPT_SW option of the CellDrxPara.DrxAlgoExtSwitch parameter is selected.
Parameter Configuration Protection Switch	eNodeBResModeAlgo.ParamCfgProtectSw	Select the DSS_CROSS_RAT_CFG_CHECK_SW option of this parameter when "cross-RAT configuration check in DSS scenarios" is required based on the network plan.

Table 4-45 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloudification Switch	NRDUCellAlgoSwitch.SpectrumCloudSwitch	- Select the LTE_NR_FDD_SPCT_SHR_SW option of this parameter if the NR cell is an NR FDD cell. - Select the LTE_NR_SDL_SPCT_SHR_SW option of this parameter if the NR cell is an NR SDL cell.

Parameter Name	Parameter ID	Setting Notes
NR Spectrum Sharing Cell Group ID ^a	NRDUCellSpctCloud.NrSpctShrCellGrpId	Set this parameter based on the network plan.
NR Spectrum Sharing Cell Group ID ^a	gNBDULteNrSpctShrCg.NrSpctShrCellGrpId	Set this parameter based on the network plan.
LTE Spectrum Sharing Cell Group ID ^a	gNBDULteNrSpctShrCg.LteSpctShrCellGrpId	Set this parameter based on the network plan. The value of the gNBDULteNrSpctShrCg.LteSpctShrCellGrpId parameter must be the same as that of the LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId parameter.
FDD LNR Spectrum Sharing Switch	gNBDULteNrSpctShrCg.FddLteNrSpctShrSwitch	Select the LTE_CSI_RS_AVOID_SW and LTE_CRS_PORT_1_RM_SW options of this parameter based on the network plan. These two options can be individually or both selected. The LTE_CSI_RS_AVOID_SW option cannot be selected when the NR cell TX/RX mode is 8T8R.
Common Control Resource RB Number	NRDUCellCoreset.CommonCtrlResRbNum	This parameter only needs to be set when the NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM . Set this parameter to RB24 or RB48 .
SSB Frequency Position Describe Method	NRDUCell.SsbDescMethod	It is recommended that this parameter be set to SSB_DESC_TYPE_NARFCN in NSA networking. In SA networking, set this parameter to SSB_DESC_TYPE_GSCN .

Parameter Name	Parameter ID	Setting Notes
SSB Frequency Position	NRDUCell.SsbFreqPos	Set this parameter to a value confirmed by Huawei engineers.
SSB Period	NRDUCell.SsbPeriod	Set this parameter based on the network plan. If the SSB period is changed, change the MBSFN subframe configuration accordingly in the corresponding LTE cell. The subframes configured for MBSFN must include the subframes for NR SSB. The MBSFN subframe configuration varies depending on the SSB period and the number of SSB beams (1, 2, or 4). • In NSA networking, set the MBSFN subframes as described in Table 4-42. • In SA networking, set the MBSFN subframes as described in Table 4-43.
SIB1 Period	NRDUCell.Sib1Period	The value MS40(40) is recommended in SA networking. The value MS160(160) is recommended in NSA networking.
Format1 RB Number	NRDUCellPucch.Format1RbNum	Set this parameter based on the network plan. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in <i>Channel Management</i> .

Parameter Name	Parameter ID	Setting Notes
Format3 RB Number	NRDUCellPucch.Format3RbNum	Set this parameter based on the network plan. For details about the cells for which this parameter takes effect, see the parameter meaning. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in <i>Channel Management</i> .
Format3 CSI-dedicated RB Number	NRDUCellPucch.CsiDedicatedRbNum	Set this parameter based on the network plan. The value RBO is not recommended. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in <i>Channel Management</i> .
Format4 RB Number	NRDUCellPucch.Format4RbNum	Set this parameter based on the network plan. For details about the cells for which this parameter takes effect, see the parameter meaning. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in <i>Channel Management</i> .
Format4 CSI-dedicated RB Number	NRDUCellPucch.Format4CsiDedicatedRbNum	Set this parameter based on the network plan. For details about the configuration suggestions for different scenarios, see data preparation for PUCCH channel management in <i>Channel Management</i> .

Parameter Name	Parameter ID	Setting Notes
Spectrum Sharing Start Symbol	NRDUCellPdcch.<i>SpctShrStartSymbol</i>	Assume that one or two LTE CRS ports are configured. When NR traffic volume requirements are greater than LTE traffic volume requirements, set this parameter to SYM1 ; when NR traffic volume requirements are lower than LTE traffic volume requirements, set this parameter to SYM2 . Otherwise, the proportion of CCE allocation failures increases in the cell with larger traffic volume requirements, affecting the average uplink and downlink UE throughputs. When four LTE CRS ports are configured, set this parameter to SYM2 .
Occupied Symbol Number	NRDUCellPdcch.<i>OccupiedSymbolNum</i>	Set this parameter to 2SYM when the NRDUCellPdcch.<i>SpctShrStartSymbol</i> parameter is set to SYM0 or SYM1 , and to 1SYM when the NRDUCellPdcch.<i>SpctShrStartSymbol</i> parameter is set to SYM2 .
Frame Offset ^b	gNodeBParam.<i>FrameOffset</i>	Set this parameter to a value confirmed by Huawei engineers.
Frame Offset ^b	gNBFreqBandConfig.<i>FrameOffset</i>	Set this parameter to a value confirmed by Huawei engineers.

Parameter Name	Parameter ID	Setting Notes
TA Offset	NRDUCell.TaOffset	Ensure that the TA offset specified by this parameter on the NR side is the same as the TA offset (specified by the CellFrameOffset.TaOffset parameter) on the LTE side. Otherwise, the actual TA offset on the LTE or NR side is inconsistent with the configured value. That is, the following configuration requirements must be met: • When the CellFrameOffset.TaOffset parameter is set to 0Ts, the NRDUCell.TaOffset parameter must be set to 0Tc. • When the CellFrameOffset.TaOffset parameter is set to 400Ts, the NRDUCell.TaOffset parameter must be set to 25600Tc. • When the CellFrameOffset.TaOffset parameter is set to 624Ts, the NRDUCell.TaOffset parameter must be set to 39936Tc. Before changing the value of this parameter, deactivate both the LTE and NR cells.
RB Reserve Mode ^c	NRDUCellRbReserve.RbRsvMode	Set this parameter to NB_GB_RB_RESERVED .

Parameter Name	Parameter ID	Setting Notes
RB Reserve Type ^c	NRDUCellRbReserve.RbRsvType	Set this parameter to UPLINK_MODE or DOWNLINK_MODE to reserve uplink and downlink RBs, respectively.
RB Reserve Start Index ^c	NRDUCellRbReserve.RbRsvStartIndex	Set this parameter to a value confirmed by Huawei engineers. The recommended configuration is described in 4.1.1.3 Simultaneous Use with Other Functions .
RB Reserve End Index ^c	NRDUCellRbReserve.RbRsvEndIndex	Set this parameter to a value confirmed by Huawei engineers. The recommended configuration is described in 4.1.1.3 Simultaneous Use with Other Functions .
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	Set this parameter based on the network plan. When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled, select the LTE_CRS_RATEMATCH_ALL_SYM_SW option if LTE performance needs to be preferentially ensured, and deselect this option if NR performance needs to be preferentially ensured.
a: In the current version, an NR spectrum sharing cell group can contain only one NR cell. b: If both the gNodeBParam.FrameOffset and gNBFreqBandConfig.FrameOffset parameters are configured, the frame offset specified by the gNBFreqBandConfig.FrameOffset parameter takes effect for the NR cell. c: These four parameters need to be configured when LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with NB-IoT deployed in LTE guard band mode.		

Table 4-46 NR parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Downlink Additional DMRS Position	NRDUCellPdsch.DlAdditionalDmrsPos	Set this parameter to POS1 .
LTE Handover Algo Switch	gNodeBParam.LteHandoverAlgoSwitch	Select the LTE_HO_TO_DSS_NR_OPT_SW option of this parameter based on the network plan.
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	Select the LTE_SS_PBCH_RM_OPT_SW option of this parameter based on the network plan.
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	Select the PWR_SPCT_DENSITY_ADAPT_SW option of this parameter based on the network plan.
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	Select the PAGING_ADJUST_SW option of this parameter based on the network plan.
Spectrum Cloud Enhancement Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	Select the SSB_ADJ_SW option of this parameter when the cell TX/RX mode is 8T8R, 16T16R, or 32T32R and four SSB beams are configured. In other scenarios, use the default setting of the SSB_ADJ_SW option.
LTE Big Event Assurance UE Number Thld	gNBDULteNrSpctShrCg.LteBigEventAssureUeNumThld	Retain the default value.
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	You are advised to select the LNR_PRACH_RES_ALLOC_OPT_SW option when the NRDUCellPrach.PrachConfigurationIndex parameter is set to a value greater than 27 .

Parameter Name	Parameter ID	Setting Notes
SSB Time Position	NRDUCell.SsbTimePos	Set this parameter to SSB23_PCI_MOD2_STAGGER when time-domain SSB staggering is required based on the network plan.
Paging Scheduling Algorithm Switch	NRDUCellPagingConfig.PagingSchAlgorithmSwitch	Select the PAGING_OSI_STAGGER_SW option when paging-OSI staggering is required based on the network plan.
Downlink Schedule Algorithm Switch	NRDUCellDISch.DISchAlgorithmSwitch	Select the TIME_DOMAIN_RMSI_STAGGER_SW option when time-domain RMSI staggering is required based on the network plan.
LNR Spectrum Share Switch	NRDUCellSpectShare.LnrSpectrumShareSwitch	Select the LNR_3MS_FRAME_OFFSET_DIFF_SW option of this parameter when LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled together with the LTE and NR flexible frame offset function based on the network plan.
Config Interception Policy Switch	gNBOamParam.CfgInterceptPolicySw	Select the DSS_CROSS_RAT_CFG_CHECK_SW option of this parameter when "cross-RAT configuration check in DSS scenarios" is required based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

This function must be activated or deactivated on both the LTE and NR sides. When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is required, parameters need to be configured for each LTE split cell on the LTE side. Some command examples for such scenarios are provided below.

Activation Command Examples (Single LTE Cell)

Activation command examples on the LTE side

```
//Deactivating the cell
DEA CELL: LocalCellId=0;
//Binding the cell that requires this function to a baseband processing unit supporting this function
MOD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=1, BaseBandEqmId=12;
//Setting the percentage of spectrum resources to be preferentially allocated to LTE MBB services, and
turning on the LTE and NR co-carrier co-CPRI data switch
ADD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSptShrLtePriResRatio=50,
LteNrCoCarrCoCpriDataSw=ON;
//(Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-1;
//(Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-0;
//Turning on the switch controlling LTE and NR flash dynamic spectrum sharing, adding the LTE cell to an
LTE spectrum sharing cell group, setting the spectrum sharing mode (assuming that SRS resources need to
be configured for LTE UEs), turning on the switch controlling power allocation optimization in spectrum
power sharing mode, turning on the WBB/MBB control policy optimization switch, and setting the LTE
downlink SPS restrict ratio
ADD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=LTE_NR_SPECTRUM_SHR,
LteNrSpectrumShrCellGrpId=0, SptShrMode=LTE_NR_PWR_DYN_SHR_WITH_SPCT,
SpectrumCloudEnhSwitch=LTE_UE_SRS_NOT_CONFIG_SW-0&LNR_PWR_WITH_SPCT_OPT_SW-1&WBB_MBB_
CONTROL_OPT_SW-1, DISpsRestrictRatio=60;
//Setting the start position of the LTE PUCCH at the lower end of the frequency domain and the end
position of the LTE PUCCH at the upper end of the frequency domain (assuming that the number of RBs
configured for the NR PUCCH exceeds the number of reserved RBs in a 10 MHz cell)
ADD CELLRBRESERVE: LocalCellId=0, Index=1, RbRsvMode=LNR_SPECTRUM_SHR_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=3, RbRsvEndIndex=46;
//Configuring MBSFN subframes (using SA networking without the SSB dual-beam function as an example)
MOD CELLMBSFNSFENHCONFIG: LocalCellId=0, SubframeAllocationMode=FOUR_FRAME,
FourFrameAllocation=10487808, RadioFrameAllocPrd=N4, RadioFrameAllocOfs=0;
//Setting the frame offset and TA offset
ADD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0,
TaOffset=0Ts;
//Activating the cell
ACT CELL: LocalCellId=0;
```

Activation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Adding an LTE spectrum sharing cell group and an NR spectrum sharing cell group
ADD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteSptShrCellGrpId=0;
//Adding the NR cell to the NR spectrum sharing cell group
ADD NRDUCELLSPCTCLOUD: NrDuCellId=0, NrSptShrCellGrpId=0;
//Setting the number of RBs for the NR PUCCH (using cells for which downlink intra-FR inter-band CA does
not take effect (INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch
parameter deselected) and cells for which the NSA TDM pattern type strategy is not set to LTE uplink
preferred (NRDUCellAlgoSwitch.NsaTdmPatternTypeStrategy set to a value other than LTE_UL_PREFERRED)
as an example)
MOD NRDUCELLPUCCH: NrDuCellId=0, Format1RbNum=RB2, Format3RbNum=RB4,
CsiDedicatedRbNum=RB4, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0;
//Configuring symbols occupied by the NR PDCCH
//When one or two LTE CRS ports are configured
MOD NRDUCELLPDCCH: NrDuCellId=0, SptShrStartSymbol=SYM1, OccupiedSymbolNum=2SYM;
//When four LTE CRS ports are configured
MOD NRDUCELLPDCCH: NrDuCellId=0, SptShrStartSymbol=SYM2, OccupiedSymbolNum=1SYM;
```

```
//Setting the frame offset
MOD GNODEBPARAM: FrameOffset=0;
//Setting the TA offset
MOD NRDUCELL: NrDuCellId=0, FrequencyBand=N1, TaOffset=0TC;

//Turning on the switch controlling LTE and NR flash dynamic spectrum sharing (with the NR cell being an
NR FDD cell)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-1;
//Turning on the switch controlling LTE and NR flash dynamic spectrum sharing (with the NR cell being an
NR SDL cell)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_SDL_SPCT_SHR_SW-1;
//Optional, required when the NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM)
Configuring the number of RBs for common control resources
MOD NRDUCELLCORESET: NrDuCellId=0, CommonCtrlResRbNum=RB24;
//Configuring the SSB frequency-domain position based on actual conditions (using SsbFreqPos set to
426860 as an example)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbDescMethod=SSB_DESC_TYPE_NARFCN,
SsbFreqPos=426860;
//Configuring the SSB and SIB1 periods
MOD NRDUCELL: NrDuCellId=0, SsbPeriod=MS20, Sib1Period=MS40;
//Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteSptShrCellGrpId=0,
FddLteNrSptShrSwitch=LTE_CSI_RS_AVOID_SW-1;
//Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteSptShrCellGrpId=0,
FddLteNrSptShrSwitch=LTE_CSI_RS_AVOID_SW-0;
//Optional, required when LTE FDD and NR Flash Dynamic Spectrum Sharing is used together with NB-IoT
deployed in LTE guard band mode) Configuring reserved RBs for NB-IoT
ADD NRDUCELLRBRESERVE: NrDuCellId=0, Index=0, RbRsvMode=NB_GB_RB_RESERVED,
RbRsvType=DOWNLINK_MODE, RbRsvStartIndex=0, RbRsvEndIndex=3;
ADD NRDUCELLRBRESERVE: NrDuCellId=0, Index=1, RbRsvMode=NB_GB_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=0, RbRsvEndIndex=2;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Activation Command Examples (Multiple LTE Cells)

Activation command examples on the LTE side

```
//Deactivating the cells
DEA CELL: LocalCellId=0;
DEA CELL: LocalCellId=3;
//Binding the cells that require this function to a baseband processing unit supporting this function
MOD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=1, BaseBandEqmId=12;
MOD EUCELLSECTOREQM: LocalCellId=3, SectorEqmId=1, BaseBandEqmId=12;
//Setting the percentage of spectrum resources to be preferentially allocated to LTE MBB services, and
turning on the LTE and NR co-carrier co-CPRI data switch
ADD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSptShrLtePriResRatio=50,
LteNrCoCarrCoCpriDataSw=ON;
//Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-1;
//Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_CSI_RS_AVOID_POLICY_SW-0;
//Turning on the switch controlling LTE and NR flash dynamic spectrum sharing, adding the LTE cells to an
LTE spectrum sharing cell group, setting the spectrum sharing mode (assuming that SRS resources need to
be configured for LTE UEs), turning on the switch controlling power allocation optimization in spectrum
power sharing mode, turning on the WBB/MBB control policy optimization switch, and setting the LTE
downlink SPS restrict ratio
ADD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=LTE_NR_SPECTRUM_SHR,
LteNrSpectrumShrCellGrpId=0, SptShrMode=LTE_NR_PWR_DYN_SHR_WITH_SPCT,
SpectrumCloudEnhSwitch=LTE_UE_SRS_NOT_CONFIG_SW-0&LNR_PWR_WITH_SPCT_OPT_SW-1&WBB_MBB_
CONTROL_OPT_SW-1, DlSpsRestrictRatio=60;
ADD SPECTRUMCLOUD: LocalCellId=3, SpectrumCloudSwitch=LTE_NR_SPECTRUM_SHR,
```

```

LteNrSpectrumShrCellGrpId=0, SpctShrMode=LTE_NR_PWR_DYN_SHR_WITH_SPCT,
SpectrumCloudEnhSwitch=LTE_UE_SRS_NOT_CONFIG_SW-0&LNR_PWR_WITH_SPCT_OPT_SW-1&WBB_MBB_
CONTROL_OPT_SW-1, DisSpsRestrictRatio=60;
//Setting the start position of the LTE PUCCH at the lower end of the frequency domain and the end
position of the LTE PUCCH at the upper end of the frequency domain (assuming that the number of RBs
configured for the NR PUCCH exceeds the number of reserved RBs in a 10 MHz cell)
ADD CELLRBRESERVE: LocalCellId=0, Index=1, RbRsvMode=LNR_SPECTRUM_SHR_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=3, RbRsvEndIndex=46;
ADD CELLRBRESERVE: LocalCellId=3, Index=1, RbRsvMode=LNR_SPECTRUM_SHR_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=3, RbRsvEndIndex=46;
//Configuring MBSFN subframes (using SA networking without the SSB dual-beam function as an example)
MOD CELLMBSFNSFENHCONFIG: LocalCellId=0, SubframeAllocationMode=FOUR_FRAME,
FourFrameAllocation=10487808, RadioFrameAllocPrd=N4, RadioFrameAllocOfs=0;
MOD CELLMBSFNSFENHCONFIG: LocalCellId=3, SubframeAllocationMode=FOUR_FRAME,
FourFrameAllocation=10487808, RadioFrameAllocPrd=N4, RadioFrameAllocOfs=0;
//Setting the frame offset
ADD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
ADD CELLFRAMEOFFSET: LocalCellId=3, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=3;

```

Activation command examples on the NR side

```

//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Adding an LTE spectrum sharing cell group and an NR spectrum sharing cell group
ADD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0;
//Adding the NR cell to the NR spectrum sharing cell group
ADD NRDUCELLSPCTCLOUD: NrDuCellId=0, NrSpctShrCellGrpId=0;
//Setting the number of RBs for the NR PUCCH (using cells for which downlink intra-FR inter-band CA does
not take effect (INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch
parameter deselected) and cells for which the NSA TDM pattern type strategy is not set to LTE uplink
preferred (NRDUCellAlgoSwitch.NsaTdmPatternTypeStrategy set to a value other than LTE_UL_PREFERRED)
as an example)
MOD NRDUCELLPUCCH: NrDuCellId=0, Format1RbNum=RB2, Format3RbNum=RB4,
CsiDedicatedRbNum=RB4, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0;
//Configuring symbols occupied by the NR PDCCH
//When one or two LTE CRS ports are configured
MOD NRDUCELLPDCCH: NrDuCellId=0, SpctShrStartSymbol=SYM1, OccupiedSymbolNum=2SYM;
//When four LTE CRS ports are configured
MOD NRDUCELLPDCCH: NrDuCellId=0, SpctShrStartSymbol=SYM2, OccupiedSymbolNum=1SYM;
//Setting the frame offset
MOD GNODEBPARAM: FrameOffset=0;
//Setting the TA offset
MOD NRDUCELL: NrDuCellId=0, FrequencyBand=N1, TaOffset=0TC;

//Turning on the switch controlling LTE and NR flash dynamic spectrum sharing (with the NR cell being an
NR FDD cell)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-1;
//Turning on the switch controlling LTE and NR flash dynamic spectrum sharing (with the NR cell being an
NR SDL cell)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_SDL_SPCT_SHR_SW-1;
//((Optional, required when the NRDUCellPdcch.OccupiedSymbolNum parameter is set to 2SYM)
Configuring the number of RBs for common control resources
MOD NRDUCELLCORESET: NrDuCellId=0, CommonCtrlResRbNum=RB24;
//Configuring the SSB frequency-domain position based on actual conditions (using SsbFreqPos set to
426860 as an example)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbDescMethod=SSB_DESC_TYPE_NARFCN,
SsbFreqPos=426860;
//Configuring the SSB and SIB1 periods
MOD NRDUCELL: NrDuCellId=0, SsbPeriod=MS20, Sib1Period=MS40;
//((Optional, configurable only when the cell TX/RX mode is not 8T8R) Configuring RE-level avoidance of the
LTE CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CSI_RS_AVOID_SW-1;
//((Optional, required when the cell TX/RX mode is 8T8R) Configuring subframe-level avoidance of the LTE
CSI-RS by NR
MOD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0,
FddLteNrSpctShrSwitch=LTE_CSI_RS_AVOID_SW-0;

```

```
//Selecting the LTE_CRS_RATEMATCH_ALL_SYM_SW option (assuming that LTE performance needs to be
preferentially ensured)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_CRS_RATEMATCH_ALL_SYM_SW-1;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Optimization Command Examples (Single LTE Cell)

Optimization command examples on the LTE side

```
//(Optional, required when NR performs CRS rate matching in the pattern of one LTE CRS port) Adjusting
the PA configuration of the cell
MOD CELLDLPCPDSPCHPA: LocalCellId=0, PaPcOff=DB0_P_A;
//Configuring the dynamic spectrum sharing frequency indication
ADD NRNFREQ: LocalCellId=0, DlArfcn=374890, AggregationAttribute=DSS_FREQ_INDICATION-1;
//Turning on the LTE and NR shared spectrum allocation adaptation switch
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LNR_RES_ALLOC_ADAPT_SW-1;
//Turning on the switch controlling inter-RAT preferential guarantee for GBR services
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSptShrSwitch=GBR_PRI_ALLOC_SW-1;
//Turning on the LTE and NR SRS resource allocation optimization switch
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_NR_SRS_ALLOC_OPT_SW-1;
//Turning on the switch for LTE and NR resource allocation direction optimization
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LNR_RES_ALLOC_DIRECTION_OPT_SW-1;
//Turning on the LTE key event assurance switch, and configuring the LTE key event UE number threshold
and LTE key event UE number threshold offset
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_BIG_EVENT_ASSURANCE_SW-1, LteBigEventUeNumThld=200,
LteBigEventUeNumThldOffset=100;
//(Optional, required when four SSB beams are configured for NR) Configuring the power offset for the LTE
PBCH and synchronization signals
MOD CELLCHPWRCFG: LocalCellId=0, PbchPwr=0, SchPwr=0;
//Enabling LTE uplink voice packet loss optimization and configuring the number of RBs guaranteed for
uplink VoLTE services
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_UPLINK_VOLTE_GTEE_SW-1;
MOD SPECTRUMCLOUD: LocalCellId=0,
SpectrumCloudEnhSwitch=LTE_UPLINK_VOLTE_ENH_SW-1&LTE_UPLINK_VOLTE_OPT_SW-1,
ULVolTEGuaranteedRbNum=3;
//Turning on the switches for DRX On Duration period position optimization for NSA- or SA-capable voice
service UEs and gap-assisted measurement position optimization in dynamic spectrum sharing scenarios
MOD CELLDLDRXPARAM: LocalCellId=0, DrxAlgoExtSwitch =
DSS_GAP_POS_OPT_SW-1&DSS_VOICE_NSA_SA_OD_POS_OPT_SW-1;
MOD CELLCQIADAPTIVECFG: LocalCellId=0, DssNsaSaVoiceApCqiPeriod=40MS;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the
LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring
the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD ENODEBRESMODEALGO: ParamCfgProtectSw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Optimization command examples on the NR side

```
//Configuring optimized handover from an LTE cell to an NR spectrum sharing cell
MOD GNODEBPARAM: FrameOffset=0, LteHandoverAlgoSwitch=LTE_HO_TO_DSS_NR_OPT_SW-1;
//Configuring the position of the downlink additional DMRS
MOD NRDUCELLPDSCH: NrDuCellId=0, DlAdditionalDmrsPos=POS1;
//Configuring CRS rate matching in the pattern of one LTE CRS port
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteSptShrCellGrpId=0,
FddLteNrSptShrSwitch=LTE_CRS_PORT_1_RM_SW-1;
//Turning on the LTE SS/PBCH rate matching optimization switch, PSD adaptation switch, and paging
adjustment switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_SS_PBCH_RM_OPT_SW-1&PWR_SPCT_DENSITY_ADAPT_SW-1&PAGING_ADJUST_SW-1;
```

```
//Configuring the LTE key event assurance UE number threshold
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteBigEventAssureUeNumThld=10;
//Optional, required when the cell TX/RX mode is 8T8R, 16T16R, or 32T32R and four SSB beams are
//configured) Turning on the SSB beam adjustment switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-1;
//Optional, recommended when the PRACH configuration index is greater than 27) Selecting the
LNR_PRACH_RES_ALLOC_OPT_SW option
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LNR_PRACH_RES_ALLOC_OPT_SW-1;
//Enabling time-domain SSB staggering
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-0;
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbTimePos=SSB23_PCI_MOD2_STAGGER;
//Enabling paging-OSI staggering
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_OSI_STAGGER_SW-1;
//Enabling time-domain RMSI staggering
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=TIME_DOMAIN_RMSI_STAGGER_SW-1;
//Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the
LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD NRDUCELLSPTSHARE: NrDuCellId=0,
LnrSpectrumShareSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring
the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD GNBFBREQBANDCONFIG: FrequencyBand=N3, FrameOffset=92160;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD GNBOAMPARAM: CfgInterceptPolicySw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Optimization Command Examples (Multiple LTE Cells)

Optimization command examples on the LTE side

```
//Configuring the dynamic spectrum sharing frequency indication
ADD NRNFREQ: LocalCellId=0, DlArfcn=374890, AggregationAttribute=DSS_FREQ_INDICATION-1;
ADD NRNFREQ: LocalCellId=3, DlArfcn=374890, AggregationAttribute=DSS_FREQ_INDICATION-1;
//Turning on the LTE and NR shared spectrum allocation adaptation switch
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LNR_RES_ALLOC_ADAPT_SW-1;
//Turning on the switch for LTE and NR resource allocation direction optimization
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LNR_RES_ALLOC_DIRECTION_OPT_SW-1;
//Turning on the LTE key event assurance switch, and configuring the LTE key event UE number threshold
and LTE key event UE number threshold offset
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LteNrSptShrSwitch=LTE_BIG_EVENT_ASSURANCE_SW-1, LteBigEventUeNumThld=200,
LteBigEventUeNumThldOffset=100;
//Enabling 3D coverage pattern
//Optional, required when four SSB beams are configured for NR) Configuring the power offset for the LTE
PBCH and synchronization signals (using the cell PA value set to -3 as an example)
MOD CELLCHPWRCFG: LocalCellId=0, PbchPwr=-600, SchPwr=-600;
//Enabling 3D coverage pattern
//Optional, required when four SSB beams are configured for NR) Configuring the power offset for the LTE
PBCH and synchronization signals (using the cell PA value set to 0 as an example)
MOD CELLCHPWRCFG: LocalCellId=0, PbchPwr=0, SchPwr=0;
//Turning on the switches for DRX On Duration period position optimization for NSA- or SA-capable voice
service UEs and gap-assisted measurement position optimization in dynamic spectrum sharing scenarios
MOD CELLDXPARA: LocalCellId=0, DrxAlgoExtSwitch =
DSS_GAP_POS_OPT_SW-1&DSS_VOICE_NSA_SA_OD_POS_OPT_SW-1;
MOD CELLQIADAPTIVECFG: LocalCellId=0, DssNsaSaVoiceApCqiPeriod=40MS;
//Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the
LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring
the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD ENODEBRESMODEALGO: ParamCfgProtectSw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Optimization command examples on the NR side


```
//Configuring optimized handover from an LTE cell to an NR spectrum sharing cell
MOD GNODEBPARAM: FrameOffset=0, LteHandoverAlgoSwitch=LTE_HO_TO_DSS_NR_OPT_SW-1;
//Configuring the position of the downlink additional DMRS
MOD NRDUCELLPDSCCH: NrDuCellId=0, DlAdditionalDmrsPos=POS1;
//Configuring CRS rate matching in the pattern of one LTE CRS port
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteSptShrCellGrpId=0,
FddLteNrSptShrSwitch=LTE_CR_S_PORT_1_RM_SW-1;
//Turning on the LTE SS/PBCH rate matching optimization switch, PSD adaptation switch, and paging
adjustment switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LTE_SS_PBCH_RM_OPT_SW-1&PWR_SPCT_DENSITY_ADAPT_SW-1&PAGING_ADJU
ST_SW-1;
//Configuring the LTE key event assurance UE number threshold
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=0, LteBigEventAssureUeNumThld=10;

//Enabling 3D coverage pattern
//(Optional, required when the cell TX/RX mode is 8T8R, 16T16R, or 32T32R and four SSB beams are
configured) Turning on the SSB beam adjustment switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-1;
//Setting the coverage scenario of broadcast beams (using SCENARIO_202 as an example)
MOD NRDUFDCELLTRPBEAM: NrDuCellTrpId=0, CoverageScenario=SCENARIO_202;
//(Optional, recommended when the PRACH configuration index is greater than 27) Selecting the
LNR_PRACH_RES_ALLOC_OPT_SW option
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=LNR_PRACH_RES_ALLOC_OPT_SW-1;
//Enabling time-domain SSB staggering
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=SSB_ADJ_SW-0;
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_FDD, SsbTimePos=SSB23_PCI_MOD2_STAGGER;
//Enabling paging-OSI staggering
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_OSI_STAGGER_SW-1;
//Enabling time-domain RMSI staggering
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=TIME_DOMAIN_RMSI_STAGGER_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Selecting the
LNR_3MS_FRAME_OFFSET_DIFF_SW option
MOD NRDUCELLSPCTSHARE: NrDuCellId=0,
LnrSpectrumShareSwitch=LNR_3MS_FRAME_OFFSET_DIFF_SW-1;
//(Optional, required only when the LTE and NR flexible frame offset function is also enabled) Configuring
the frame offset for the cell with the LNR_3MS_FRAME_OFFSET_DIFF_SW option selected
MOD GNBFREQBANDCONFIG: FrequencyBand=N3, FrameOffset=92160;
//Turning on the switch for "cross-RAT configuration check in DSS scenarios"
MOD GNBOAMPARAM: CfgInterceptPolicySw=DSS_CROSS_RAT_CFG_CHECK_SW-1;
```

Deactivation Command Examples (Single LTE Cell)

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivation command examples on the LTE side

```
//Deactivating the cell
DEA CELL: LocalCellId=0;
//Removing cell RB reservation configurations
RMV CELLRBRESERVE: LocalCellId=0, Index=1;
//(Optional, required when MBSFN subframes are not configured before function activation and are
configured upon function activation) Rolling back the configurations of MBSFN subframes
MOD CELLMBSFNSFENHCONFIG: LocalCellId=0, SubframeAllocationMode=NOT_CONFIG;
//Rolling back spectrum sharing enhancement (All functions that depend on LTE FDD and NR Flash
Dynamic Spectrum Sharing must be rolled back. The following uses uplink VoLTE enhancement as an
example.)
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=LNR_UPLINK_VOLTE_GTEE_SW-0;
MOD SPECTRUMCLOUD: LocalCellId=0,
SpectrumCloudEnhSwitch=LTE_UPLINK_VOLTE_ENH_SW-0&LTE_UPLINK_VOLTE_OPT_SW-0,
ULVolTEGuaranteedRbNum=0;
//Turning off the switch controlling LTE FDD and NR Flash Dynamic Spectrum Sharing
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=OFF;
//Removing the LTE cell from the LTE spectrum sharing cell group
RMV SPECTRUMCLOUD: LocalCellId=0;
```

```
RMV LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0;
//Activating the cell
ACT CELL: LocalCellId=0;
```

Deactivation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Turning off the switch controlling LTE FDD and NR Flash Dynamic Spectrum Sharing
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-0;
//Removing the NR cell from the NR spectrum sharing cell group
RMV NRDUCELLSPCTCLOUD: NrDuCellId=0;
RMV GNB DULTENRSPCTSHRCG: NrSpctShrCellGrpId=0;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Deactivation Command Examples (Multiple LTE Cells)

Deactivation command examples on the LTE side

```
//Deactivating the cells
DEA CELL: LocalCellId=0;
DEA CELL: LocalCellId=3;
//Removing cell RB reservation configurations
RMV CELLRBRESERVE: LocalCellId=0, Index=1;
RMV CELLRBRESERVE: LocalCellId=3, Index=1;
//Optional, required when MBSFN subframes are not configured before function activation and are
//configured upon function activation) Rolling back the configurations of MBSFN subframes
MOD CELLMBSFNSFENHCONFIG: LocalCellId=0, SubframeAllocationMode=NOT_CONFIG;
MOD CELLMBSFNSFENHCONFIG: LocalCellId=3, SubframeAllocationMode=NOT_CONFIG;
//Turning off the switch controlling LTE FDD and NR Flash Dynamic Spectrum Sharing
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=OFF;
MOD SPECTRUMCLOUD: LocalCellId=3, SpectrumCloudSwitch=OFF;
//Removing the LTE cells from the LTE spectrum sharing cell group
RMV SPECTRUMCLOUD: LocalCellId=0;
RMV SPECTRUMCLOUD: LocalCellId=3;
RMV LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=3;
```

Deactivation command examples on the NR side

```
//Deactivating the cell
DEA NRCELL: NrCellId=0;
//Turning off the switch controlling LTE FDD and NR Flash Dynamic Spectrum Sharing
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudSwitch=LTE_NR_FDD_SPCT_SHR_SW-0;
//Removing the NR cell from the NR spectrum sharing cell group
RMV NRDUCELLSPCTCLOUD: NrDuCellId=0;
RMV GNB DULTENRSPCTSHRCG: NrSpctShrCellGrpId=0;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Whether LTE FDD and NR Flash Dynamic Spectrum Sharing has taken effect can be queried using MML commands or by tracing signaling messages. Whether the

LTE key event assurance function is enabled and the TA offsets that take effect on the LTE and NR sides can be queried using MML commands.

When LTE FDD and NR Flash Dynamic Spectrum Sharing (multiple LTE cells) is enabled, activation verification needs to be performed in both LTE cells.

Using MML Commands

The basic functions of LTE FDD and NR Flash Dynamic Spectrum Sharing take effect when both of the following conditions are met:

- The value of the **Spectrum Sharing Status** parameter in the output of the eNodeB MML command **DSP LTENRSPCTSHRCELLGRP** is **DYNAMIC_SHARING**.
- The value of the **Spectrum Sharing Status** parameter in the output of the gNodeB MML command **DSP GNBULTENRSPCTSHRCG** is **Dynamic Sharing**.

The enhanced function LTE key event assurance takes effect when both of the following conditions are met:

- The value of **Heavy Traffic Status** in the output of the eNodeB command **DSP LTENRSPCTSHRCELLGRP** is **LTE_HEAVY_TRAFFIC** or **LTE_NON_HEAVY_TRAFFIC**.
- The value of **Heavy Traffic Status** in the output of the gNodeB command **DSP GNBULTENRSPCTSHRCG** is **LTE Heavy Traffic Assurance** or **Not LTE Heavy Traffic Assurance**. The value **LTE Heavy Traffic Assurance** indicates that this function is in use.

Run the following commands to observe the TA offsets that take effect on the LTE and NR sides:

- On the eNodeB side, run the **DSP LTENRSPCTSHRCELLGRP** command. **Actual TA Offset** in the command output indicates the actually effective TA offset in the current LTE spectrum sharing cell group.
- On the gNodeB side, run the **DSP GNBULTENRSPCTSHRCG** command. **Actual TA Offset** in the command output indicates the actually effective TA offset in the current NR spectrum sharing cell group.

Tracing Signaling Messages

NOTICE

Before activating this function in LTE and NR cells, start signaling tracing on the MAE-Access. Otherwise, messages regarding the activation of this function cannot be observed after this function is activated.

LTE FDD and NR Flash Dynamic Spectrum Sharing takes effect when the following conditions are met on both the eNodeB and gNodeB sides:

Perform the following steps to start eNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **LTE > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. View the value of the **lnss-group-status-change-info > nr-ul-cell-information/nr-dl-cell-information > spectrum-sharing-status** IE in the **LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION** message. The value is **ss-dynamic-sharing**.

----End

Perform the following steps to start gNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **NR > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. View the value of the **lnss-group-status-change-info > nr-ul-cell-information/nr-dl-cell-information > spectrum-sharing-status** IE in the **LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION** message. The value is **ss-dynamic-sharing**.

----End

4.4.3 Network Monitoring

This function increases cell downlink throughput as follows:

- In the case of LTE-to-NR spectrum sharing, the average downlink throughput of the relevant NR cell is increased. Average downlink throughput of an NR cell = **N.ThpVol.DL.Cell/N.ThpTime.DL.Cell**
- In the case of NR-to-LTE spectrum sharing, the average downlink throughput of the relevant LTE cell is increased. Average downlink throughput of an LTE cell = **L.Thrp.bits.DL/L.Thrp.Time.Cell.DL.HighPrecision**

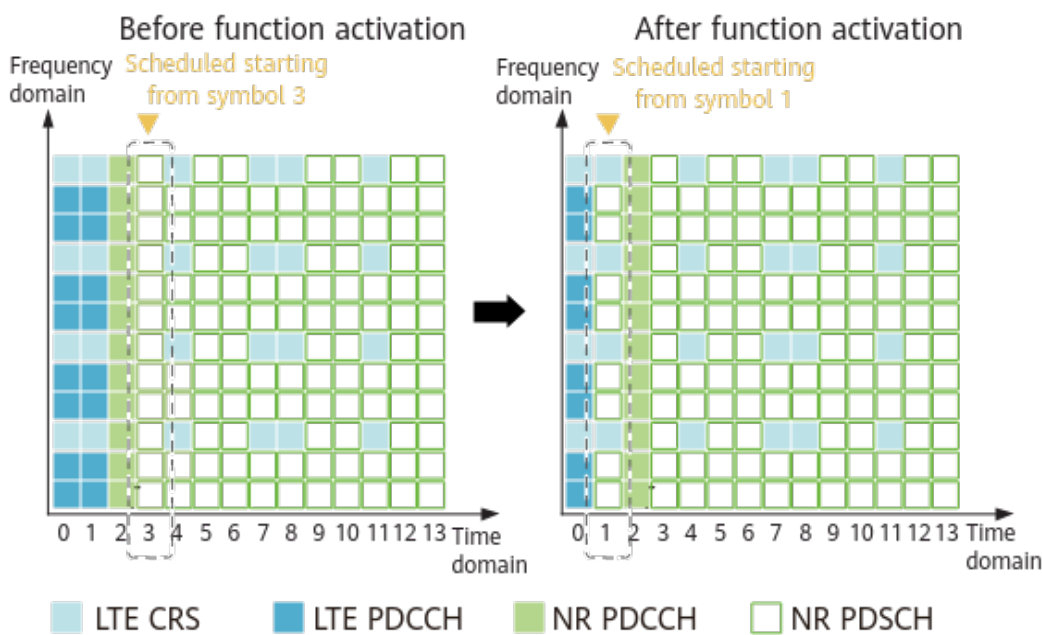
5 DSS-based Flexible PDSCH Scheduling

5.1 Principles

When LTE CRS needs to occupy symbols 0 and 1, the NR PDCCH can only be deployed in symbol 2. The NR PDSCH must be scheduled following the NR PDCCH. After DSS-based Flexible PDSCH Scheduling is enabled, **if the LTE PDCCH does not occupy symbol 0 or 1, the NR PDSCH can be scheduled starting from symbol 0 or 1 in rate matching mode, instead of being scheduled following the NR PDCCH.** This increases the spectrum resources available for the NR cell.

Figure 5-1 illustrates an example in which the NR PDSCH is scheduled from symbol 1 as the LTE PDCCH does not occupy it. This function is supported when the NR parameter `NRDUCellIDISchRes.SpectShrPdschStartSymbol` is set to **SYM0** or **SYM1**.

Figure 5-1 Flexible scheduling between the NR PDSCH and LTE PDCCH



After this function takes effect, more spectrum resources are available for the NR PDSCH. However, as LTE CRS is transmitted in the corresponding symbols, this function also brings CRS interference and negative impacts when UEs are located in overlapping areas of intra-frequency LTE DSS cells with different PCIs and different PCI-mod-3 results. Therefore, this function is not recommended when intra-frequency LTE cells have different PCI-mod-3 results, a large number of UEs are located in overlapping areas, and the downlink PRB usage of neighboring cells is low. The **N.User.RRConn.Avg.CellOverlap** counter is used to measure the average number of UEs in RRC_CONNECTED mode in overlapping areas. The measurement can be enabled by selecting the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter.

After this function is enabled, access failures or service drops may occur due to PDCCH blind detection failures. Such UE compatibility issues can be solved by enabling the PDCCH rate matching UE compatibility rectification function (controlled by the PDCCH_RATEMATCH_COMPATIBLE_SW option of the NR parameter NRDUCellPdsch.RateMatchSwitch). For details, see *Downlink Scheduling*.

This function applies when LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.

5.2 Network Analysis

5.2.1 Benefits

When the LTE cell is lightly loaded, this function increases the **User Downlink Average Throughput (DU)** of NR UEs.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has the following impacts:

- The average uplink and downlink throughputs of the LTE cell decrease, so do the average uplink and downlink throughputs of LTE UEs.
 - Average uplink cell throughput = **Cell Uplink Average Throughput**
 - Average downlink cell throughput = **Cell Downlink Average Throughput**
 - Average uplink UE throughput = **User Uplink Average Throughput**
 - Average downlink UE throughput = **User Downlink Average Throughput**
- The CFI value on the LTE side changes. Therefore, the CCE usage, which is calculated using the following formula, changes: CCE usage = $(L.ChMeas.CCE.CommUsed + L.ChMeas.CCE.ULUsed + L.ChMeas.CCE.DLUsed) / L.ChMeas.CCE.Avail$.

UE compatibility issues may occur when DSS-based Flexible PDSCH Scheduling is enabled. If such issues occur, the access success rate decreases and certain indicators (such as those related to PDU sessions and service drops) deteriorate on the NR side. To prevent this, you are advised to select the

DSS_FLEX_PDSCH_SCH_SW_OFF option of the **gNBUECompatSw.BlacklistCtrlSwitch** parameter, so that DSS-based Flexible PDSCH Scheduling does not take effect for blacklisted UEs.

5.3 Requirements

5.3.1 Licenses

The following license is required only when 3900 or 5900 series base stations are used.

RAT	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
NR	FOFD-051250	DSS-based Flexible PDSCH Scheduling (NR FDD)	NR0S0DSSSF00	DSS-based Flexible PDSCH Scheduling (NR FDD)	gNodeB	per Cell
For the license control item NR0S0DSSSF00: - If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. - If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: - When the license sales number pre-check function is enabled, the NRDUCellIDISchRes.SpctShrPdschStartSymbol parameter value cannot be changed to SYM0 or SYM1 due to license insufficiency, preventing cell activation failures. - When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.						

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
NR FDD	CRS interference avoidance for Distributed MIMO	D_MIMO_CRS_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Static Multiple Beam	None	<i>Massive MIMO Basics (FDD)</i>	When DSS-based Flexible PDSCH Scheduling is used together with Static Multiple Beam, the proportion of times DSS-based Flexible PDSCH Scheduling takes effect decreases.

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD)</i>	When DSS-based Flexible PDSCH Scheduling is used together with Static Multiple Beam, the proportion of times DSS-based Flexible PDSCH Scheduling takes effect decreases.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

As this function is used in LTE FDD and NR spectrum sharing scenarios, the base station model requirements are the same as those for LTE FDD and NR spectrum sharing.

Boards

As this function is used in LTE FDD and NR spectrum sharing scenarios, the board requirements are the same as those for LTE FDD and NR spectrum sharing.

RF Modules

As this function is used in LTE FDD and NR spectrum sharing scenarios, the RF module requirements are the same as those for LTE FDD and NR spectrum sharing.

Cells

As this function is used in LTE FDD and NR spectrum sharing scenarios, the cell requirements are the same as those for LTE FDD and NR spectrum sharing.

5.3.4 Others

NR UEs must support the RB- and symbol-level semi-persistent rate matching capability. If the *Phy-Parameters* IE includes the *rateMatchingResrcSetSemi-Static* sub-IE with the value of "supported", the UE supports this capability. For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Sharing PDSCH Start Symbol	NRDUCellDLSchRes.SpctShrPdschStartSymbol	Set this parameter to SYM0 or SYM1 based on the network plan when DSS-based Flexible PDSCH Scheduling is required.
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	When DSS-based Flexible PDSCH Scheduling is required according to the network plan, select the PDCCH_RATEMATCH_COMPATIBLE_SW option of this parameter to resolve UE compatibility issues.

5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling DSS-based Flexible PDSCH Scheduling (based on the assumption that the start symbol for the PDSCH in the case of spectrum sharing is set to SYM1)
MOD NRDUCELLDLSCHRES: NrDuCellId=0, SpctShrPdschStartSymbol=SYM1;
//Turning on the PDCCH rate matching compatibility switch
MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=PDCCH_RATEMATCH_COMPATIBLE_SW-1;
```

Deactivation Command Examples

```
//Disabling DSS-based Flexible PDSCH Scheduling  
MOD NRDUCELLDLSCHRES: NrDuCellId=0, SpctShrPdschStartSymbol=NOT_CONFIG;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Use counters to monitor the activation of the DSS-based Flexible PDSCH Scheduling function. If the downlink spectral efficiency of the NR cell increases after this function is enabled, this function has taken effect. The downlink spectral efficiency of the NR cell equals $\text{N.ThpVol.DL.Cell} / (\text{N.PRBDL.DrbUsed.Avg} \times \text{N.DLPDSCH.Tti.Num})$ in the unit of kbit per (TTI x PRB).

NOTE

The downlink spectral efficiency of the NR cell can be expressed as bits per second per Hz by using the following formula: $\text{N.ThpVol.DL.Cell} / (\text{N.PRBDL.DrbUsed.Avg} \times \text{N.DLPDSCH.Tti.Num}) \times 1000 \times 1000 / 180000$. The idea behind this formula is as follows: 1000 is introduced for a kbit-to-bit conversion, because the unit of the **N.ThpVol.DL.Cell** counter is kbit. 1000 is introduced for a TTI-to-second conversion, because one second has 1000 TTIs (1s = 1000 ms, 1 TTI = 1 ms). 180000 is introduced for a PRB-to-Hz conversion, because one PRB spans 180,000 Hz in the frequency domain (1 PRB = 12 subcarriers, 1 subcarrier = 15,000 Hz).

5.4.3 Network Monitoring

After this function is enabled, monitor the network gains and impacts of this function using the counters listed in [5.2 Network Analysis](#).

6 DSS/Hybrid DSS Performance Boost

6.1 Principles

To further improve spectrum utilization when LTE FDD and NR share spectrum, DSS/Hybrid DSS Performance Boost is introduced. This feature includes the following functions: LTE PDCCH overhead optimization, LTE CRS overhead optimization, LTE polarization-based power assistance, optimized scheduling of LTE and NR UEs, DSS preferential-resource percentage adaptation, DSS inter-RAT PDCCH spatial multiplexing, DSS PDCCH resource sharing optimization, and LNR independent configuration of uplink and downlink resource percentages. [Table 6-1](#) describes whether the individual functions are compatible with LTE FDD and NR Flash Dynamic Spectrum Sharing and Hybrid DSS Based on Asymmetric Bandwidth. For details about Hybrid DSS Based on Asymmetric Bandwidth, see *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*.

Table 6-1 Functions in DSS/Hybrid DSS Performance Boost

Function	LTE FDD and NR Flash Dynamic Spectrum Sharing	Hybrid DSS Based on Asymmetric Bandwidth
LTE PDCCH overhead optimization	Compatible	Incompatible
LTE CRS overhead optimization	Compatible	Compatible
LTE polarization-based power assistance	Compatible	Compatible
Optimized scheduling of LTE and NR UEs	Compatible	Compatible
DSS preferential-resource percentage adaptation	Compatible	Compatible
DSS inter-RAT PDCCH spatial multiplexing	Compatible	Compatible

Function	LTE FDD and NR Flash Dynamic Spectrum Sharing	Hybrid DSS Based on Asymmetric Bandwidth
DSS PDCCH resource sharing optimization	Compatible	Compatible
LNR independent configuration of uplink and downlink resource percentages	Incompatible	Compatible

6.1.1 LTE PDCCH Overhead Optimization

LTE PDCCH overhead optimization is applicable to scenarios where DSS-based Flexible PDSCH Scheduling is enabled by setting the NR parameter **NRDUCellIDISchRes.SpectShrPdschStartSymbol** to **SYM0**. After DSS-based Flexible PDSCH Scheduling is enabled, if there are no UEs on the LTE side and the LTE PDCCH does not occupy symbol 0, the NR PDSCH can be scheduled starting from symbol 0 in rate matching mode so that the NR cell can obtain more available spectrum resources. For details about DSS-based Flexible PDSCH Scheduling, see [5 DSS-based Flexible PDSCH Scheduling](#).

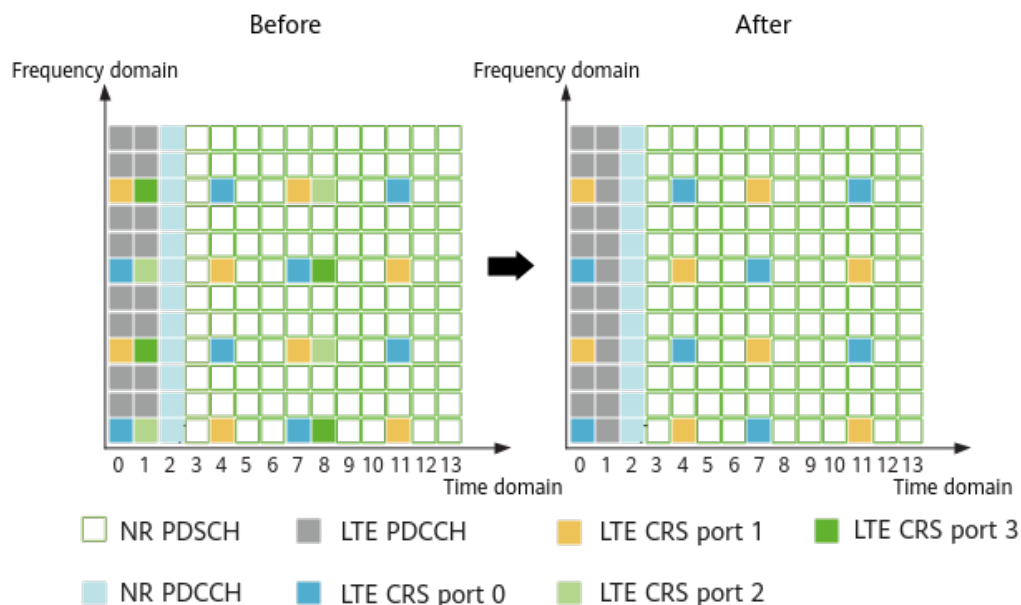
LTE PDCCH overhead optimization allows the symbols that are not occupied by the LTE PDCCH to be allocated to the NR PDSCH when there are UEs on the LTE side but scheduling is not performed. Specifically, when there are UEs on the LTE side, the eNodeB determines the amount of LTE PDCCH resources required in each TTI. If no LTE PDCCH resource is required and no feedback is to be sent on the PHICH in a TTI, all LTE PDCCH symbols in this TTI can be used by the NR PDSCH.

LTE PDCCH overhead optimization can be enabled by selecting the **LTE_PDCCH_OVERHEAD_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter on the LTE side.

6.1.2 LTE CRS Overhead Optimization

When four CRS ports are configured for LTE, fewer resources are available for NR. **LTE CRS overhead optimization allows the CRS of ports 2 and 3 to be dynamically muted in the space domain when the LTE traffic volume is low and the CRS of ports 2 and 3 is not required. This function reduces the LTE CRS resource consumption and increases the amount of resources available for NR.** [Figure 6-1](#) shows resource consumption before and after this function is enabled.

Figure 6-1 Resource consumption before and after LTE CRS overhead optimization is enabled



This function can be enabled by turning on the LTE CRS overhead optimization switch on both the LTE and NR sides. When the threshold for triggering LTE CRS overhead optimization is reached, space-domain CRS dynamic muting is performed. The details are as follows:

- LTE side
 - This function is controlled by the **LTE_CRS_OVERHEAD_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter.
 - The **LteNrSpctShrCellGrp.CrsOverheadOptThld** parameter specifies the threshold for triggering LTE CRS overhead optimization. When the average number of UEs in RRC_CONNECTED mode in the LTE cell is less than or equal to this threshold, the CRS of ports 2 and 3 is muted, and newly admitted UEs in the NR cell can use the time-frequency resources corresponding to the muted CRS. In other cases, LTE CRS is not muted.
- NR side
 - This function is controlled by the **CRS_OVERHEAD_OPT_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** parameter.
 - When the **TRS_RM_PERIOD_OPT_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter is also selected, the TRS rate matching period optimization function is enabled to improve the benefits of LTE CRS overhead optimization. If this option is selected and TRS rate matching is enabled, the TRS rate matching period is the same as the TRS period specified by the **NRDUCellCsirs.TrsPeriod** parameter. If this option is deselected and TRS rate matching is enabled, the TRS rate matching period is always 20 ms.

After space-domain CRS dynamic muting is triggered, the eNodeB performs related procedures to ensure its effect. For details about the procedures, see the descriptions of eNodeB actions in section "CRS Dynamic Muting" in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.

NOTICE

LTE CRS overhead optimization involves only space-domain CRS dynamic muting and does not involve time-domain CRS dynamic muting.

6.1.3 LTE Polarization-based Power Assistance

When the power allocation mode is set to the spectrum power sharing mode (the LTE parameter **SpectrumCloud.SpctShrMode** is set to **LTE_NR_PWR_DYN_SHR_WITH_SPCT**), power resources can be saved, but the capacity performance may deteriorate. When the LTE parameter **CellDlpcPdschPa.PaPcOff** is set to **DB_3_P_A** or a smaller value, LTE CRS occupies too much power. As a result, the power that can be shared with the NR PDSCH and LTE PDSCH is limited on the symbols where LTE CRS appears, reducing the spectrum resources available for the NR PDSCH.

To address this issue, LTE polarization-based power assistance is introduced. This function is controlled by the **POWER_POLARIZATION_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter on the LTE side. When this option is selected, **the eNodeB balances the LTE CRS transmit power between channels to improve the power utilization of carriers. This addresses the issue that NR cannot use the entire bandwidth due to power insufficiency in spectrum power sharing mode.**

6.1.4 Optimized Scheduling of LTE and NR UEs

When LTE FDD and NR Flash Dynamic Spectrum Sharing or Hybrid DSS Based on Asymmetric Bandwidth is enabled, the downlink services of LTE and NR UEs are scheduled in the same TTI. If the UEs on either side are large-packet UEs and the UEs on the other side are small-packet UEs, the large-packet UEs are scheduled in two or more TTIs. As a result, the downlink scheduling duration increases and user experience of large-packet UEs deteriorates. Large-packet UEs and small-packet UEs on LTE and NR networks are defined as follows:

- LTE small-packet UEs: The measured valid downlink packet sizes are all within ranges corresponding to **L.Traffic.DL.PktSize.Samp.Index0** to **L.Traffic.DL.PktSize.Samp.Index3**.
- LTE large-packet UEs: The measured valid downlink packet sizes are all within ranges corresponding to **L.Traffic.DL.PktSize.Samp.Index7** to **L.Traffic.DL.PktSize.Samp.Index9**.
- NR small-packet UEs: The measured valid downlink packet sizes are all within ranges corresponding to **N.Traffic.DL.PktSize.Samp.Index0** to **N.Traffic.DL.PktSize.Samp.Index3**.
- NR large-packet UEs: The measured valid downlink packet sizes are all within ranges corresponding to **N.Traffic.DL.PktSize.Samp.Index7** to **N.Traffic.DL.PktSize.Samp.Index9**.

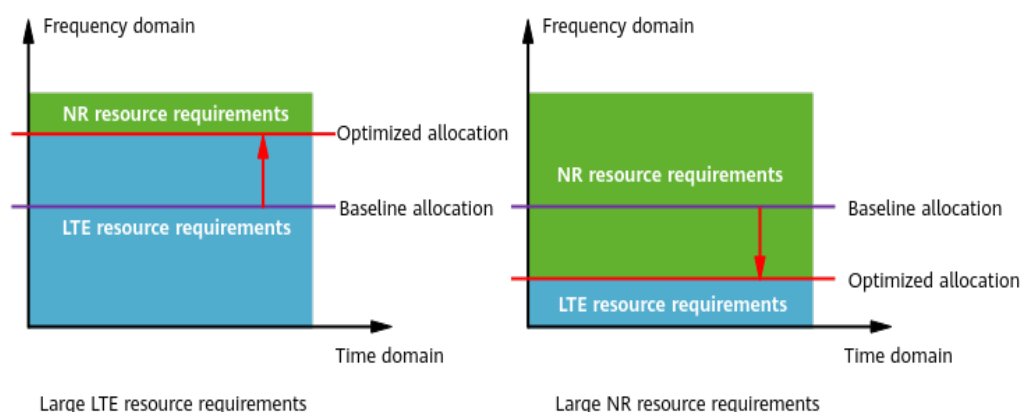
The function of optimized scheduling of LTE and NR UEs enables large-packet UEs and small-packet UEs to be scheduled in different TTIs. This way, large-packet UEs are scheduled in a single TTI, and their scheduling duration will not increase and the average downlink UE throughput will not decrease either.

The function of optimized scheduling of LTE and NR UEs is controlled by the **LNR_SPCT_SHR_SCH_OPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter on the LTE side.

6.1.5 DSS Preferential-Resource Percentage Adaptation

The manually configured percentages of resources preferentially allocated may not match the percentages of resource requirements, in which case LTE and NR UE rates are affected as UEs do not have sufficient resources. DSS preferential-resource percentage adaptation is introduced to prevent this issue. This function is controlled by the **DSS_PREFER_RES_RATIO_ADAPT_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter on the LTE side. If this option is selected, the percentage of resources preferentially allocated to LTE equals a percentage that is first adaptively determined based on the percentages of traffic volume requirements of LTE and NR and then adjusted by the factor specified by **SpectrumCloud.LtePrefResRatioAdaptFactor**. The percentage of resources preferentially allocated to NR equals 100% minus the percentage for LTE. These percentages apply to PUSCH and PDSCH resource allocation. **Figure 6-2** illustrates how this function works. The percentages obtained through this function are used in the same way as those obtained by setting the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** as detailed in **4.1.1.2.2 Flexible Spectrum Priority Mode**.

Figure 6-2 DSS preferential-resource percentage adaptation



The **SpectrumCloud.LtePrefResAdaptUlResReqThd** and **SpectrumCloud.LtePrefResAdaptDlResReqThd** parameters specify the uplink and downlink thresholds, respectively, for DSS preferential-resource percentage adaptation to take effect. This function takes effect only if the amounts of requested uplink and downlink resources are both less than or equal to the corresponding thresholds.

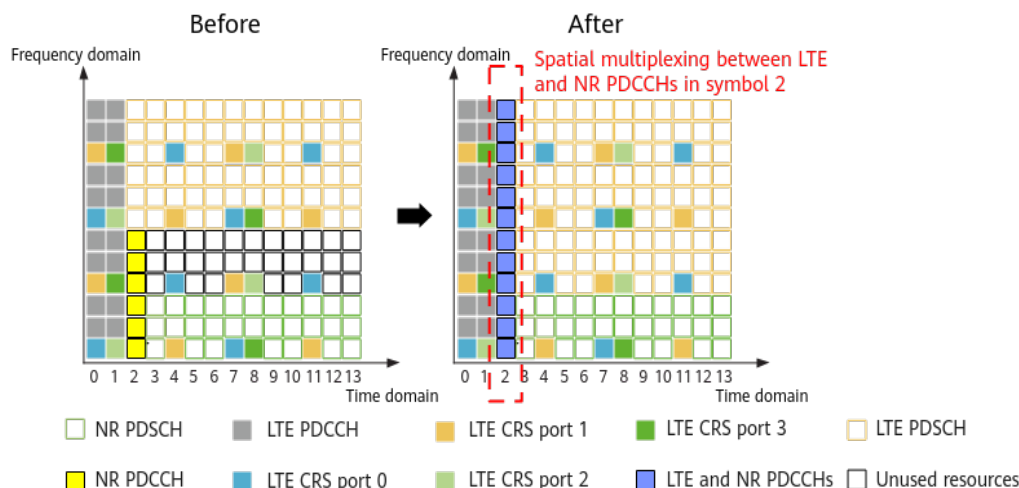
It is recommended that this function be enabled when LTE and NR differ greatly in load and the percentages of resources allocated to LTE and NR are expected to adapt to changes in traffic volume requirements.

6.1.6 DSS Inter-RAT PDCCH Spatial Multiplexing

DSS inter-RAT PDCCH spatial multiplexing allows spatial multiplexing between LTE and NR PDCCHs in symbol 2 when four CRS ports are configured

for LTE (with the **Cell.CrsPortNum** parameter set to **CRS_PORT_4**). This gives the LTE PDSCH access to RBs that would otherwise be left unused, so that resource utilization increases and LTE UEs achieve higher downlink user-perceived rates. Figure 6-3 illustrates how this function works. This function applies to scenarios where the downlink LTE load is heavy and the downlink NR load is light (but not extremely light).

Figure 6-3 Resource consumption before and after DSS inter-RAT PDCCH spatial multiplexing is enabled



This function requires turning on the switch for DSS inter-RAT PDCCH spatial multiplexing on both the LTE and NR sides. That is, the following options must be selected:

- LTE: **DSS_INTER_RAT_PDCCH_SPC_MUX_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter
- NR: **DSS_INTER_RAT_PDCCH_SPC_MUX_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** parameter

For this function to take effect in a higher proportion of cases, set the **LteNrSpctShrCellGrp.LtePdcchSym2PsdThld** parameter to **0** and the **LteNrSpctShrCellGrp.LtePdcchAvailCceNumUpprLmt** parameter to **255**. In addition, to further reduce the interference on NR, set the **LteNrSpctShrCellGrp.LtePdcchSym2PsdRdcFactor** parameter to **REDUCE_TO_1_4**.

This function takes effect only when an NR cell shares spectrum with no more than two LTE cells.

6.1.7 DSS PDCCH Resource Sharing Optimization

Resources for the NR PDCCH may be insufficient when four CRS ports are configured for LTE (that is, the **Cell.CrsPortNum** parameter is set to **CRS_PORT_4**). Insufficient resources for the NR PDCCH will result in a high downlink CCE allocation failure rate on the NR side and consequently a poor NR experience. This issue can be resolved by the DSS PDCCH resource sharing optimization function. This function allows resource multiplexing between LTE CRS

and NR PDCCH to increase the number of available CCEs for the NR PDCCH. With more CCE resources, the downlink CCE allocation success rate and downlink user-perceived rate will increase on the NR side.

This function applies to scenarios where the downlink LTE load is light, the downlink NR load is heavy, and the downlink CCE allocation failure rate on the NR side is greater than 20%.

This function requires turning on the switch for DSS PDCCH resource sharing optimization on both the LTE and NR sides. That is, the following options must be selected:

- LTE: **DSS_PDCCH_SHARING_OPT_SW** option of the **SpectrumCloud.SpectrumCloudExtSwitch** parameter
- NR: **DSS_PDCCH_SHARING_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter

For this function to take effect in a higher proportion of cases, you are advised to set the **LteNrSpctShrCellGrp.LteCrssym1PwrReduceFactor** parameter to **REDUCE_TO_1_2** to further reduce the interference of LTE CRS on NR. (Setting this parameter to a value other than **OFF** decreases the LTE uplink and downlink user-perceived rates; the smaller the value, the larger the decrease. Therefore, the value **REDUCE_TO_1_4** is not recommended.) In addition, to further reduce the interference on NR, you are advised to set the **LteNrSpctShrCellGrp.LtePdcchSym2PsdThld** parameter to **0**, the **LteNrSpctShrCellGrp.LtePdcchAvailCceNumUpprLmt** parameter to **255**, and the **LteNrSpctShrCellGrp.LtePdcchSym2PsdRdcFactor** parameter to **REDUCE_TO_1_4**.

It is recommended that this function be enabled only when the LTE parameter **CellDlpcPdschPa.PaPcOff** is set to **DB_3_P_A**. Enabling this function when this parameter is set to **DB_0_P_A** will bring significant negative impacts to LTE.

This function can take effect only when an NR cell shares spectrum with no more than two LTE cells.

NOTICE

Due to protocol restrictions, enabling DSS PDCCH resource sharing optimization requires the LTE parameter **CellCsiRsParaCfg.CsiRsSwitch** to be set to **NOT_CFG**.

If this **CellCsiRsParaCfg.CsiRsSwitch** parameter is set to **FIXED_CFG** or **ADAPTIVE_CFG**, enabling DSS PDCCH resource sharing optimization increases the BLER and service drop rate on the NR side, provided that the **LTE_CSI_RS_AVOID_POLICY_SW** option of the LTE parameter **LteNrSpctShrCellGrp.LteNrSpctShrSwitch** is selected and the **LTE_CSI_RS_AVOID_SW** option of the NR parameter **gNBDULteNrSpctShrCfg.FddLteNrSpctShrSwitch** is also selected. To address these issues, DSS PDCCH resource sharing policy optimization must be enabled (controlled by the **DSS_PDCCH_SHR_POLICY_OPT_SW** option of the **CellAlgoExtSwitch.LnrCellAlgoSwitch** parameter). This optimization prevents DSS PDCCH resource sharing optimization from taking effect, thereby preventing the increases in the BLER and service drop rate.

6.2 Network Analysis

6.2.1 Benefits

LTE PDCCH Overhead Optimization and LTE CRS Overhead Optimization

After these two functions are both enabled, the average downlink throughput of NR UEs and the downlink spectral efficiency of the NR cell increase. The average downlink throughput of NR UEs equals $(\text{N.ThpVol.DL} - \text{N.ThpVol.DL.LastSlot}) / \text{N.ThpTime.DL.RmvLastSlot}$. The downlink spectral efficiency of the NR cell equals $\text{N.ThpVol.DL.Cell} / (\text{N.PR.B.DL.DrbUsed.Avg} \times \text{N.DL.PDSCH.Tti.Num})$ in the unit of kbit per (TTI x PRB).

NOTE

The downlink spectral efficiency of the NR cell can be expressed as bits per second per Hz by using the following formula: $\text{N.ThpVol.DL.Cell} / (\text{N.PR.B.DL.DrbUsed.Avg} \times \text{N.DL.PDSCH.Tti.Num}) \times 1000 \times 1000 / 180000$. The idea behind this formula is as follows: 1000 is introduced for a kbit-to-bit conversion, because the unit of the **N.ThpVol.DL.Cell** counter is kbit. 1000 is introduced for a TTI-to-second conversion, because one second has 1000 TTIs (1s = 1000 ms, 1 TTI = 1 ms). 180000 is introduced for a PRB-to-Hz conversion, because one PRB spans 180,000 Hz in the frequency domain (1 PRB = 12 subcarriers, 1 subcarrier = 15,000 Hz).

The benefits can be achieved when the LTE cell is lightly loaded.

LTE Polarization-based Power Assistance

After this function is enabled, if the power allocation mode is set to the spectrum power sharing mode and the **CellDlpcPdschPa.PaPcOff** parameter is set to a value less than or equal to **DB_3_P_A** (-3 dB), the average downlink throughput of NR UEs increases.

Optimized Scheduling of LTE and NR UEs

This function can change the average number of available downlink PRBs on the LTE and NR sides (indicated by **L.ChMeas.PR.B.DL.Actual.Avail** and **N.PR.B.DL.Actual.Avail.Avg**, respectively), thereby increasing the downlink UE throughput. The details are as follows:

- If LTE UEs are small-packet UEs, the average downlink throughput of NR UEs increases.
- If NR UEs are small-packet UEs, the average downlink throughput of LTE UEs increases. Average downlink throughput of LTE UEs = $(\text{L.Thrp.bits.DL} - \text{L.Thrp.bits.DL.LastTTI}) / \text{L.Thrp.Time.DL.RmvLastTTI}$.

DSS Preferential-Resource Percentage Adaptation

This function changes the average numbers of available uplink and downlink PRBs on the LTE and NR sides (indicated by **L.ChMeas.PR.B.UL.Actual.Avail**, **N.PR.B.UL.Actual.Avail.Avg**, **L.ChMeas.PR.B.DL.Actual.Avail**, and **N.PR.B.DL.Actual.Avail.Avg**). As a result, the average uplink and downlink UE throughputs change.

Average uplink throughput of LTE UEs = $(L.Thrp.bits.UL - L.Thrp.bits.UE.UL.SmallPkt) / L.Thrp.Time.UE.UL.RmvSmallPkt$

Average uplink throughput of NR UEs = $(N.ThpVol.UL - N.ThpVol.UE.UL.SmallPkt) / N.ThpTime.UE.UL.RmvSmallPkt$

Compared with a baseline scenario where the percentage of resources preferentially allocated to LTE is set to 50%, this function increases the overall carrier-level throughput of LTE and NR if the overall load of LTE and NR is light.

Overall downlink carrier-level throughput of LTE and NR = $((L.Thrp.bits.DL - L.Thrp.bits.DL.LastTTI) \times 1000 + N.ThpVol.DL - N.ThpVol.DL.LastSlot) / (L.Thrp.Time.DL.RmvLastTTI + N.ThpTime.DL.RmvLastSlot \times 1000)$

DSS Inter-RAT PDCCH Spatial Multiplexing

It is recommended that this function be enabled when the NR uplink CCE usage is greater than 10%. When four LTE CRS ports are configured, this function increases the average downlink throughput of LTE UEs. The higher the NR PDSCH RB waste rate and the higher the NR uplink CCE usage, the greater the benefits.

NR uplink CCE usage = $N.CCE.ULDCI.Used.Avg / N.CCE.Avail.Avg$

NR PDSCH RB waste rate = $(N.PR.BD.Actual.Avail.Avg - N.PR.BD.Used.Avg) / N.PR.BD.Avail.Avg$

DSS PDCCH Resource Sharing Optimization

This function increases the **User Downlink Average Throughput (DU)**.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

- **LTE PDCCH overhead optimization**
 After this function takes effect, the CFI value on the LTE side changes. As a result, the CCE usage changes, which is calculated using the following formula: $CCE\ usage = (L.ChMeas.CCE.CommUsed + L.ChMeas.CCE.ULUsed + L.ChMeas.CCE.DLUsed) / L.ChMeas.CCE.Avail$. If the CFI is not transmitted on the LTE side, demodulation of certain LTE UEs may be affected.
 UE compatibility issues may occur when LTE PDCCH overhead optimization is enabled. If such issues occur, the access success rate decreases and certain indicators (such as those related to PDU sessions and service drops) deteriorate on the NR side. To prevent this, you are advised to select the **DSS_FLEX_PDSCH_SCH_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlSwitch** parameter, so that DSS-based Flexible PDSCH Scheduling does not take effect for blacklisted UEs.
- **LTE CRS overhead optimization**
 The network impacts of LTE CRS overhead optimization are the same as those of LTE CRS dynamic muting. For details about the impacts on LTE networks, see descriptions of network impacts of CRS dynamic muting in *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.
 After this function takes effect, the number of REs where LTE CRS appears decreases, and consequently the amount of resources available for the NR

PDSCH increases. The downlink spectral efficiency of the NR cell and the average downlink MCS index on the NR side increase.

UE compatibility issues may occur when LTE CRS overhead optimization is enabled. If such issues occur, the access success rate decreases and certain indicators (such as those related to PDU sessions and service drops) deteriorate on the NR side. To prevent this, you are advised to select the **CRS_OVERHEAD_OPT_OFF** option of the **gNBUECompatSw.BlacklistCtrlSwitch** parameter, so that LTE CRS overhead optimization does not take effect for blacklisted UEs.

- LTE polarization-based power assistance

After this function takes effect, the RF module power usage increases and the base station power consumption slightly increases. As inter-channel power balancing is performed among LTE CRS ports, the quality of signals received by UEs may change, the distribution of LTE UEs may slightly change, and the average LTE power spectral density may be affected.

- Optimized scheduling of LTE and NR UEs

After this function takes effect, the value of $(L.Thrp.Time.DL - L.Thrp.Time.DL.RmvLastTTI)/L.Thrp.Time.DL$ on the LTE side and the value of $(N.ThpTime.DL - N.ThpTime.DL.RmvLastSlot)/N.ThpTime.DL$ on the NR side increase and the uplink and downlink PDCCH CCE allocation success rates change. In addition, the average number of available uplink PRBs and the average number of available downlink PRBs change on the LTE and NR sides (indicated by **N.PRB.UL.Actual.Avail.Avg**, **N.PRB.DL.Actual.Avail.Avg**, **L.ChMeas.PRB.UL.Actual.Avail**, and **L.ChMeas.PRB.DL.Actual.Avail**).

This function affects uplink resource allocation. As a result, the average uplink cell throughput and the average uplink UE throughput fluctuate in both LTE and NR cells.

- Average uplink throughput of an LTE cell = **Cell Uplink Average Throughput**
- Average uplink throughput of LTE UEs = **User Uplink Average Throughput**
- Average uplink throughput of an NR cell = **Cell Uplink Average Throughput (DU)**
- Average uplink throughput of NR UEs = **User Uplink Average Throughput (DU)**

- DSS preferential-resource percentage adaptation

Compared with the baseline scenario where the percentage of resources preferentially allocated to LTE is set to 50%, enabling this function increases the resources allocated to the heavier-load RAT and decreases the resources allocated to the lighter-load RAT. As a result, the uplink and downlink CCE allocation failure rates may change. In addition, the values of the following indicators may increase for the heavier-load RAT and may decrease for the lighter-load RAT:

- Average uplink throughput of LTE UEs = **User Uplink Average Throughput**
- Average uplink throughput of NR UEs = **User Uplink Average Throughput (DU)**

- Average downlink throughput of LTE UEs = **User Downlink Average Throughput**
- Average downlink throughput of NR UEs = **User Downlink Average Throughput (DU)**
- If the configured percentage of resources preferentially allocated to LTE is high (80%, for example) in the baseline scenario, the percentage for LTE adaptively obtained through this function may be lower than this baseline percentage. As a result, the LTE rate decreases, but the NR rate increases.
- If the configured percentage of resources preferentially allocated to LTE is low (20%, for example) in the baseline scenario, the percentage for LTE adaptively obtained through this function may be much higher than this baseline percentage. As a result, the LTE rate increases significantly, but the NR rate decreases.

When DSS preferential-resource percentage adaptation is enabled, resources are preferentially allocated to the heavier-load RAT. As a result, the user-perceived rate of the lighter-load RAT may decrease.

- DSS inter-RAT PDCCH spatial multiplexing

After this function takes effect, the LTE PDCCH DTX proportion, LTE PRB usage, and values of the following indicators in both the uplink and downlink of LTE and NR sides change: CCE allocation failure rate, IBLER, average PDCCH CCE aggregation level, PDCCH CCE usage, and average number of available PRBs.

- $$\text{LTE PDCCH DTX proportion} = \frac{(\text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl1} + \text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl2} + \text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl4} + \text{L.ChMeas.PDCCH.DL.DTXNum.AggLvl8})}{(\text{L.ChMeas.PDCCH.AggLvl1Num} + \text{L.ChMeas.PDCCH.AggLvl2Num} + \text{L.ChMeas.PDCCH.AggLvl4Num} + \text{L.ChMeas.PDCCH.AggLvl8Num})}$$
- $$\text{LTE downlink CCE allocation failure rate} = \frac{\text{L.ChMeas.CCE.DL.AllocFail}}{(\text{L.ChMeas.CCE.DL.AllocFail} + \text{L.ChMeas.PDCCH.AggLvl1Num} + \text{L.ChMeas.PDCCH.AggLvl2Num} + \text{L.ChMeas.PDCCH.AggLvl4Num} + \text{L.ChMeas.PDCCH.AggLvl8Num})}$$
- $$\text{LTE uplink CCE allocation failure rate} = \frac{(\text{L.ChMeas.CCE.UL.AllocFail})}{(\text{L.ChMeas.CCE.UL.AllocFail} + \text{L.ChMeas.PDCCH.UL.AggLvl1Num} + \text{L.ChMeas.PDCCH.UL.AggLvl2Num} + \text{L.ChMeas.PDCCH.UL.AggLvl4Num} + \text{L.ChMeas.PDCCH.UL.AggLvl8Num})}$$
- $$\text{NR downlink CCE allocation failure rate} = \frac{(\text{N.CCE.DL.AllocReq.Num} - (\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num}))}{\text{N.CCE.DL.AllocReq.Num}}$$
- $$\text{NR uplink CCE allocation failure rate} = \frac{(\text{N.CCE.UL.AllocReq.Num} - (\text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl4Num} + \text{N.CCE.UL.AggLvl8Num} + \text{N.CCE.UL.AggLvl16Num}))}{\text{N.CCE.UL.AllocReq.Num}}$$
- $$\text{LTE uplink IBLER} = \frac{(\text{L.Traffic.UL.SCH.QPSK.ErrTB.Ibler} + \text{L.Traffic.UL.SCH.16QAM.ErrTB.Ibler} + \text{L.Traffic.UL.SCH.64QAM.ErrTB.Ibler} + \text{L.Traffic.UL.SCH.256QAM.ErrTB.Ibler})}{(\text{L.Traffic.UL.SCH.QPSK.TB} + \text{L.Traffic.UL.SCH.16QAM.TB} + \text{L.Traffic.UL.SCH.64QAM.TB} + \text{L.Traffic.UL.SCH.256QAM.TB})}$$

- L.Traffic.UL.SCH.16QAM.TB + L.Traffic.UL.SCH.64QAM.TB + L.Traffic.UL.SCH.256QAM.TB)**
- LTE downlink IBLER = **(L.Traffic.DL.SCH.QPSK.ErrTB.Ibler + L.Traffic.DL.SCH.16QAM.ErrTB.Ibler + L.Traffic.DL.SCH.64QAM.ErrTB.Ibler + L.Traffic.DL.SCH.256QAM.ErrTB.Ibler)/(L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB)**
 - NR uplink IBLER = **(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)/(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)**
 - NR downlink IBLER = **(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)/(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)**
 - LTE average uplink PDCCH CCE aggregation level = **(L.ChMeas.PDCCH.UL.AggLvl1Num x 1 + L.ChMeas.PDCCH.UL.AggLvl2Num x 2 + L.ChMeas.PDCCH.UL.AggLvl4Num x 4 + L.ChMeas.PDCCH.UL.AggLvl8Num x 8)/(L.ChMeas.PDCCH.UL.AggLvl1Num + L.ChMeas.PDCCH.UL.AggLvl2Num + L.ChMeas.PDCCH.UL.AggLvl4Num + L.ChMeas.PDCCH.UL.AggLvl8Num)**
 - LTE average downlink PDCCH CCE aggregation level = **(L.ChMeas.PDCCH.AggLvl1Num x 1 + L.ChMeas.PDCCH.AggLvl2Num x 2 + L.ChMeas.PDCCH.AggLvl4Num x 4 + L.ChMeas.PDCCH.AggLvl8Num x 8)/(L.ChMeas.PDCCH.AggLvl1Num + L.ChMeas.PDCCH.AggLvl2Num + L.ChMeas.PDCCH.AggLvl4Num + L.ChMeas.PDCCH.AggLvl8Num)**
 - NR average uplink PDCCH CCE aggregation level = **(N.CCE.UL.AggLvl1Num x 1 + N.CCE.UL.AggLvl2Num x 2 + N.CCE.UL.AggLvl4Num x 4 + N.CCE.UL.AggLvl8Num x 8 + N.CCE.UL.AggLvl16Num x 16)/(N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)**
 - NR average downlink PDCCH CCE aggregation level = **(N.CCE.DL.AggLvl1Num x 1 + N.CCE.DL.AggLvl2Num x 2 + N.CCE.DL.AggLvl4Num x 4 + N.CCE.DL.AggLvl8Num x 8 + N.CCE.DL.AggLvl16Num x 16)/(N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num)**
 - LTE PDCCH downlink CCE usage = **L.ChMeas.CCE.DLUsed/L.ChMeas.CCE.Avail**
 - LTE PDCCH uplink CCE usage = **L.ChMeas.CCE.ULUsed/L.ChMeas.CCE.Avail**
 - NR PDCCH downlink CCE usage = **N.CCE.DLDCI.Used.Avg/N.CCE.Avail.Avg**

- NR PDCCH uplink CCE usage = **N.CCE.ULDCI.Used.Avg/N.CCE.Avail.Avg**
- LTE PRB usage = **L.ChMeas.PRB.DL.Used.Avg/L.ChMeas.PRB.DL.Avail**

In addition, DSS inter-RAT PDCCH spatial multiplexing affects uplink resource allocation, which changes the average uplink UE and cell throughputs on the LTE side as well as the uplink and downlink UE and cell throughputs on the NR side.

- Average uplink throughput of an LTE cell = **Cell Uplink Average Throughput**
- Average uplink throughput of an NR cell = **Cell Uplink Average Throughput (DU)**
- Average downlink throughput of an NR cell = **Cell Downlink Average Throughput (DU)**

- DSS PDCCH resource sharing optimization

This function changes the values of the following indicators on both the LTE and NR sides: proportions of uplink and downlink PDCCH DTXs, BLER, PDCCH CCE allocation success rate, average uplink and downlink PDCCH CCE aggregation levels, uplink and downlink PDCCH CCE usages, and average numbers of available uplink and downlink PRBs. In addition, on the NR side, the downlink MCS index decreases, and the downlink PRB usage may increase. On the LTE side, CRS measurement is affected, and the uplink and downlink user-perceived rates decrease.

In addition, DSS PDCCH resource sharing optimization affects uplink resource allocation, which changes the average the uplink and downlink cell throughputs as well as the average uplink UE throughput on both the NR and LTE sides.

If the **Cell.CrsPortNum** parameter is set to **CRS_PORT_4**, DSS PDCCH resource sharing optimization involves an IE related to zero-power CSI-RS (ZP CSI-RS). If UEs do not support this IE, compatibility issues will occur. It is recommended that the **DSS_PDCCH_SHARING_OPT_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter be selected so that DSS PDCCH resource sharing optimization does not take effect for blacklisted UEs.

6.3 Requirements

6.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE FDD	MRFD-19 0221	DSS/Hybrid DSS Performance Boost(LTE FDD)	LT1SDHDPBF01	eNodeB	per Cell

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR FDD	MRFD-190261	DSS/Hybrid DSS Performance Boost(NR FDD)	NR0SDHDPBF01	gNodeB	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

- LTE PDCCH overhead optimization

RAT	Function Name	Function Switch	Reference
NR FDD	DSS-based Flexible PDSCH Scheduling	NRDUCellDISchRes.S pctShrPdschStart- Symbol set to SYM0	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

- LTE CRS overhead optimization

RAT	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.Spectr umCloudSwitch set to LTE_NR_SPECTRUM_S HR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	NRDUCellAlgoSwitch.SpectrumCloudSwitch LTE_NR_FDD_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR FDD cell LTE_NR_SDL_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR SDL cell	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>
LTE FDD	DRX parameter group ^a	DrxParaGroup.OnDurationTimer and DrxParaGroup.LongDrxCycle	<i>DRX and Signaling Control</i>
a: To ensure that all UEs in a cell can complete data transmission in the On Duration period after alignment, the On Duration period must be set to a value greater than or equal to 1/64 of the long DRX cycle.			

- LTE polarization-based power assistance

RAT	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	NRDUCellAIgoSwitch.SpectrumCloudSwitch • LTE_NR_FDD_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR FDD cell • LTE_NR_SDL_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR SDL cell	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

- Optimized scheduling of LTE and NR UEs

RAT	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	NRDUCellAlgoSwitch.SpectrumCloudSwitch • LTE_NR_FDD_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR FDD cell • LTE_NR_SDL_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR SDL cell	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>
NR FDD	Number of Guaranteed RBs	NRDUCellPdcch.GuaranteedRbNum set to 0	None

- DSS preferential-resource percentage adaptation

RAT	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	NRDUCellAlgoSwitch.SpectrumCloudSwitch • LTE_NR_FDD_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR FDD cell • LTE_NR_SDL_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR SDL cell	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

• DSS inter-RAT PDCCH spatial multiplexing

RAT	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	NRDUCellAIgoSwitch.SpectrumCloudSwitch • LTE_NR_FDD_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR FDD cell • LTE_NR_SDL_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR SDL cell	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

- DSS PDCCH resource sharing optimization

RAT	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	NRDUCellAlgoSwitch.SpectrumCloudSwitch LTE_NR_FDD_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR FDD cell LTE_NR_SDL_SPCT_SHR_SW option of this parameter selected if the NR cell is an NR SDL cell	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>
NR FDD	UE-specific PDCCH full-bandwidth configuration ^a	UE_PDCCH_FULL_BANDWIDTH_CFG_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	None
a: This dependency relationship exists only if Hybrid DSS Based on Asymmetric Bandwidth is enabled. If UE-specific PDCCH full-bandwidth configuration is enabled but DSS PDCCH resource sharing optimization is disabled, the NR downlink user-perceived rate may decrease.			

6.3.2.2 Mutually Exclusive Functions

- LTE PDCCH overhead optimization
None
- LTE CRS overhead optimization

RAT	Function Name	Function Switch	Reference
LTE FDD	Coordinated pilot scheduling	EmimoCpsSwitch option of the CellAlgoSwitch.EmimoSwitch parameter	<i>eMIMO (FDD)</i>
LTE FDD	Breathing Pilot	BreathingPilotSwitch option of the CellDlschAlgo.BreathingPilotAlgoSwitch parameter	<i>Breathing Pilot</i>
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>

RA T	Function Name	Function Switch	Reference
NR FD D	LTE CRS full-symbol rate matching	LTE_CRS_RATEMATCH_ALL_SYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	None
NR FD D	CRS interference avoidance for Distributed MIMO	D_MIMO_CRS_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>
NR FD D	Optimization of the start symbol of the PDSCH in MBSFN subframes	MBSFN_PDSCH_STARTSYM_OPT_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	None

- LTE polarization-based power assistance

RA T	Function Name	Function Switch	Reference
LTE FD D	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPwrSaving.SymbolShutDownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>
LTE FD D	CRS port-channel mapping adjustment	PDSCHCfg.LogicalPortSwapSwitch	<i>MIMO Smart 8T8R (FDD)</i>
LTE FD D	Flexible TM4 MU-MIMO	FLEX_TM4_MU_MIMO_SW option of the CellMimoParaCfg.MimoSwitch parameter	<i>Low-Band Booster</i>
LTE FD D	WTTx MU-MIMO	Tm9FourLayerPairSwitch option of the CellAlgoExtSwitch.WttxMuMimoSwitch parameter	<i>WTTx MU-MIMO (FDD)</i>
LTE FD D	Downlink MU-MIMO in TM4	EmimoMuMimoSwitch option of the CellAlgoSwitch.EmimoSwitch parameter	<i>eMIMO (FDD)</i>

RA T	Function Name	Function Switch	Reference
LTE FDD	Downlink MU-MIMO in TM9	4TxTM9MuMimoSwitch option of the CellAlgoSwitch . <i>EmimoSwitch</i> parameter	<i>eMIMO (FDD)</i>

- Optimized scheduling of LTE and NR UEs

RA T	Function Name	Function Switch	Reference
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell . <i>GigaBandCellAlgoSw</i> parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>

- DSS preferential-resource percentage adaptation

RA T	Function Name	Function Switch	Reference
LTE FDD	LTE FDD and NR Uplink Spectrum Sharing	SpectrumCloud . <i>SpectrumCloudSwitch</i> set to LTE_NR_UPLINK_SPECTRUM_SHR	<i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i>

- DSS inter-RAT PDCCH spatial multiplexing

RA T	Function Name	Function Switch	Reference
NR FDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch . <i>RateMatchSwitch</i> parameter	<i>Downlink Scheduling</i>
NR FDD	Rate-matching-pattern-configuration-free PDCCH rate matching	PDCCH_NO_PATTERN_RATE_MATCH_SW option of the NRDUCellPdcch . <i>PdcchAlgoSwitch</i> parameter	<i>Downlink Scheduling</i>

RA T	Function Name	Function Switch	Reference
NR FD D	Downlink power aggregation	DL_POWER_AGG_SW option of the NRDUCellChnPwr.LteNrSpctShr PwrAggSwitch parameter	None

- DSS PDCCH resource sharing optimization

RA T	Function Name	Function Switch	Reference
LTE FD D	LTE cell PA configuratio n	CellDlpcPdschPa.PaPcOff set to DB_4DOT77_P_A or DB_6_P_A	None
NR FD D	Rate- matching- pattern- configuratio n-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwit ch parameter	<i>Downlink Scheduling</i>
NR FD D	Rate- matching- pattern- configuratio n-free PDCCH rate matching	PDCCH_NO_PATTERN_RATE_M ATCH_SW option of the NRDUCellPdcch.PdcchAlgoSwitc h parameter	<i>Downlink Scheduling</i>
NR FD D	Downlink power aggregation	DL_POWER_AGG_SW option of the NRDUCellChnPwr.LteNrSpctShr PwrAggSwitch parameter	None
NR FD D	Non-Slot symbol- level scheduling for low latency	NON_SLOT_SW option of the NRDUCellAlgoSwitch.LowLaten cySwitch parameter	None
NR FD D	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaEnhance dAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>
NR FD D	Intra-base- station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwit ch parameter	<i>CoMP</i>

RA T	Function Name	Function Switch	Reference
NR FD D	Inter-base- station downlink joint transmissio n	INTER_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwit ch parameter	None
NR FD D	Flexible and dynamic configuratio n of CSI-RS resources in a cell	NRDUCellCsirs.CsirsCellResour- ceNum set to FD_RESOURCE	None
NR FD D	Enhanced Horizontal Mode	NRDUFddCellTrpBeam.Beamfor mingCapability set to HORIZONTAL_ENH_MODE	None
NR FD D	Inter- frequency CSI-RS interference avoidance	INTER_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommCh nIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>
NR FD D	Intra- frequency CSI-RS interference avoidance	INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommCh nIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>
NR FD D	Optimizatio n of the start symbol of the PDSCH in MBSFN subframes	MBSFN_PDSCH_STARTSYM_OPT _SW option of the NRDUCellAlgoSwitch.Spectrum CloudEnhSwitch parameter	None

6.3.2.3 Function Impacts

- LTE PDCCH overhead optimization

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	DSS inter-RAT PDCCH spatial multiplexing	DSS_INTER_RAT_PDCCH_SPC_MUX_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	Enabling both LTE PDCCH overhead optimization and DSS inter-RAT PDCCH spatial multiplexing will decrease the gains from these two functions.
LTE FDD	DSS PDCCH resource sharing optimization	- LTE: DSS_PDCCH_SHARING_OPT_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter - NR: DSS_PDCCH_SHARING_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	Only DSS PDCCH resource sharing optimization takes effect when it is enabled together with LTE PDCCH overhead optimization. Considering this, DSS PDCCH resource sharing optimization is not recommended when LTE PDCCH overhead optimization has been enabled. This is because such function activation may generate negative impacts.

- LTE CRS overhead optimization

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	LTE CRS overhead optimization does not take effect in inter-gNodeB CA scenarios.

RA T	Function Name	Function Switch	Reference	Description
NR FD D	RB Reserve Mode	NRDUCellRbReserve.RbRsvMode	None	LTE CRS overhead optimization may take effect in a lower proportion of cases and therefore offer lower or even no gains if NRDUCellRbReserve.RbRsvMode is set to NB_GB_RB_RESERVED , LNR_AVAIL_RB_RESERVED , GU_RB_RESERVED , SPECIFIC_RB_RESERVED , or NB_SA_RB_RESERVED .
NR FD D	Flexible bandwidth	FLEXIBLE_BANDWIDTH_SW option of the NRDUCellFlexBw.FlexBwAlgoSwitch parameter	<i>Flexible Bandwidth (FDD)</i>	LTE CRS overhead optimization may take effect in a lower proportion of cases and therefore offer lower or even no gains if it is enabled together with flexible bandwidth.
NR FD D	Compact bandwidth	NRDUCell.CustomizedBwConfigInd set to CONFIG	<i>Scalable Bandwidth</i>	LTE CRS overhead optimization may take effect in a lower proportion of cases and therefore offer lower or even no gains if it is enabled together with compact bandwidth.

The other function impacts of LTE CRS overhead optimization are the same as those of LTE CRS dynamic muting. For details, see descriptions of function impacts of CRS dynamic muting in *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.

- LTE polarization-based power assistance

RAT	Function Name	Function Switch	Reference	Description
NR FDD	PSD adaptation	PWR_SPCT_DENSITY_ADAPT_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	None	When both PSD adaptation and LTE polarization-based power assistance are enabled, gains from these two functions will decrease but no negative impacts will occur.

- Optimized scheduling of LTE and NR UEs
None
- DSS preferential-resource percentage adaptation
None
- DSS inter-RAT PDCCH spatial multiplexing

RAT	Function Name	Function Switch	Reference	Description
NR FDD	PSD adaptation	PWR_SPCT_DENSITY_ADAPT_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	None	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with PSD adaptation. As a result, the gains from these two functions will decrease.
LTE FDD	PDCCH boundary power control optimization	PDCCH_BOUNDARY_PWR_CTRL_OPT_SW option of the CellQciPara.QciAlgorithmSwitch parameter	None	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with PDCCH boundary power control optimization. As a result, the gains from these two functions will decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	PDCCH spare power allocation	CellPdcchAlgo.PdcchSparePowerAllocation set to WEAKCOVERAGEUSER or BIGPKTUSER	None	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with PDCCH spare power allocation. As a result, the gains from these two functions will decrease.
LTE FDD	PDCCH power control improvement	CellPdcchAlgo.PdcchPowerEnhancedSwitch set to ON_ALL , ON_EXPANDED_RANGE , or ON_BASIC_RANGE	None	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with PDCCH power control improvement. As a result, the gains from these two functions will decrease.
LTE FDD	PDCCH capacity improvement	CellPdcchAlgo.PdcchCapacityImprovementSwitch	None	Enabling both PDCCH capacity improvement and DSS inter-RAT PDCCH spatial multiplexing will decrease the gains from these two functions.
LTE FDD	Uplink coverage enhancement	CellAlgoExtSwitch.UlCoverageEnhancementSwitch	None	Enabling both uplink coverage enhancement and DSS inter-RAT PDCCH spatial multiplexing will decrease the gains from these two functions.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with Dynamic Power Sharing Between LTE Carriers. As a result, the gains from these two functions will decrease.
LTE FDD	Extreme power sharing	LTE_EXTREME_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with extreme power sharing. As a result, the gains from these two functions will decrease.
LTE FDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with RF channel intelligent shutdown. As a result, the gains from these two functions will decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynMut-ing.RfChnDynMut-ingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with RF channel dynamic muting. As a result, the gains from these two functions will decrease.
LTE FDD	LTE PDCCH overhead optimization	LTE_PDCCH_OVERHEAD_OPT_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	Enabling both LTE PDCCH overhead optimization and DSS inter-RAT PDCCH spatial multiplexing will decrease the gains from these two functions.
LTE FDD NR FDD	DSS and NR Flexible Refarming	- LTE: DSS_FLEX_REFARM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter - NR: DSS_FLEX_REFARM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with DSS and NR Flexible Refarming. As a result, the gains from DSS inter-RAT PDCCH spatial multiplexing will decrease.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D NR FD D	LNR coordinated channel shutdown	- LTE: CellRfShutdown. MultiRatJointCh nShutdownSw set to ON - NR: MULTI_RAT_RF_ SHUTDOWN_S W option of the NRDUCellAlgoS witch.PowerSavi ngSwitch parameter	<i>Multi-RAT Coordinate d Energy Saving</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with LNR coordinated channel shutdown. As a result, the gains from these two functions will decrease.
LTE FD D	Flexible power allocation in RF channel dynamic muting	RF_DYN_MUTE_FLE X_PWR_ALLOC_SW option of the CellRfChnDynMut- ing.RfChnDynMut ingAlgoSwitch parameter	None	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with flexible power allocation in RF channel dynamic muting. As a result, the gains from these two functions will decrease.
LTE FD D	Downlink intelligent joint power control	DL_INTEL_JOINT_P C_SW option of the CellDLSchExtPar- am.DLSchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with downlink intelligent joint power control. As a result, the gains from these two functions will decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink coordinated power control	DL_COORD_PC_SW option of the CellMimoParaCfg.MimoSwitch parameter	<i>Low-Band Booster</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with downlink coordinated power control. As a result, the gains from these two functions will decrease.

- DSS PDCCH resource sharing optimization

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	CSI-RS	CellCsiRsParaCfg.Cs iRsSwitch	None	Due to protocol restrictions, enabling DSS PDCCH resource sharing optimization requires the LTE parameter CellCsiRsParaCfg.CsiRsSwitch to be set to NOT_CFG. If this CellCsiRsParaCfg.CsiRsSwitch parameter is set to FIXED_CFG or ADAPTIVE_CFG, enabling DSS PDCCH resource sharing optimization increases the BLER and service drop rate on the NR side, provided that the LTE_CSI_RS_AVOID_POLICY_SW option of the LTE parameter LteNrSpctShrCellGrp.LteNrSpctShrSwitch is selected and the LTE_CSI_RS_AVOID_SW option of the NR parameter gNBULteNrSpctShrCfg.FddLteNrSpctShrSwitch is also selected. To address these issues, the DSS_PDCCH_SHR_POLICY_OPT_SW option of the CellAlgoExtSwitch.LnrCellAlgoSwitch parameter must

RA T	Function Name	Function Switch	Reference	Description
				be selected to prevent DSS PDCCH resource sharing optimization from taking effect.
NR FD D	PSD adaptation	PWR_SPCT_DENSITY_ADAPT_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter	None	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with PSD adaptation. As a result, the gains from these two functions will decrease.
LTE FD D	PDCCH boundary power control optimization	PDCCH_BOUNDARY_PWR_CTRL_OPT_SW option of the CellQciPara.QciAlgorithmSwitch parameter	None	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with PDCCH boundary power control optimization. As a result, the gains from these two functions will decrease.
LTE FD D	PDCCH spare power allocation	CellPdcchAlgo.PdcchSparePowerAllocation set to WEAKCOVERAGEUSER or BIGPKTUSER	None	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with PDCCH spare power allocation. As a result, the gains from these two functions will decrease.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	PDCCH power control improvement	CellPdcchAlgo.PdcccPowerEnhancedSwitch set to ON_ALL , ON_EXPANDED_RANGE , or ON_BASIC_RANGE	None	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with PDCCH power control improvement. As a result, the gains from these two functions will decrease.
LTE FDD	PDCCH capacity improvement	CellPdcchAlgo.PdcccCapacityImproveSwitch	None	Enabling both PDCCH capacity improvement and DSS PDCCH resource sharing optimization will decrease the gains of these two functions.
LTE FDD	Uplink coverage enhancement	CellAlgoExtSwitch.UlCoverageEnhancementSw	None	Enabling both uplink coverage enhancement and DSS PDCCH resource sharing optimization will decrease the gains of these two functions.
LTE FDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with Dynamic Power Sharing Between LTE Carriers. As a result, the gains from these two functions will decrease.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	Extreme power sharing	LTE_EXTREME_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with extreme power sharing. As a result, the gains from these two functions will decrease.
LTE FDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with RF channel intelligent shutdown. As a result, the gains from these two functions will decrease.
LTE FDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with RF channel dynamic muting. As a result, the gains from these two functions will decrease.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D NR FD D	DSS and NR Flexible Refarming	• LTE: DSS_FLEX_REFAR M_SW option of the SpectrumCloud. SpectrumCloudE nhSwitch parameter • NR: DSS_FLEX_REFAR M_SW option of the NRDUCellAlgoS witch.Spectrum CloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with DSS and NR Flexible Refarming. As a result, the gains from DSS PDCCH resource sharing optimization will decrease.
LTE FD D NR FD D	LNR coordinated channel shutdown	• LTE: CellRfShutdown. MultiRatJointCh nShutdownSw set to ON • NR: MULTI_RAT_RF_ SHUTDOWN_S W option of the NRDUCellAlgoS witch.PowerSavi ngSwitch parameter	<i>Multi-RAT Coordinate d Energy Saving</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with LNR coordinated channel shutdown. As a result, the gains from these two functions will decrease.
LTE FD D	Flexible power allocation in RF channel dynamic muting	RF_DYN_MUTE_FLE X_PWR_ALLOC_SW option of the CellRfChnDynMut- ing.RfChnDynMut- ingAlgoSwitch parameter	None	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with flexible power allocation in RF channel dynamic muting. As a result, the gains from these two functions will decrease.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDLSchExtParam.DLSchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with downlink intelligent joint power control. As a result, the gains from these two functions will decrease.
LTE FDD	Downlink coordinated power control	DL_COORD_PC_SW option of the CellMimoParaCfg.MimoSwitch parameter	<i>Low-Band Booster</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with downlink coordinated power control. As a result, the gains from these two functions will decrease.
NR FDD	Inter- gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	Inter-gNodeB CA will take effect for a lower proportion of UEs when enabled together with DSS PDCCH resource sharing optimization. This is because an SCell of a UE has to be deactivated if the corresponding PCell works in a version earlier than this SCell.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	LTE PDCCH overhead optimization	LTE_PDCCH_OVERHEAD_OPT_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	Only DSS PDCCH resource sharing optimization takes effect when it is enabled together with LTE PDCCH overhead optimization. Considering this, DSS PDCCH resource sharing optimization is not recommended when LTE PDCCH overhead optimization has been enabled. This is because such function activation may generate negative impacts.
NR FDD	RB Reserve Mode	NRDUCellRbReserve.RbRsvMode	None	DSS PDCCH resource sharing optimization may take effect in a lower proportion of cases and therefore offer lower or even no gains if NRDUCellRbReserve.RbRsvMode is set to NB_GB_RB_RESERVED , LNR_AVAIL_RB_RESERVED , GU_RB_RESERVED , SPECIFIC_RB_RESERVED , or NB_SA_RB_RESERVED .

RA T	Function Name	Function Switch	Reference	Description
NR FDD	Flexible bandwidth	FLEXIBLE_BANDWIDTH_SW option of the NRDUCellFlexBw.FlexibleBwAlgoSwitch parameter	<i>Flexible Bandwidth (FDD)</i>	DSS PDCCH resource sharing optimization may take effect in a lower proportion of cases and therefore offer lower or even no gains if it is enabled together with flexible bandwidth.
NR FDD	Compact bandwidth	NRDUCell.CustomizedBwConfigInd set to CONFIG	<i>Scalable Bandwidth</i>	DSS PDCCH resource sharing optimization may take effect in a lower proportion of cases and therefore offer lower or even no gains if it is enabled together with compact bandwidth.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE PDCCH overhead optimization
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.
- LTE CRS overhead optimization
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.
- LTE polarization-based power assistance
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.
- Optimized scheduling of LTE and NR UEs

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS preferential-resource percentage adaptation

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS inter-RAT PDCCH spatial multiplexing

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, LampSite base stations cannot be used.

- DSS PDCCH resource sharing optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, LampSite base stations cannot be used.

For the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing, see [4.3.3 Hardware](#).

Boards

- LTE PDCCH overhead optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- LTE CRS overhead optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of both the LTE FDD and NR Flash Dynamic Spectrum Sharing function and the LTE CRS dynamic muting function must be met. For the board requirements of CRS dynamic muting, see the related descriptions in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.

- LTE polarization-based power assistance

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, only UBBPe and later boards can be used among the boards supporting LTE FDD and NR Flash Dynamic Spectrum Sharing.

- Optimized scheduling of LTE and NR UEs

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS preferential-resource percentage adaptation

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS inter-RAT PDCCH spatial multiplexing

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS PDCCH resource sharing optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

For the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing, see [4.3.3 Hardware](#).

RF Modules

- LTE PDCCH overhead optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- LTE CRS overhead optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of both the LTE FDD and NR Flash Dynamic Spectrum Sharing function and the LTE CRS dynamic muting function must be met. For the RF module requirements of CRS dynamic muting, see the related descriptions in *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.

- LTE polarization-based power assistance

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- Optimized scheduling of LTE and NR UEs

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS preferential-resource percentage adaptation

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS inter-RAT PDCCH spatial multiplexing

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

- DSS PDCCH resource sharing optimization

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.

For the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing, see [4.3.3 Hardware](#).

Cells

- LTE PDCCH overhead optimization
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.
- LTE CRS overhead optimization
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, the TX/RX mode of each LTE cell must be 4T4R or 8T8R, the TX/RX mode of the NR cell must be 4T4R, 8T8R, or 4T, and the **Cell.CrsPortNum** parameter must be set to **CRS_PORT_4** for each LTE cell.
- LTE polarization-based power assistance
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, the cell TX/RX mode must be 4T4R and the **Cell.CrsPortNum** parameter must be set to **CRS_PORT_4** for each LTE cell.
- Optimized scheduling of LTE and NR UEs
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.
- DSS preferential-resource percentage adaptation
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met.
- DSS inter-RAT PDCCH spatial multiplexing
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, the TX/RX mode of each LTE cell must be 4T4R/16T16R/8T8R/32T32R, the TX/RX mode of the NR cell must be 4T/4T4R/8T8R/16T16R/32T32R, there cannot be more than two LTE split cells, and the **Cell.CrsPortNum** parameter must be set to **CRS_PORT_4** for each LTE cell.
- DSS PDCCH resource sharing optimization
To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, the following requirements must be met:
 - On the NR side:
 - The NR cell bandwidth does not exceed 30 MHz.
 - **NRDUCell.NrDuCellNetworkingMode** is set to **NORMAL_CELL**.
 - On the LTE side:
 - **Cell.MultiRruCellFlag** is set to **BOOLEAN_FALSE** or **Cell.MultiRruCellMode** is set to a value other than **SFN**.
 - **CellDlpcPdschPa.PaPcOff** is set to a value other than **DB_6_P_A** and **DB_4DOT77_P_A**.

- **Cell.CrsPortNum** is set to **CRS_PORT_4**. Alternatively, **Cell.CrsPortNum** is set to **CRS_PORT_2**, **NRDUCELLPDCCH.SpctShrStartSymbol** is set to **SYM2**, and **NRDUCELLPDCCH.OccupiedSymbolNum** is set to **1SYM**.

For the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing, see [4.3.3 Hardware](#).

6.3.4 Others

LTE CRS overhead optimization and DSS PDCCH resource sharing optimization require that NR UEs support at least 32 PDSCH RE mapping patterns per TTl. If the *Phy-Parameters* IE contains the *pdsch-RE-MappingFR1-PerSlot* sub-IE with a value greater than or equal to n32, the UE supports the two functions. For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

[Table 6-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Spectrum Cloud Enhancement Switch	SpectrumCloud.Sp ectrumCloudEnhS witch	LTE_PDCCH_O VERHEAD_OP T_SW	Select this option when LTE PDCCH overhead optimization is required based on the network plan.
Spectrum Cloud Enhancement Switch	SpectrumCloud.Sp ectrumCloudEnhS witch	LTE_CRS_OVER HEAD_OPT_S W	Select this option when LTE CRS overhead optimization is required based on the network plan.
DRX Algorithm Switch	CellDrxPara.DrxAl gSwitch	None	It is recommended that DRX be enabled to prolong the CRS dynamic muting duration.

Parameter Name	Parameter ID	Option	Setting Notes
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	TRS_RM_PERIOD_OPT_SW	It is recommended that this option be selected or deselected for the entire network based on the network plan. When LTE CRS overhead optimization is enabled, select this option.
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	TRS_RATEMATCH_SW	It is recommended that this option be selected or deselected for the entire network based on the network plan. When LTE CRS overhead optimization is enabled, select this option.
CRS Overhead Optimization Threshold	LteNrSpctShrCellGrp.CrsOverheadOptThld	None	Set this parameter to an appropriate value when LTE CRS overhead optimization is required based on the network plan. You are advised to set this parameter to its recommended value.
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	CRS_OVERHEAD_OPT_SW	Select this option when LTE CRS overhead optimization is required based on the network plan.
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	POWER_POLARIZATION_SW	Select this option when LTE polarization-based power assistance is required based on the network plan, that is, when the power allocation mode is set to the spectrum power sharing mode and the LTE parameter CellDlpcPdschPa.PaPcOff is set to DB_3_P_A or a smaller value.

Parameter Name	Parameter ID	Option	Setting Notes
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	LNR_SPCT_SHR_SCH_OPT_SW	Select this option when optimized scheduling of LTE and NR UEs is required based on the network plan.
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	DSS_PREFER_RES_RATIO_ADAPT_SW	It is recommended that this option be selected when the overall rates of LTE and NR are expected to be the highest based on the network plan.
LTE Prefer Resource Ratio Adapt Factor	SpectrumCloud.LTEPrefResRatioAdaptFactor	None	It is recommended that this parameter be set to 1 when the overall rates of LTE and NR are expected to be the highest based on the network plan. To increase the NR rate, set this parameter to a smaller value. To increase the LTE rate, set this parameter to a larger value.
LTE Prefer Res Ratio Adapt UL Res Req Amount Thld	SpectrumCloud.LTEPrefResAdaptULResReqThd	None	Set this parameter based on the network plan. For DSS preferential-resource percentage adaptation to take effect in a larger proportion of cases, set this parameter to a larger value. For this function to take effect in a smaller proportion of cases, set this parameter to a smaller value.

Parameter Name	Parameter ID	Option	Setting Notes
LTE Prefer Res Ratio Adapt DL Res Req Amount Thld	SpectrumCloud.LtePrefResAdaptDL-ResReqThd	None	Set this parameter based on the network plan. For DSS preferential-resource percentage adaptation to take effect in a larger proportion of cases, set this parameter to a larger value. For this function to take effect in a smaller proportion of cases, set this parameter to a smaller value.
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	DSS_INTER_RAT_PDCCH_SPC_MUX_SW	Select this option when DSS inter-RAT PDCCH spatial multiplexing is required based on the network plan.
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	DSS_INTER_RAT_PDCCH_SPC_MUX_SW	Select this option when DSS inter-RAT PDCCH spatial multiplexing is required based on the network plan.
LTE PDCCH Sym 2 PSD Reduction Factor	LteNrSptcShrCellGrp.LtePdcchSym2PsdRdcFactor	None	Set this parameter to REDUCE_TO_1_4 when DSS inter-RAT PDCCH spatial multiplexing is required based on the network plan. This value is also recommended when DSS PDCCH resource sharing optimization is required.

Parameter Name	Parameter ID	Option	Setting Notes
LTE PDCCH Sym 2 PSD Thld	LteNrSpctShrCellGrp.LtePdcchSym2PsdThld	None	Set this parameter to 0 when DSS inter-RAT PDCCH spatial multiplexing is required based on the network plan. This value is also recommended when DSS PDCCH resource sharing optimization is required.
LTE PDCCH Available CCE Num Upper Limit	LteNrSpctShrCellGrp.LtePdcchAvailCceNumUpprLmt	None	Set this parameter to 255 when DSS inter-RAT PDCCH spatial multiplexing is required based on the network plan. This value is also recommended when DSS PDCCH resource sharing optimization is required.
Spectrum Cloud Extend Switch	SpectrumCloud.SpectrumCloudExtSwitch	DSS_PDCCH_SHARING_OPT_SW	Select this option when DSS PDCCH resource sharing optimization is required based on the network plan.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	DSS_PDCCH_SHARING_OPT_SW	Select this option when DSS PDCCH resource sharing optimization is required based on the network plan.
LTE CRS Sym 1 Pwr Reduction Factor	LteNrSpctShrCellGrp.LteCrssym1PwrReduceFactor	None	The value REDUCE_TO_1_2 is recommended when DSS PDCCH resource sharing optimization is required based on the network plan. This value takes effect only when the Cell.CrsPortNum parameter is set to CRS_PORT_4 .

6.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- **LTE PDCCH overhead optimization**
//Enabling LTE PDCCH overhead optimization
MOD SPECTRUMCLOUD: LocalCellId=0,
SpectrumCloudEnhSwitch=LTE_PDCCH_OVERHEAD_OPT_SW-1;
- **LTE CRS overhead optimization**
//Setting the DRX algorithm switch
MOD CELLDXPARAM: LocalCellId=0, DrxAlgSwitch=ON;
//Enabling LTE CRS overhead optimization on the LTE side
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=LTE_CRS_OVERHEAD_OPT_SW-1;
//Setting the CRS overhead optimization threshold
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, CrsOverheadOptThld=5;

//Enabling LTE CRS overhead optimization on the NR side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=CRS_OVERHEAD_OPT_SW-1;
MOD NRDUCELLPDSCH: NrDuCellId=0,
RateMatchSwitch=TRS_RM_PERIOD_OPT_SW-1&TRS_RATEMATCH_SW-1;
- **LTE polarization-based power assistance**
//Enabling LTE polarization-based power assistance on the LTE side
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=POWER_POLARIZATION_SW-1;
- **Optimized scheduling of LTE and NR UEs**
//Enabling optimized scheduling of LTE and NR UEs
MOD NRDUCELLPDCCH: NrDuCellId=0, GuaranteedRbNum=0;
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=LNR_SPCT_SHR_SCH_OPT_SW-1;
- **DSS preferential-resource percentage adaptation**
//Enabling DSS preferential-resource percentage adaptation
MOD SPECTRUMCLOUD: LocalCellId=1,
SpectrumCloudEnhSwitch=DSS_PREFER_RES_RATIO_ADAPT_SW-1;
//Changing the settings of LtePrefResRatioAdaptFactor, LtePrefResAdaptUIResReqThd, and
LtePrefResAdaptDIResReqThd
MOD SPECTRUMCLOUD: LocalCellId=1, LtePrefResRatioAdaptFactor=10,
LtePrefResAdaptUIResReqThd=100, LtePrefResAdaptDIResReqThd=100;
- **DSS inter-RAT PDCCH spatial multiplexing**
//Enabling DSS inter-RAT PDCCH spatial multiplexing
//Turning on the LTE switch for DSS inter-RAT PDCCH spatial multiplexing
MOD SPECTRUMCLOUD: LocalCellId=0,
SpectrumCloudEnhSwitch=DSS_INTER_RAT_PDCCH_SPC_MUX_SW-1;
//Turning on the NR switch for DSS inter-RAT PDCCH spatial multiplexing
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SpectrumCloudEnhSwitch=DSS_INTER_RAT_PDCCH_SPC_MUX_SW-1;
//Setting LtePdcchSym2PsdRdcFactor, LtePdcchSym2PsdThld, and LtePdcchAvailCceNumUpplmt
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0,
LtePdcchSym2PsdRdcFactor=REDUCE_TO_1_4, LtePdcchSym2PsdThld=0,
LtePdcchAvailCceNumUpplmt=255;

- **DSS PDCCH resource sharing optimization**
 //Enabling DSS PDCCH resource sharing optimization
 //Turning on the LTE switch for DSS PDCCH resource sharing optimization
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=DSS_PDCCH_SHARING_OPT_SW-1;
 //Turning on the NR switch for DSS PDCCH resource sharing optimization
 MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=DSS_PDCCH_SHARING_OPT_SW-1;
 //Setting LtePdcchSym2PsdRdcFactor, LtePdcchSym2PsdThld, LtePdcchAvailCceNumUpprLmt, and LteCrsSym1PwrReduceFactor
 MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LtePdcchSym2PsdRdcFactor=REDUCE_TO_1_4, LtePdcchSym2PsdThld=0, LtePdcchAvailCceNumUpprLmt=255, LteCrsSym1PwrReduceFactor=REDUCE_TO_1_2;
 //(Optional) Disabling DSS PDCCH resource sharing optimization for specified UEs. For actual UE models, contact Huawei engineers.
 ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE, UeFeatureValueContent="YYYYYYYYYYY", UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
 ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=FDD, BlacklistCtrlExtSwitch=DSS_PDCCH_SHARING_OPT_SW_OFF-1;

Deactivation Command Examples

- **LTE PDCCH overhead optimization**
 //Disabling LTE PDCCH overhead optimization
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=LTE_PDCCH_OVERHEAD_OPT_SW-0;
- **LTE CRS overhead optimization**
 //Disabling LTE CRS overhead optimization
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=LTE_CRS_OVERHEAD_OPT_SW-0;
 MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=CRS_OVERHEAD_OPT_SW-0;
 MOD NRDUCELLPDSCH: NrDuCellId=0, RateMatchSwitch=TRS_RM_PERIOD_OPT_SW-0;
 MOD CELLDXPARAM: LocalCellId=0, DrxAlgSwitch=OFF;
- **LTE polarization-based power assistance**
 //Disabling LTE polarization-based power assistance
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=POWER_POLARIZATION_SW-0;
- **Optimized scheduling of LTE and NR UEs**
 //Disabling optimized scheduling of LTE and NR UEs
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=LNR_SPCT_SHR_SCH_OPT_SW-0;
- **DSS preferential-resource percentage adaptation**
 //Disabling DSS preferential-resource percentage adaptation
 MOD SPECTRUMCLOUD: LocalCellId=1, SpectrumCloudEnhSwitch=DSS_PREFER_RES_RATIO_ADAPT_SW-0;
- **DSS inter-RAT PDCCH spatial multiplexing**
 //Disabling DSS inter-RAT PDCCH spatial multiplexing
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=DSS_INTER_RAT_PDCCH_SPC_MUX_SW-0;
 MOD NRDUCELLALGOSWITCH: NrDuCellId=0, SpectrumCloudEnhSwitch=DSS_INTER_RAT_PDCCH_SPC_MUX_SW-0;
 MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LtePdcchSym2PsdRdcFactor=REDUCE_TO_1_4, LtePdcchSym2PsdThld=0, LtePdcchAvailCceNumUpprLmt=255;
- **DSS PDCCH resource sharing optimization**
 //Disabling DSS PDCCH resource sharing optimization
 MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudExtSwitch=DSS_PDCCH_SHARING_OPT_SW-0;
 MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=DSS_PDCCH_SHARING_OPT_SW-0;
 MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LtePdcchSym2PsdRdcFactor=REDUCE_TO_1_4, LtePdcchSym2PsdThld=0, LtePdcchAvailCceNumUpprLmt=255, LteCrsSym1PwrReduceFactor=OFF;
 //(Optional) Disabling differentiated handling on UEs
 RMV GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=FDD;
 RMV GNBUEINFO: UeInfoIndex=1;

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Use counters and inter-RAT Huawei-proprietary interface trace to check whether DSS/Hybrid DSS Performance Boost has taken effect. For details about the benefits of this function, see [6.2.1 Benefits](#).

LTE PDCCH Overhead Optimization

Observe the downlink spectral efficiency of the NR cell. If the downlink spectral efficiency increases after this function is enabled, LTE PDCCH overhead optimization has taken effect.

LTE CRS Overhead Optimization

The LTE CRS overhead optimization function has taken effect when the following conditions on the eNodeB and gNodeB sides are met:

Perform the following steps to start eNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **LTE > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. The value of the lte-dl-cell-information > lte-crs-ext-information > lte-crs-overhead-opt-enable IE in the LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION message is 1.

----End

Perform the following steps to start gNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **NR > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. The value of the NrCellCommonInformation > nr-cell-function-switch > lte-crs-overhead-opt IE in the LNR_INTER_RAT_SPECTRUM_NEGOTIATION_REQUEST message is 1.

----End

LTE Polarization-based Power Assistance

Observe the value of the NR counter **N.PRB.DL.Actual.Avail.Avg**. If the counter value increases after this function is enabled, LTE polarization-based power assistance has taken effect.

When only this function is enabled among the functions in DSS/Hybrid DSS Performance Boost, an increase in the average number of available downlink PRBs on the NR side can be observed.

Optimized Scheduling of LTE and NR UEs

The function of optimized scheduling of LTE and NR UEs has taken effect when the following condition on the eNodeB side is met:

Perform the following steps to start eNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **LTE > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR cells, messages are traced over the inter-RAT Huawei-proprietary interface. The value of the lnrss-group-information > lte-spectrum-sharing--group-information > lnr-sch-opt-switch IE in the LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION message is 1.

----End

DSS Preferential-Resource Percentage Adaptation

After turning on the function switch, observe the following counters on the LTE and NR sides: **L.ChMeas.PRB.UL.Actual.Avail**, **N.PRB.UL.Actual.Avail.Avg**, **L.ChMeas.PRB.DL.Actual.Avail**, and **N.PRB.DL.Actual.Avail.Avg**.

Assume that the baseline percentage of resources preferentially allocated to LTE is 50%. If the number of RBs available for the heavier-load RAT (LTE or NR) increases after this function is enabled, this function has taken effect.

DSS Inter-RAT PDCCH Spatial Multiplexing

After turning on the function switches on both the LTE and NR sides, observe the number of times the PDCCH occupies three symbols in a measurement period in the LTE cell (indicated by **L.ChMeas.PDCCH.SymNum.3**). If the number increases after this function is enabled, this function has taken effect.

DSS PDCCH Resource Sharing Optimization

After turning on the functions switches on both the LTE and NR sides, observe the average number of available PDCCH CCEs in the NR cell (indicated by **N.CCE.Avail.Avg**). If the number increases after this function is enabled, this function has taken effect.

6.4.3 Network Monitoring

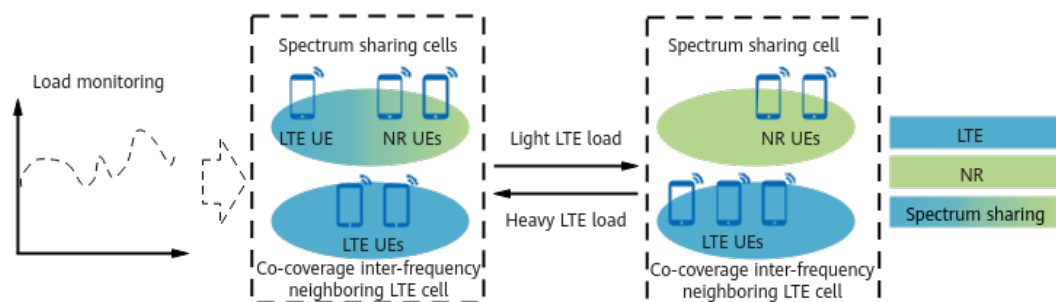
After this function is enabled, monitor the network gains using the counters listed in [6.2.1 Benefits](#).

7 DSS and NR Flexible Refarming

7.1 Principles

The LTE load on a network changes with time. According to 3GPP specifications, when LTE FDD and NR share spectrum, resources are still consumed by LTE control channels and signals even if there is no LTE load. To further improve spectrum utilization in LTE FDD and NR spectrum sharing scenarios, DSS and NR Flexible Refarming is introduced. **With this function, when an LTE spectrum sharing cell and its co-coverage inter-frequency neighboring cells are under light load, UEs in the LTE spectrum sharing cell are transferred to these neighboring cells; the shared spectrum is refarmed to NR-only, improving spectral efficiency. When these co-coverage neighboring LTE cells come under heavy load, the NR-only spectrum is refarmed back to spectrum sharing to ensure LTE capacity. Figure 7-1 shows how this function works.**

Figure 7-1 Refarming mechanism



This function is enabled by turning on function switches and adding co-coverage inter-frequency neighboring LTE cells.

- Turning on function switches
 - LTE: Select the **DSS_FLEX_REFARM_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter.
 - NR: Select the **DSS_FLEX_REFARM_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch** parameter.

- Adding a maximum of 14 co-coverage inter-frequency neighboring LTE cells to the spectrum sharing cell group
 - Intra-base-station inter-frequency neighboring cells:
 - For cells with **SpectrumCloud.SpectrumCloudSwitch** set to **GL_SPECTRUM_CONCURRENCY**, **UL_SPECTRUM_SHARING**, **UL_SPECTRUM_SHR_PH2_WITH_GSM**, or **DC_HSDPA_BASED_UL_SPECTRUM_SHR**: Set the LTE parameter **SpectrumCloud.MultiRatShrLnrSptCoGrpld** to the ID of the spectrum sharing cell group to which the LTE spectrum sharing cell belongs. The co-coverage inter-frequency neighboring cells with this setting will take over services of the LTE spectrum sharing cell after refarming to NR.
 - For other cells: Set the LTE parameter **SpectrumCloud.LteNrSpectrumShrCellGrpld** to the ID of the spectrum sharing cell group to which the LTE spectrum sharing cell belongs, and set the **SpectrumCloud.SpectrumCloudSwitch** parameter to **LTE_NR_COORD_SPECTRUM_SHR**. The co-coverage inter-frequency neighboring cells with these settings will take over services of the LTE spectrum sharing cell after refarming to NR.
 - Inter-base-station inter-frequency neighboring cells: Configure the LTE MO **SptCloudAuxEnodeb** to add co-coverage inter-frequency neighboring cells that will take over services of the LTE spectrum sharing cell after refarming to NR. When adding a co-coverage inter-frequency neighboring cell, configure the mobile network code (MNC) and mobile country code (MCC) to those that form the public land mobile network (PLMN) ID of this cell's primary operator.

After DSS and NR Flexible Refarming is enabled, the eNodeB monitors the load of cells in the spectrum sharing cell group over a period of 30 minutes. These cells include the LTE spectrum sharing cell, co-coverage inter-frequency neighboring LTE cells, and NR cell. The load is indicated by the number of UEs, downlink PRB usage, and uplink PRB usage.

When the spectrum sharing cells in the cell group are sharing spectrum, the eNodeB calculates the overall number of UEs and overall PRB usages that involve both the LTE spectrum sharing cell and its co-coverage inter-frequency neighboring cells.

- Overall number of UEs = (Number of UEs in the spectrum sharing cell + Total number of UEs in the co-coverage inter-frequency neighboring cells)/Number of the co-coverage inter-frequency neighboring cells
- Overall uplink PRB usage = (Number of uplink RBs used in the spectrum sharing cell + Total number of uplink RBs used in the co-coverage inter-frequency neighboring cells)/Total number of available uplink RBs in the co-coverage inter-frequency neighboring cells
- Overall downlink PRB usage = (Number of downlink RBs used in the spectrum sharing cell + Total number of downlink RBs used in the co-coverage inter-frequency neighboring cells)/Total number of available downlink RBs in the co-coverage inter-frequency neighboring cells

When the spectrum sharing cells in the cell group are sharing spectrum, refarming to NR is triggered if all the following conditions are met:

- Overall uplink PRB usage $\leq \text{LteNrSpctShrCellGrp.UlPrbUsageThld} - \text{LteNrSpctShrCellGrp.UlPrbUsageThldOffset}$
- Overall downlink PRB usage $\leq \text{LteNrSpctShrCellGrp.PrbUsageThld} - \text{LteNrSpctShrCellGrp.PrbUsageThldOffset}$
- Overall number of UEs $\leq \text{LteNrSpctShrCellGrp.RrcConnUeNumThld} - \text{LteNrSpctShrCellGrp.RrcConnUeNumThldOffset}$

Refarming involves transferring UEs from the LTE spectrum sharing cell to its inter-frequency neighboring cells based on inter-frequency measurements. Specifically, the eNodeB delivers measurement configurations related to event A4 to RRC_CONNECTED UEs, where the RSRP threshold for event A4 equals $\min(\text{InterFreqHoGroup.InterFreqHoA4ThdRsrp}, -105)$ and the other measurement configurations are the same as those for coverage-based inter-frequency handover (detailed in section "Measurement Configuration Delivery" of "Coverage-based Inter-Frequency Handover" in *Mobility Management in Connected Mode in eRAN Feature Documentation*). After receiving an event A4 measurement report from a UE, the eNodeB hands over the UE to an inter-frequency neighboring cell that meets the event A4 threshold.

Some UEs may fail to be transferred out of the LTE spectrum sharing cell. If the number of such UEs is less than or equal to the forced refarming threshold (specified by the **SpectrumCloud.ForcedRefarmingUeNumThld** parameter), these UEs will be redirected to target frequencies and the LTE spectrum sharing cell will be forcibly shut down, accompanied by spectrum refarming to NR. For details about how to select a target frequency for redirection, see section "Measurement Configuration Delivery" of "Coverage-based Inter-Frequency Handover" in *Mobility Management in Connected Mode in eRAN Feature Documentation*.

When only NR is using the spectrum of the spectrum sharing cells in the cell group, triggering of refarming to spectrum sharing can be determined with or without considering the NR load, depending on the setting of the **LteNrSpctShrCellGrp.DssFlexRefarmPolicy** parameter:

- If the **LteNrSpctShrCellGrp.DssFlexRefarmPolicy** parameter is set to **LTE_NR_FAIR**, refarming from NR to spectrum sharing is triggered when any of the following conditions is met:
 - Uplink PRB usage of a co-coverage inter-frequency neighboring cell $> \text{LteNrSpctShrCellGrp.UlPrbUsageThld} + \text{LteNrSpctShrCellGrp.UlPrbUsageThldOffset}$
 - Downlink PRB usage of a co-coverage inter-frequency neighboring cell $> \text{LteNrSpctShrCellGrp.PrbUsageThld} + \text{LteNrSpctShrCellGrp.PrbUsageThldOffset}$
 - Number of UEs in a co-coverage inter-frequency neighboring cell $> \text{LteNrSpctShrCellGrp.RrcConnUeNumThld} + \text{LteNrSpctShrCellGrp.RrcConnUeNumThldOffset}$

In addition, there is a protection mechanism against refarming from spectrum sharing to NR. This mechanism mitigates the network impacts of signaling reconfiguration, which is required for NR UEs during refarming, when the NR spectrum sharing cell is heavily loaded. Specifically, the NR cell is considered to be under heavy load and refarming from spectrum sharing to NR is suspended for three days if the NR cell meets any of the following conditions:

- Uplink PRB usage of the NR cell >
gNBDULteNrSpctShrCg.UlPrbUsageThld
- Downlink PRB usage of the NR cell >
gNBDULteNrSpctShrCg.PrbUsageThld
- Number of UEs in the NR cell >
gNBDULteNrSpctShrCg.RrcConnUeNumThld
- If the **LteNrSpctShrCellGrp.DssFlexRefarmPolicy** parameter is set to **NR_PRI**, refarming from NR to spectrum sharing is triggered only when all the following NR load conditions are also met:
 - Uplink PRB usage of the NR cell ≤
gNBDULteNrSpctShrCg.UlPrbUsageThld
 - Downlink PRB usage of the NR cell ≤
gNBDULteNrSpctShrCg.PrbUsageThld
 - Number of UEs in the NR cell ≤
gNBDULteNrSpctShrCg.RrcConnUeNumThld

Frequent spectrum refarming causes frequent UE transfer, affecting network performance. To mitigate this issue, the **LteNrSpctShrCellGrp.LteNrSpctHoProtDuration** parameter can be used to specify the minimum interval between two spectrum refarming procedures. Once spectrum refarming to NR is triggered, another refarming procedure (from NR to spectrum sharing) will not occur within the interval specified by **LteNrSpctShrCellGrp.LteNrSpctHoProtDuration** unless the co-coverage inter-frequency neighboring cells become abnormal. Likewise, after being refarmed from NR to spectrum sharing, the spectrum will not be refarmed again to NR within this interval.

DSS and NR Flexible Refarming allows multiple sets of spectrum sharing cells to be added to the same refarming group using the **LteNrSpctShrCellGrp.DssFlexRefarmGrpId** parameter. (A set of spectrum sharing cells refers to cells engaged in spectrum sharing in the same spectrum sharing cell group.) Synchronous refarming works between different sets of spectrum sharing cells in the same refarming group as follows:

- Refarming from spectrum sharing to NR is performed only if all sets of spectrum sharing cells meet the conditions for the refarming.
- Refarming from NR to spectrum sharing is performed as long as one set of spectrum sharing cells meets the conditions for the refarming.

NOTICE

The shared spectrum will not be refarmed to NR if the LTE spectrum sharing cell meets any of the following conditions:

- UEs running voice services camp on the cell.
 - UEs engaged in emergency calls camp on the cell.
 - Relay nodes (RNs) have accessed the cell.
 - The cell is in the key event assurance state.
-

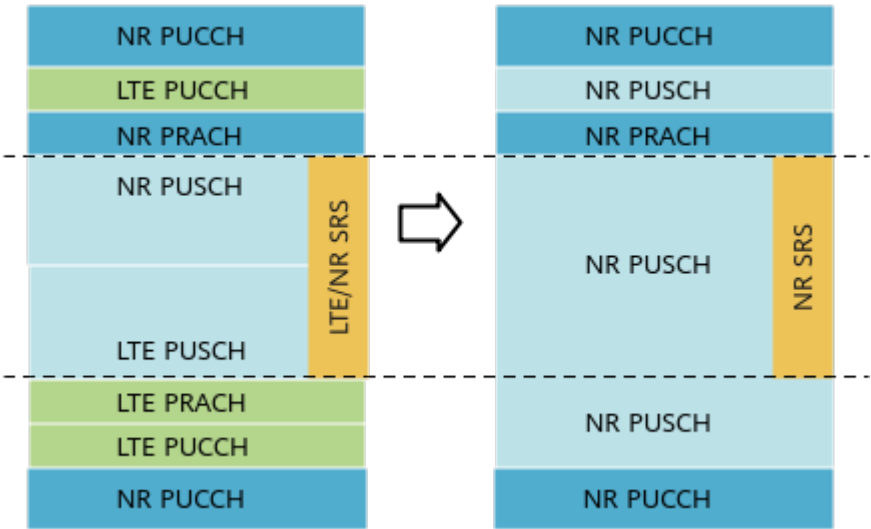
Uplink Physical Channels

When LTE FDD and NR share spectrum, NR and LTE use the same uplink spectrum resources through frequency division multiplexing (FDM). After refarming from spectrum sharing to NR, NR occupies all the uplink spectrum resources.

The following descriptions take LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios as an example.

- Before refarming from spectrum sharing to NR, the total available shared PUSCH resources for LTE and NR equal their total spectrum resources minus the resources occupied by the PUCCHs and PRACHs of LTE and NR.
The total available shared PUSCH resources for LTE and NR are dynamically shared between them based on traffic requirements.
- After refarming from spectrum sharing to NR, the resources available for the NR PUSCH equal the total spectrum resources minus the resources occupied by the NR PUCCH and PRACH.

Figure 7-2 Uplink physical channels

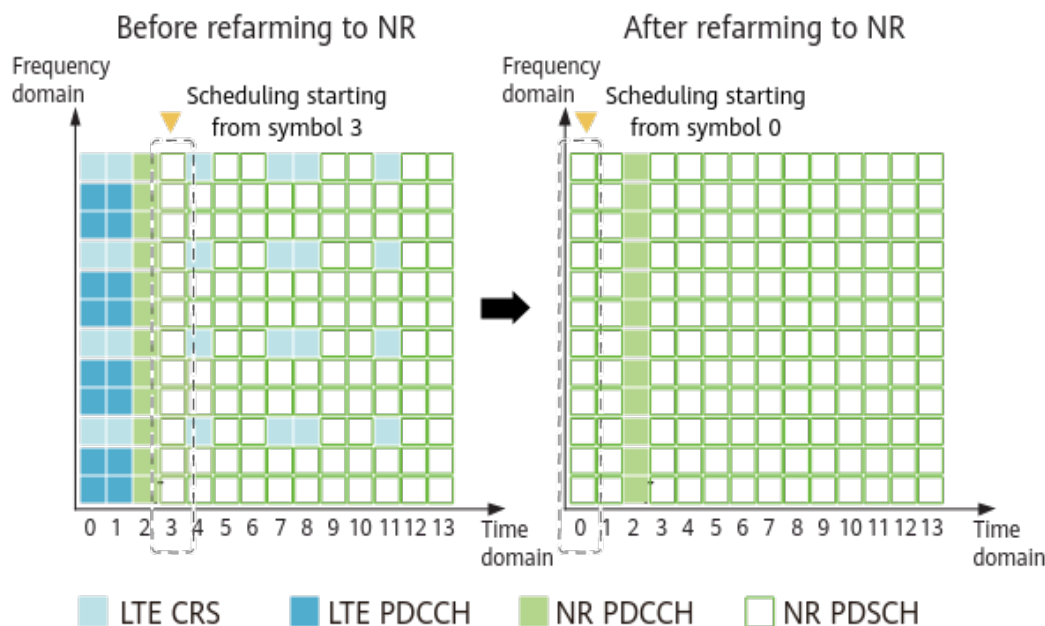


Downlink Physical Channels

When LTE FDD and NR share spectrum and LTE CRS needs to occupy symbols 0 and 1, the NR PDCCH can only be deployed in symbol 2. The NR PDSCH must be scheduled following the NR PDCCH. In addition, rate matching is performed for the NR PDSCH on the REs where LTE CRS appears to avoid interference from LTE CRS. After refarming from spectrum sharing to NR, the NR PDSCH can be scheduled starting from symbol 0, and LTE CRS rate matching is not needed.

Figure 7-3 shows the scheduling.

Figure 7-3 NR PDSCH scheduling before and after refarming



When this function is enabled, access failures or service drops may occur due to PDCCH blind detection failures. This UE compatibility issue can be solved by enabling the PDCCH rate matching UE compatibility rectification function (controlled by the **PDCCH_RATEMATCH_COMPATIBLE_SW** option of the NR parameter **NRDUCellPdsch.RateMatchSwitch**). For details, see *Downlink Scheduling*.

When this function is enabled, the service drop rate may increase due to changes in the configurations for CRS rate matching. This UE compatibility issue can be resolved by selecting the **CRS_RM_CONFIG_COMPAT_SW** option of the **NRDUCellPdsch.RateMatchSwitch** parameter. With this setting, the *lte-CRS-ToMatchAround* IE can be configured in the *ServingCellConfig* IE but not in the *ServingCellConfigCommon* IE.

7.2 Network Analysis

7.2.1 Benefits

This function increases the average downlink throughput of an NR cell and the uplink and downlink perceived throughputs of NR UEs when LTE cells are lightly loaded.

- Average downlink throughput of an NR cell = **Cell Downlink Average Throughput (DU)**
- Uplink perceived throughput of NR UEs = **User Uplink Average Throughput (DU)**
- Downlink perceived throughput of NR UEs = **User Downlink Average Throughput (DU)**

NR UEs must have the RB- and symbol-level semi-persistent rate matching capability; otherwise, the increase in the downlink perceived throughput of NR UEs

will be reduced. A UE has this capability if the *Phy-Parameters* IE includes the *rateMatchingResrcSetSemi-Static* IE with the value "supported". For details, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.5.0.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

The average uplink and downlink throughputs of LTE cells decrease, so do the average uplink and downlink throughputs of LTE UEs.

- Average uplink cell throughput = **Cell Uplink Average Throughput**
- Average downlink cell throughput = **Cell Downlink Average Throughput**
- Average uplink UE throughput = **User Uplink Average Throughput**
- Average downlink UE throughput = **User Downlink Average Throughput**

Refarming from spectrum sharing to NR has the following impacts:

- The uplink and downlink CCE allocation failure rates decrease in the NR cell because more resources are available for the PDCCH.
- Given the same NR traffic volume, the downlink PRB usage of the NR cell and the air interface delay in the cell decrease.
- The uplink PRB usage, average downlink and uplink MCS indexes, and downlink and uplink BLERs of the NR cell change.
- The number of uplink preallocations, scheduling chances, and padding data will increase in the NR cell. Therefore, the average uplink throughput of the NR cell (indicated by **Cell Uplink Average Throughput (DU)**) may change.

During refarming from spectrum sharing to NR, UEs in the LTE spectrum sharing cell are transferred to its co-coverage inter-frequency neighboring cells. As a result, the values of the counters listed in [Table 7-1](#) and [Table 7-2](#) increase in all these cells.

In addition, to promptly decrease the number of UEs in the LTE spectrum sharing cell, new UE access to the cell is prohibited, and the barred state of the cell is broadcast through SIB1. However, a small number of UEs may still send RRC connection setup requests to access the cell, causing an increase in the **L.RRC.SetupFail.Rej** counter value and a decrease in the **RRC Setup Success Rate**. The number of such access failures is indicated by the **L.RRC.SetupFail.ResFail.CellBarred** counter.

Table 7-1 Counters of the LTE spectrum sharing cell

Networking Scenario	Counter ID	Counter Name
An inter-frequency neighboring LTE cell is served by the same base station.	1526726998	L.HHO.IntraeNB.InterFreq.Prep AttOut
	1526726999	L.HHO.IntraeNB.InterFreq.Exec AttOut

Networking Scenario	Counter ID	Counter Name
	1526727000	L.HHO.IntraeNB.InterFreq.Exec SuccOut
An inter-frequency neighboring LTE cell is served by another base station.	1526727004	L.HHO.IntereNB.InterFreq.Prep AttOut
	1526727005	L.HHO.IntereNB.InterFreq.Exec AttOut
	1526727006	L.HHO.IntereNB.InterFreq.Exec SuccOut

Table 7-2 Counters of a co-coverage inter-frequency neighboring cell

Networking Scenario	Counter ID	Counter Name
The inter-frequency neighboring LTE cell is served by the same base station.	1526727253	L.HHO.IntraeNB.PrepAttIn
	1526727254	L.HHO.IntraeNB.ExecAttIn
	1526727255	L.HHO.IntraeNB.ExecSuccIn
The inter-frequency neighboring LTE cell is served by another base station.	1526727256	L.HHO.IntereNB.PrepAttIn
	1526727257	L.HHO.IntereNB.ExecAttIn
	1526727258	L.HHO.IntereNB.ExecSuccIn

After DSS and NR Flexible Refarming takes effect, cells in an area may be in different spectrum usage states: some in the spectrum sharing state, while others in the NR state. To mitigate the impacts of these coexisting states on LTE performance, it is recommended that CQI-based inter-frequency handover be enabled. For details, see *Mobility Management in Connected Mode* in *eRAN Feature Documentation*.

It is also recommended that intra-RAT mobility load balancing be enabled, which helps to quickly reduce the heavy load of co-coverage neighboring LTE cells when the NR-only spectrum is refarmed back to spectrum sharing. For details, see *Intra-RAT Mobility Load Balancing* in *eRAN Feature Documentation*.

7.3 Requirements

7.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE FDD	MRFD-200222	DSS and NR Flexible Refarming (LTE FDD)	LT1SDSSRELO0	eNodeB	per Cell
NR FDD	MRFD-200262	DSS and NR Flexible Refarming (NR FDD)	NR0SDSSREN01	gNodeB	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

- Prerequisite functions on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

- Prerequisite functions on the NR side

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPCT_SH R_SW or LTE_NR_SDL_SPCT_SH R_SW option of the NRDUCellAlgoSwitch. SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

7.3.2.2 Mutually Exclusive Functions

- Mutually exclusive functions on the LTE side

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FD D	Static Multiple Beam in Smart 8T8R	None	<i>Smart 8T8R (FDD)</i>
LTE FD D	Massive MIMO Static Multiple Beam	None	<i>Massive MIMO Basics (FDD)</i>

- Mutually exclusive functions on the NR side

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FD D	LTE CRS full- symbol rate matching	LTE_CRS_RATEMATCH _ALL_SYM_SW option of the NRDUCellAlgoSwitch. SpectrumCloudEnhS- witch parameter	None
NR FD D	LTE 4-port CRS full-symbol rate matching	LTE_CRS_RM_ALL_SY M_4PORT_SW option of the NRDUCellAlgoSwitch. SpectrumCloudEnhS- witch parameter	None

7.3.2.3 Function Impacts

- Function impacts on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	DSS and NR Flexible Refarming cannot take effect in spectrum sharing cells that are configured as basic cells for intelligent power-off of carriers in the same coverage when both functions are enabled.
LTE FDD	Dynamic carrier shutdown	CARRIER_DYN_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	DSS and NR Flexible Refarming cannot take effect in spectrum sharing cells that are configured as basic cells for dynamic carrier shutdown when both functions are enabled.
LTE FDD	Multi-carrier coordinated energy saving	CellShutdown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIER_MANUAL_SHUTDOWN , or ON_CARRIERPRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	DSS and NR Flexible Refarming cannot take effect in spectrum sharing cells that are configured as basic cells for multi-carrier coordinated energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	DSS and NR Flexible Refarming cannot take effect in spectrum sharing cells that are configured as basic cells for Cell Switch-off Based on Automatic Co-coverage Identification when both functions are enabled.
LTE FDD	Uplink CA	CaUL2CCSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The number of uplink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD uplink CA takes effect.
LTE FDD	Downlink CA	None	<i>Carrier Aggregation</i>	The number of downlink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD downlink CA takes effect.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Intra-RAT mobility load balancing	InterFreqMlbSwitch option of the CellAlgoSwitch.MlbAlgoSwitch parameter	<i>Intra-RAT Mobility Load Balancing</i>	Refarming from spectrum sharing to NR transfers LTE UEs to co-coverage inter-frequency neighboring cells, increasing the load of these neighboring cells. In addition, as the number of available LTE carriers decreases by one, the PRB usage and the number of UEs may change.
LTE FDD	Multi-band optimal carrier selection	MbfcsSwitch option of the CellAlgoSwitch.MlbAlgoSwitch parameter	<i>Multi-band Optimal Carrier Selection</i>	After refarming from spectrum sharing to NR, the number of target frequencies decreases and the optimal carrier may change.
LTE FDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SWITCH option of the MultiCarrierUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling</i>	After refarming from spectrum sharing to NR, the building progress and application of virtual grid models may be affected and the selected carrier combination may change.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	NSA networking based on EPC	NSA_DC_CAPABILITY_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	After refarming from spectrum sharing to NR, the number of LTE frequencies decreases by one and therefore the number of NSA UEs may decrease.
LTE FDD	NSA carrier combination selection based on user experience	NSA_LTE_FDD_SEL_OPT_SW option of the EnodebAlgoExtSwitch.MultiNetworkWorkingOptionOptSw parameter	<i>EPC-based NSA Experience Improvement</i>	After refarming from spectrum sharing to NR, the number of RBs available for NR may change and the optimal carrier combination may also change.
LTE FDD	LTE spectrum coordination	SpectrumCoordinationSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>LTE Spectrum Coordination</i>	After the carrier of an LTE spectrum sharing cell is shut down, fewer cells can be selected for spectrum coordination, which therefore may take effect for fewer UEs.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Dedicated carrier for TM9	Tm9DedicatedCarrier.Tm9CcDLEarfcn Tm9DedicatedCarrier.WorkMode	<i>Dedicated Carrier for TM9</i>	If the co-coverage inter-frequency neighboring cells of an LTE spectrum sharing cell work on TM9-dedicated carriers, UE transfer may fail. As a result, the LTE rate will be adversely affected by inter-frequency handovers and exclusive spectrum use by NR will produce no gains. If the co-coverage inter-frequency neighboring cells are distributed on TM9-dedicated carriers and non-TM9-dedicated carriers, LTE load imbalance may occur. This is because TM4 UEs can be transferred only to cells on non-TM9-dedicated carriers, while TM9 UEs can be transferred to cells on either type of carriers. To address the imbalance, you are also advised to enable adaptive TM4/TM9 layering.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC</i>	If eMTC UEs camp on an LTE spectrum sharing cell, the shared spectrum will not be refarmed to NR. If eMTC introduction is enabled in an LTE spectrum sharing cell but disabled in all its co-coverage inter-frequency neighboring cells, the shared spectrum will not be refarmed to NR.
LTE FDD	Out-of-band relay	OutOfBandRelaySwitch option of the CellAlgoSwitch.RelaySwitch parameter	<i>Relay</i>	If a relay cell with a relay subframe ratio of 10/10 is configured as a co-coverage inter-frequency neighboring cell, LTE UEs may fail to be transferred to this cell. As a result, DSS and NR Flexible Refarming takes effect in a lower proportion of cases or does not take effect at all.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	WBB User Steering Management	Cell.SpecifiedCellFlag	<i>WBB</i>	The types of UEs that are allowed to camp on an LTE spectrum sharing cell must also be allowed to camp on its co-coverage inter-frequency neighboring cells. Otherwise, the shared spectrum will not be refarmed to NR.
LTE FDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENodeBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENodeBSharingMode set to HYBRID_SHARED	<i>RAN Sharing</i>	In RAN sharing scenarios, the PLMN IDs of an LTE spectrum sharing cell must be a subset of the PLMN IDs of its co-coverage inter-frequency neighboring cells. If the preceding condition cannot be met in network planning, you are not advised to enable DSS and NR Flexible Refarming.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	DSS inter-RAT PDCCH spatial multiplexing	DSS_INTER_RAT_PDCCH_SPC_MUX_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with DSS and NR Flexible Refarming. As a result, the gains from DSS inter-RAT PDCCH spatial multiplexing will decrease.
LTE FDD	DSS PDCCH resource sharing optimization	DSS_PDCCH_SHARING_OPT_SW option of the SpectrumCloud.SpectrumCloudExtSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with DSS and NR Flexible Refarming. As a result, the gains from DSS PDCCH resource sharing optimization will decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>	If a cell with GSM and LTE Spectrum Concurrency enabled is configured as a co-coverage inter-frequency neighboring cell, changes in the GSM load may affect the LTE PRB usage and, consequently, the proportion of occasions where DSS and NR Flexible Refarming takes effect. When the GSM load increases, the amount of resources available for LTE may decrease, the LTE PRB usage may increase, and the proportion of occasions where DSS and NR Flexible Refarming takes effect may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	GSM and LTE Spectrum Concurrency Phase 2	GL_SPCT_CONCURRENCY_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>GSM and LTE Spectrum Concurrency</i>	If a cell with GSM and LTE Spectrum Concurrency Phase 2 enabled is configured as a co-coverage inter-frequency neighboring cell, changes in the GSM load may affect the LTE PRB usage and, consequently, the proportion of occasions where DSS and NR Flexible Refarming takes effect. When the GSM load increases, the amount of resources available for LTE may decrease, the LTE PRB usage may increase, and the proportion of occasions where DSS and NR Flexible Refarming takes effect may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>	If a cell with UMTS and LTE Spectrum Sharing (LTE FDD) enabled is configured as a co-coverage inter-frequency neighboring cell, changes in the UMTS load may affect the LTE PRB usage and, consequently, the proportion of occasions where DSS and NR Flexible Refarming takes effect. When the UMTS load increases, the amount of resources available for LTE may decrease, the LTE PRB usage may increase, and the proportion of occasions where DSS and NR Flexible Refarming takes effect may decrease. In turn, whether DSS and NR Flexible Refarming takes effect also affects the amount of resources available for UMTS. When DSS and NR Flexible Refarming takes effect, the LTE load may increase, the amount of resources available

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentatation)	Description
				for UMTS may decrease, and the UMTS UE throughput may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	If a cell with UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD) enabled is configured as a co-coverage inter-frequency neighboring cell, changes in the UMTS load may affect the LTE PRB usage and, consequently, the proportion of occasions where DSS and NR Flexible Refarming takes effect. When the UMTS load increases, the amount of resources available for LTE may decrease, the LTE PRB usage may increase, and the proportion of occasions where DSS and NR Flexible Refarming takes effect may decrease. In turn, whether DSS and NR Flexible Refarming takes effect also affects the amount of resources available for UMTS. When DSS and NR Flexible Refarming takes effect, the LTE load may increase, the amount of

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				resources available for UMTS may decrease, and the UMTS UE throughput may decrease.
LTE FDD	CMAS	CellAlgoSwitch. <i>CmasSwitch</i>	<i>CMAS</i>	The following applies when DSS and NR Flexible Refarming is in effect and the shared spectrum of spectrum sharing cells is refarmed to NR-only: • With "PWS message-triggered exit of cells from energy saving" disabled, CMAS warning notifications cannot be sent in the spectrum sharing cells. • With "PWS message-triggered exit of cells from energy saving" enabled, the NR-only spectrum of the spectrum sharing cells is refarmed back to spectrum sharing upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	ETWS	None	ETWS	The following applies when DSS and NR Flexible Refarming is in effect and the shared spectrum of spectrum sharing cells is refarmed to NR-only: • With "PWS message-triggered exit of cells from energy saving" disabled, ETWS warning notifications cannot be sent in the spectrum sharing cells. • With "PWS message-triggered exit of cells from energy saving" enabled, the NR-only spectrum of the spectrum sharing cells is refarmed back to spectrum sharing upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	DSS and NR Flexible Refarming cannot take effect in spectrum sharing cells that are configured as basic cells for RF module fast deep dormancy when both functions are enabled.
LTE FDD	Cell RX channel interference and noise query	STR CELLRXINTRFOIQUERY	<i>MIMO</i>	If DSS and NR Flexible Refarming is in effect, the shared spectrum of a spectrum sharing cell is refarmed to NR-only, and reporting of invalid uplink interference counter values is enabled (controlled by the UL_IF_INVALID_IN_CARROFF_SW option of the EnodebCounterParaGrp.EnodebCounterAlgoSwitch parameter), then RX channel interference and noise query for the cell will not produce any results.

- Function impacts on the NR side

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	Spectrum refarming will trigger activation or deactivation of CRS rate matching. If the involved cell serves as an SCell for inter-gNodeB CA, the SCell will be removed.
NR FDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	A cell whose spectrum has been refarmed to NR and its intra-frequency cells engaged in spectrum sharing cannot serve as cooperating cells of each other.
NR FDD	Virtual Grid-based Multi-Frequency Coordination	VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation.AggregationAttribute parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	After an LTE cell is shut down, interference to its intra-frequency neighboring cells in the overlapping areas decreases and spectral efficiency increases, which may affect the building and use of spectral efficiency grids.
NR FDD	PDCCH symbol number adaptation	UE_PDCCH_SYMBOL_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	When DSS and NR Flexible Refarming is enabled, spectrum sharing and PDCCH symbol number adaptation can both be enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When DSS and NR Flexible Refarming is enabled together with LTE and NR intelligent carrier shutdown, an LTE spectrum sharing cell cannot serve as an overlapping neighboring cell of the associated NR spectrum sharing cell.
NR FDD	Distributed MIMO	INTRA_GNB_DL_D_MIMO_SW and INTER_GNB_DL_D_MIMO_SW options of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Distributed Antenna Solutions (FDD)</i>	Coherent joint transmission cannot take effect in spectrum sharing cells.
NR FDD	DSS PDCCH resource sharing optimization	DSS_PDCCH_SHARING_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with DSS and NR Flexible Refarming. As a result, the gains from DSS PDCCH resource sharing optimization will decrease.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. For details, see [4.3.3 Hardware](#).

Boards

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the board requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. For details, see [4.3.3 Hardware](#).

RF Modules

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. For details, see [4.3.3 Hardware](#).

Cells

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. In addition, the cell TX/RX mode cannot be 16T16R or 32T32R. For the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing, see [4.3.3 Hardware](#).

The shared spectrum of an LTE spectrum sharing cell will not be refarmed to NR if each of its co-coverage inter-frequency neighboring cells meets any of the following conditions:

- The cell is a downlink-only cell (with **Cell.WorkMode** set to **DL_ONLY**). For details, see *Cell Management*.
- The cell is a licensed assisted access (LAA) cell (with **Cell.WorkMode** set to **LAA**). For details, see *Licensed Assisted Access (TDD)*.
- The cell is an eMTC-only cell (with the **EMTC_ONLY_CELL_SWITCH** option of the **CellEmtcAlgo.EmtcAlgoSwitch** parameter selected). For details, see *eMTC*.
- Access to the cell is barred.

7.3.4 Others

None

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

[Table 7-3](#) and [Table 7-4](#) describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-3 LTE parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	DSS_FLEX_REF ARM_SW	Select this option when DSS and NR Flexible Refarming is required based on the network plan.
UL PRB Usage Threshold	LteNrSpectShrCellGroup.UlPrbUsageThld	None	Set this parameter based on the network plan. If this parameter is set to an excessively small value, DSS and NR Flexible Refarming may not be triggered. If this parameter is set to an excessively large value, LTE performance may deteriorate.
UL PRB Usage Threshold Offset	LteNrSpectShrCellGroup.UlPrbUsageThldOffset	None	Set this parameter based on the network plan. If this parameter is set to an excessively small value, ping-pong switching between spectrum usage states may occur. If this parameter is set to an excessively large value, the current spectrum usage state tends to be retained.
PRB Usage Threshold	LteNrSpectShrCellGroup.PrbUsageThld	None	Set this parameter based on the network plan. If this parameter is set to an excessively small value, DSS and NR Flexible Refarming may not be triggered. If this parameter is set to an excessively large value, LTE performance may deteriorate.

Parameter Name	Parameter ID	Option	Setting Notes
PRB Usage Threshold Offset	LteNrSpectShrCellGroup.PrbUsageThldOffset	None	Set this parameter based on the network plan. If this parameter is set to an excessively small value, ping-pong switching between spectrum usage states may occur. If this parameter is set to an excessively large value, the current spectrum usage state tends to be retained.
RRC Connected UE Number Threshold	LteNrSpectShrCellGroup.RrcConnUeNumThld	None	Set this parameter based on the network plan. If this parameter is set to an excessively small value, DSS and NR Flexible Refarming may not be triggered. If this parameter is set to an excessively large value, LTE performance may deteriorate.
RRC Connected UE Number Threshold Offset	LteNrSpectShrCellGroup.RrcConnUeNumThldOffset	None	Set this parameter based on the network plan. If this parameter is set to an excessively small value, ping-pong switching between spectrum usage states may occur. If this parameter is set to an excessively large value, the current spectrum usage state tends to be retained.

Parameter Name	Parameter ID	Option	Setting Notes
DSS Flexible Refarming Group ID	LteNrSpctShrCellGroup.DssFlexRefarmGrpId	None	If intra-base-station intra-frequency LTE spectrum sharing cells have the same PCI mod 3 value, set this parameter to the same value for the spectrum sharing cell groups to which these cells belong. Refarming will be simultaneously triggered in these cell groups.
DSS Flexible Refarming Policy	LteNrSpctShrCellGroup.DssFlexRefarmingPolicy	None	To preferentially guarantee LTE performance, set this parameter to LTE_NR_FAIR . To preferentially guarantee NR performance, set this parameter to NR_PRI .
LTE and NR Spectrum Handover Protect Duration	LteNrSpctShrCellGroup.LteNrSpctHoProtectDuration	None	Set this parameter to 60 .
Forced Refarming UE Number Threshold	SpectrumCloud.ForcedRefarmingUeNumThld	None	The value 3 is recommended. A larger value of this parameter results in a higher probability that forced refarming is triggered and a higher probability that the service drop rate increases. A smaller value of this parameter results in the opposite effects. If this parameter is set to 0 , the service drop rate is not affected.

Parameter Name	Parameter ID	Option	Setting Notes
Spectrum Cloud Switch	<i>SpectrumCloud.SpectrumCloudSwitch</i>	None	Set this parameter to LTE_NR_COORD_SPECTRUM_SHR if a co-coverage inter-frequency neighboring LTE cell needs to be configured based on the network plan.
LTE and NR Spectrum Sharing Cell Group ID	<i>SpectrumCloud.LteNrSpectrumShrCellGrpId</i>	None	Set this parameter based on the network plan. Set this parameter when configuring co-coverage inter-frequency neighboring LTE cells.
Multi-RAT Sharing LNR Spt Coord Group ID	<i>SpectrumCloud.MultiRatShrLnrSpt-CoGrpId</i>	None	Set this parameter based on the network plan. Set this parameter if a co-coverage inter-frequency neighboring LTE cell needs to be configured and the <i>SpectrumCloud.SpectrumCloudSwitch</i> parameter has been set to GL_SPECTRUM_CONCURRENCY , UL_SPECTRUM_SHARING , UL_SPECTRUM_SHR_P H2_WITH_GSM , or DC_HSDPA_BASED_UL_SPECTRUM_SHR for the cell.

Table 7-4 NR parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Spectrum Cloudification Enhance Switch	NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch	DSS_FLEX_REF_ARM_SW	Select this option when DSS and NR Flexible Refarming is required based on the network plan.
UL PRB Usage Threshold	gNBDULteNrSpctShrCg.UlPrbUsageThld	None	Set this parameter based on the network plan. When LteNrSpctShrCellGrp.DssFlexRefarmPolicy is set to NR_PRI on the LTE side, an excessively small value of this parameter will lead to a low probability of triggering refarming from NR to spectrum sharing, affecting LTE performance; an excessively large value of this parameter will cause triggering of refarming from NR to spectrum sharing even when the NR load is heavy, affecting NR performance.

Parameter Name	Parameter ID	Option	Setting Notes
PRB Usage Threshold	gNBDULteNrSpctShrCg.PrbUsageThld	None	Set this parameter based on the network plan. When LteNrSpctShrCellGrp.DssFlexRefarmPolicy is set to NR_PRI on the LTE side, an excessively small value of this parameter will lead to a low probability of triggering refarming from NR to spectrum sharing, affecting LTE performance; an excessively large value of this parameter will cause triggering of refarming from NR to spectrum sharing even when the NR load is heavy, affecting NR performance.
RRC_CONNECTED UE Number Threshold	gNBDULteNrSpctShrCg.RrcConnUeNumThld	None	Set this parameter based on the network plan. When LteNrSpctShrCellGrp.DssFlexRefarmPolicy is set to NR_PRI on the LTE side, an excessively small value of this parameter will lead to a low probability of triggering refarming from NR to spectrum sharing, affecting LTE performance; an excessively large value of this parameter will cause triggering of refarming from NR to spectrum sharing even when the NR load is heavy, affecting NR performance.

Parameter Name	Parameter ID	Option	Setting Notes
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	PDCCH_RATE_MATCH_COMPATIBLE_SW	Select this option to resolve the accompanying UE compatibility issue when DSS and NR Flexible Refarming is required based on the network plan.
Rate Match Switch	NRDUCellPdsch.RateMatchSwitch	CRS_RM_CONFIG_COMPAT_SW	Select this option to resolve the accompanying UE compatibility issue when DSS and NR Flexible Refarming is required based on the network plan.

7.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Activation command examples on the LTE side

```
//Configuring either an intra-base-station (recommended) or inter-base-station co-coverage inter-frequency neighboring cell
//(Optional) Configuring an intra-base-station co-coverage inter-frequency neighboring cell with none of the following functions enabled: GSM and LTE Spectrum Concurrency, UMTS and LTE Spectrum Sharing, and UMTS and LTE Spectrum Sharing Based on DC-HSDPA
ADD SPECTRUMCLOUD: LocalCellId=2, SpectrumCloudSwitch=LTE_NR_COORD_SPECTRUM_SHR, LteNrSpectrumShrCellGrpId=1;
//(Optional) Configuring an intra-base-station co-coverage inter-frequency neighboring cell with GSM and LTE Spectrum Concurrency (this function is used as an example) enabled
MOD SPECTRUMCLOUD: LocalCellId=2, SpectrumCloudSwitch=GL_SPECTRUM_CONCURRENCY, MultiRatShrLnrSptCoGrpId=1;
//(Optional) Configuring an inter-base-station co-coverage inter-frequency neighboring cell when there are no intra-base-station co-coverage inter-frequency neighboring cells
ADD SPCTCLOUDAUXNODEB: LteNrSpectrumShrCellGrpId=1, Mcc="450", Mnc="06", EnodebId=12345, CellId=2;

//(Optional) Adding spectrum sharing cell groups to the same refarming group if the intra-base-station intra-frequency LTE spectrum sharing cells in these groups have the same PCI mod 3 value
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=1, DssFlexRefarmGrpId=0;
```

```
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=2, DssFlexRefarmGrpId=0;  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=3, DssFlexRefarmGrpId=0;  
  
//Enabling DSS and NR Flexible Refarming  
MOD SPECTRUMCLOUD: LocalCellId=1, SpectrumCloudEnhSwitch=DSS_FLEX_REFARM_SW-1;  
  
//The following optimization command examples involve only one set of spectrum sharing cells. You are  
advised to perform the same settings for spectrum sharing cell sets added to the same refarming group.  
//Modifying the PRB usage thresholds, state protection duration, and UE number threshold with setting  
PrbUsageThldOffset and UPrbUsageThldOffset to 15 used as an example  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=1, PrbUsageThld=40, PrbUsageThldOffset=15,  
UPrbUsageThld=40, UPrbUsageThldOffset=15, RrcConnUeNumThld=80, RrcConnUeNumThldOffset=30,  
LteNrSptHoProtDuration=60;  
//((Optional) Setting a priority policy for LTE and NR to preferentially guarantee LTE performance  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=1, DssFlexRefarmPolicy=LTE_NR_FAIR;  
//((Optional) Setting a priority policy for LTE and NR to preferentially guarantee NR performance  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=1, DssFlexRefarmPolicy=NR_PRI;  
//Configuring the UE number threshold for forced refarming  
MOD SPECTRUMCLOUD: LocalCellId=1, ForcedRefarmingUeNumThld=3;
```

Activation command examples on the NR side

```
//Enabling DSS and NR Flexible Refarming  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, SpectrumCloudEnhSwitch=DSS_FLEX_REFARM_SW-1;  
//Turning on the PDCCH rate matching compatibility switch  
MOD NRDUCELLPDSCH: NrDuCellId=1, RateMatchSwitch=PDCCH_RATEMATCH_COMPATIBLE_SW-1;  
//((Optional) Running the following command when configurations on the LTE side preferentially guarantee  
NR performance  
MOD GNBDULTENRSPCTSHRCG: NrSptShrCellGrpId=1, UPrbUsageThld=30, PrbUsageThld=30,  
RrcConnUeNumThld=80;  
//((Optional) Turning on the CRS rate matching configuration compatibility switch to prevent UE  
compatibility issues that lead to an increased service drop rate when the configurations for CRS rate  
matching change  
MOD NRDUCELLPDSCH: NrDuCellId=1, RateMatchSwitch=CRS_RM_CONFIG_COMPAT_SW-1;
```

Deactivation Command Examples

Deactivation command examples on the LTE side

```
//Disabling DSS and NR Flexible Refarming  
MOD SPECTRUMCLOUD: LocalCellId=1, SpectrumCloudEnhSwitch=DSS_FLEX_REFARM_SW-0;  
//((Optional) Removing the refarming group if the spectrum sharing cell groups that include intra-frequency  
cells have been added to this refarming group  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=1, DssFlexRefarmGrpId=255;  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=2, DssFlexRefarmGrpId=255;  
MOD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=3, DssFlexRefarmGrpId=255;  
//Removing the configurations of intra- or inter-base-station co-coverage inter-frequency neighboring cells  
that have been activated  
//((Optional) Removing the configurations of an intra-base-station co-coverage inter-frequency neighboring  
cell with none of the following functions enabled: GSM and LTE Spectrum Concurrency, UMTS and LTE  
Spectrum Sharing, and UMTS and LTE Spectrum Sharing Based on DC-HSDPA  
MOD SPECTRUMCLOUD: LocalCellId=2, SpectrumCloudSwitch=OFF;  
//((Optional) Removing the configurations of an intra-base-station co-coverage inter-frequency neighboring  
cell with GSM and LTE Spectrum Concurrency (this function is used as an example) enabled  
MOD SPECTRUMCLOUD: LocalCellId=2, SpectrumCloudSwitch=GL_SPECTRUM_CONCURRENCY,  
MultiRatShrLnrSptCoGrpId=255;  
//((Optional) Removing the configurations of an inter-base-station co-coverage inter-frequency neighboring  
cell when there are no intra-base-station co-coverage inter-frequency neighboring cells  
RMV SPTCLOUDAUXNODEB: LteNrSpectrumShrCellGrpId=1, Mcc="450", Mnc="06", EnodebId=12345,  
CellId=2;
```

Deactivation command examples on the NR side

```
//Disabling DSS and NR Flexible Refarming  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, SpectrumCloudEnhSwitch=DSS_FLEX_REFARM_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Use MML commands or counters to monitor the activation of DSS and NR Flexible Refarming.

Using MML Commands

DSS and NR Flexible Refarming has been enabled and the spectrum has been refarmed from spectrum sharing to NR when both of the following conditions are met:

- The value of the **Spectrum Sharing Status** parameter in the output of the eNodeB MML command **DSP LTENRSPCTSHRCELLGRP** is **FLEXIBLE_REFARMING_NR**.
- The value of the **Spectrum Sharing Status** parameter in the output of the gNodeB MML command **DSP GNBULTENRSPCTSHRCG** is **Flexible Refarming from DSS**.

NOTE

After an LTE cell is shut down, the value of **Cell instance state** in the output of the eNodeB command **DSP CELL** is **Barred**.

Using Counters

DSS and NR Flexible Refarming has been enabled and the spectrum has been refarmed from spectrum sharing to NR when both of the following conditions are met:

- On the LTE side, the values of all the following counters are zero: **L.ChMeas.PRB.UL.Avail**, **L.ChMeas.PRB.UL.Actual.Avail**, **L.ChMeas.PRB.DL.Avail**, and **L.ChMeas.PRB.DL.Actual.Avail**. This indicates that LTE is not occupying the spectrum and spectrum sharing is not in effect.
- On the NR side, the **N.PRB.UL.Actual.Avail.Avg** and **N.PRB.UL.Avail.Avg** counters have the same value and the **N.PRB.DL.Actual.Avail.Avg** and **N.PRB.DL.Avail.Avg** counters have the same value. This indicates that NR now has exclusive use of the spectrum and spectrum sharing is not in effect.

7.4.3 Network Monitoring

After this function is enabled, monitor the network gains and impacts of this function using the counters listed in [7.2 Network Analysis](#).

8 LTE and NR Uplink Spectrum Pooling

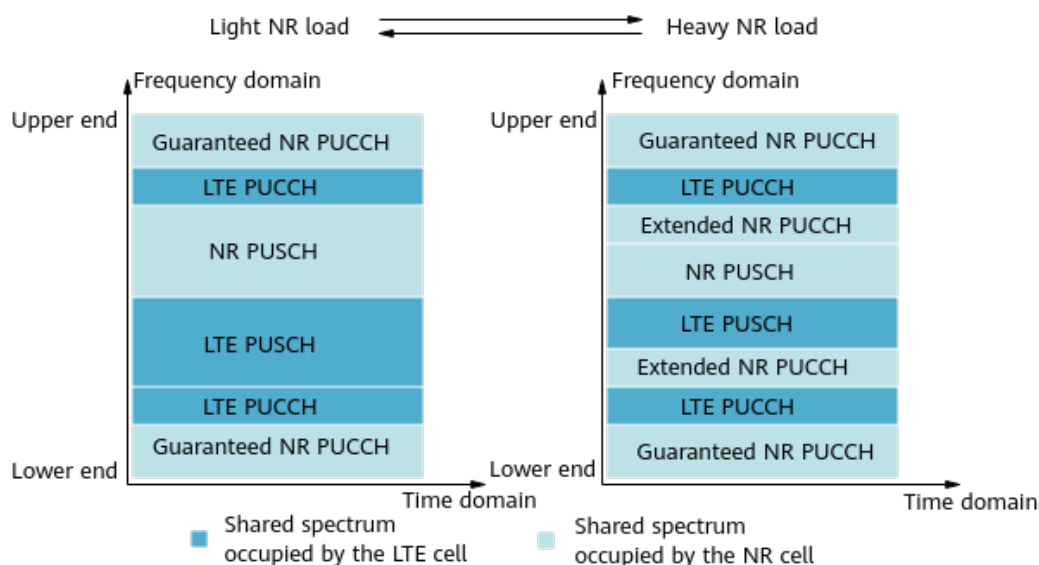
8.1 Principles

LTE and NR Uplink Spectrum Pooling aims to enhance uplink spectrum utilization when LTE FDD and NR share spectrum. The feature comprises two functions: LNR uplink control channel pooling and LNR uplink data channel pooling. These functions are applicable to both LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios and Hybrid DSS Based on Asymmetric Bandwidth scenarios. For detailed implementation in Hybrid DSS Based on Asymmetric Bandwidth scenarios, see *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*.

8.1.1 LNR Uplink Control Channel Pooling

When LTE FDD and NR share spectrum, their PUCCHs are deployed in frequency division mode, with the number of RBs for the NR PUCCH being configurable only through NR parameters. This approach leads to inefficient use of RB resources under low-load conditions. To mitigate this inefficiency, LNR PUCCH RB resource adaptation has been introduced to enable LNR uplink control channel pooling. [Figure 8-1](#) shows how this function works. With this function, two types of RBs are available for the NR PUCCH: guaranteed RBs and extended RBs. Guaranteed RBs are symmetrically positioned at both ends of the bandwidth, while extended RBs are adjusted adaptively based on the number of NR UEs and the load. **Under low-load conditions, guaranteed RBs alone are sufficient for the PUCCH to meet service demands. This leaves more available RBs for LTE and NR PUSCHs, and thereby increases the uplink UE throughput. As the load increases, extended RBs are also needed for the PUCCH to accommodate additional service demands. NR and LTE collaboratively determine the position of extended RBs for the NR PUCCH to ensure that the PUCCHs of the two RATs occupy consecutive RBs. This means that when extended RBs start or stop being used by the NR PUCCH, the LTE PUCCH is adjusted accordingly.**

Figure 8-1 LNR PUCCH RB resource adaptation



PUCCH resource adjustments change LTE system information. To avoid frequent changes, the **NRDUCellGigaBand.LnrPucchRbAdjProtDuration** parameter can be set to specify the minimum interval between NR PUCCH resource adjustments. Once the NR PUCCH is adjusted to use extended RBs, it will not be readjusted to stop using these RBs within the minimum interval.

Related configurations are as follows:

- Guaranteed NR PUCCH RBs

The number of guaranteed NR PUCCH RBs varies depending on the networking as follows:

- NSA networking: Number of guaranteed NR PUCCH RBs = **NRDUCellPucch.Format1RbNum** + **NRDUCellPucch.Format3RbNum** + **NRDUCellPucch.Format4RbNum** + **NRDUCellPucch.CsiDedicatedRbNum** + **NRDUCellPucch.Format4CsiDedicatedRbNum**
- NSA and SA hybrid networking and SA networking: Number of guaranteed NR PUCCH RBs = **NRDUCellPucch.Format1RbNum** + **NRDUCellPucch.Format3RbNum** + **NRDUCellPucch.Format4RbNum** + **NRDUCellPucch.CsiDedicatedRbNum** + **NRDUCellPucch.Format4CsiDedicatedRbNum** + Number of NR common PUCCH RBs

The **NRDUCellPucch.Format3RbNum** and **NRDUCellPucch.Format4RbNum** parameters need to be set only when downlink intra-band CA (controlled by the **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter) or downlink intra-FR inter-band CA (controlled by the **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter) is enabled. The number of NR common PUCCH RBs depends on the setting of the **COMM_PUCCH_RB_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter. For details about the parameters involved in the preceding formulas and the number of NR common PUCCH RBs, see descriptions of the PUCCH in *Channel Management* in *5G RAN Feature Documentation*.

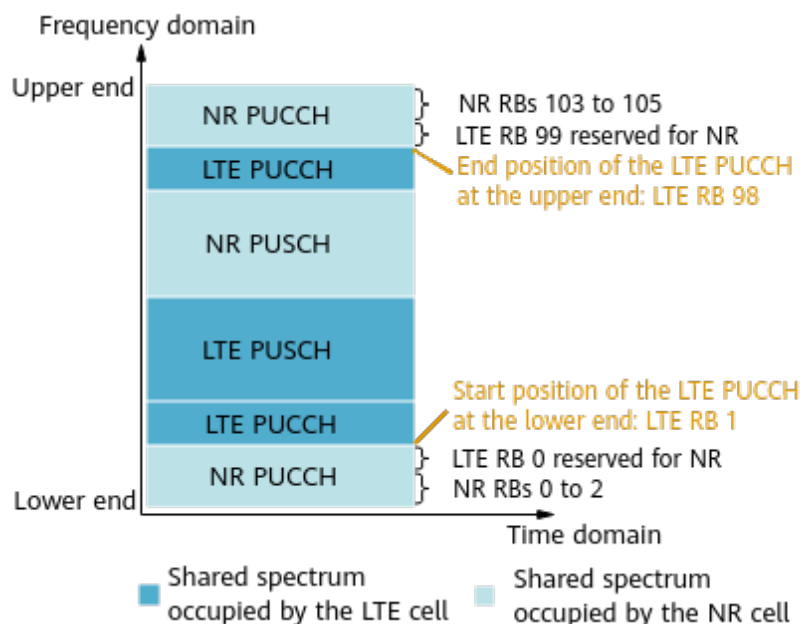
- Extended NR PUCCH RBs
The number of extended NR PUCCH RBs is specified by the LTE parameter **PUCCHCfg.NrPucchExtendedRbNum**.
- LTE PUCCH
The LTE PUCCH is symmetrically positioned adjacent to guaranteed NR PUCCH RBs to ensure that NR and LTE PUCCHs occupy consecutive RBs. The position of the LTE PUCCH therefore depends on the position and number of guaranteed NR PUCCH RBs, and is configured using LTE parameters listed in [Table 8-1](#). The number of LTE PUCCH RBs is dynamically adjusted based on the load. For details, see descriptions of the PUCCH in *Physical Channel Resource Management* in *eRAN Feature Documentation*.

Table 8-1 Parameters for configuring the LTE PUCCH position

Parameter	Function
CellRbReserve.RbRsvMode	Specifies the RB reservation mode.
CellRbReserve.RbRsvType	Specifies the RB reservation type.
CellRbReserve.RbRsvStartIndex	Specifies the start index of reserved RBs.
CellRbReserve.RbRsvEndIndex	Specifies the end index of reserved RBs.

The following description uses a 20 MHz bandwidth as an example to illustrate how to determine the start and end positions of the LTE PUCCH. In a 20 MHz bandwidth, NR has 106 available RBs in the uplink, and LTE has 100 available RBs in the uplink. Assume that eight RBs are guaranteed for the NR PUCCH, then there are four guaranteed NR PUCCH RBs at each end of the frequency domain: RBs 0 to 3 at the lower end and RBs 102 to 105 at the upper end (as illustrated in [Figure 8-2](#)). As NR has six more RBs (106 – 100) than LTE in the frequency domain, that is, NR has three more RBs than LTE at each end of the frequency domain, LTE needs to reserve one RB at each end (RB 0 at the lower end and RB 99 at the upper end) for NR so that the NR PUCCH can occupy four RBs at each end of the frequency domain. In this case, you need to set the **CellRbReserve.RbRsvStartIndex** parameter to **1** (RB 0 is reserved for NR), and the **CellRbReserve.RbRsvEndIndex** parameter to **98** (RB 99 is reserved for NR). [Figure 8-2](#) illustrates this configuration.

Figure 8-2 Example of the start and end positions of the LTE PUCCH



LNR PUCCH RB resource adaptation is controlled by the **LNR_PUCCH_RB_RES_ADAPT_SW** option of the **NRDUCellGigaBand.GigaBandCellAlgoSw** parameter. When this function is enabled, you are advised to enable the following functions as well:

- LNR PUCCH resource optimization (controlled by the **LNR_PUCCH_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter), which improves the accuracy of resource allocation to LTE and NR PUCCHs
- Optimized paging triggered by SIB updates (controlled by the **SibPagingOptCtrlSwitch** option of the **CellSiMap.SibUpdOptSwitch** parameter), which increases the LTE system information update success rate
- To improve uplink resource utilization, you are advised to turn on the switch for format-3 and format-4 ACK RB resource adaptation (controlled by the **F3_AND_F4_ACK_RB_RES_ADAPT_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter). For details about format-3 and format-4 ACK RB resource adaptation, see *Channel Management* in *5G RAN Feature Documentation*.

8.1.2 LNR Uplink Data Channel Pooling

LNR uplink data channel pooling enables precise uplink scheduling, thereby increasing uplink UE throughput in LTE FDD and NR spectrum sharing scenarios.

[Table 8-2](#) describes its included functions.

Table 8-2 Functions of LNR uplink data channel pooling

Function Name	Function Switch	Principle
LNR uplink precise RB estimation	LNR_UL_RB_PRECISE_ESTIMATE_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter on the LTE side	This function estimates uplink resources in real time in LTE and NR spectrum sharing scenarios. To achieve this, the gNodeB corrects NR uplink RB estimates based on the latest obtained NR service requirements. The correction coefficient is specified by the NRDUCellGigaBand.LnrULRbEstCorrectCoeff parameter. By applying this correction, the information used to estimate uplink RBs approximates more closely to the conditions at the moment of scheduling, resulting in more accurate RB estimation and accurate alignment of cross-RAT resource allocation with service requirements.
Optimized uplink preallocation trigger	UL_PREALLOCATION_TRIG_OPT_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter on the LTE side	This function adjusts the occasions for triggering uplink preallocation on the LTE side based on services. As a result, fewer RBs are reserved for preallocation and resource allocation accurately aligns with service requirements. The GRP_NO_PREALLOCATE_TRIGGER_OPT_SW option of the AsParaGroup.AccelerationServiceOptExtSw parameter can be set to forbid optimized uplink preallocation trigger for UEs in a specified group.

Function Name	Function Switch	Principle
Uplink differentiated scheduling based on LTE and NR flexible spectrum	LNR_FLEX_SPCT_UL_DIFF_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter on the LTE side	When uplink large-packet services are identified on the NR side, this function allocates more uplink resources to NR UEs and activates DSS inter-RAT PDCCH spatial multiplexing, improving the performance of uplink large-packet services during NR drive tests. When uplink large-packet services are identified on the LTE side, this function allocates more uplink resources to LTE UEs, improving the performance of uplink large-packet services during LTE drive tests. When this function is enabled, the UL_LARGE_PKT_SERV_DECT_SW option of the EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw parameter must be selected to enable UE identification based on uplink services.
Uplink differentiated power control and scheduling	UL_PWR_CTRL_SCH_DIFF_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter on the LTE side	This function enables differentiated power control and scheduling based on uplink services of UEs, increasing the uplink throughput of identified large-packet LTE UEs. The GigaBandCell.ServiceBasedOpt-MaxEffTime parameter specifies the maximum duration during which this function can take effect. When this function is enabled, the UL_LARGE_PKT_SERV_DECT_SW option of the EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw parameter must be selected to enable UE identification based on uplink services.

In "uplink differentiated scheduling based on LTE and NR flexible spectrum" and "uplink differentiated power control and scheduling", large-packet LTE UEs are identified based on the average amount of remaining data in the uplink buffer of UEs. Specifically:

- A UE is identified as an uplink large-packet UE if the UE's average amount of remaining data in the uplink buffer is greater than or equal to the threshold specified by **CellServiceDiffCfg.UllargePacketDataVoThd** for at least n (n equals the value of **CellServiceDiffCfg.JudgeUllargePacketCntThd**) consecutive evaluation periods.
- A UE is identified as a non-uplink-large-packet UE if the UE's average amount of remaining data in the uplink buffer is less than the threshold specified by **CellServiceDiffCfg.UllargePacketDataVoThd** for at least n (n equals the value of **CellServiceDiffCfg.ExitUllargePacketCntThd**) consecutive evaluation periods.

In "uplink differentiated scheduling based on LTE and NR flexible spectrum", a UE can be identified as an uplink large-packet UE only when both of the following conditions are met in an evaluation period specified by **NRDUCellServExp.FullBufferUeEvalPrd**: (1) The average PUSCH PRB usage of the UE is greater than or equal to the value of **NRDUCellServExp.FullBufferUeUlPrbUsageThld**. (2) The proportion of TTIs in which uplink scheduling is performed for the UE is greater than or equal to the value of **NRDUCellServExp.FullBufferUeUlTtiRatioThld**.

8.2 Network Analysis

8.2.1 Benefits

When all sub-functions of LTE and NR Uplink Spectrum Pooling take effect, the average uplink throughputs of certain identified LTE and NR UEs and the uplink LTE UE throughput will increase in LTE FDD and NR spectrum sharing scenarios.

The benefits of this feature are affected as follows:

- The benefits will increase if SA networking is used and either of the following NR CA functions is enabled: downlink intra-band CA (controlled by the **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter) and downlink intra-FR inter-band CA (controlled by the **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter).
- The benefits will increase if the network load is neither extremely light nor extremely heavy. The larger the ratio of LTE PRB usage to NR PRB usage, the higher the benefits.

You are advised to enable this feature for LTE and NR spectrum sharing cells that meet the conditions listed in [Table 8-3](#).

Table 8-3 Performance indicator value ranges within which the feature is recommended

Performance Indicator	Value in Percentage
LTE PUSCH PRB usage	20% to 30%
NR PUSCH PRB usage	5% to 10%
LTE downlink PRB usage	30% to 40%

Performance Indicator	Value in Percentage
NR downlink PRB usage	20% to 30%

- The higher the proportion of uplink small packets on the LTE side, the greater the benefits.
Proportion of uplink small packets = $\frac{\text{L.Thrp.bits.UE.UL.SmallPkt}}{\text{L.Thrp.bits.UL}} \times 100\%$
- The benefits will increase if the number of RRC_CONNECTED UEs on the LTE side (indicated by **L.Traffic.User.Avg**) is less than or equal to the value of **CellUlschAlgo.PreallocationUserCntThld**.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

LNR PUCCH RB Resource Adaptation

LNR PUCCH RB resource adaptation affects the values of performance indicators as follows:

- Due to the adjustment of LTE and NR PUCCH resources, the values of the following counters change: **L.ChMeas.PR.B.UL.Actual.Avail**, **L.ChMeas.PR.B.UL.PUSCH.Actual.Avail**, **N.PR.B.UL.Actual.Avail.Avg**, **N.PR.B.UL.PUSCH.Actual.Avail.Avg**, and **N.PR.B.UL.PUSCH.Avail.Avg**.
- Due to possible changes in the LTE and NR PUCCH RB positions, the values of the following counters may change: **L.UL.RSSI.PUCCH.IndexX**, **L.UL.RSRP.PUCCH.IndexX**, **L.UL.SINR.PUCCH.IndexX**, **L.UL.Interference.PUCCH.IndexX**, and **N.UL.SINR.PUCCH.IndexX**.
- PUCCH resource adjustments change LTE system information. This leads to more LTE paging messages related to system information changes, as indicated by higher values of counters such as **L.Paging.UU.SIUpdate** and **L.ChMeas.CCE.CommUsed**. In addition, as UEs need to re-acquire system information, their access or handover delay may increase. If UEs fail to receive the updated system information, the service drop rate may increase and the RRC connection setup success rate may decrease on the LTE side.
RRC connection setup success rate = $\frac{\text{L.RRC.ConnReq.Succ}}{\text{L.RRC.ConnReq.Att}} \times 100\%$
- The average uplink MCS index on the LTE side may change.
Average uplink MCS index on the LTE side = $\frac{\sum(n \times \text{L.ChMeas.PUSCH.MCS}.n)}{\sum \text{L.ChMeas.PUSCH.MCS}.n}$, where n ranges from 0 to 28
- The average uplink MCS index on the NR side may change.
Average uplink MCS index on the NR side = $\frac{\sum(n \times \text{N.ChMeas.PUSCH.MCS}.n)}{\sum \text{N.ChMeas.PUSCH.MCS}.n}$, where n ranges from 0 to 28

The setting of the **PUCCHCfg.NrPucchExtendedRbNum** parameter has the following impacts:

- A smaller value of this parameter results in fewer resources available to extend the NR PUCCH, more available PUSCH resources in heavy-load scenarios, higher LTE and NR uplink UE throughputs, but possibly a lower NR downlink UE throughput.
- A larger value of this parameter results in more resources available to extend the NR PUCCH, fewer available PUSCH resources in heavy-load scenarios, lower LTE and NR uplink UE throughputs, but possibly a higher NR downlink UE throughput.

The setting of the **NRDUCellGigaBand.LnrPucchRbAdjProtDuration** parameter has the following impacts:

- A smaller value of this parameter results in a faster decrease in NR PUCCH RBs, a larger increase in available LTE and NR PUSCH resources, but more LTE system information changes.
- A larger value of this parameter results in a slower decrease in NR PUCCH RBs, a smaller increase in available LTE and NR PUSCH resources, but fewer LTE system information changes.

LNR Uplink Precise RB Estimation

LNR uplink precise RB estimation affects the values of performance indicators as follows:

- The **N.PR.B.UL.PUSCH.Actual.Avail.Avg** counter value decreases.
- The **L.ChMeas.PR.B.UL.PUSCH.Actual.Avail** counter value increases.
- The uplink user-perceived throughput on the NR side may slightly change.
- The average uplink MCS index on the LTE side may change.

$$\text{Average uplink MCS index on the LTE side} = \frac{\sum (n \times \text{L.ChMeas.PUSCH.MCS}.n)}{\sum \text{L.ChMeas.PUSCH.MCS}.n}$$
 where n ranges from 0 to 28
- The average uplink MCS index on the NR side may change.

$$\text{Average uplink MCS index on the NR side} = \frac{\sum (n \times \text{N.ChMeas.PUSCH.MCS}.n)}{\sum \text{N.ChMeas.PUSCH.MCS}.n}$$
 where n ranges from 0 to 28

The setting of the **NRDUCellGigaBand.LnrULRbEstCorrectCoeff** parameter has the following impacts:

- A smaller value of this parameter results in a milder correction for NR uplink RB resource estimation and a smaller increase in the uplink LTE UE throughput.
- A larger value of this parameter results in a stronger correction for NR uplink RB resource estimation, a larger increase in the uplink LTE UE throughput, but possible changes in the uplink NR UE throughput.

Optimized Uplink Preallocation Trigger

Optimized uplink preallocation trigger affects the values of performance indicators as follows:

- The **L.ChMeas.PR.B.UL.PreSch.Used.Avg** counter value decreases.
- The **User Downlink Average Throughput** may slightly decrease.

- The **L.Thrp.bits.UL** counter value may slightly change.

When the **GRP_NO_PREALLOCATE_TRIG_OPT_SW** option of the **AsParaGroup.AccelerationServOptExtSw** parameter is selected for a group, UEs in this group will experience a decrease in the average uplink throughput and possibly an increase in the average downlink throughput.

8.3 Requirements

8.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE FDD	MRFD-221 225	LTE and NR Uplink Spectrum Pooling (LTE FDD)	LT1SLNULSP00	eNodeB	per Cell
NR FDD	MRFD-221 265	LTE and NR Uplink Spectrum Pooling (NR FDD)	NR0SLNULSP00	gNodeB	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

- Prerequisite functions on the LTE side

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud. SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	Functions for LTE and NR Uplink Spectrum Pooling can be enabled only if LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.
LT E FD D	PUCCH resource adjustment	PucchSwitch option of the CellAlgoSwitch. PucchAlgoSwitch parameter	<i>Physical Channel Resource Management</i>	The PUCCHCfg.NrPucchExtendedRbNum parameter can be set to a value other than 0 only if PUCCH resource adjustment is enabled.
LT E FD D	Uplink scheduler- controlled power	SchedulerCtrlPowerSwitch option of the CellAlgoSwitch. ULSchSwitch parameter	<i>Uplink Scheduling</i>	Uplink differentiated power control and scheduling can be enabled only if uplink scheduler-controlled power is enabled.

- Prerequisite functions on the NR side

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentati on)	Description
NR FD D	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPECT_SHR_SW or LTE_NR_SDL_SPECT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	Functions for LTE and NR Uplink Spectrum Pooling can be enabled only if LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.

8.3.2.2 Mutually Exclusive Functions

- Mutually exclusive functions on the LTE side

RA T	Function Name	Function Switch	Reference (eRAN Feature Documentati on)	Description
LT E FD D	Static Multiple Beam in Smart 8T8R	None	<i>Smart 8T8R (FDD)</i>	The PUCCHCfg.NrPuc chExtendedRbNu m parameter can be set to a value other than 0 only if Static Multiple Beam in Smart 8T8R is disabled.
LT E FD D	Massive MIMO Static Multiple Beam	None	<i>Massive MIMO Basics (FDD)</i>	The PUCCHCfg.NrPuc chExtendedRbNu m parameter can be set to a value other than 0 only if massive MIMO Static Multiple Beam is disabled.
LT E FD D	Downlink massive CA	DLMassiveCaSw itch option of the CaMgtCfg.CellC aAlgoSwitch parameter	<i>Carrier Aggregation</i>	The PUCCHCfg.NrPuc chExtendedRbNu m parameter can be set to a value other than 0 only if downlink massive CA is disabled.
LT E FD D	DS3 delta shift	PUCCHCfg.Delt aShift set to DS3_DELTA_SHI FT	<i>Physical Channel Resource Management</i>	The PUCCHCfg.NrPuc chExtendedRbNu m parameter can be set to a value other than 0 only if Delta shift is not set to DS3_DELTA_SHIF T .

- Mutually exclusive functions on the NR side

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentatio n)	Description
NR FD D	Ultimate PUCCH compression	PUCCH_COMPRESSION_SW option of the NRDUCellPucch. PucchAlgoSwitch parameter	<i>Channel Management</i>	LNR PUCCH RB resource adaptation can be enabled only if ultimate PUCCH compression is disabled.
NR FD D	Compact bandwidth	NRDUCell.CustomizedBwConfigInd set to CONFIG_LEFT_ALIGN or CONFIG_RIGHT_ALIGN	<i>Scalable Bandwidth</i>	When Customized Bandwidth Config Indicator is set to CONFIG_LEFT_ALIGN or CONFIG_RIGHT_ALIGN , LNR PUCCH RB resource adaptation must be disabled.
NR FD D	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp. BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	With LTE FDD and NR Flash Dynamic Spectrum Sharing enabled and Hybrid DSS Based on Asymmetric Bandwidth disabled, LNR PUCCH RB resource adaptation can be enabled and the NRDUCellGigaBand.LnrUIRbEstCorrectCoeff parameter can be set to a value other than 0 only if UE bandwidth adaptation is disabled.

8.3.2.3 Function Impacts

- Function impacts on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	VoLTE	None	<i>VoLTE</i>	Optimized uplink preallocation trigger does not take effect for UEs with voice service bearers established.
LTE FDD	Video	None	None	Optimized uplink preallocation trigger does not take effect for UEs with video service bearers established.
LTE FDD	PTT	None	<i>PTT</i>	Optimized uplink preallocation trigger does not take effect for UEs with PTT service bearers established.
LTE FDD	Low Latency Flag	CellQciPara.LowLatencyFlag set to TRUE	<i>Air Interface Latency Optimization</i>	Optimized uplink preallocation trigger does not take effect for UEs with low-latency service bearers established.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	SFN	Cell.MultiRruCellMode set to SFN or MPRU_AGGREGATION	<i>SFN</i>	If the PUCCHCfg.NrPucchExtendedRbNum parameter is set to a value other than 0 , all physical cells in an SFN cell must be deployed on the UBBPe or a later board. If this requirement is not met, the values of the L.UL.RSSI.PUCCH.IndexX and L.UL.Interference.PUCCH.IndexX counters will change.
LTE FDD	Air interface acceleration guarantee for network services	CellAlgoSwitch.AsAlgoSwitch	<i>Video Experience Optimization</i>	The GRP_NO_PREALLOCATE_TRIG_OPT_SW option of the AsParaGroup.AccelerationServOptExtSw parameter specifies whether optimized uplink preallocation trigger can take effect for identified acceleration guarantee UEs.
LTE FDD	Enhanced TCP link setup acceleration	CellAlgoSwitch.ServiceDiffSwitch	<i>Video Experience Optimization</i>	

- Function impacts on the NR side

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR FDD	Compact bandwidth	NRDUCell.CustomizedBwConfigInd set to CONFIG , CONFIG_LEFT_ALIGN , or CONFIG_RIGHT_ALIGN	<i>Scalable Bandwidth</i>	When compact bandwidth is enabled, LNR uplink precise RB estimation does not take effect.
NR FDD	Flexible bandwidth	FLEXIBLE_BANDWIDTH_SWITCH option of the NRDUCellFlexBw.FlexBwAlgorithm parameter	<i>Flexible Bandwidth (FDD)</i>	When flexible bandwidth is enabled, LNR uplink precise RB estimation does not take effect.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the base station model requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. For details, see [4.3.3 Hardware](#).

Boards

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, only UBBPe and later series boards can be used among the boards supporting LTE FDD and NR Flash Dynamic Spectrum Sharing.

RF Modules

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the RF module requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. For details, see [4.3.3 Hardware](#).

Cells

To enable this function in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, the cell requirements of LTE FDD and NR Flash Dynamic Spectrum Sharing must be met. For details, see [4.3.3 Hardware](#).

If **NRDUCellSulRelation.SulType** is set to **DYN_SUL**,
NRDUCellGigaBand.LnrUIRbEstCorrectCoeff must be set to **0**.

8.3.4 Others

None

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

[Table 8-4](#) and [Table 8-5](#) describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-4 LTE parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
GigaBand Cell Algorithm Switch	GigaBandCell.GigaBandCellAlgoSw	LNR_UL_RB_PRECISE_ESTIMATE_SW	Select this option when LNR uplink precise RB estimation is required based on the network plan.
GigaBand Cell Algorithm Switch	GigaBandCell.GigaBandCellAlgoSw	UL_PREALLOCATION_TRIGGER_OPT_SW	Select this option when optimized uplink preallocation trigger is required based on the network plan.
GigaBand Cell Algorithm Switch	GigaBandCell.GigaBandCellAlgoSw	LNR_FLEX_SPECT_UL_DIFF_SCHED_SW	Select this option when "uplink differentiated scheduling based on LTE and NR flexible spectrum" is required based on the network plan.
GigaBand Cell Algorithm Switch	GigaBandCell.GigaBandCellAlgoSw	UL_PWR_CTRL_SCHED_DIFF_SW	Select this option when "uplink differentiated power control and scheduling" is required based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
NR PUCCH Extended RB Number	PUCCHCfg.NrPucchExtendedRbNum	None	- With eMTC introduction (controlled by the EMTC_SWITCH option of CellEmtcAlgo.EmtcAlgoSwitch) disabled, PUCCHCfg.NrPucchExtendedRbNum must be set to 5 or less if PUCCHCfg.DeltaShift is set to DS1_DELTA_SHIFT, and must be set to 4 or less if PUCCHCfg.DeltaShift is set to DS2_DELTA_SHIFT. - With eMTC introduction enabled, PUCCHCfg.NrPucchExtendedRbNum must be set to 3 or less if PUCCHCfg.DeltaShift is set to DS1_DELTA_SHIFT, and must be set to 2 or less if PUCCHCfg.DeltaShift is set to DS2_DELTA_SHIFT.
SIB Update Optimization Switch	CellSiMap.SibUpdateSwitch	SibPagingOptCtrlSwitch	You are advised to select this option when LNR PUCCH RB resource adaptation is enabled, to increase the success rate of LTE system information update.

Parameter Name	Parameter ID	Option	Setting Notes
Service Based Optimization Max Effective Time	GigaBandCell.ServiceBasedOptMaxEfTime	None	Set this parameter when "uplink differentiated power control and scheduling" is required based on the network plan.
Acceleration Service Optimization Ext Sw	AsParaGroup.AccelerationServOptExtSw	GRP_NO_PREALLOCATE_TRIGGER_OPT_SW	Select this option when optimized uplink preallocation trigger is enabled but needs to be forbidden for UEs in a specified group.
Multi Networking Option Opt Switch	EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw	UL_LARGE_PACKET_SERV_DECT_SW	Select this option when "uplink differentiated power control and scheduling" is enabled.
UL Large Packet Data Volume Threshold	CellServiceDiffCfg.UllargePacketDataVoThd	None	Set this parameter based on the network plan.
Judge UL Large Packet Count Threshold	CellServiceDiffCfg.JudgeUllargePacketCntThd	None	Set this parameter based on the network plan.
Exit UL Large Packet Count Threshold	CellServiceDiffCfg.ExitUllargePacketCntThd	None	Set this parameter based on the network plan.

Table 8-5 NR parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
GigaBand Cell Algorithm Switch	NRDUCellGigaBand.GigaBandCellAlgoSw	LNR_PUCCH_RB_RES_ADAPT_SW	Select this option when LNR PUCCH RB resource adaptation is required based on the network plan.
LNR PUCCH RB Adjustment Protection Duration	NRDUCellGigaBand.LnrPucchRbAdjProtDuration	None	Set this parameter when LNR PUCCH RB resource adaptation is required based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	LNR_PUCCH_REES_OPT_SW	You are advised to select this option when LNR PUCCH RB resource adaptation is required based on the network plan.
LNR UL RB Estimate Correction Coefficient	NRDUCellGigaBand.LnrULRbEstCorrectCoeff	None	Set this parameter when LNR uplink precise RB estimation is required based on the network plan.
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	COMM_PUCCH_RB_RES_OPT_SW	Set this option based on the network plan.
Full Buffer UE Evaluation Period	NRDUCellServExp.FullBufferUeEvalPeriod	None	Set this parameter based on the network plan.
Full Buffer UE UL PRB Usage Threshold	NRDUCellServExp.FullBufferUeULPrbUsageThld	None	Set this parameter based on the network plan.
Full Buffer UE UL TTI Ratio Threshold	NRDUCellServExp.FullBufferUeULTTIRatioThld	None	Set this parameter based on the network plan.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgoExtSwitch	F3_AND_F4_ACK_RB_RES_ADAPT_SW	You are advised to select this option when LNR PUCCH RB resource adaptation is required based on the network plan.

8.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Activation command examples on the LTE side

```
//Configuring the thresholds for identifying large-packet LTE UEs
MOD CELLSERVICEDIFFCFG: LocalCellId=1, ULLargePacketDataVoThd=25, JudgeULLargePacketCntThd=3,
ExitULLargePacketCntThd=3;
//Enabling LNR PUCCH RB resource adaptation (using a bandwidth of 20 MHz as an example)
DEA CELL: LocalCellId=1;
MOD PUCCHCFG: LocalCellId=1, NrPucchExtendedRbNum=1;
//Adjusting the start and end RBs based on the configured number of NR PUCCH RBs and bandwidth
MOD CELLRBRESERVE: LocalCellId=1, Index=1, RbRsvMode=LNR_SPECTRUM_SHR_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=1, RbRsvEndIndex=98;
MOD CELLSIMAP: LocalCellId=1, SibUpdOptSwitch=SibPagingOptCtrlSwitch-1;
ACT CELL: LocalCellId=1;
//Enabling LNR uplink precise RB estimation, optimized uplink preallocation trigger, uplink differentiated
scheduling based on LTE and NR flexible spectrum, and uplink differentiated power control and scheduling
MOD GIGABANDCELL: LocalCellId=1,
GigaBandCellAlgoSw=LNR_UL_RB_PRECISE_ESTIMATE_SW-1&UL_PREALLOCATION_TRIG_OPT_SW-1&LNR_F
LEX_SPCT_UL_DIFF_SCH_SW-1&UL_PWR_CTRL_SCH_DIFF_SW-1, ServiceBasedOptMaxEffTime=120;
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=UL_LARGE_PKT_SERV_DECT_SW-1;
```

Activation command examples on the NR side

```
//Configuring the thresholds for identifying large-packet NR UEs
MOD NRDUCELLSERVEEXP: NrDuCellId=1, FullBufferUeEvalPrd=3, FullBufferUeUlPrbUsageThld=15,
FullBufferUeULTiRatioThld=20;
//Enabling LNR PUCCH RB resource adaptation (using a bandwidth of 20 MHz as an example)
DEA NRCELL: NrCellId=1;
MOD NRDUCELLPUCCH: NrDuCellId=1, Format1RbNum=RB2, Format3RbNum=RB2,
CsiDedicatedRbNum=RB2, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0,
PucchAlgoSwitch=COMM_PUCCH_RB_RES_OPT_SW-1, PucchAlgoExtSwitch=LNR_PUCCH_RES_OPT_SW-1;
MOD NRDUCELLGIGABAND: NrDuCellId=1, GigaBandCellAlgoSw=LNR_PUCCH_RB_RES_ADAPT_SW-1,
LnrPucchRbAdjProtDuration=30;
ACT NRCELL: NrCellId=1;
//Setting the LnrUlRbEstCorrectCoeff parameter
MOD NRDUCELLGIGABAND: NrDuCellId=1, LnrUlRbEstCorrectCoeff=60;
//Turning on the switch for format-3 and format-4 ACK RB resource adaptation
MOD NRDUCELLPUCCH: NrDuCellId=1, PucchAlgoExtSwitch=F3_AND_F4_ACK_RB_RES_ADAPT_SW-1;
```

Deactivation Command Examples

Deactivation command examples on the LTE side

```
//Disabling LNR PUCCH RB resource adaptation
DEA CELL: LocalCellId=1;
MOD PUCCHCFG: LocalCellId=1, NrPucchExtendedRbNum=0;
MOD CELLRBRESERVE: LocalCellId=1, Index=1, RbRsvMode=LNR_SPECTRUM_SHR_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=4, RbRsvEndIndex=95;
MOD CELLSIMAP: LocalCellId=1, SibUpdOptSwitch=SibPagingOptCtrlSwitch-0;
ACT CELL: LocalCellId=1;
//Disabling LNR uplink precise RB estimation, optimized uplink preallocation trigger, uplink differentiated
scheduling based on LTE and NR flexible spectrum, and uplink differentiated power control and scheduling
MOD GIGABANDCELL: LocalCellId=1,
GigaBandCellAlgoSw=LNR_UL_RB_PRECISE_ESTIMATE_SW-0&UL_PREALLOCATION_TRIG_OPT_SW-0&LNR_F
LEX_SPCT_UL_DIFF_SCH_SW-0&UL_PWR_CTRL_SCH_DIFF_SW-0;
```

Deactivation command examples on the NR side

```
//Disabling LNR PUCCH RB resource adaptation
DEA NRCELL: NrCellId=1;
MOD NRDUCELLPUCCH: NrDuCellId=1, Format1RbNum=RB2, Format3RbNum=RB4,
CsiDedicatedRbNum=RB4, Format4RbNum=RB0, Format4CsiDedicatedRbNum=RB0,
PucchAlgoSwitch=COMM_PUCCH_RB_RES_OPT_SW-0;
MOD NRDUCELLGIGABAND: NrDuCellId=1, GigaBandCellAlgoSw=LNR_PUCCH_RB_RES_ADAPT_SW-0;
ACT NRCELL: NrCellId=1;
//Setting the LnrUlRbEstCorrectCoeff parameter
MOD NRDUCELLGIGABAND: NrDuCellId=1, LnrUlRbEstCorrectCoeff=0;
```

```
//Turning off the switch for format-3 and format-4 ACK RB resource adaptation  
MOD NRDUCELLPUCCH: NrDuCellId=1, PucchAlgoExtSwitch=F3_AND_F4_ACK_RB_RES_ADAPT_SW-0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

After LTE and NR Uplink Spectrum Pooling is enabled, observe the average uplink throughputs of certain identified LTE and NR UEs and the uplink LTE UE throughput. If these throughputs increase, this function has taken effect.

8.4.3 Network Monitoring

After this function is enabled, monitor the network gains and impacts of this function using the counters listed in [8.2 Network Analysis](#).

9 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- Feature parameter description documents in *eRAN Feature Documentation*
 - *Carrier Aggregation*
 - *Uplink Scheduling*
 - *Downlink Scheduling*
 - *Physical Channel Resource Management*
 - *VoLTE*
 - *MIMO*
 - *eMIMO (FDD)*
 - *Massive MIMO Basics (FDD)*
 - *Massive MIMO Enhancements (FDD)*
 - *UL CoMP*
 - *DRX and Signaling Control*
 - *Intra-RAT Mobility Load Balancing*
 - *LCS*
 - *Extended CP*
 - *Random Access Control*
 - *SFN*
 - *Cell Management*
 - *Flexible Bandwidth based on Overlap Carriers (FDD)*
 - *GSM and LTE Buffer Zone Optimization*
 - *UL Refarming Zero Bufferzone*
 - *UMTS and LTE Zero Bufferzone*
 - *CDMA and LTE Zero Bufferzone*
 - *Uplink Coordinated Scheduling*
 - *CSPC*
 - *Relay*
 - *eMTC*

- *NB-IoT Basics (FDD)*
- *Extended Cell Range*
- *Superior Uplink Coverage (FDD)*
- *eMBMS*
- *Virtual 4T4R (FDD)*
- *DL CoMP (FDD)*
- *GSM and LTE Dynamic Power Sharing*
- *Super Combined Cell (FDD)*
- *Energy Conservation and Emission Reduction*
- *TDM eICIC (FDD)*
- *ICIC*
- *Short TTI (FDD)*
- *Smart 8T8R (FDD)*
- *PTT*
- *On-Demand TX Power Allocation Under EME*
- *Dedicated Carrier for TM9*
- *Seamless Intra-Band Carrier Joining (FDD)*
- *Compact Bandwidth (FDD)*
- *High Speed Mobility*
- *CPRI Compression*
- *pRRU Uplink Interference Suppression*
- *RAN Sharing*
- *Synchronization*
- *Low-Band Booster*
- *Physical Resource Slicing (FDD)*
- *Licensed Assisted Access (TDD)*
- *LumiLink Solution*
- *WTTx MU-MIMO (FDD)*
- *WBB*
- *LTE Flexible Spectrum (FDD)*
- *QoS Management*
- *Air Interface Latency Optimization*
- *Video Experience Optimization*
- *CMAS*
- *ETWS*
- Feature parameter description documents in *5G RAN Feature Documentation*
 - *RuralLink Solution*
 - *Flexible Bandwidth (FDD)*
 - *Admission Control*
 - *Carrier Aggregation*

- *CA Experience Improvement*
- *Downlink Scheduling*
- *Channel Management*
- *Energy Conservation and Emission Reduction*
- *Power Control*
- *Multi-Operator Sharing*
- *Beam Management*
- *VoNR*
- *Network Slicing*
- *Resource Control in NSA Networking*
- *CoMP*
- *Indoor Coverage*
- *Cell Combination*
- *Hyper Cell*
- *Scalable Bandwidth*
- *Super Uplink*
- *UL and DL Decoupling*
- *High Speed Mobility*
- *CPRI Compression*
- *Cell Management*
- *Synchronization*
- *Smart Coverage (FDD)*
- *Smart 8T8R (FDD)*
- *Smart Massive MIMO (FDD)*
- *Turbo Coverage (FDD)*
- *Distributed Antenna Solutions (FDD)*
- *URLLC*
- *RedCap*
- *Interference Avoidance*
- *LumiLink Solution*
- *5G Networking and Signaling*
- *Multi-Frequency Smart Aggregation*
- *Uplink Flexible Spectrum Scheduling*
- *NR Inter-Carrier Dynamic Power Allocation*
- *QCI-based Resource Control*
- *Ultimate Cloud X Experience*
- *Dynamic Power Sharing Between LTE Carriers*
- *LTE and NR Power Allocation*
- *LTE Spectrum Coordination*
- *Multi-RAT Coordinated Energy Saving*

- *GSM and LTE Spectrum Concurrency*
- *UMTS and LTE Dynamic Power Sharing*
- *UMTS and LTE Spectrum Sharing*
- *UMTS and LTE Spectrum Sharing Based on DC-HSDPA*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *EPC-based NSA Experience Improvement*
- *LTE FDD and NR Inter-RAT Unified Scheduling*
- *LTE and NR Zero Bufferzone*